

# **AIRCRAFT CHARACTERISTICS AIRPORT AND MAINTENANCE PLANNING**

## **AC**

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**HIGHLIGHTS**
**Revision No. 6 - Dec 01/25**

| LOCATIONS                                                                                                                                               | CHG<br>CODE | DESCRIPTIONS OF CHANGE                                                                                                                                          |
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| <u>CHAPTER 2</u><br>Section 2-3<br>Subject 2-3-0<br><br>FIGURE Ground Clearances - Ground<br>Clearances for WV000 at MRW and 135 000<br>kg (297 624 lb) | R           | ADDED THE FLAG NOTE "EMPTY WEIGHT<br>FOR MAINTENANCE (DEPENDING ON<br>THE AIRCRAFT CONFIGURATION)" IN<br>THE ILLUSTRATION.                                      |
| Section 2-9<br>Subject 2-9-0<br><br>FIGURE Tow Truck Power - NLG Servicing 3                                                                            | R           | DELETED THE NOTE FROM THE<br>ILLUSTRATION.<br>UPDATED THE ILLUSTRATION TO SHOW<br>THE CONNECTION BETWEEN THE<br>CONNECTOR (5GCB) AND THE ELECTRIC<br>BOX (8GH). |
| Section 2-11<br>Subject 2-11-0<br><br>FIGURE Antennas and Probes - Location                                                                             | R           | ADDED THE DETAILED VIEW OF THE<br>NOSE SECTION IN THE ILLUSTRATION.                                                                                             |

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**SCOPE****1-1-0 Purpose****\*\*ON A/C A330-700L****Introduction**

1. The A330-700L AIRCRAFT CHARACTERISTICS - AIRPORT AND MAINTENANCE PLANNING (AC) manual is issued for the A330-700L to give necessary data to airport operators, airlines and Maintenance/Repair Organizations (MRO) for airport and maintenance facilities planning.

The A330-700L is designed to replace the existing A300-600ST known as Beluga, to modernize and improve Airbus aircraft parts transportation. The A330-700L will offer a large volume in its main deck cargo compartment, which can carry a pair of A350 wings.

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## 1-2-1 Glossary

### \*\*ON A/C A330-700L

#### Glossary

#### 1. List of Abbreviations

|      |                                           |
|------|-------------------------------------------|
| A/C  | Aircraft                                  |
| AC   | Aircraft Characteristics manual           |
| ACN  | Aircraft Classification Number            |
| ACR  | Aircraft Classification Rating            |
| AMM  | Aircraft Maintenance Manual               |
| APU  | Auxiliary Power Unit                      |
| C/L  | Center Line                               |
| CBR  | California Bearing Ratio                  |
| CC   | Cargo Compartment                         |
| CG   | Center of Gravity                         |
| CKPT | Cockpit                                   |
| E    | Young's Modulus                           |
| ELEC | Electric, Electrical, Electricity         |
| ESWL | Equivalent Single Wheel Load              |
| FAA  | Federal Aviation Administration           |
| FDL  | Fuselage Datum Line                       |
| FR   | Frame                                     |
| FSTE | Full Size Trolley Equivalent              |
| FWD  | Forward                                   |
| GPU  | Ground Power Unit                         |
| GSE  | Ground Support Equipment                  |
| HYD  | Hydraulic                                 |
| ICAO | International Civil Aviation Organisation |
| IDG  | Integrated Drive Generator                |
| ISA  | International Standard Atmosphere         |
| L    | Radius of relative stiffness              |
| LCN  | Load Classification Number                |
| LD   | Load Device                               |
| LD   | Lower Deck                                |
| LDG  | Landing Gear                              |
| LH   | Left Hand                                 |

|      |                                |
|------|--------------------------------|
| LMLG | Left Main Landing Gear         |
| MAC  | Mean Aerodynamic Chord         |
| MAX  | Maximum                        |
| MD   | Main Deck                      |
| MDCC | Main Deck Cargo Compartment    |
| MIN  | Minimum                        |
| MLG  | Main Landing Gear              |
| NLG  | Nose Landing Gear              |
| OAT  | Outside Air Temperature        |
| PCA  | Portland Cement Association    |
| PCN  | Pavement Classification Number |
| PCR  | Pavement Classification Rating |
| PLU  | Power Locking Unit             |
| RH   | Right Hand                     |
| RMLG | Right Main Landing Gear        |
| ULD  | Unit Load Device               |
| WV   | Weight Variant                 |

## 2. Design Weight Terminology

- Maximum Design Ramp Weight (MRW):  
Maximum weight for ground maneuver (including weight of taxi and run-up fuel) as limited by aircraft strength and airworthiness requirements. It is also called Maximum Design Taxi Weight (MTW).
- Maximum Design Landing Weight (MLW):  
Maximum weight for landing as limited by aircraft strength and airworthiness requirements.
- Maximum Design Take-Off Weight (MTOW):  
Maximum weight for take-off as limited by aircraft strength and airworthiness requirements. (This is the maximum weight at start of the take-off run).
- Maximum Design Zero Fuel Weight (MZFW):  
Maximum permissible weight of the aircraft without usable fuel.
- Usable Volume:  
Usable volume available for cargo, pressurized fuselage, passenger compartment and cockpit.
- Water Volume:  
Maximum volume of cargo compartment.
- Usable Fuel:  
Fuel available for aircraft propulsion.

**AIRCRAFT DESCRIPTION**
**2-1-1 General Aircraft Characteristics Data**
**\*\*ON A/C A330-700L**
General Aircraft Characteristics Data

1. The following table gives characteristics of A330-700L model, these data are specific to each weight variant:

| Aircraft Characteristics        |                            |                            |
|---------------------------------|----------------------------|----------------------------|
|                                 | WV000                      | WV001                      |
| Maximum Ramp Weight (MRW)       | 227 900 kg<br>(502 433 lb) | 205 900 kg<br>(453 932 lb) |
| Maximum Take-Off Weight (MTOW)  | 227 000 kg<br>(500 449 lb) | 205 000 kg<br>(451 948 lb) |
| Maximum Landing Weight (MLW)    | 187 000 kg<br>(412 264 lb) | 187 000 kg<br>(412 264 lb) |
| Maximum Zero Fuel Weight (MZFW) | 178 000 kg<br>(392 423 lb) | 178 000 kg<br>(392 423 lb) |
| Estimated Maximum Payload       | 50 500 kg<br>(111 333 lb)  | 50 500 kg<br>(111 333 lb)  |
| Operating Weight Empty (OWE)    | 127 500 kg<br>(281 089 lb) | 127 500 kg<br>(281 089 lb) |

2. The following table gives characteristics of A330-700L model, these data are common to each weight variant:

| Aircraft Characteristics                       |                                                |
|------------------------------------------------|------------------------------------------------|
| Seats in courier area                          | 4                                              |
| Usable Fuel Capacity<br>(density = 0.785 kg/l) | 73 000 kg<br>(160 937 lb)                      |
| Pressurized Fuselage Volume                    | 87.4 m <sup>3</sup><br>(3087 ft <sup>3</sup> ) |
| Cockpit Volume                                 | 5.75 m <sup>3</sup>                            |

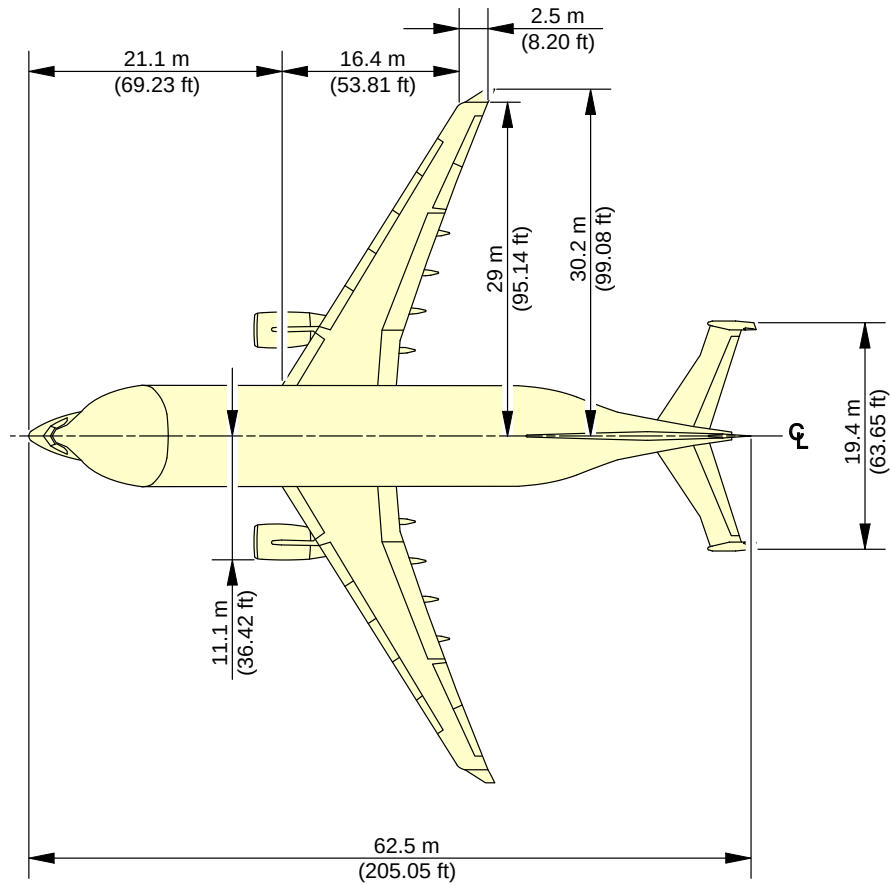


| Aircraft Characteristics                 |                                                   |
|------------------------------------------|---------------------------------------------------|
|                                          | (203 ft <sup>3</sup> )                            |
| Main-Deck Cargo-Compartment Water Volume | 2 209 m <sup>3</sup><br>(78 010 ft <sup>3</sup> ) |
| Usable Volume, AFT CC                    | 60.7 m <sup>3</sup><br>(2 144 ft <sup>3</sup> )   |
| Usable Volume, Bulk CC                   | 19.7 m <sup>3</sup><br>(696 ft <sup>3</sup> )     |
| Water Volume, AFT CC                     | 85.7 m <sup>3</sup><br>(3 026 ft <sup>3</sup> )   |
| Water Volume, Bulk CC                    | 22.7 m <sup>3</sup><br>(802 ft <sup>3</sup> )     |

**2-2-0 General Aircraft Dimensions****\*\*ON A/C A330-700L**General Aircraft Dimensions

1. This section gives general aircraft dimensions.

**\*\*ON A/C A330-700L**

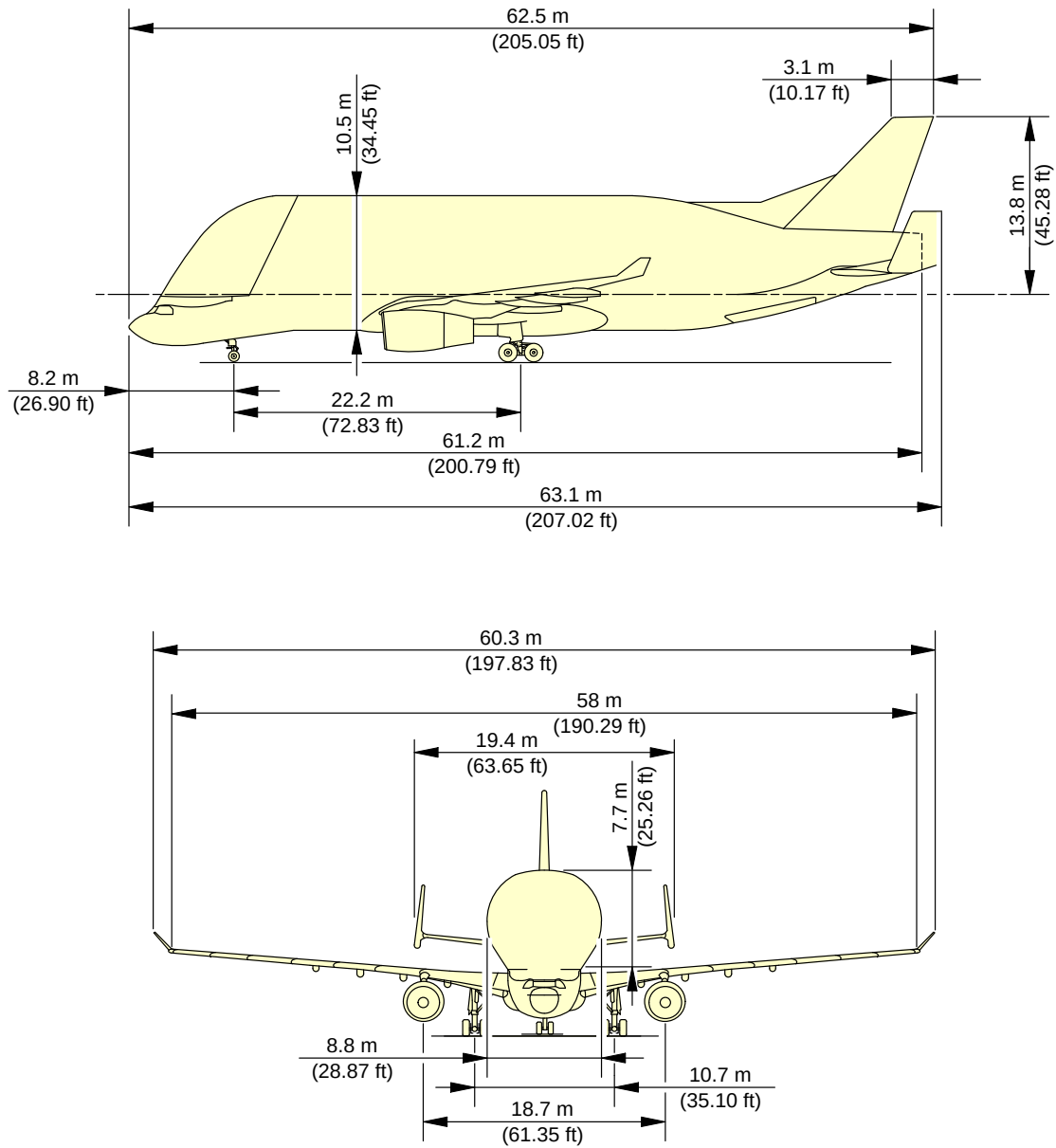


**NOTE:**  
RELATED TO AIRCRAFT ATTITUDE AND WEIGHT.

F\_AC\_020200\_1\_0130101\_01\_00

General Aircraft Dimensions  
(Sheet 1 of 2)  
FIGURE-2-2-0-991-013-A01

**\*\*ON A/C A330-700L**



**NOTE:**  
RELATED TO AIRCRAFT ATTITUDE AND WEIGHT.

F\_AC\_020200\_1\_0130102\_01\_00

General Aircraft Dimensions  
(Sheet 2 of 2)  
FIGURE-2-2-0-991-013-A01

**2-3-0 Ground Clearances****\*\*ON A/C A330-700L**Ground Clearances

1. This section gives the height of various points of the aircraft, above the ground, for different aircraft configurations.

Dimensions in the tables can change with tire type, weight and balance and other special conditions.

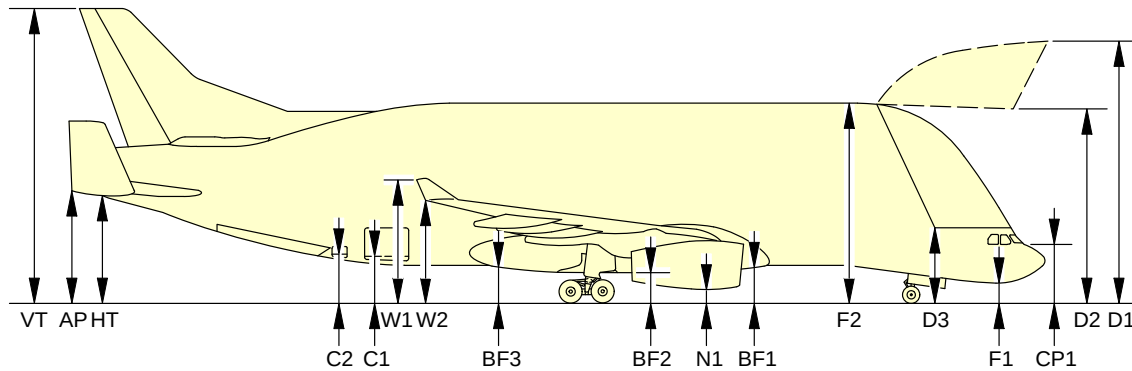
NOTE : Tire pressure and shock absorbers are fixed in the standard condition.

The dimensions are given for the weight variant WV000:

- A light weight 135 000 kg (297 624 lb), with a FWD CG and an AFT CG,
- An aircraft at MRW 227 000 kg (500 449 lb) with a FWD CG and an AFT CG,
- Aircraft on jacks, FDL at 6.515 m (21.37 ft).

NOTE : Cargo door ground clearances are measured from the floor level.

**\*\*ON A/C A330-700L**



| AIRCRAFT CONFIGURATION FOR WV000 |     | MRW 227 000 kg (500 450 lb) |       |              |       | 135 000 kg (297 625 lb) <span style="border: 1px solid blue; padding: 0 2px;">01</span> |       |              |       |
|----------------------------------|-----|-----------------------------|-------|--------------|-------|-----------------------------------------------------------------------------------------|-------|--------------|-------|
|                                  |     | FORWARD CG (20.5%)          |       | AFT CG (28%) |       | FORWARD CG (18%)                                                                        |       | AFT CG (25%) |       |
|                                  |     | m                           | ft    | m            | ft    | m                                                                                       | ft    | m            | ft    |
| FUSELAGE                         | F2  | 12.90                       | 42.32 | 12.94        | 42.45 | 13.02                                                                                   | 42.72 | 13.06        | 42.85 |
|                                  | F1  | 1.20                        | 3.94  | 1.27         | 4.17  | 1.28                                                                                    | 4.20  | 1.35         | 4.43  |
|                                  | D1  | 17.10                       | 56.10 | 17.18        | 56.36 | 17.18                                                                                   | 56.36 | 17.25        | 56.59 |
|                                  | D2  | 12.28                       | 40.29 | 12.35        | 40.52 | 12.37                                                                                   | 40.58 | 12.43        | 40.78 |
|                                  | D3  | 5.01                        | 16.44 | 5.09         | 16.70 | 5.09                                                                                    | 16.70 | 5.16         | 16.93 |
|                                  | CP1 | 3.69                        | 12.11 | 3.76         | 12.34 | 3.77                                                                                    | 12.37 | 3.84         | 12.60 |
|                                  | BF1 | 1.94                        | 6.36  | 1.95         | 6.40  | 2.11                                                                                    | 6.92  | 2.12         | 6.96  |
|                                  | BF2 | 1.93                        | 6.33  | 1.93         | 6.33  | 2.11                                                                                    | 6.92  | 2.11         | 6.92  |
|                                  | BF3 | 1.95                        | 6.40  | 1.94         | 6.36  | 2.15                                                                                    | 7.05  | 2.14         | 7.02  |
|                                  | HT  | 6.84                        | 22.44 | 6.74         | 22.11 | 7.16                                                                                    | 23.49 | 7.06         | 23.16 |
| DOORS                            | AP  | 7.39                        | 24.25 | 7.28         | 23.88 | 7.71                                                                                    | 25.30 | 7.60         | 24.93 |
|                                  | VT  | 18.89                       | 61.98 | 18.79        | 61.65 | 19.21                                                                                   | 63.02 | 19.10        | 62.66 |
|                                  | C1  | 3.07                        | 10.07 | 3.03         | 9.94  | 3.31                                                                                    | 10.86 | 3.27         | 10.73 |
| WINGS                            | C2  | 3.10                        | 10.17 | 3.05         | 10.01 | 3.36                                                                                    | 11.02 | 3.30         | 10.83 |
|                                  | W1  | 7.94                        | 26.05 | 7.91         | 25.95 | 8.18                                                                                    | 26.84 | 8.14         | 26.71 |
| ENGINE                           | W2  | 6.58                        | 21.59 | 6.54         | 21.46 | 6.81                                                                                    | 22.34 | 6.77         | 22.21 |
|                                  | N1  | 0.88                        | 2.89  | 0.89         | 2.92  | 1.04                                                                                    | 3.41  | 1.05         | 3.44  |

**NOTE:**

CARGO DOOR GROUND CLEARANCES ARE MEASURED FROM FLOOR LEVEL.

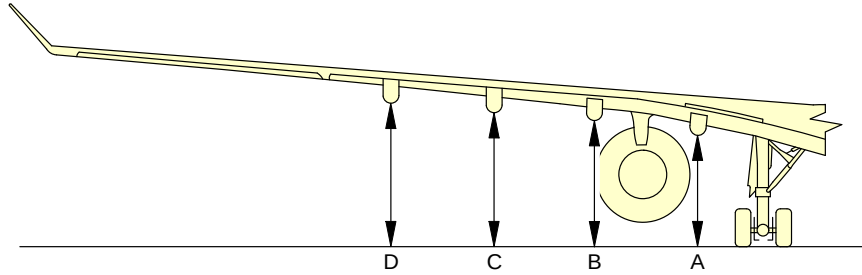
THE VALUES GIVEN IN THE TABLE DEPEND ON THE POSITION OF THE CENTER OF GRAVITY (CG) AND ON THE AIRCRAFT WEIGHT.

01 EMPTY WEIGHT FOR MAINTENANCE (DEPENDING ON THE AIRCRAFT CONFIGURATION).

F\_AC\_020300\_1\_0450101\_01\_02

Ground Clearances  
Ground Clearances for WV000 at MRW and 135 000 kg (297 624 lb)  
FIGURE-2-3-0-991-045-A01

**\*\*ON A/C A330-700L**



| FLAP TRACKS RETRACTED |              |   |              |       |            |       |
|-----------------------|--------------|---|--------------|-------|------------|-------|
| AIRCRAFT TYPE         | DESCRIPTION  |   | MRW CG 20.5% |       | MRW CG 28% |       |
|                       |              |   | m            | ft    | m          | ft    |
| A330-700L             | FLAP TRACK 2 | A | 3.75         | 12.30 | 3.74       | 12.27 |
|                       | FLAP TRACK 3 | B | 4.26         | 13.98 | 4.25       | 13.94 |
|                       | FLAP TRACK 4 | C | 4.51         | 14.80 | 4.49       | 14.73 |
|                       | FLAP TRACK 5 | D | 4.88         | 16.01 | 4.85       | 15.91 |

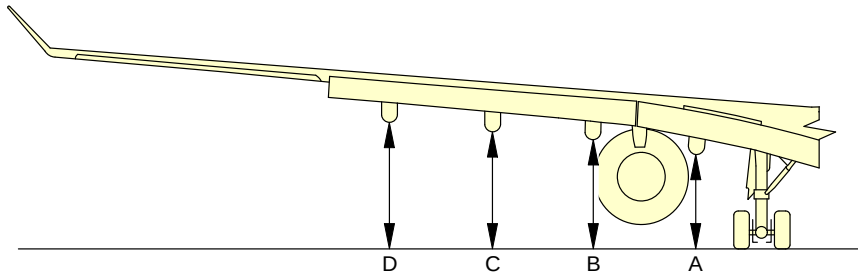
**NOTE:**

THE VALUES GIVEN IN THE TABLE DEPEND ON THE POSITION OF THE CENTER OF GRAVITY (CG) AND ON THE AIRCRAFT WEIGHT.

F\_AC\_020300\_1\_0470101\_01\_01

Ground Clearances  
Ground Clearances for Flaps Retracted with WV000  
FIGURE-2-3-0-991-047-A01

**\*\*ON A/C A330-700L**



| FLAP TRACKS 1+F |              |   |                 |       |               |       |
|-----------------|--------------|---|-----------------|-------|---------------|-------|
| AIRCRAFT TYPE   | DESCRIPTION  |   | MRW<br>CG 20.5% |       | MRW<br>CG 28% |       |
|                 |              |   | m               | ft    | m             | ft    |
| A330-700L       | FLAP TRACK 2 | A | 3.48            | 11.42 | 3.37          | 11.06 |
|                 | FLAP TRACK 3 | B | 3.99            | 13.09 | 3.88          | 12.73 |
|                 | FLAP TRACK 4 | C | 4.24            | 13.91 | 4.12          | 13.52 |
|                 | FLAP TRACK 5 | D | 4.61            | 15.12 | 4.48          | 14.70 |

**NOTE:**

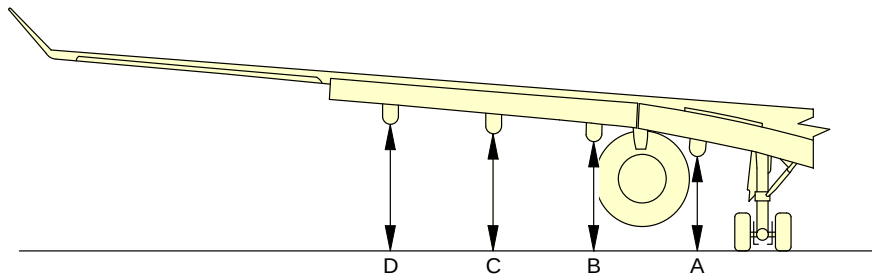
THE VALUES GIVEN IN THE TABLE DEPEND ON THE POSITION OF THE CENTER OF GRAVITY (CG) AND ON THE AIRCRAFT WEIGHT.

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Ground Clearances  
Ground Clearances for Flaps in Intermediate Position with WV000  
FIGURE-2-3-0-991-048-A01



**\*\*ON A/C A330-700L**



| FLAP TRACKS EXTENDED |              |   |                 |       |               |       |
|----------------------|--------------|---|-----------------|-------|---------------|-------|
| AIRCRAFT TYPE        | DESCRIPTION  |   | MRW<br>CG 20.5% |       | MRW<br>CG 28% |       |
|                      |              |   | m               | ft    | m             | ft    |
| A330-700L            | FLAP TRACK 2 | A | 2.82            | 9.25  | 2.81          | 9.22  |
|                      | FLAP TRACK 3 | B | 3.33            | 10.93 | 3.32          | 10.89 |
|                      | FLAP TRACK 4 | C | 3.58            | 11.75 | 3.56          | 11.68 |
|                      | FLAP TRACK 5 | D | 3.95            | 12.96 | 3.92          | 12.86 |

**NOTE:**

THE VALUES GIVEN IN THE TABLE DEPEND ON THE POSITION OF THE CENTER OF GRAVITY (CG) AND ON THE AIRCRAFT WEIGHT.

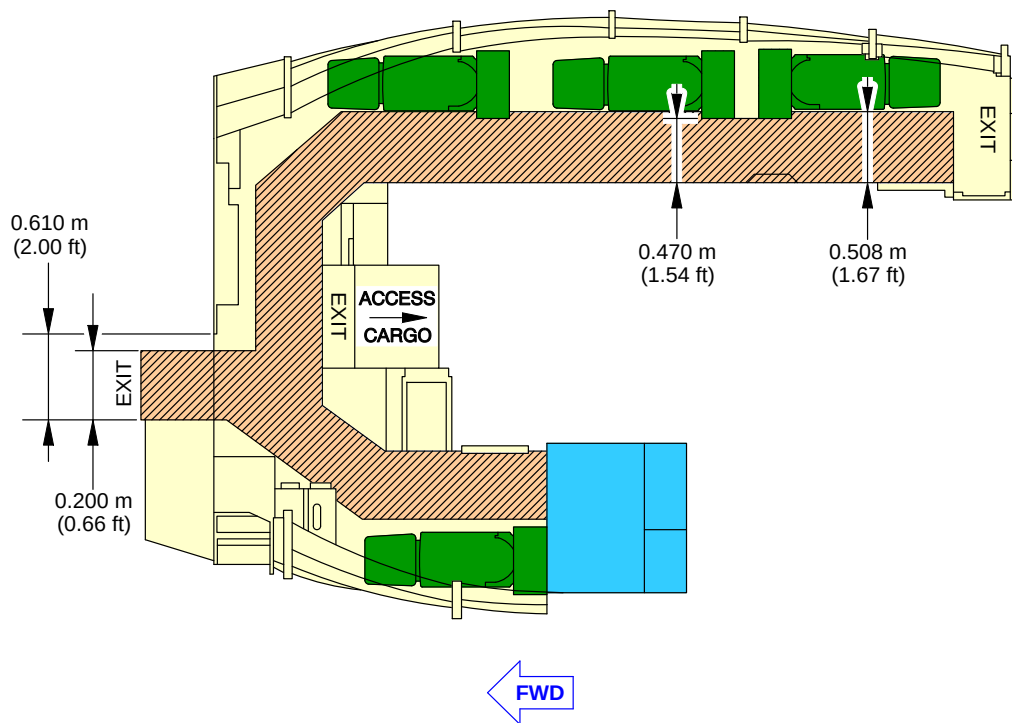
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Ground Clearances  
Ground Clearances for Flaps Fully Extended with WV000  
FIGURE-2-3-0-991-049-A01

**2-4-1 Interior Arrangements - Plan View****\*\*ON A/C A330-700L**Interior Arrangements - Plan View

1. This section gives the interior configuration of courier area showing the width of passway.

**\*\*ON A/C A330-700L**



**NOTE:**

- SEAT
- LAVATORY
- COURIER AREA

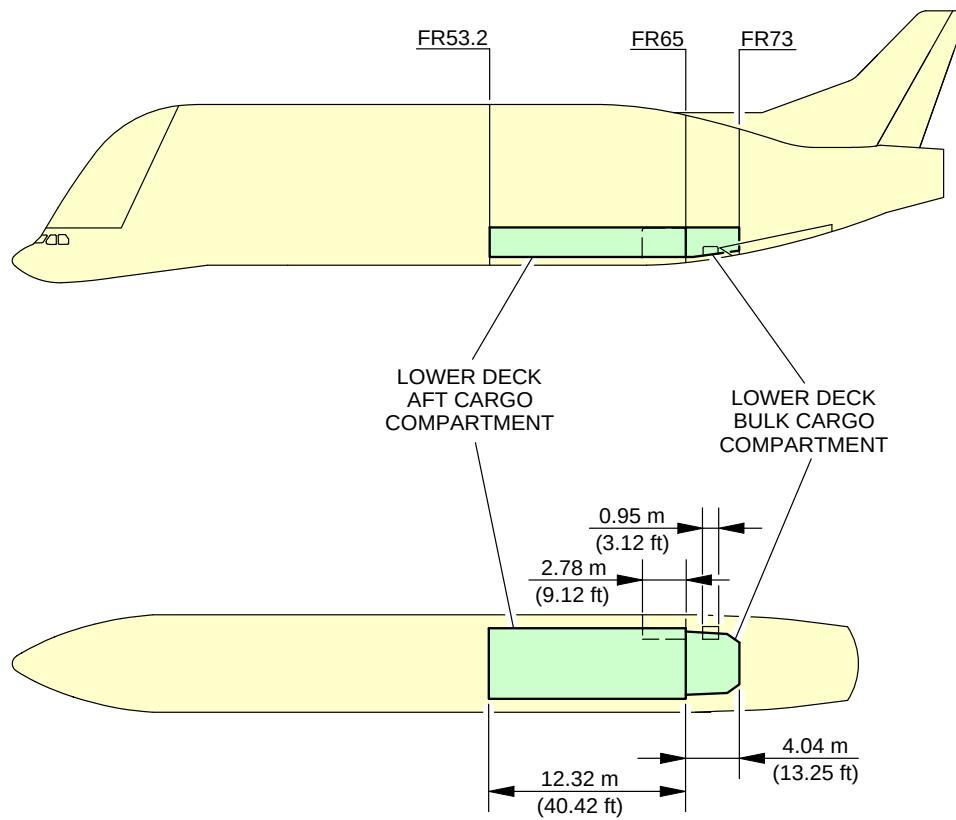
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Interior Arrangements - Plan View  
FIGURE-2-4-1-991-010-A01

**2-6-1 Lower Deck Cargo Compartments****\*\*ON A/C A330-700L**Lower Deck Cargo Compartments

1. This section gives the following data about lower deck cargo compartments:
  - Location and dimensions
  - Loading combinations.

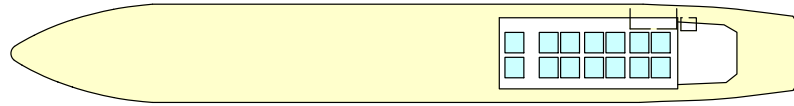
**\*\*ON A/C A330-700L**



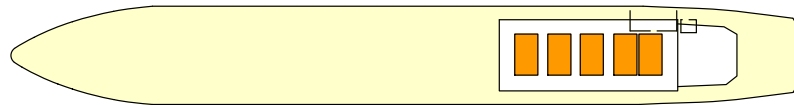
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Lower Deck Cargo Compartments  
Location and Dimensions (Sheet 1 of 2)  
FIGURE-2-6-1-991-013-A01

**\*\*ON A/C A330-700L**



14 LD3 60.4 in X 61.5 in



5 PALLETS 88 in X 125 in



4 PALLETS 96 in X 125 in

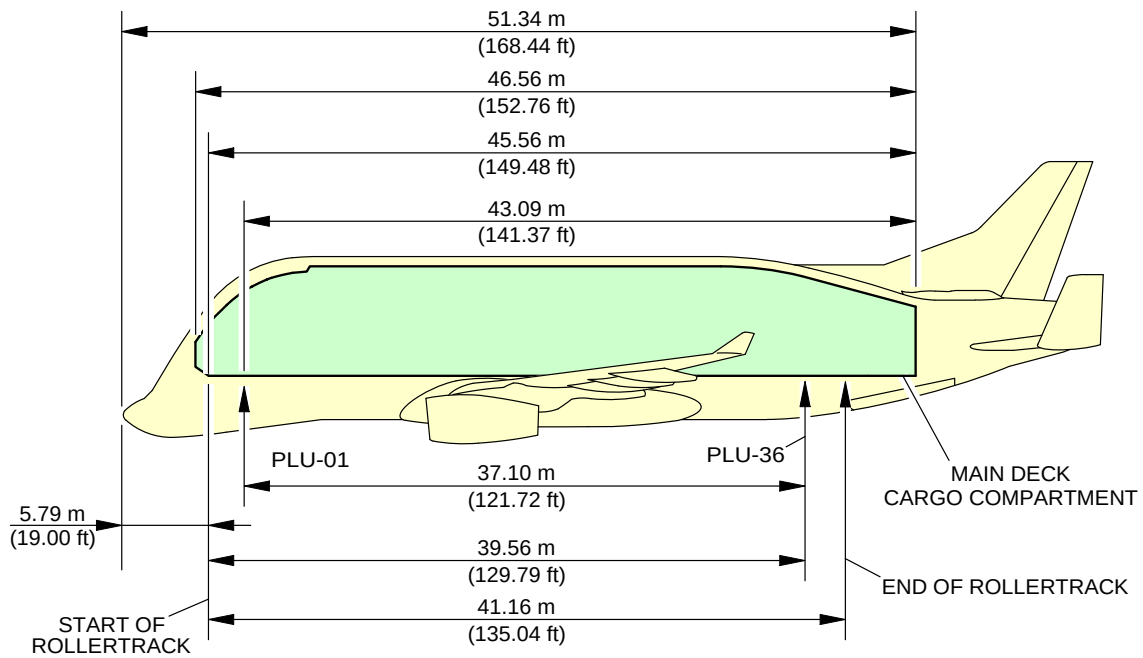
F\_AC\_020601\_1\_0130102\_01\_00

Lower Deck Cargo Compartments  
Loading Combinations (Sheet 2 of 2)  
FIGURE-2-6-1-991-013-A01

**2-6-2 Main Deck Cargo Compartments****\*\*ON A/C A330-700L**Main Deck Cargo Compartment

1. This section gives the following data about the main-deck cargo compartment:
  - Location and dimensions.

**\*\*ON A/C A330-700L**



F\_AC\_020602\_1\_0070101\_01\_00

Main-Deck Cargo Compartment  
Location and Dimensions  
FIGURE-2-6-2-991-007-A01

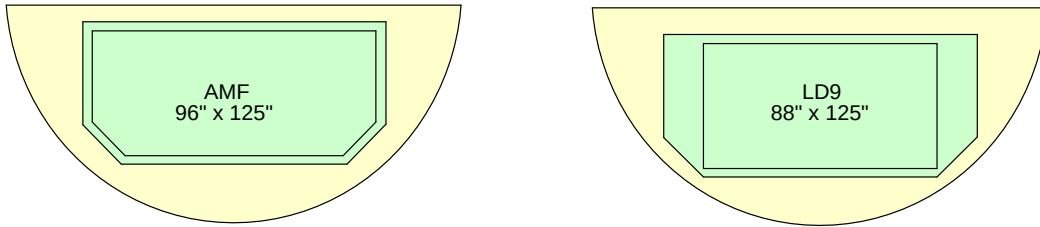


**2-6-3 Main and Lower Deck Cross-sections****\*\*ON A/C A330-700L**Main and Lower Deck Cross-sections

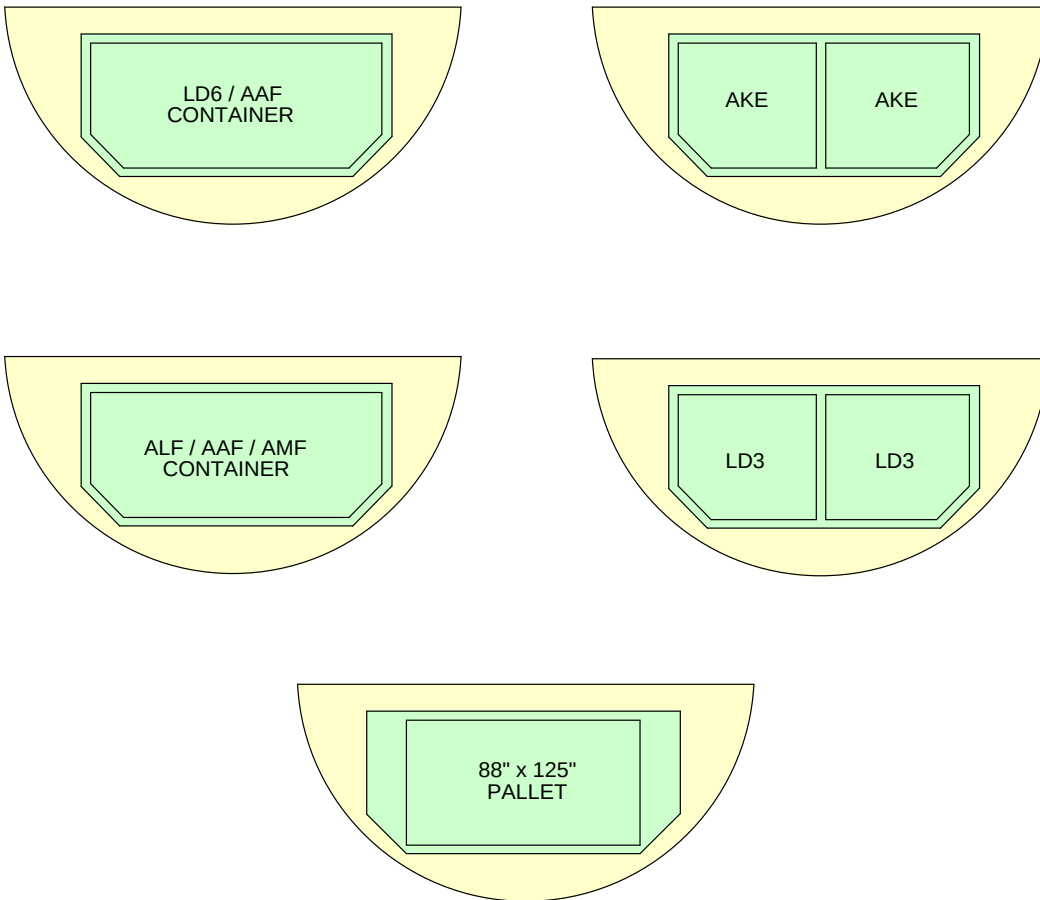
1. This section gives main and lower deck cross-sections for cargo version.

**\*\*ON A/C A330-700L**

## REFERENCE CARGO CONFIGURATION LAYOUT



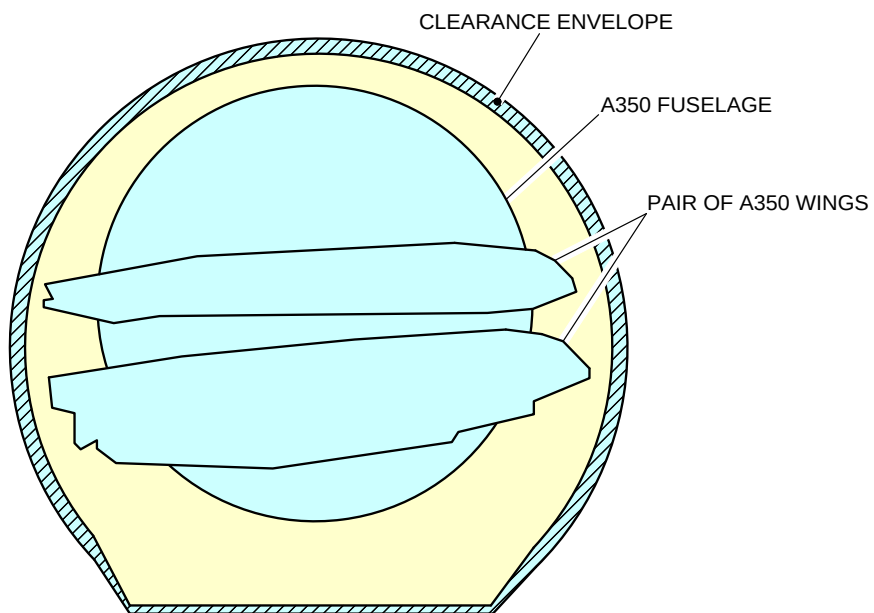
## OPTIONAL CARGO CONFIGURATIONS



F\_AC\_020603\_1\_0020101\_01\_00

Lower-Deck Cargo Cross-sections  
FIGURE-2-6-3-991-002-A01

**\*\*ON A/C A330-700L**



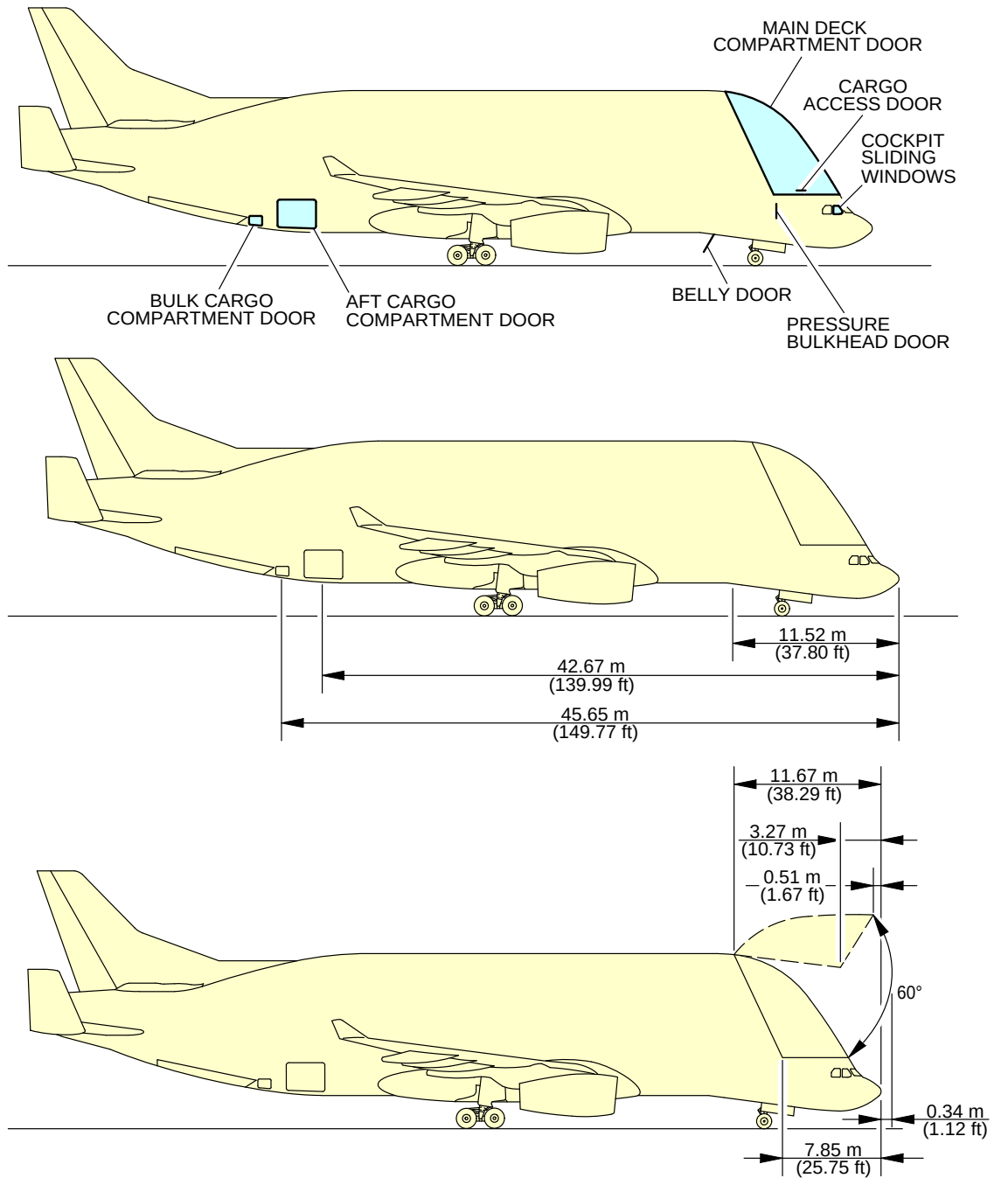
F\_AC\_020603\_1\_0030101\_01\_00

Main-Deck-Cargo Arrangement Cross-sections  
FIGURE-2-6-3-991-003-A01

**2-7-0 Door Clearances****\*\*ON A/C A330-700L**Door Clearances

1. This section gives door location, identification and clearances.

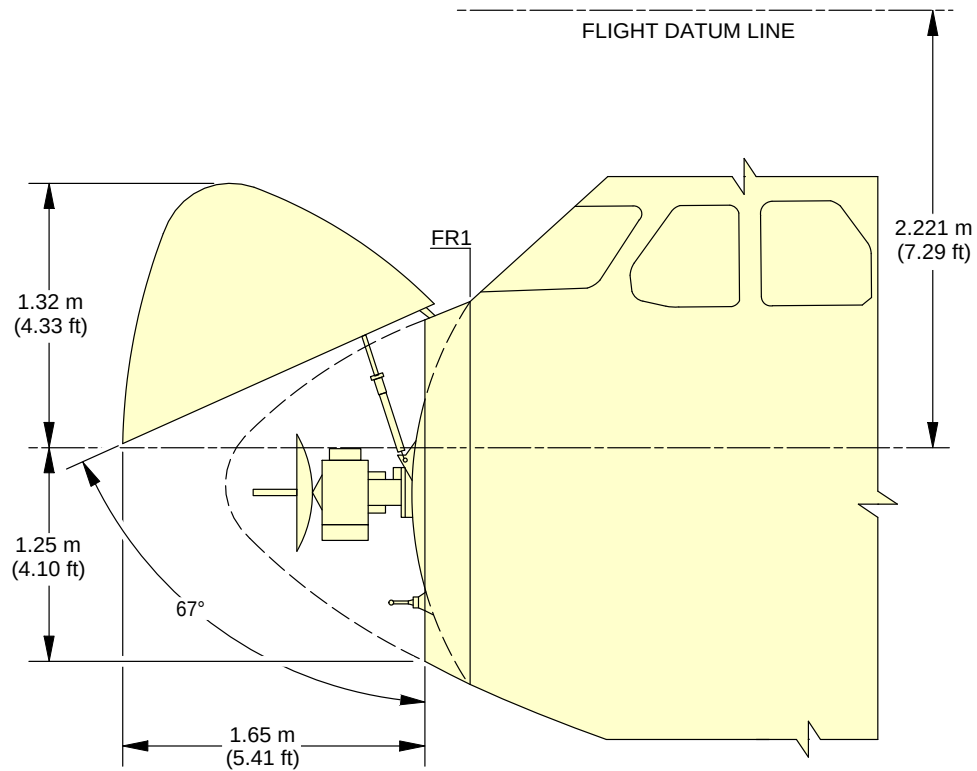
**\*\*ON A/C A330-700L**



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Lateral Position of Doors from Aircraft Nose  
FIGURE-2-7-0-991-055-A01

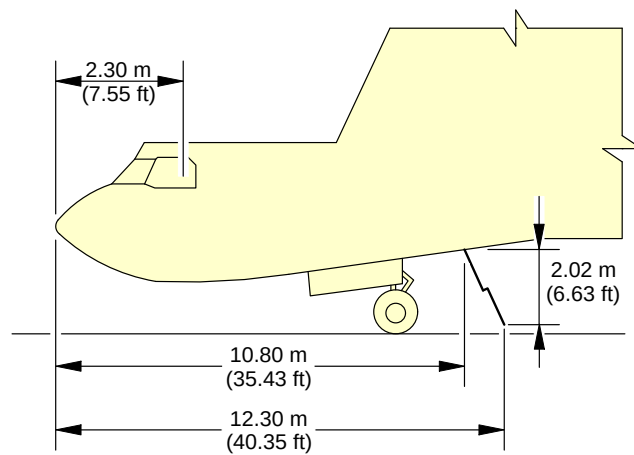
**\*\*ON A/C A330-700L**



F\_AC\_020700\_1\_0560101\_01\_01

Radome  
FIGURE-2-7-0-991-056-A01

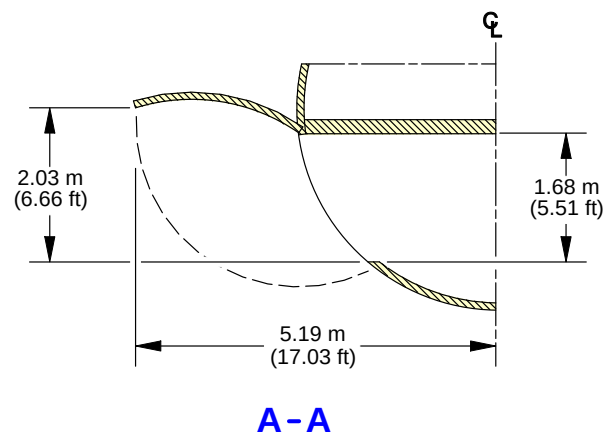
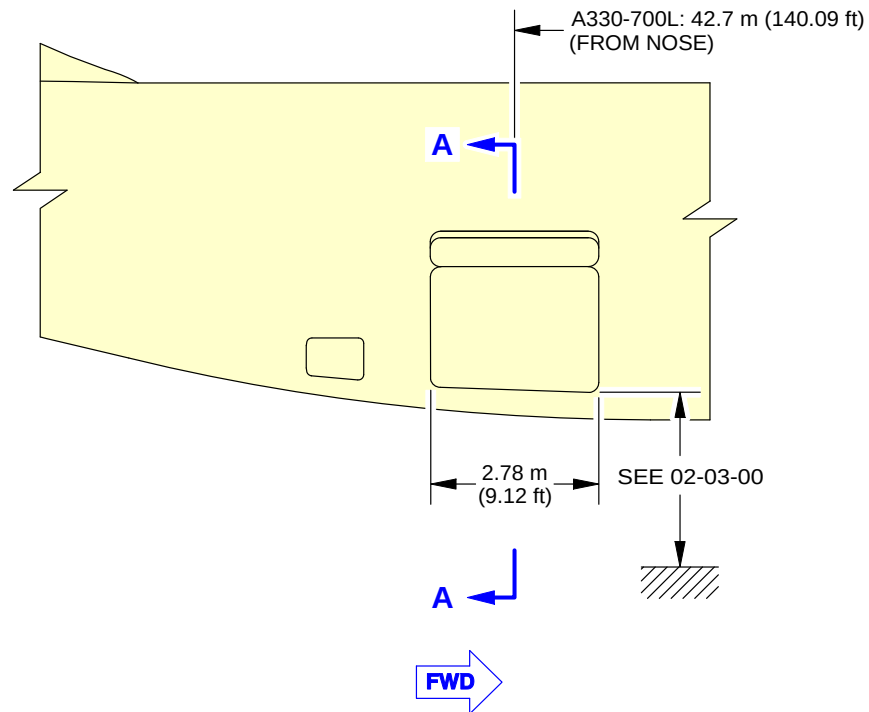
**\*\*ON A/C A330-700L**



F\_AC\_020700\_1\_0600101\_01\_00

Belly Door and Sliding Windows  
FIGURE-2-7-0-991-060-A01

**\*\*ON A/C A330-700L**

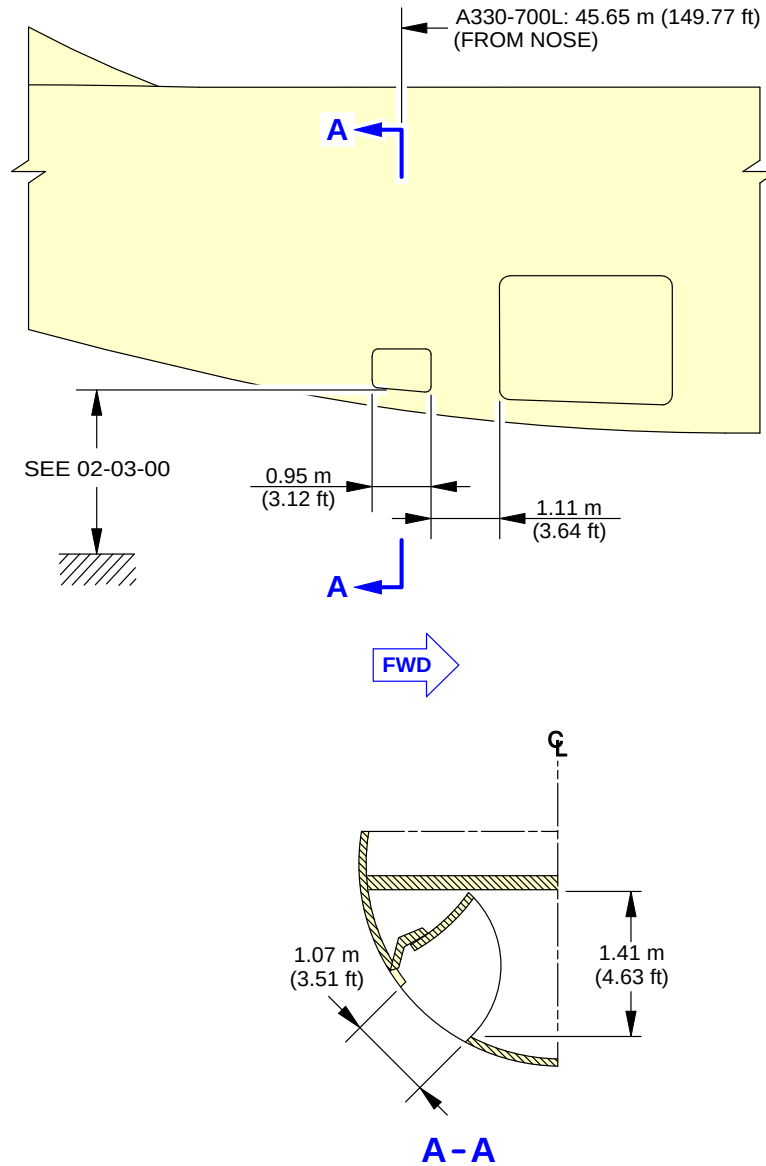


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Aft Cargo-Compartment Door  
FIGURE-2-7-0-991-057-A01



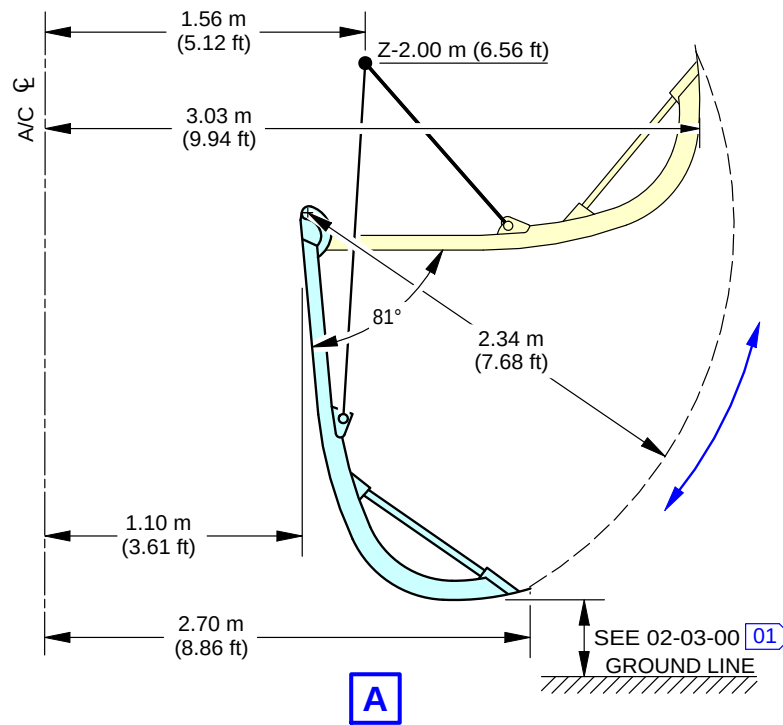
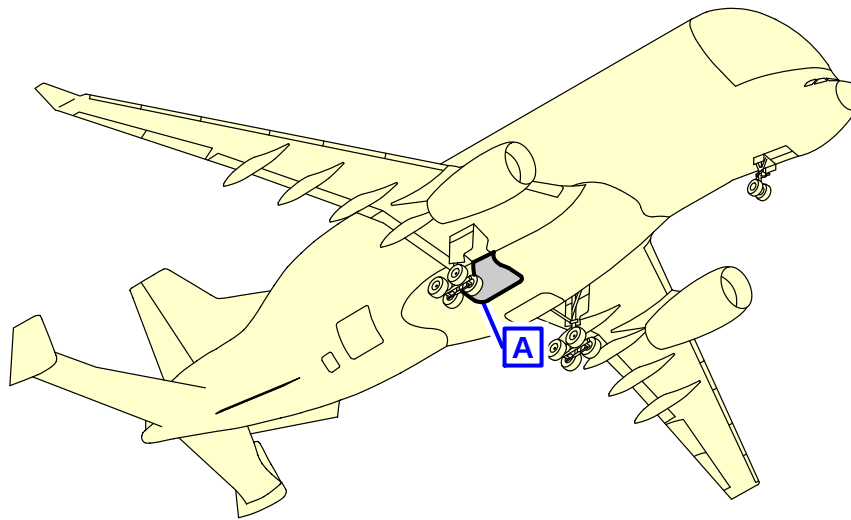
**\*\*ON A/C A330-700L**



F\_AC\_020700\_1\_0580101\_01\_00

Bulk Cargo-Compartment Door  
FIGURE-2-7-0-991-058-A01

**\*\*ON A/C A330-700L**



RH SHOWN  
LH SYMMETRICAL

**NOTE:**

01 DEPENDING ON CG POSITION AND AIRCRAFT WEIGHT.

F\_AC\_020700\_1\_0590101\_01\_00

MLG Doors  
FIGURE-2-7-0-991-059-A01

**2-9-0 Landing Gear****\*\*ON A/C A330-700L**Landing Gear**1. General**

All dimensions shown are minimum dimensions with zero clearances.  
Dimensions for elevators and related mechanisms must be added to the following figures.

**A. Elevators**

These can be either mechanical or hydraulic. Elevators are used to:

- Let easy movement of persons and equipment around the main landing gears
- Lift and remove the landing gear assemblies out of the pits.

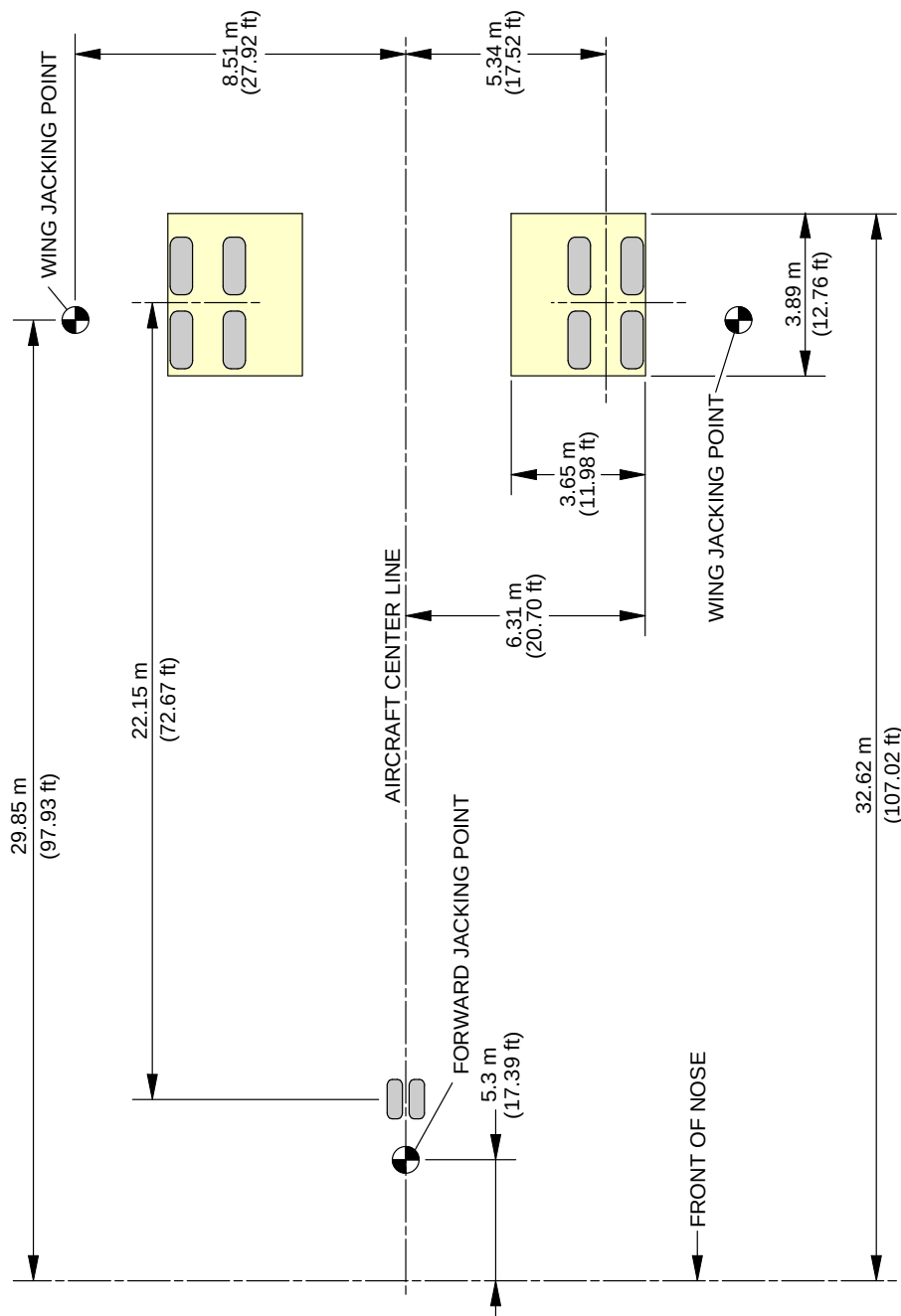
**B. Jacking**

The aircraft must be in position over the pits to put the gear on the elevators. Jacks must be installed and engaged with all the jacking points (Ref. Section 2-14 for Jacking).

Jacks must support the total aircraft weight i.e. when the landing gears do not touch the elevators on retraction/extension tests.

When tripod support jacks are used, the tripod-base circle radius must be small because the locations required for positioning the jacks are close to the sides of the pits.

**\*\*ON A/C A330-700L**

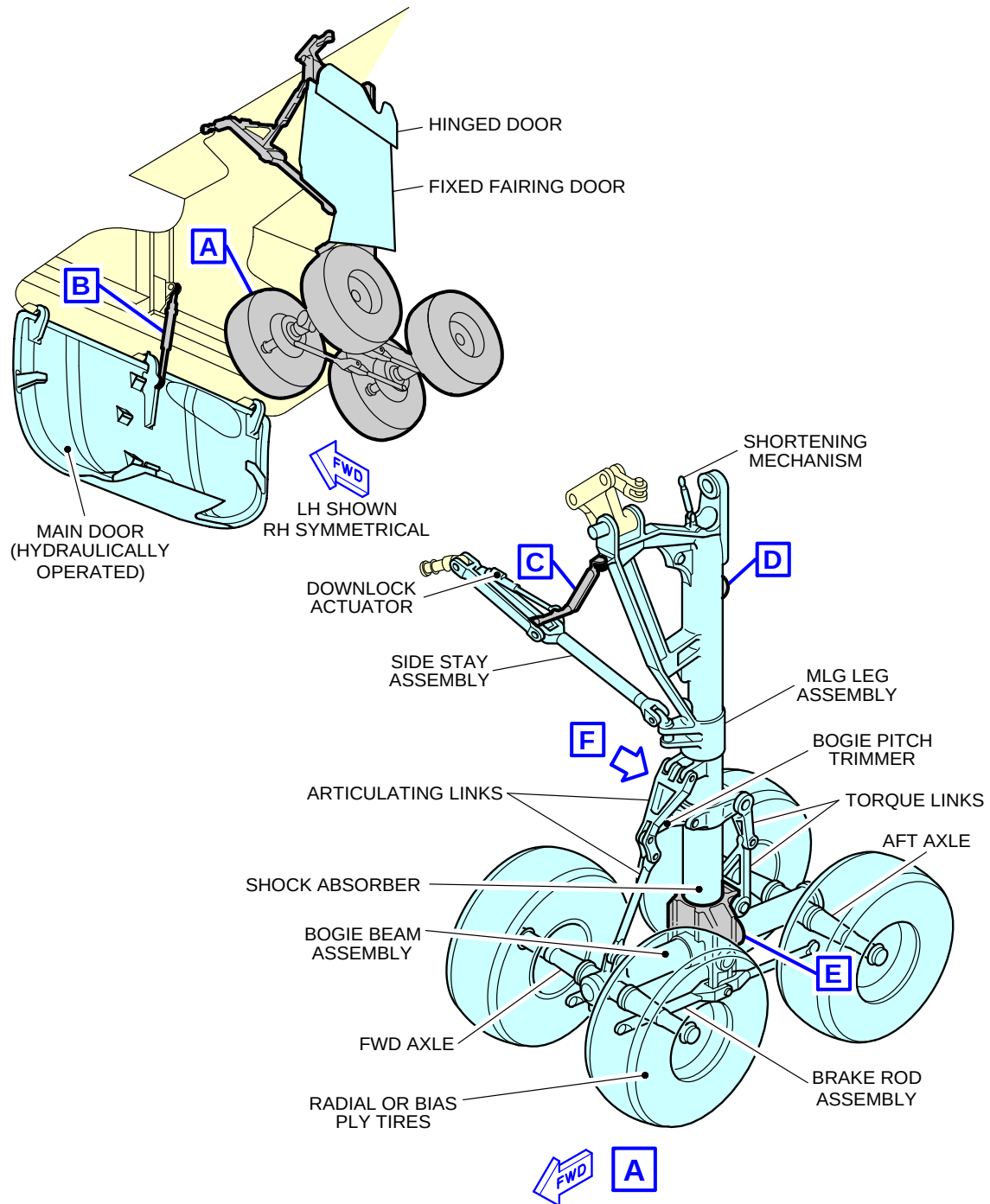


**NOTE:**  
ENVELOPES SHOWN WITH ZERO CLEARANCE TO OUTSIDE EDGE OF TIRES.

F\_AC\_020900\_1\_0210101\_01\_01

Landing Gear Position  
FIGURE-2-9-0-991-021-A01

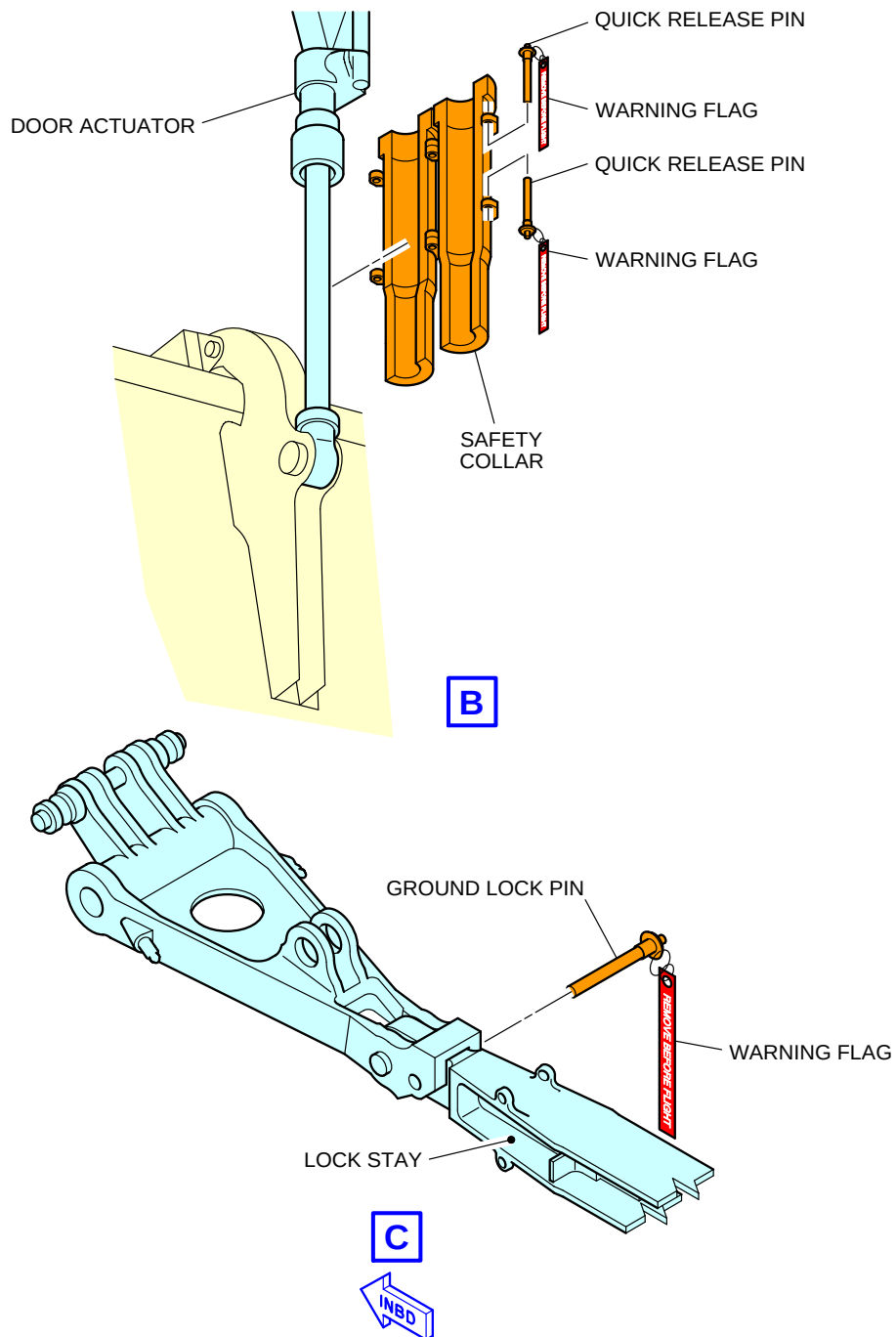
**\*\*ON A/C A330-700L**



F\_AC\_020900\_1\_0220101\_01\_00

Main Landing Gear  
(Sheet 1 of 3)  
FIGURE-2-9-0-991-022-A01

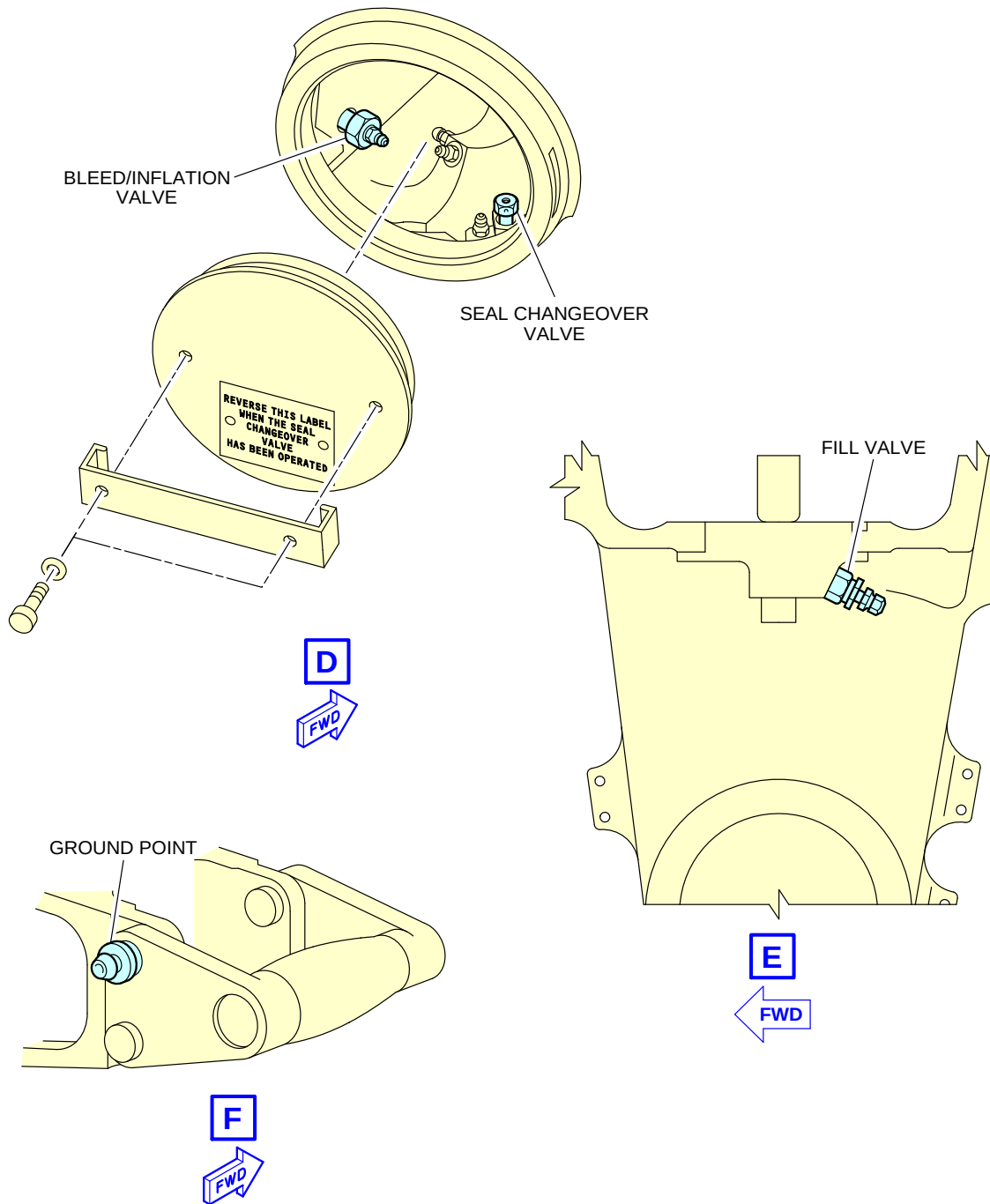
**\*\*ON A/C A330-700L**



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Main Landing Gear  
Mechanical locking of MLG (Sheet 2 of 3)  
FIGURE-2-9-0-991-022-A01

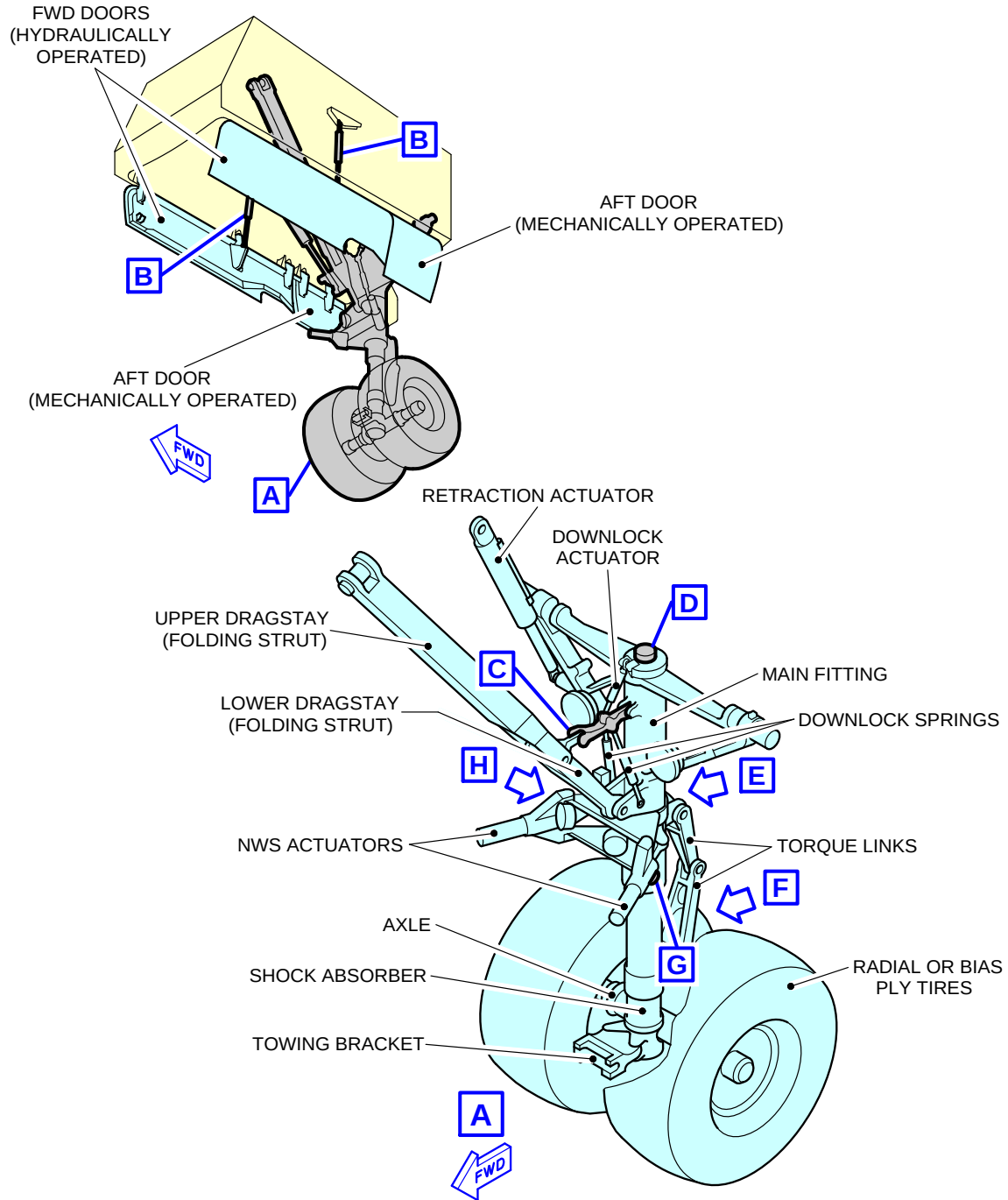
**\*\*ON A/C A330-700L**



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Main Landing Gear  
MLG Servicing (Sheet 3 of 3)  
FIGURE-2-9-0-991-022-A01

**\*\*ON A/C A330-700L**

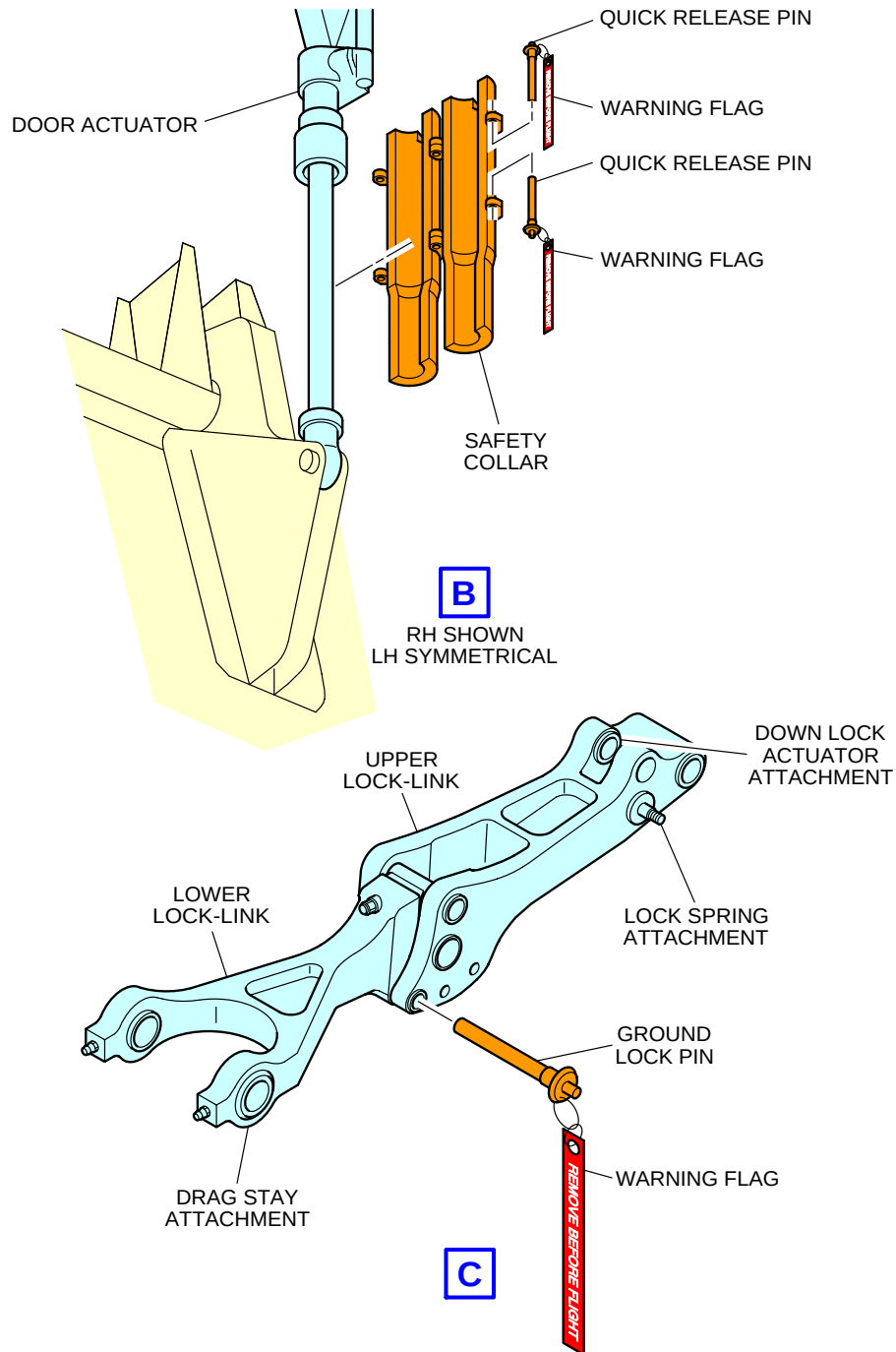


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Nose Landing Gear  
(Sheet 1 of 4)  
FIGURE-2-9-0-991-023-A01



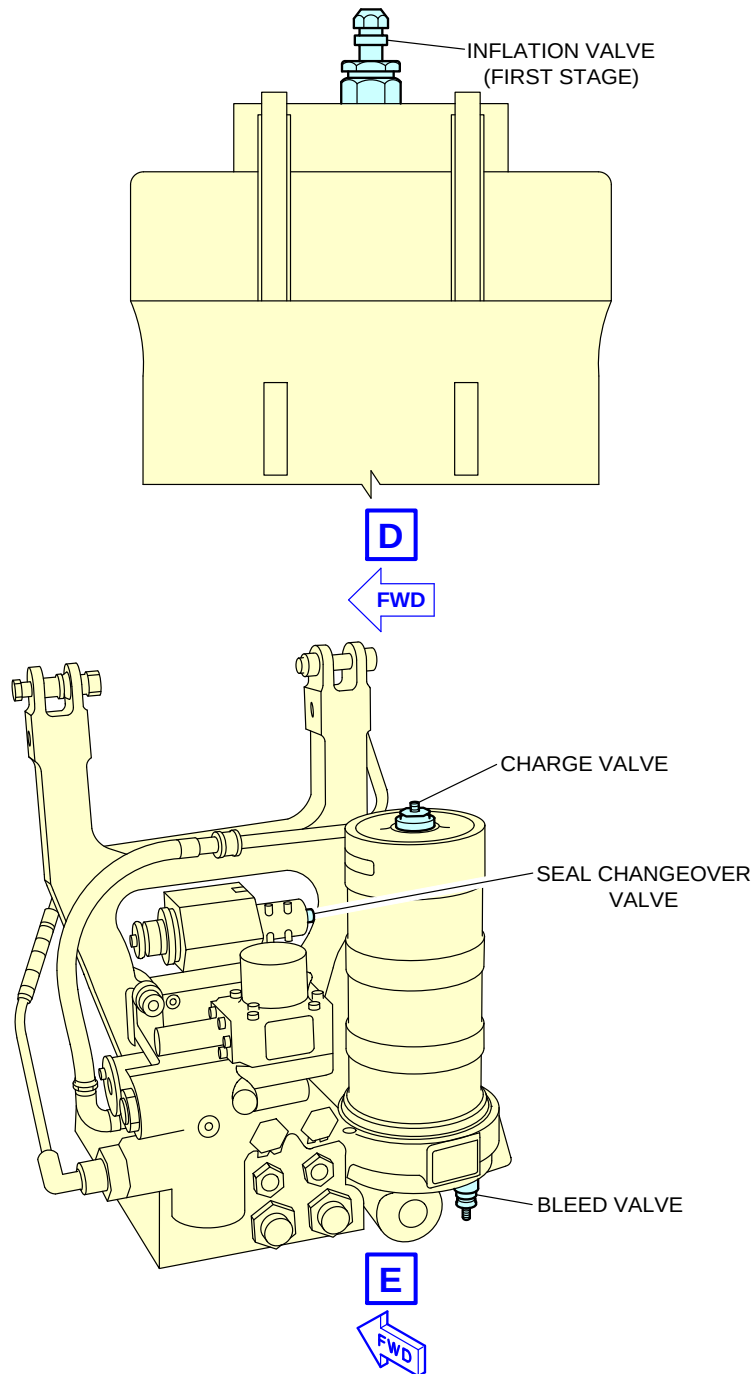
**\*\*ON A/C A330-700L**



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Nose Landing Gear  
Mechanical Locking of NLG (Sheet 2 of 4)  
FIGURE-2-9-0-991-023-A01

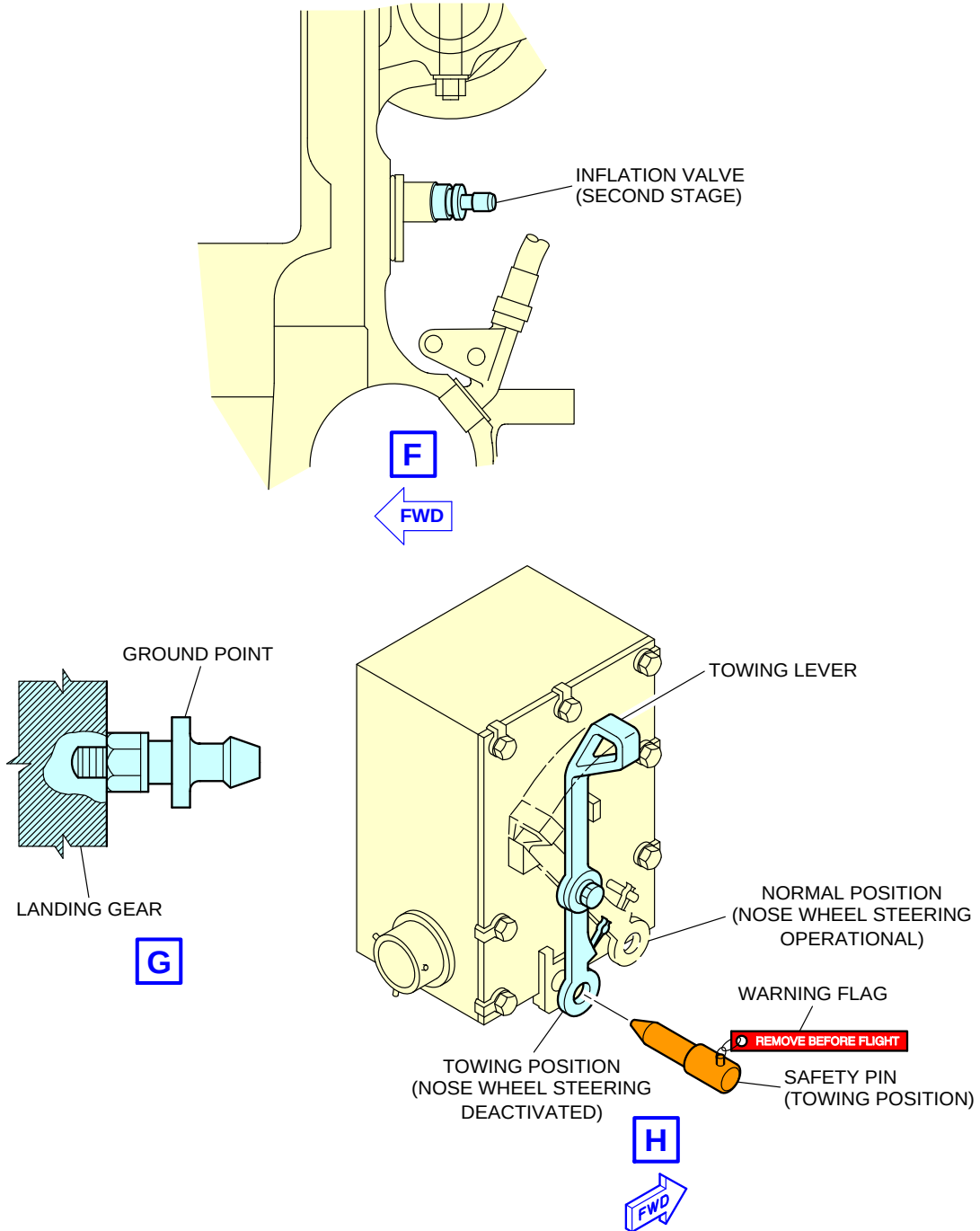
**\*\*ON A/C A330-700L**



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Nose Landing Gear  
NLG Servicing 1 (Sheet 3 of 4)  
FIGURE-2-9-0-991-023-A01

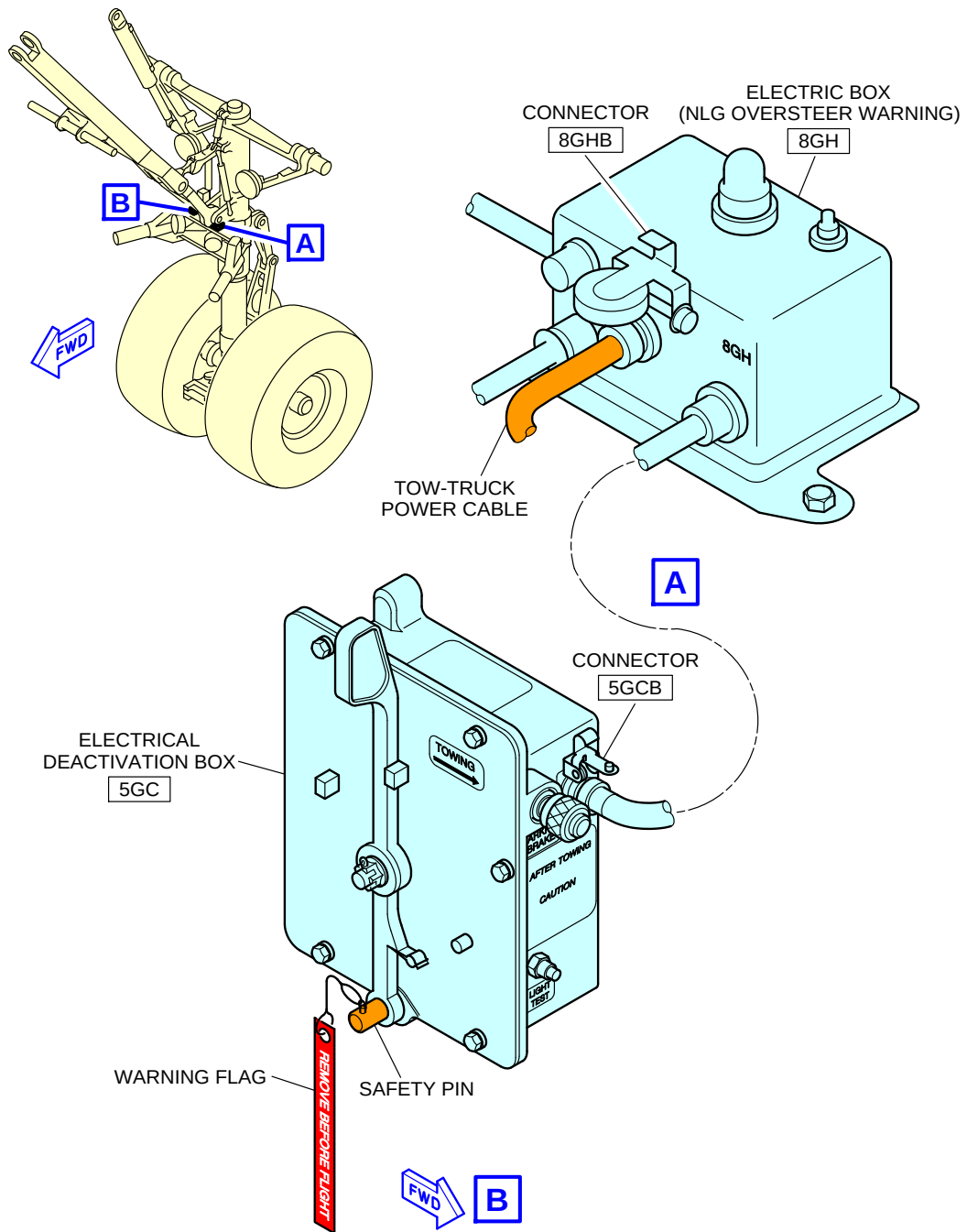
**\*\*ON A/C A330-700L**



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Nose Landing Gear  
NLG Servicing 2 (Sheet 4 of 4)  
FIGURE-2-9-0-991-023-A01

**\*\*ON A/C A330-700L**



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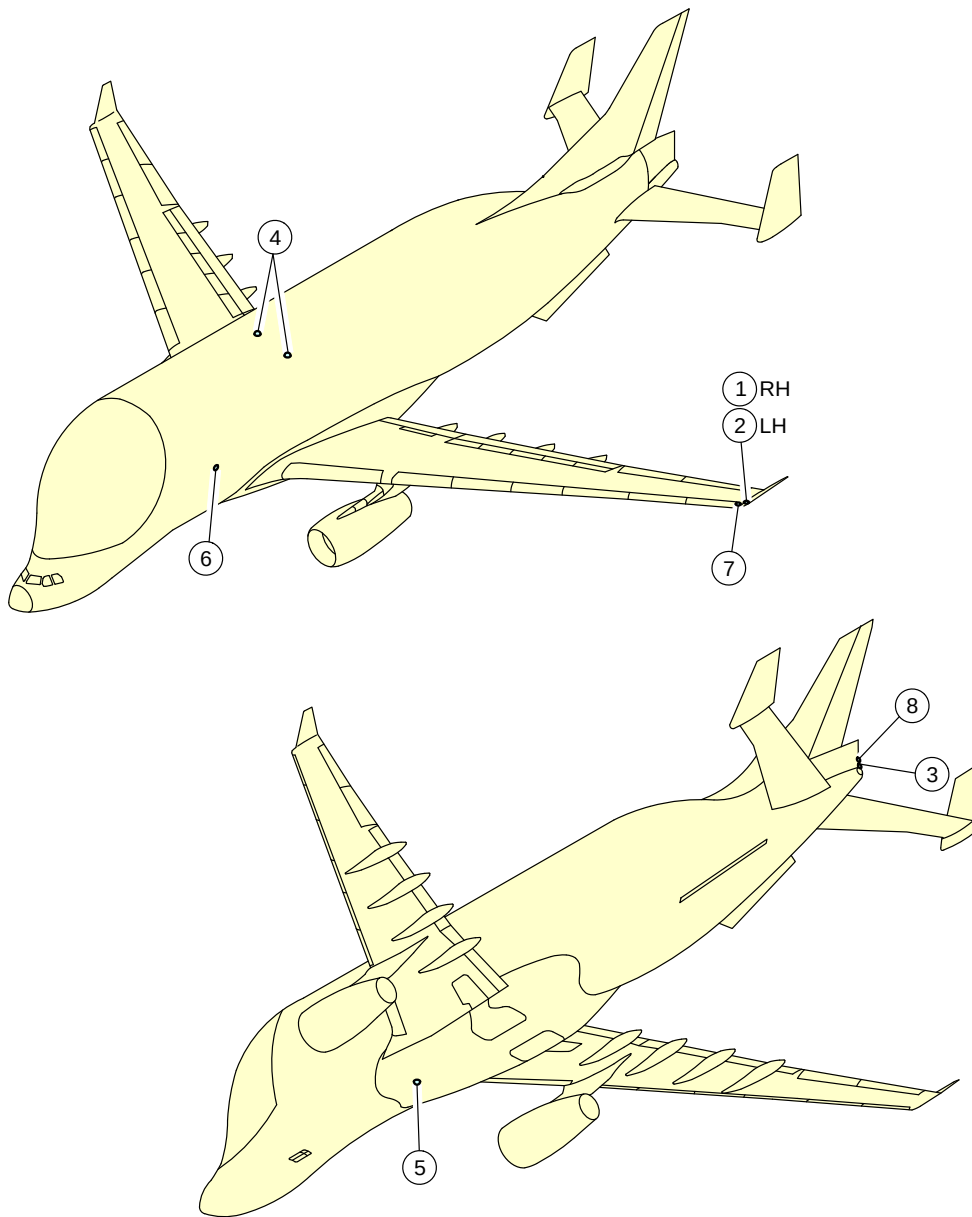
Tow Truck Power  
NLG Servicing 3  
FIGURE-2-9-0-991-024-A01

**2-10-0 Exterior Lighting**
**\*\*ON A/C A330-700L**
Exterior Lighting
**1. General**

This section gives the location of the aircraft exterior lighting.

| EXTERIOR LIGHTING |                                           |
|-------------------|-------------------------------------------|
| ITEM              | DESCRIPTION                               |
| 1                 | RIGHT NAVIGATION LIGHT (GREEN)            |
| 2                 | LEFT NAVIGATION LIGHT (RED)               |
| 3                 | TAIL NAVIGATION LIGHT (WHITE)             |
| 4                 | UPPER ANTI-COLLISION LIGHT/BEACON (RED)   |
| 5                 | LOWER ANTI-COLLISION LIGHT/BEACON (RED)   |
| 6                 | WING SCAN LIGHTS                          |
| 7                 | WING STROBE LIGHT (HIGH INTENSITY, WHITE) |
| 8                 | TAIL STROBE LIGHT (HIGH INTENSITY, WHITE) |
| 9                 | LANDING LIGHTS                            |
| 10                | RUNWAY TURN-OFF LIGHTS                    |
| 11                | TAXI LIGHTS                               |
| 12                | TAKE-OFF LIGHTS                           |
| 13                | CARGO-COMPARTMENT FLOOD LIGHTS            |

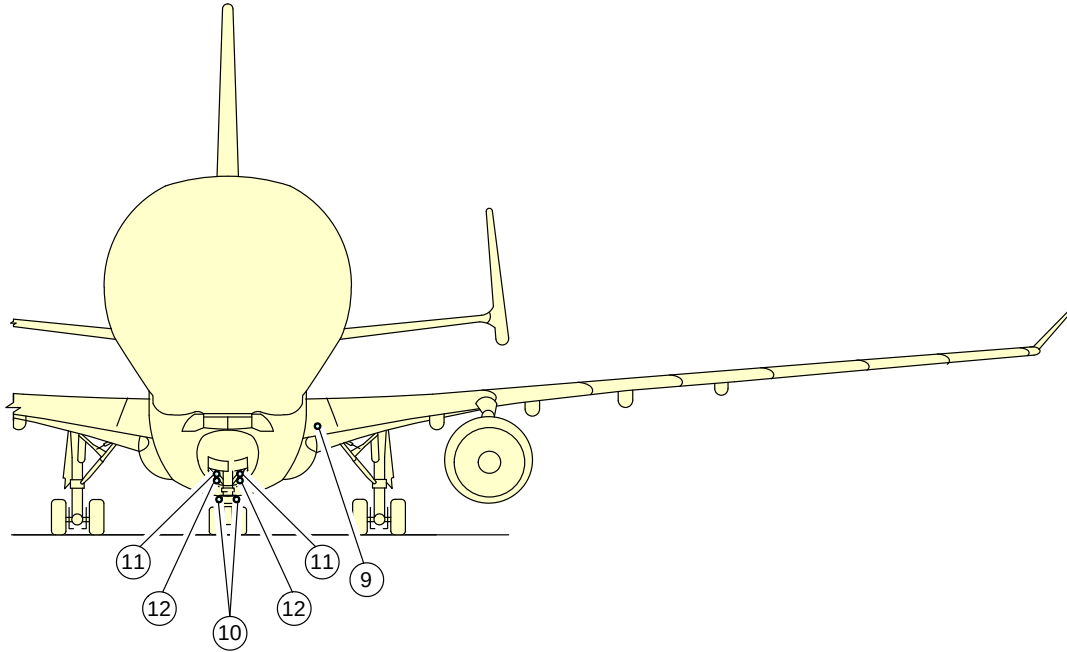
**\*\*ON A/C A330-700L**



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Exterior Lighting  
Lights General Layout (Sheet 1 of 5)  
FIGURE-2-10-0-991-010-A01

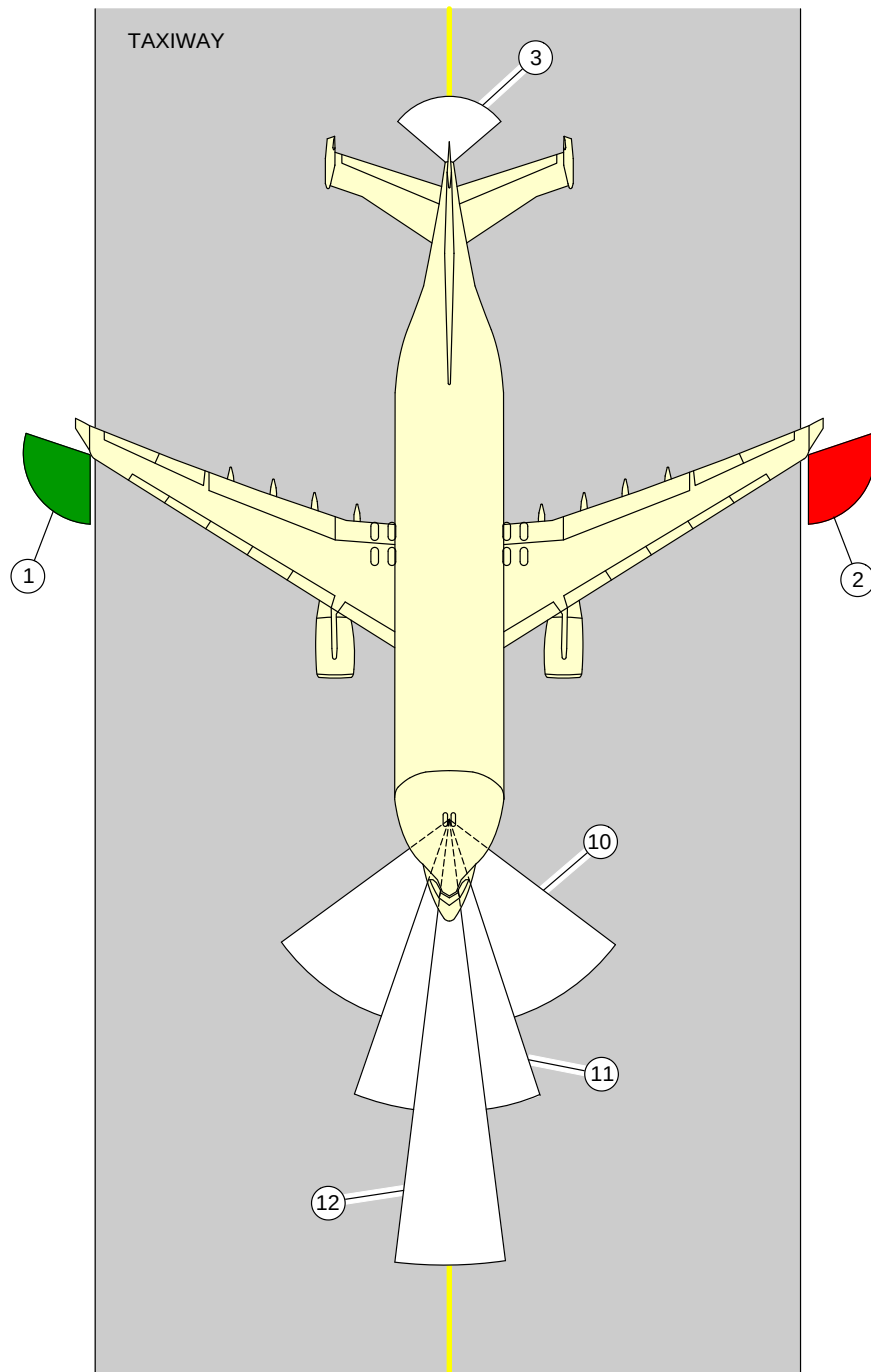
**\*\*ON A/C A330-700L**



F\_AC\_021000\_1\_0100102\_01\_00

Exterior Lighting  
Lights General Layout (Sheet 2 of 5)  
FIGURE-2-10-0-991-010-A01

**\*\*ON A/C A330-700L**

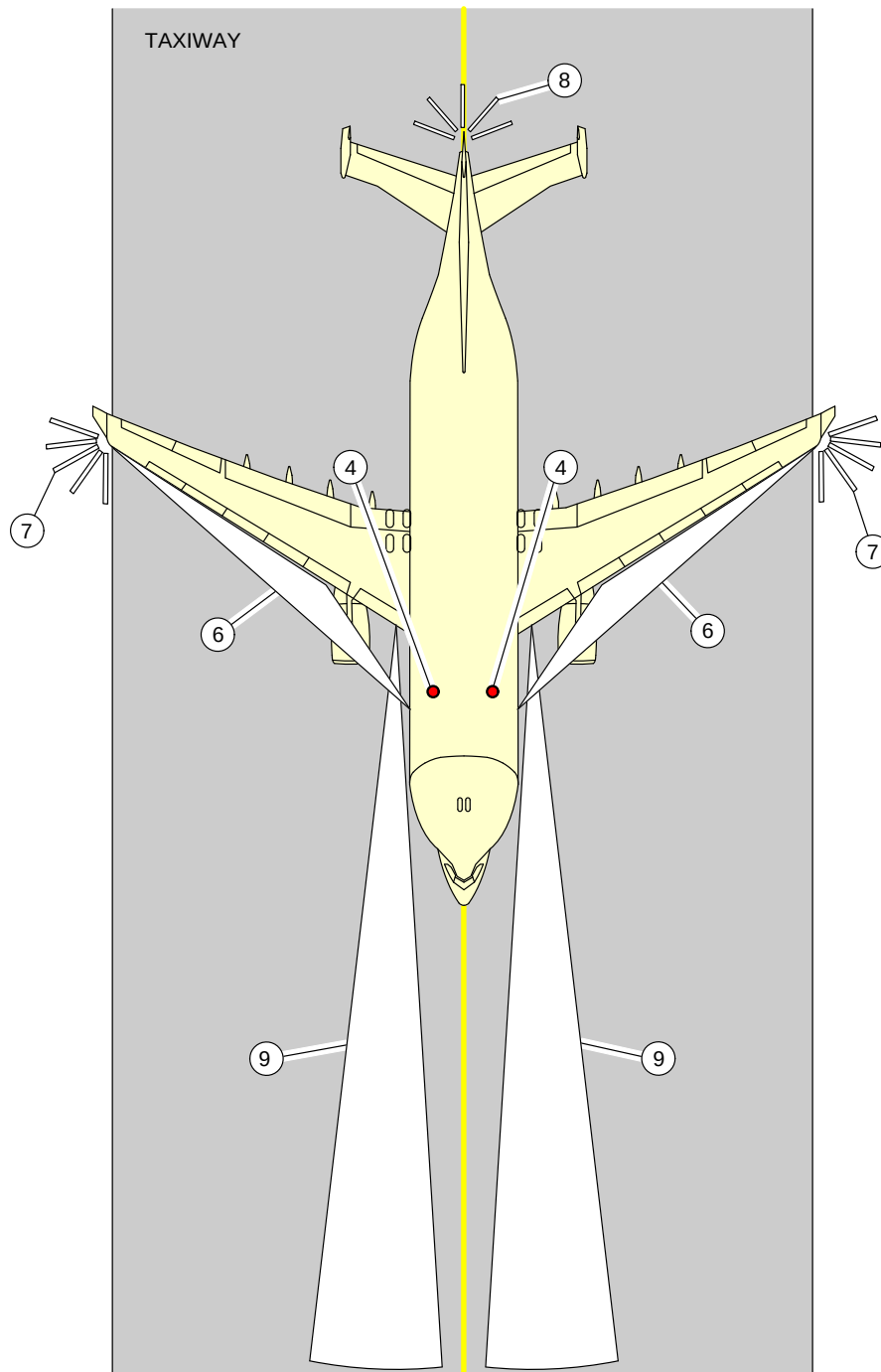


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Exterior Lighting  
(Sheet 3 of 5)  
FIGURE-2-10-0-991-010-A01



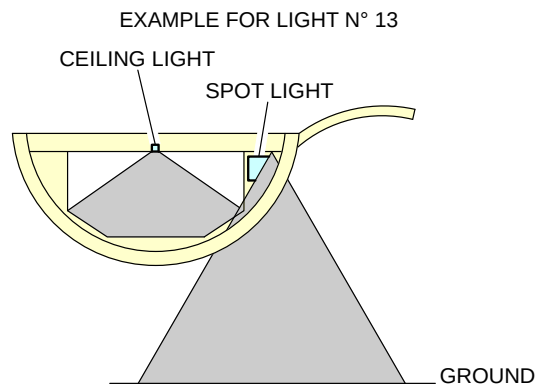
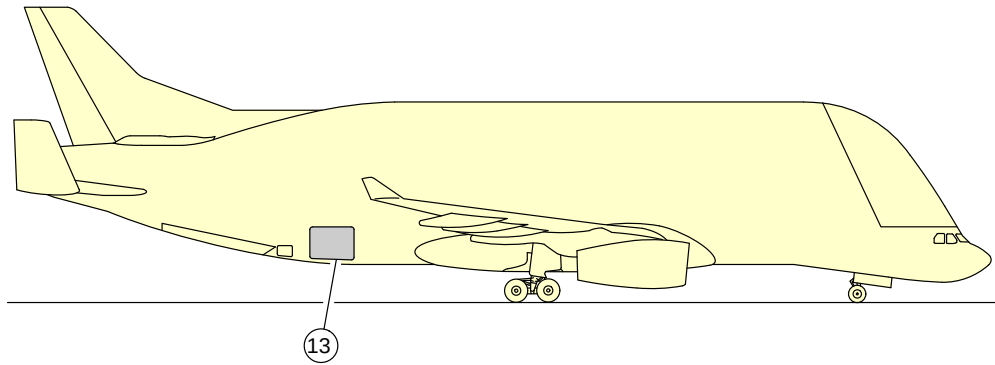
**\*\*ON A/C A330-700L**



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Exterior Lighting  
(Sheet 4 of 5)  
FIGURE-2-10-0-991-010-A01

**\*\*ON A/C A330-700L**



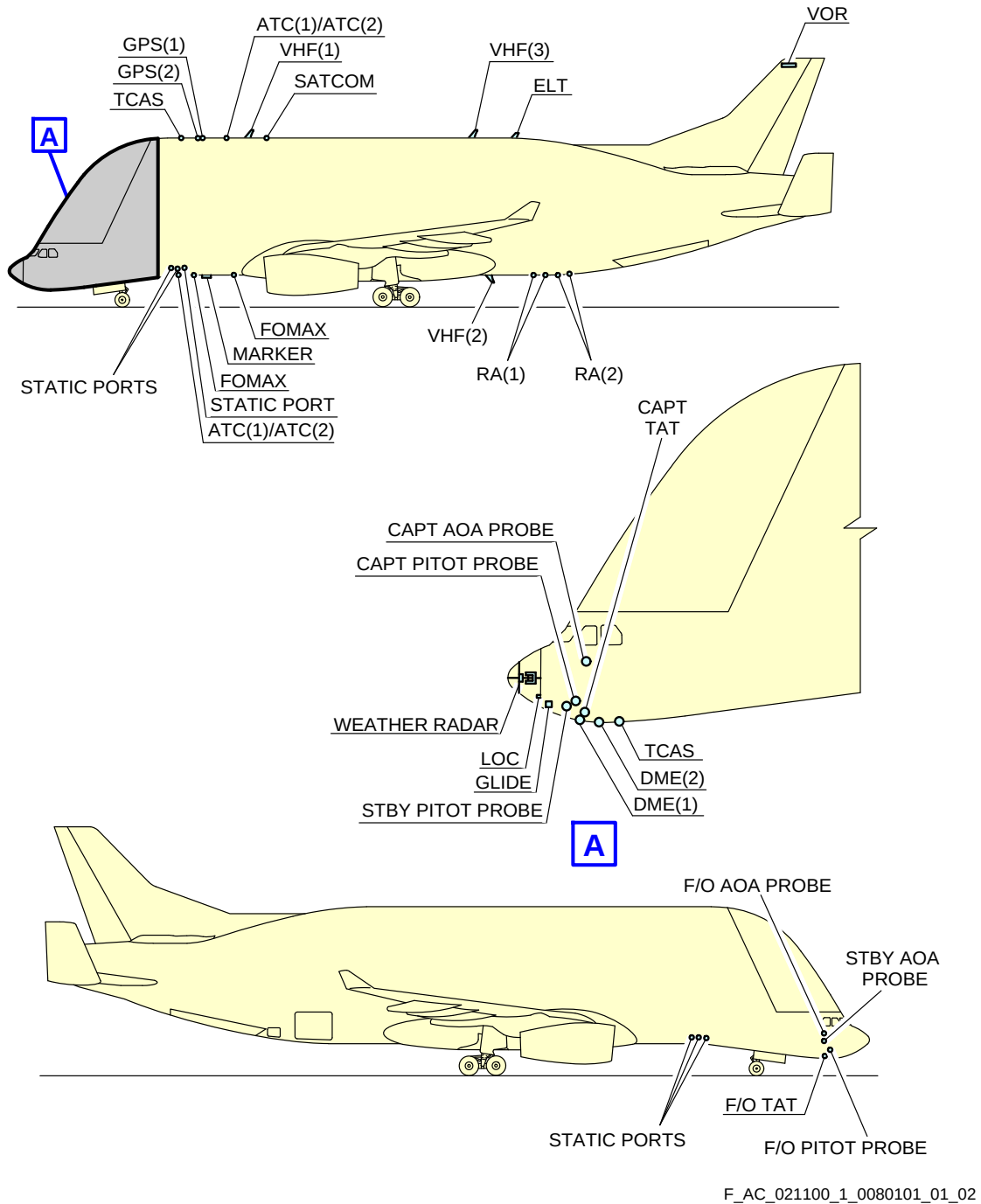
F\_AC\_021000\_1\_0100105\_01\_00

Exterior Lighting  
Aft Lower-Cargo light (Sheet 5 of 5)  
FIGURE-2-10-0-991-010-A01

**2-11-0      Antennas and Probes Location****\*\*ON A/C A330-700L**Antennas and Probes Location

1. This section gives the location of antennas and probes.

**\*\*ON A/C A330-700L**



Antennas and Probes  
Location  
FIGURE-2-11-0-991-008-A01

**2-12-0 Engine and Nacelle****\*\*ON A/C A330-700L**Engine and Nacelle**1. Engine and Nacelle - RR TRENT 700 Engine****A. Engine**

The RB211-TRENT 700 engine is a high bypass ratio, triple spool turbofan.

The principal modules of the engine are:

- The Low Pressure Compressor (LPC) rotor
- The Intermediate Pressure (IP) compressor
- The intermediate case
- The HP system (this includes the High Pressure Compressor (HPC), the combustion system and the High Pressure Turbine (HPT))
- The Intermediate Pressure Turbine (IPT)
- The external gearbox
- The LPC case
- The Low Pressure Turbine (LPT).

The compressor system has three axial flow compressors in a triple spool configuration.

The compressors are turned independently by their related turbines, each at its most satisfactory speed.

The LP system has a single-stage compressor installed at the front of the engine. A shaft connects the compressor to a four-stage turbine at the rear of the gas generator. The gas generator also includes an eight-stage IP compressor, a six-stage HPC and a combustion system.

Each of the compressors in the gas generator is connected to, and turned by, a different single stage turbine. Between the HPC and the HPT is the annular combustion system which burns a mixture of fuel and air to supply energy as heat. Behind the LPT there is a common nozzle assembly which mixes the cold air and hot gas exhaust flows. The external gearbox module is installed below the rear case of the fan case. It has a gear train that decreases and increases the speed to meet the specified drive requirements of each accessory.

**B. Nacelle**

The nacelle gives the engine an aerodynamic shape. Each engine is housed in a nacelle suspended from a pylon attached below the wing. The nacelle consists of the following major components:

- (1) Air Intake Cowl

The air intake cowl is attached to the forward flange of the front LPC case. Its function is to supply inlet air in a satisfactory condition for the engine compressors.

(2) Fan Cowl Doors

The fan cowl doors hang on the aircraft wing pylon and are closed around the LPC cases. They can be opened during ground maintenance to give access to the components installed on the cases and to let the thrust reverser cowl doors be opened.

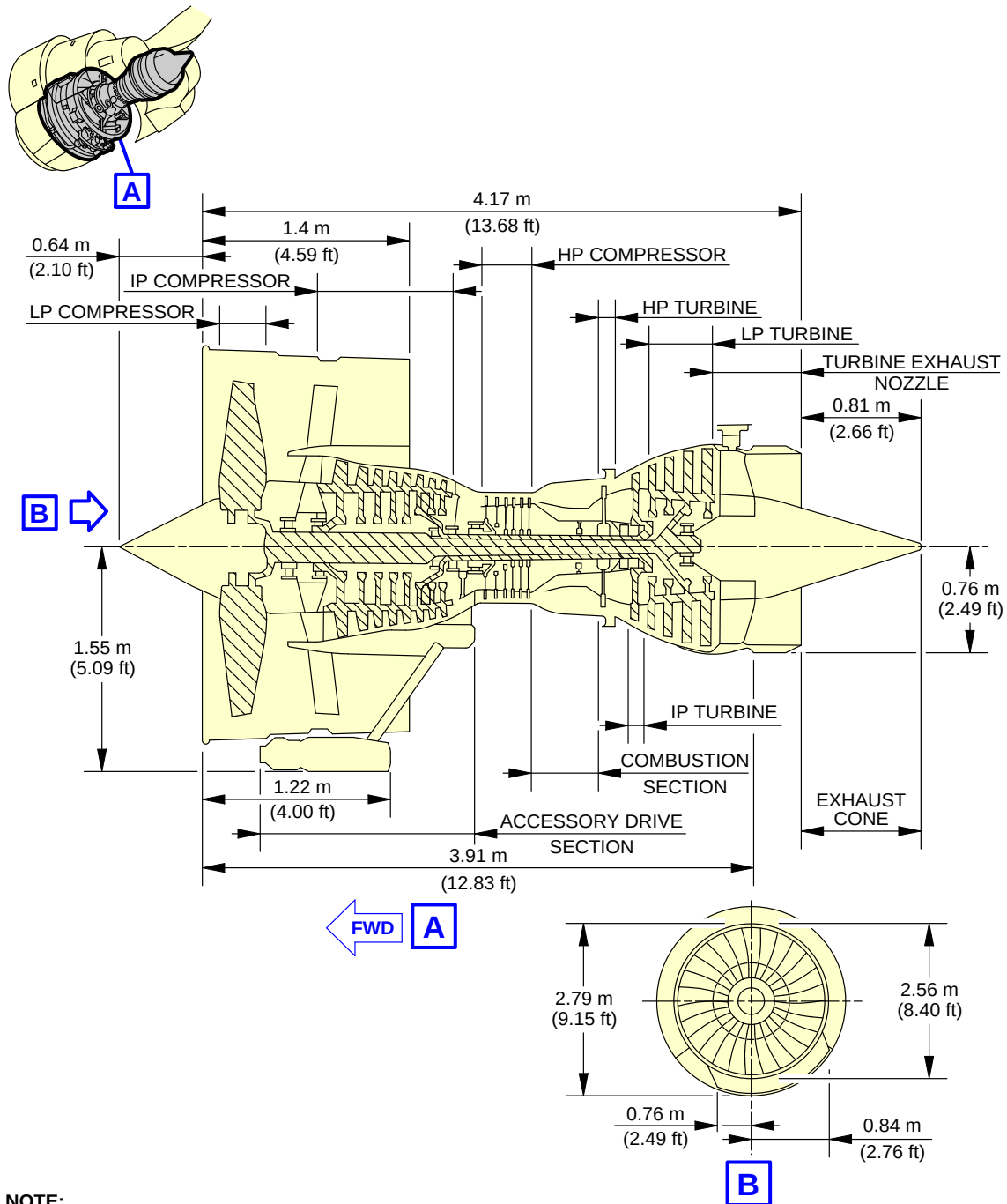
(3) Thrust Reverser

The thrust reverser is a component of the aircraft engine nacelle. The thrust reverser is a twin thrust reverser cowl door ('C' duct) construction providing a fan duct inner wall fairing for the core engine between the top and bottom bifurcation walls. The thrust reverser incorporates hydraulically-powered actuators to operate four pivoting doors which redirect the fan air flow in reverse thrust. Hydraulic power is provided from the aircraft hydraulic system to position the doors in a "stowed" position for forward thrust and "deployed" position for reverse thrust.

(4) Common Nozzle Assembly (CNA)

The CNA is attached to the aft flange of the exhaust case. The function of the CNA is to mix the core engine exhaust with the LPC outlet air.

**\*\*ON A/C A330-700L**

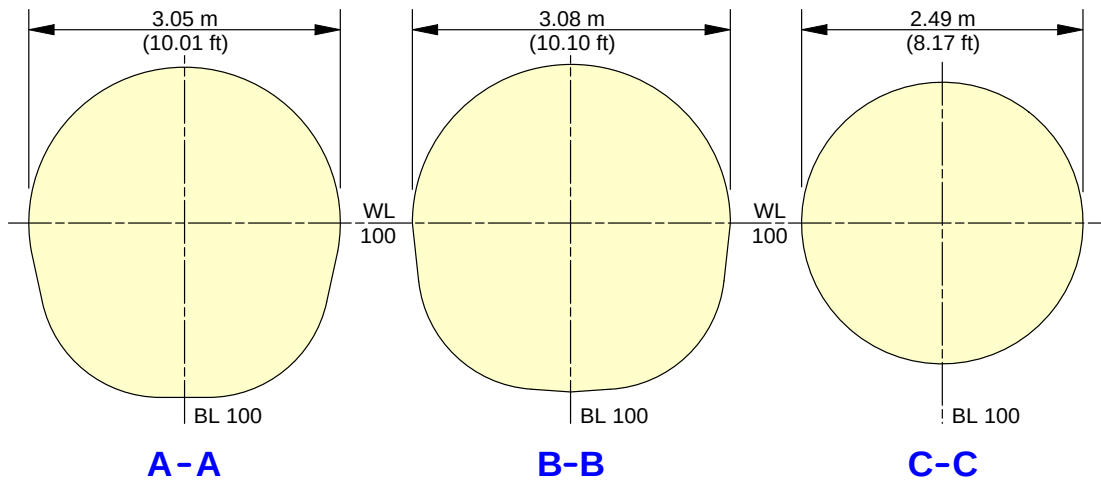
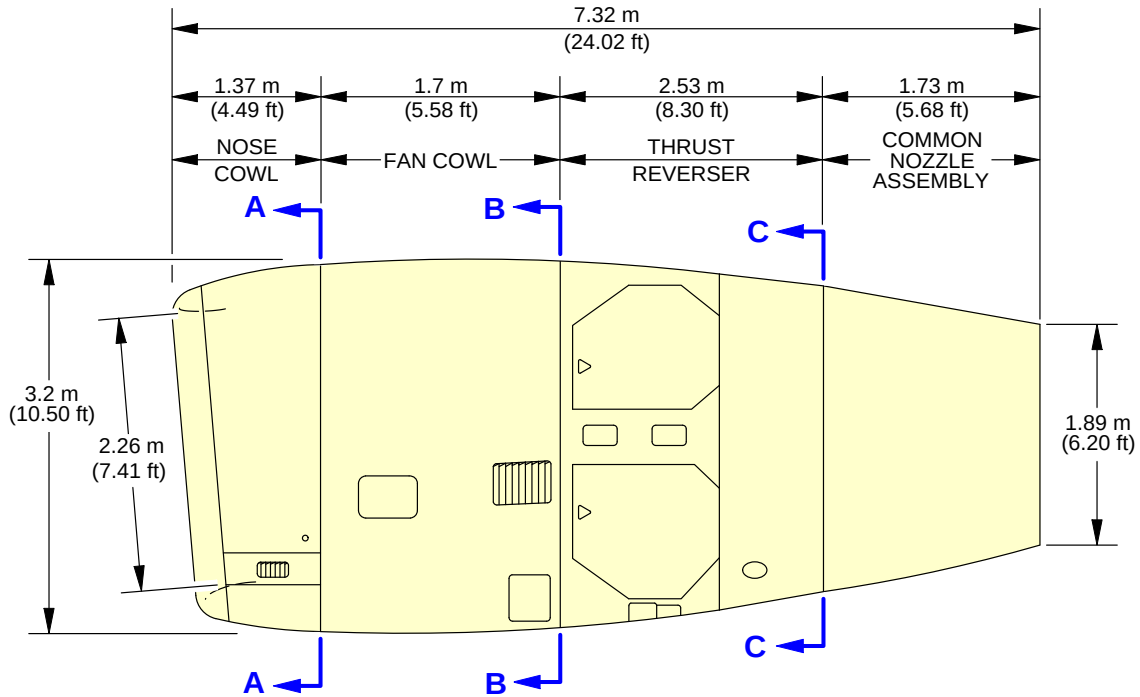


**NOTE:**  
APPROXIMATE DIMENSIONS.

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Engine and Nacelle  
Engine Dimensions - RR TRENT 700  
FIGURE-2-12-0-991-030-A01

**\*\*ON A/C A330-700L**



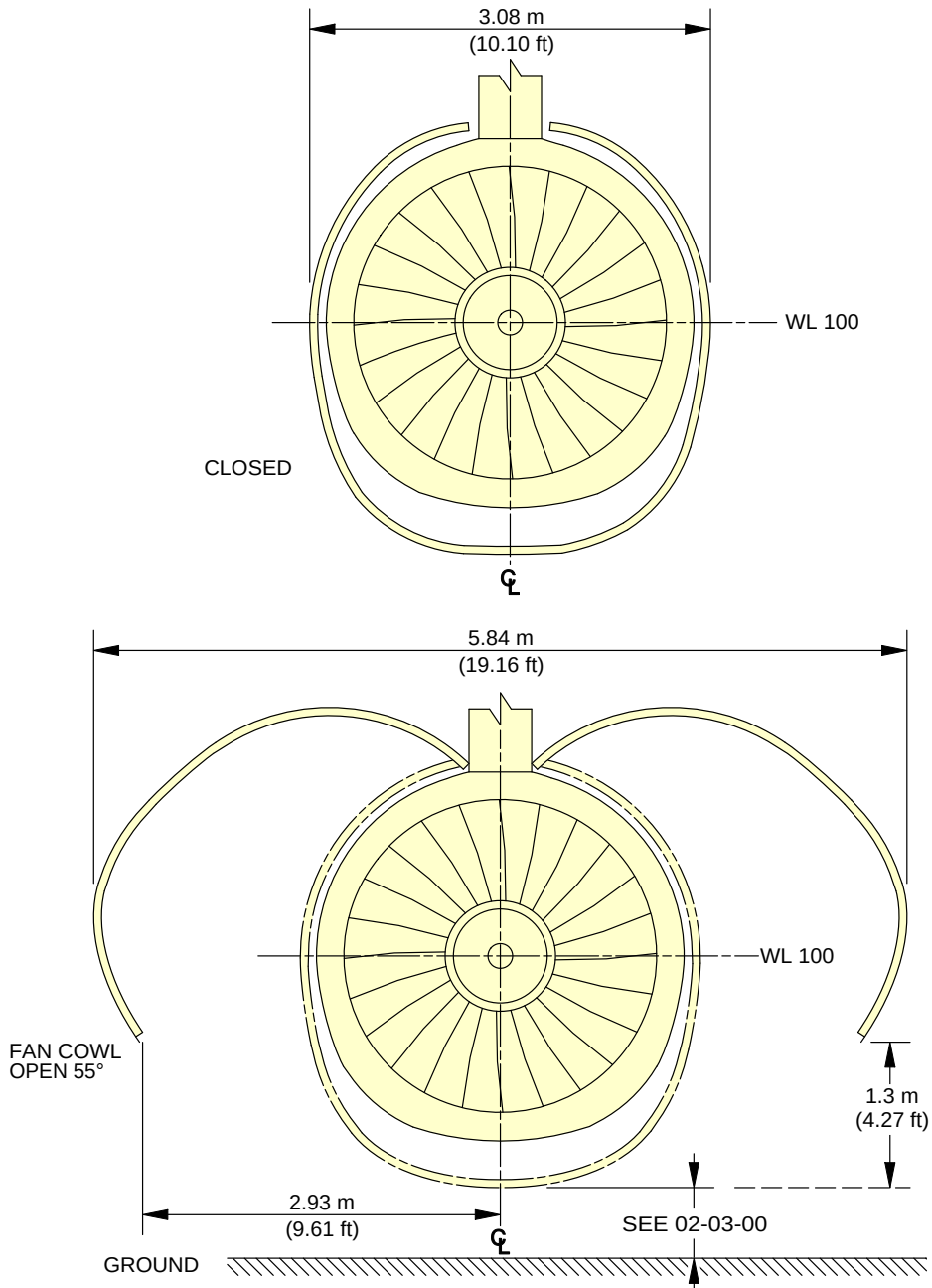
**NOTE:**  
APPROXIMATE DIMENSIONS.

F\_AC\_021200\_1\_0310101\_01\_00

Engine and Nacelle  
Nacelle Dimensions - RR TRENT 700  
FIGURE-2-12-0-991-031-A01



**\*\*ON A/C A330-700L**

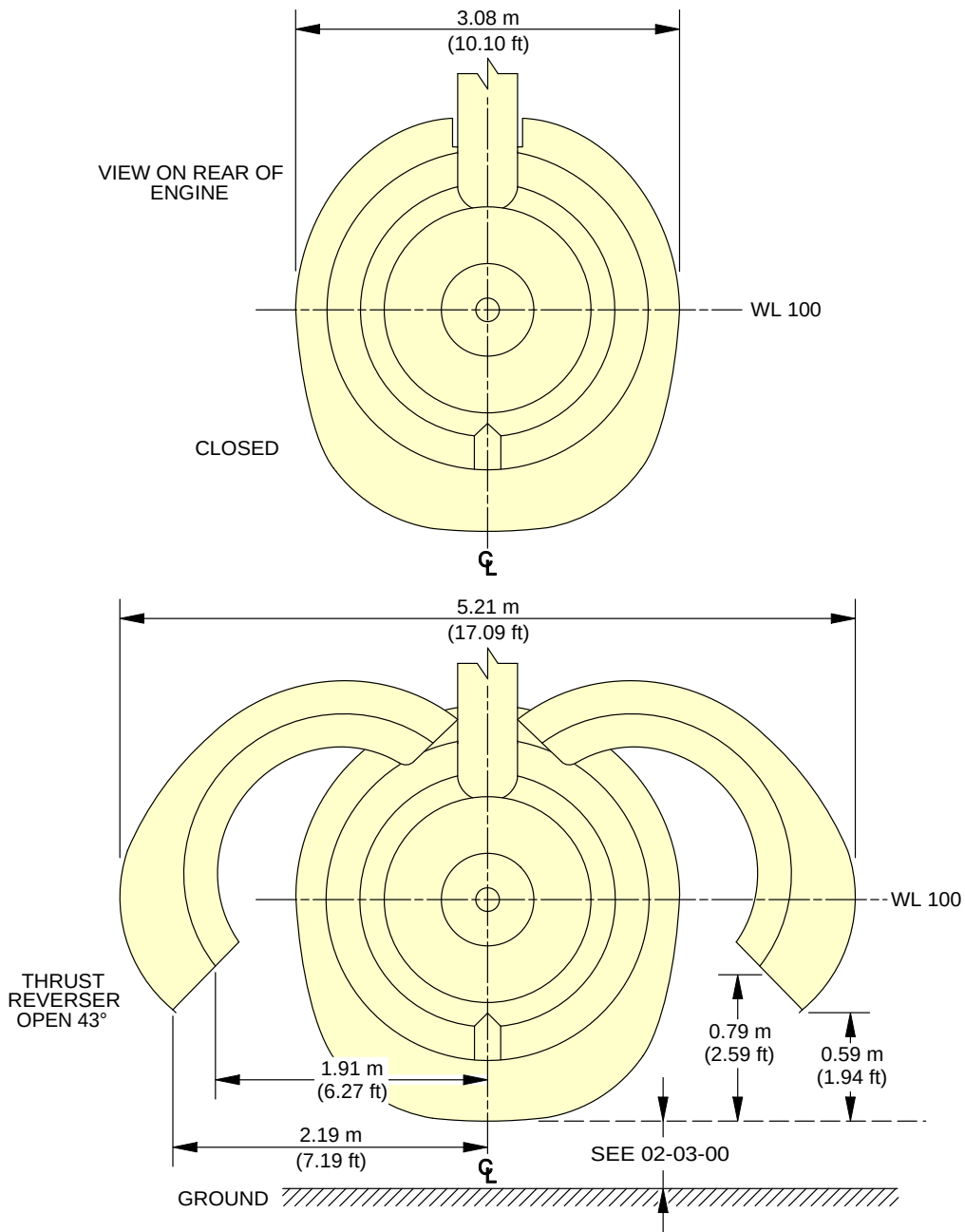


**NOTE:**  
APPROXIMATE DIMENSIONS.

F\_AC\_021200\_1\_0320101\_01\_00

Engine and Nacelle  
Fan Cowls - RR TRENT 700  
FIGURE-2-12-0-991-032-A01

**\*\*ON A/C A330-700L**



**NOTE:**  
APPROXIMATE DIMENSIONS.

F\_AC\_021200\_1\_0330101\_01\_00

Engine and Nacelle  
Thrust Reverser Cowls - RR TRENT 700  
FIGURE-2-12-0-991-033-A01

**2-12-1 Auxiliary Power Unit****\*\*ON A/C A330-700L**Auxiliary Power Unit**1. General**

The Auxiliary Power Unit (APU) and its related mechanical components are installed at the rear part of the fuselage in the tailcone section. The APU compartment is a fireproof area (identified as the Fire Zone).

The APU is a pneumatic and shaft-power gas-turbine engine and is used for the ground and in-flight power supply of the aircraft.

The APU supplies:

- Mechanical shaft-power to operate a generator
- Bleed-air to the Main Engine Start (MES) and the Environmental Control System (ECS).

A part of the automatic system, with the pneumatic and the electromechanical controls, operates the start and the acceleration functions of the APU.

An air intake system with a flap-type door is installed in front of the APU compartment. The exhaust gases pass overboard at the end of the fuselage cone.

**2. Powerplant**

The APU is the Garrett Gas-Turbine Compressor Power-unit (GTCP) 331-350C with a single shaft engine.

The engine is the primary component of the APU, which is of the modular design. The modules of the engine are:

- The power section
- The load compressor
- The accessory drive gearbox with LRU(s).

The power section has a two-stage centrifugal compressor, a reverse-flow annular combustion chamber and a three-stage axial turbine. The power section directly operates the one-stage centrifugal load-compressor which supplies the bleed-air to the pneumatic system. The inlet guide vanes as part of the load compressor, control the airflow.

The power section also operates the gearbox which is attached to the load compressor. The following LRU's are mounted on the gearbox :

- The APU generator,
- The starter motor,
- The oil pump,
- The Fuel Control Unit (FCU),
- The cooling air fan.

The APU has a gearbox-driven oil-cooled AC generator.

The cooling air and ventilation system of the APU supplies the air for cooling of the APU and the equipment on the APU. It also supplies the air for ventilation of the APU compartment.

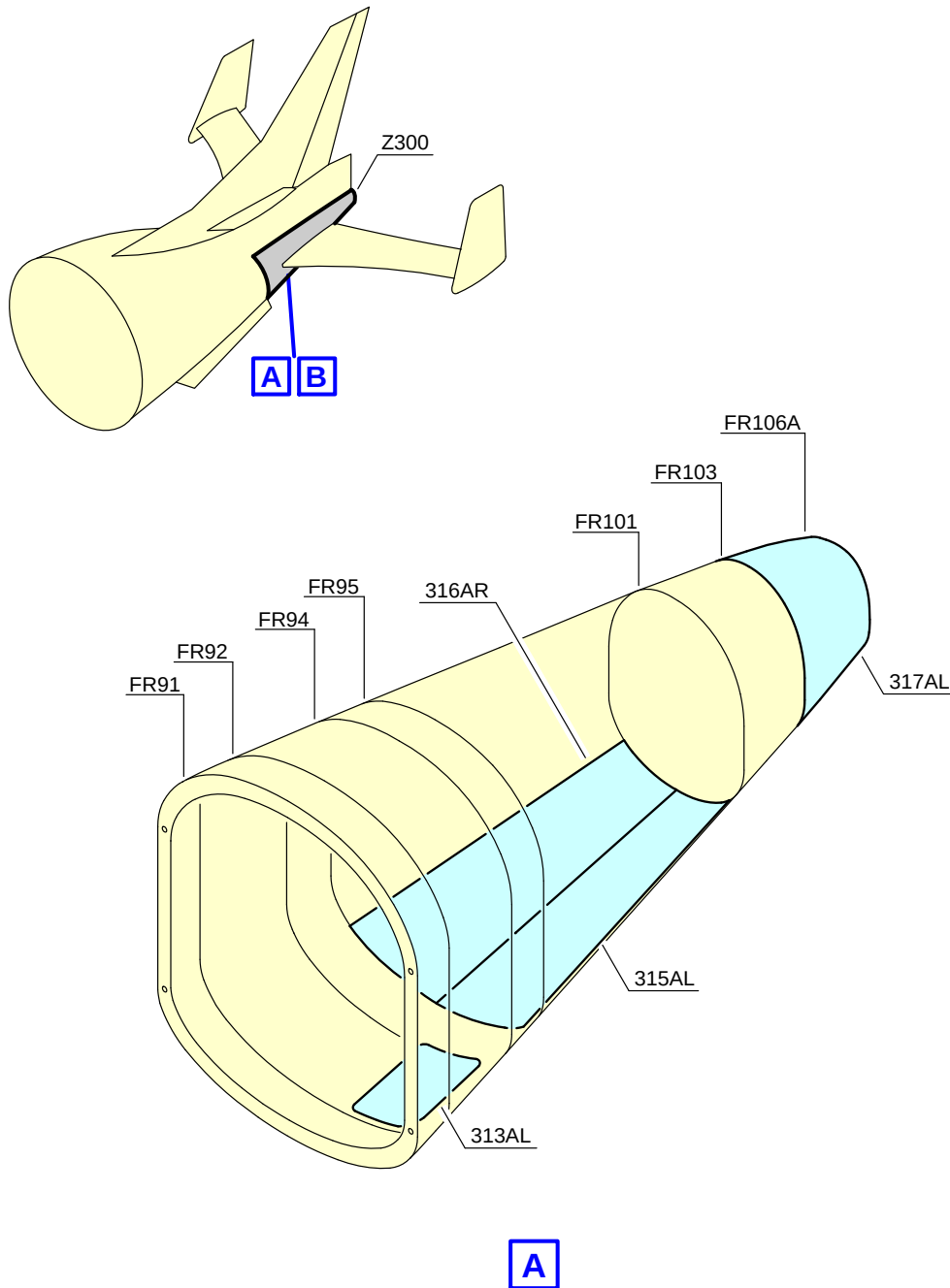
3. Control Circuit

The Electronic Control Box (ECB), which controls the FCU and the Inlet Guide Vanes (IGV), keeps the APU at a constant speed. The control circuit is used to start the APU, to shut it down, to control it and to prevent internal failure.

4. Controls and Indication

The primary APU controls and indications are installed in the overhead panel, on the center pedestal panel and on the forward center panel. External APU panels are also installed on the nose landing gear and on the refuel/defuel panel, to initiate an APU emergency shut-down.

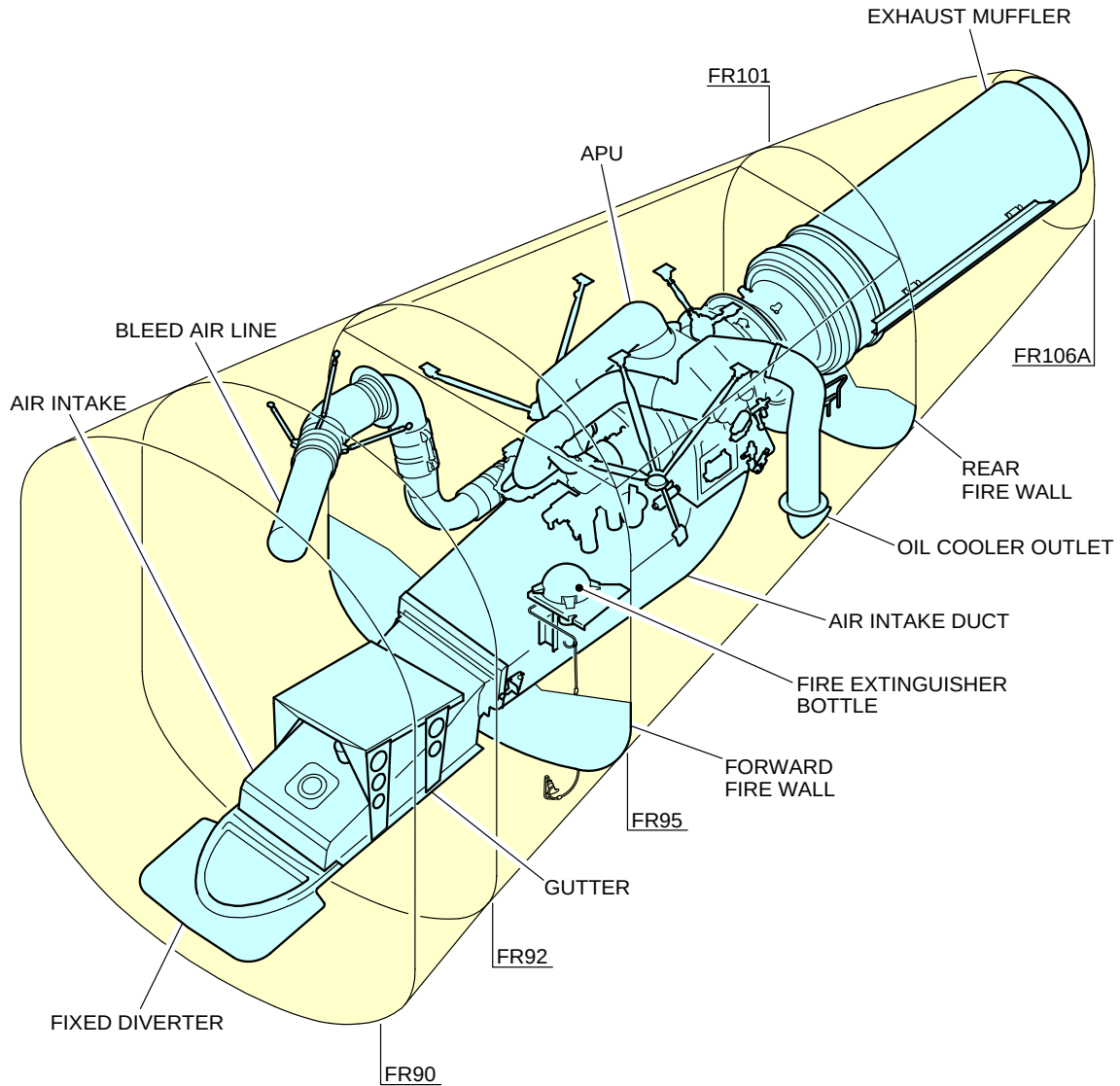
**\*\*ON A/C A330-700L**



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Auxiliary Power Unit  
Access Doors (Sheet 1 of 2)  
FIGURE-2-12-1-991-004-A01

**\*\*ON A/C A330-700L**



**B**

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Auxiliary Power Unit  
General Layout (Sheet 2 of 2)  
FIGURE-2-12-1-991-004-A01

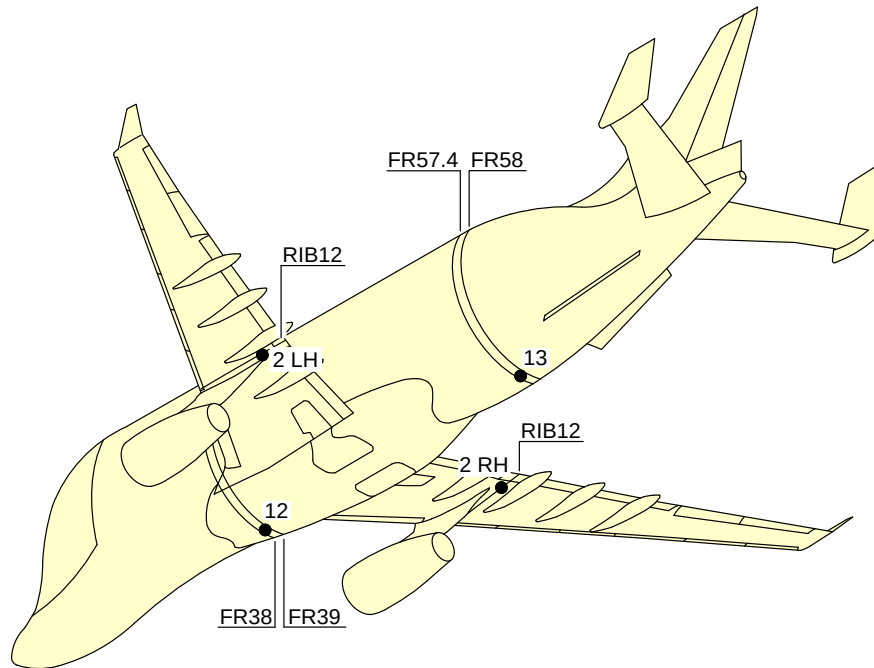
**2-13-0 Levelling, symmetry and Alignment****\*\*ON A/C A330-700L**Leveling and Symmetry**1. Quick Leveling**

The quick leveling procedures are documented in AMM.

**2. Precision Leveling**

For precise leveling, it is necessary to install sighting rods in the receptacles located under the fuselage (points 12 and 13 for longitudinal leveling) and under the wings (points 2LH and 2RH for lateral leveling) and use a sighting tube. With the aircraft on jacks, adjust the jacks until the reference marks on the sighting rods are aligned in the sighting plane (aircraft level).

**\*\*ON A/C A330-700L**



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Location of Leveling Points  
FIGURE-2-13-0-991-008-A01



**2-14-0 Jacking for Maintenance****\*\*ON A/C A330-700L**Jacking for Maintenance**1. Aircraft Jacking Points for Maintenance****A. General**

(1) The A330-700L can be jacked:

- At not more than 152 000 kg (335 103 lb),
- Within the limits of the permissible wind speed when the aircraft is jacked outside a closed environment.

**B. Primary Jacking Points**

(1) The aircraft is provided with three primary jacking points:

- One located under the forward fuselage (after FR11),
- Two located under the wings (one under each wing), at the intersection of RIB10 and the rear of the spar-datum.

(2) Three jack adapters (ground equipment) are used as intermediary parts between the aircraft jacking points and the jacks:

- One female spherical jack adapter at the forward fuselage,
- Two female spherical jack pad adapters at the wings (one at each wing).

**C. Auxiliary Jacking Point (Safety Stay)**

(1) When the aircraft is on jacks, a safety stay is placed under the fuselage at FR87 to prevent tail tipping caused by accidental displacement of the aircraft center of gravity.

(2) The safety point must not be used for lifting the aircraft.

(3) One male spherical stay adapter (ground equipment) is used as an intermediary part between the aircraft safety point and the stay.

**2. Jacks and Safety Stay****A. Jack Design**

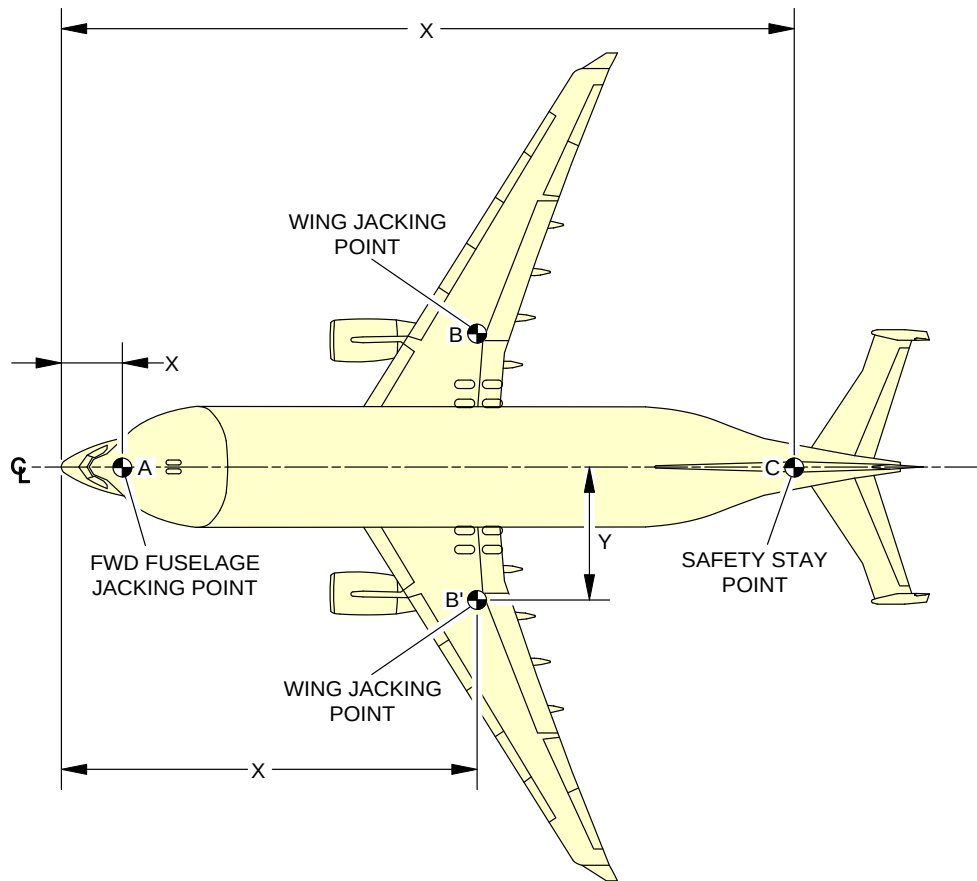
(1) The maximum eligible loads given in the table are the maximum loads applicable on jack fittings.

- (2) In fully retracted position (jack stroke at minimum), the height of the jack is such that the jack may be placed beneath the aircraft under the most adverse conditions, namely, tires deflated and shock absorbers depressurized, with sufficient clearance between the aircraft jacking point and the jack upper end.
- (3) The lifting jack stroke enables the aircraft to be jacked up so that the Fuselage Datum Line (FDL) may be positioned up to 7.2 m (23.62 ft) from the ground to allow all required maintenance procedures and in particular, the removal/installation of the landing-gear shock absorbers.

**B. Safety Stay**

- (1) The stay stroke enables the aircraft tail to be supported up to the FDL positioned 7.2 m (23.62 ft) from the ground.

**\*\*ON A/C A330-700L**



|                                |    | X     |        | Y     |        | MAXIMUM JACKING FORCE |
|--------------------------------|----|-------|--------|-------|--------|-----------------------|
|                                |    | m     | ft     | m     | ft     | daN                   |
| FORWARD FUSELAGE JACKING POINT | A  | 5.3   | 17.39  | 0     | 0      | 12 686.4              |
| WING JACKING POINT             | B  | 29.85 | 97.93  | 8.51  | 27.92  | 71 923.3              |
|                                | B' | 29.85 | 97.93  | -8.51 | -27.92 | 71 923.3              |
| SAFETY STAY                    | C  | 55.17 | 181.00 | 0     | 0      | 4 500                 |

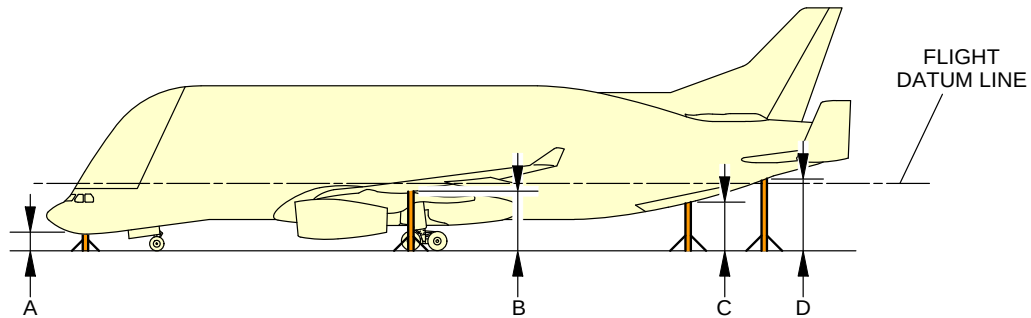
**NOTE:**

SAFETY STAY IS NOT USED FOR JACKING.

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Jacking for Maintenance  
Jacking Points Layout and Maximum Jacking Force  
FIGURE-2-14-0-991-023-A01

## \*\*ON A/C A330-700L



|                                                                                                                                                                                                                                                                                                         | HEIGHT IN m (ft) |                         |                         |                 |                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------------------------|-------------------------|-----------------|-----------------|
|                                                                                                                                                                                                                                                                                                         | A                | B                       |                         | C               | D               |
| AIRCRAFT ON WHEELS WITH STANDARD TIRES <a href="#">01</a> , MAX. JACK WEIGHT<br>152 000 kg (335 103 lb)<br>(BALANCE 18% MEAN AERODYNAMIC CHORD)                                                                                                                                                         | 1.43<br>(4.69)   | 4.62<br>(15.16)         |                         | 3.82<br>(12.53) | 5.27<br>(17.29) |
| AIRCRAFT ON WHEELS WITH STANDARD TIRES <a href="#">01</a> , MAX. JACK WEIGHT<br>152 000 kg (335 103 lb)<br>(BALANCE 34% MEAN AERODYNAMIC CHORD)                                                                                                                                                         | 1.56<br>(5.12)   | 4.60<br>(15.09)         |                         | 3.68<br>(12.07) | 5.09<br>(16.70) |
| AIRCRAFT ON WHEELS WITH STANDARD TIRES <a href="#">01</a> , OEW 127 000 kg (279 987 lb)                                                                                                                                                                                                                 | 1.52<br>(4.99)   | 4.68<br>(15.35)         |                         | 3.87<br>(12.70) | 5.31<br>(17.42) |
| AIRCRAFT ON WHEELS, NOSE LANDING GEAR SHOCK ABSORBER DEFLATED AND FLAT TIRES                                                                                                                                                                                                                            | 1.0<br>(3.28)    | 4.68<br>(15.35)         |                         | 4.29<br>(14.07) | 5.83<br>(19.13) |
| AIRCRAFT ON WHEELS, BOTH MAIN LANDING GEAR SHOCK ABSORBERS DEFLATED AND FLAT TIRES                                                                                                                                                                                                                      | 1.61<br>(5.28)   | 4.10<br>(13.45)         |                         | 2.73<br>(8.96)  | 4.04<br>(13.25) |
| AIRCRAFT ON WHEELS, NOSE LANDING GEAR AND LEFT MAIN LANDING GEAR SHOCK ABSORBERS DEFLATED AND FLAT TIRES                                                                                                                                                                                                | 1.06<br>(3.48)   | (LH)<br>3.96<br>(12.99) | (RH)<br>4.81<br>(15.78) | 3.71<br>(12.17) | 5.18<br>(16.99) |
| AIRCRAFT ON JACKS, FUSELAGE DATUM REFERENCE PARALLEL TO GROUND AT 6.51 m (21.36 ft), SHOCK-ABSORBER RELAXED, CLEARANCE OF MAIN LANDING GEAR WHEELS: 0.24 m (0.79 ft) (STANDARD TIRES <a href="#">01</a> ), CLEARANCE OF NOSE LANDING GEAR WHEELS: 0.96 m (3.15 ft) (STANDARD TIRES <a href="#">01</a> ) | 2.66<br>(8.73)   | 5.81<br>(19.06)         |                         | 4.98<br>(16.34) | 6.42<br>(21.06) |

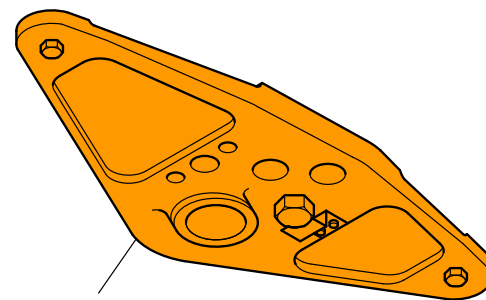
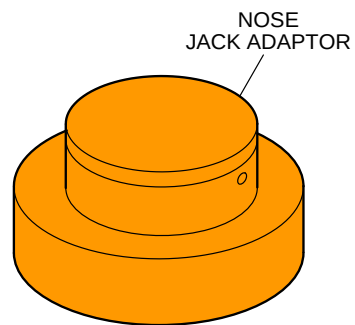
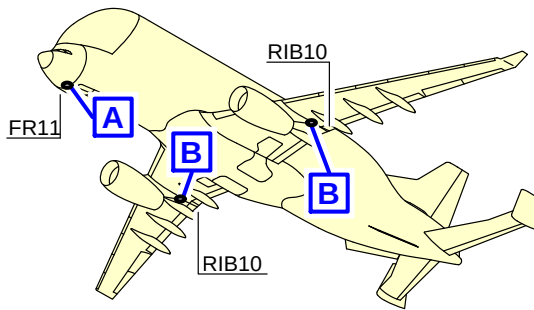
### NOTE:

[01](#) STANDARD TIRES: NOSE LANDING GEAR = 1 050 x 395 R16  
MAIN LANDING GEAR = 1 400 x 530 R23

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Jacking for Maintenance  
Ground Clearance on Jacks  
FIGURE-2-14-0-991-024-A01

**\*\*ON A/C A330-700L**



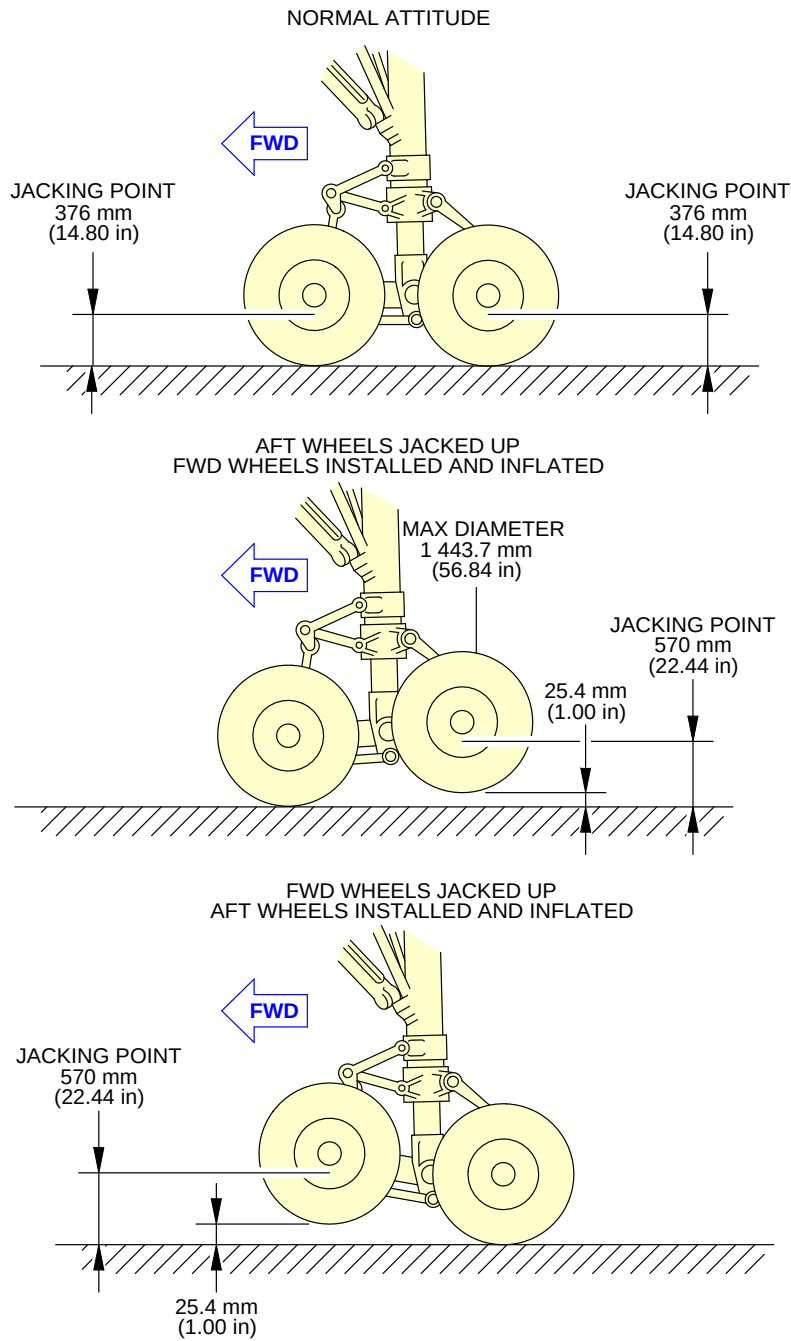
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Jacking for Maintenance  
Jacking Point Adaptors  
FIGURE-2-14-0-991-025-A01

**2-14-1 Jacking of the Landing Gear****\*\*ON A/C A330-700L**Jacking of the Landing Gear

1. Not applicable.

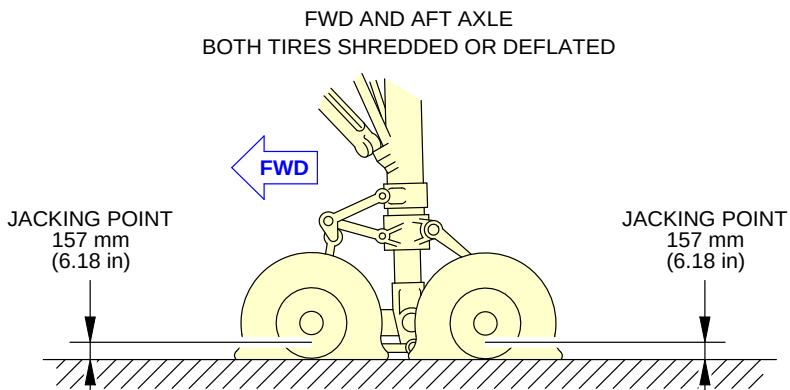
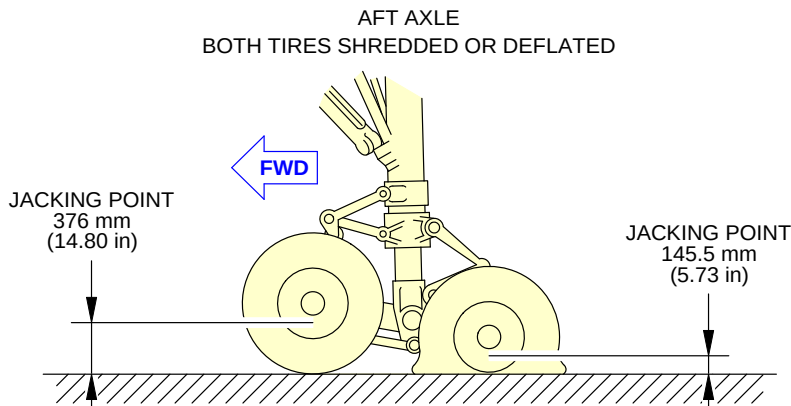
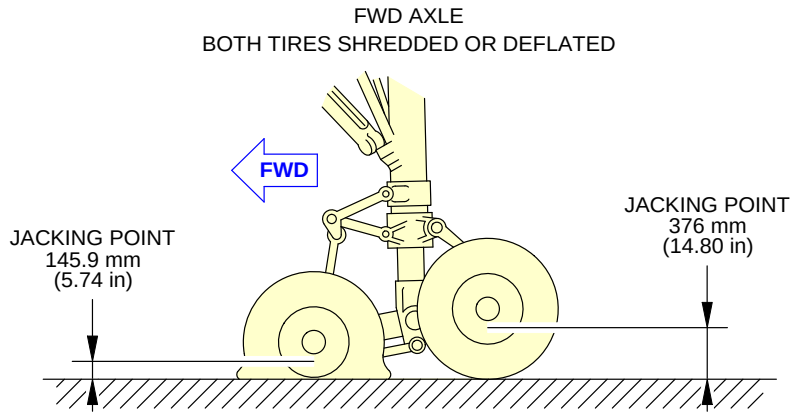
**\*\*ON A/C A330-700L**



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Jacking of the Landing Gear  
Jacking of the Landing Gear (Sheet 1 of 3)  
FIGURE-2-14-1-991-027-A01

**\*\*ON A/C A330-700L**

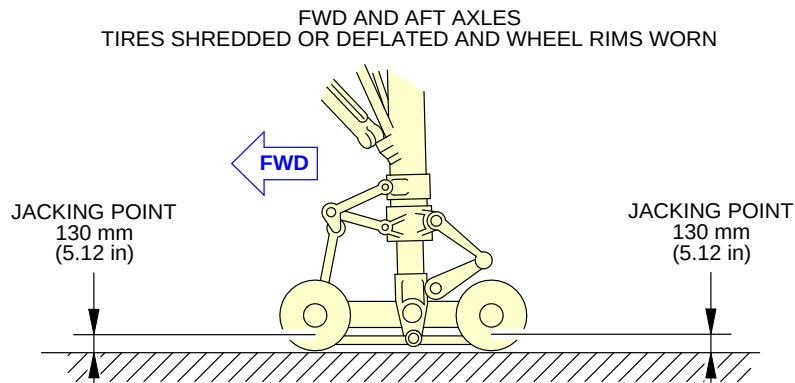
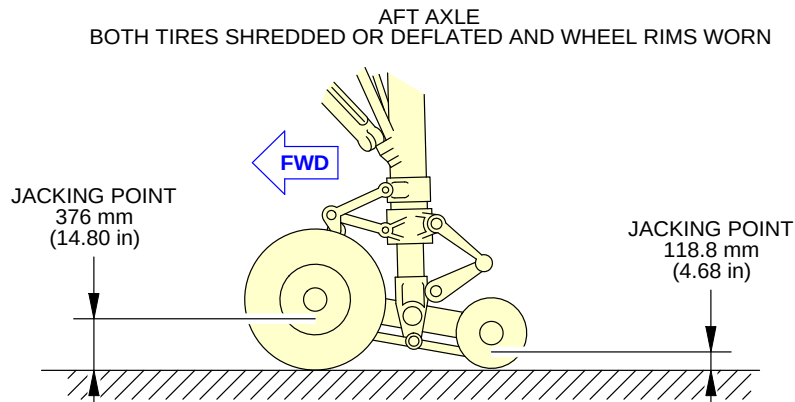
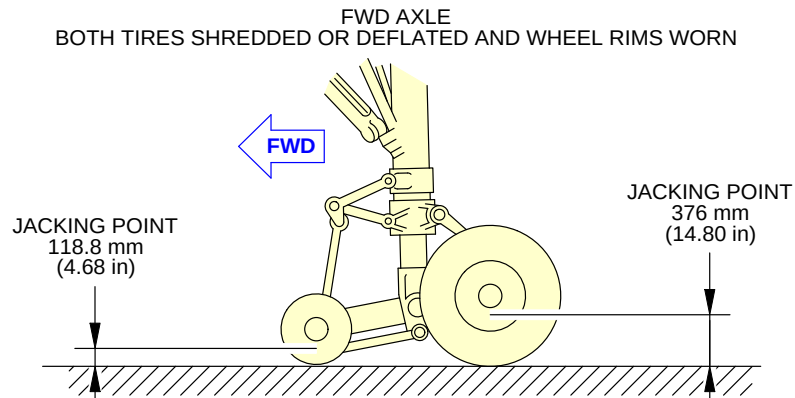


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Jacking of the Landing Gear  
Jacking of the Landing Gear (Sheet 2 of 3)  
FIGURE-2-14-1-991-027-A01



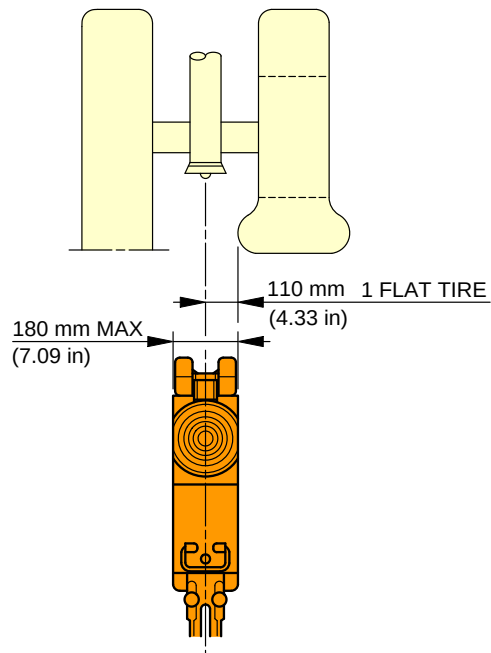
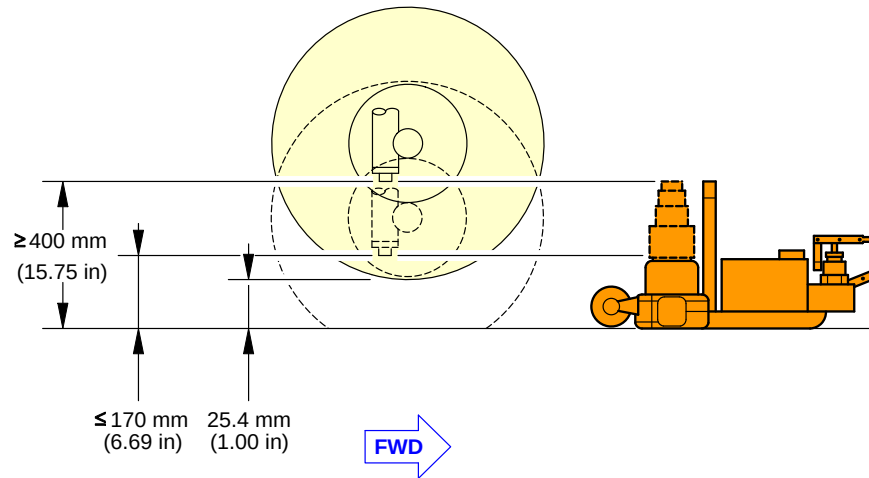
**\*\*ON A/C A330-700L**



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Jacking of the Landing Gear  
Jacking of the Landing Gear (Sheet 3 of 3)  
FIGURE-2-14-1-991-027-A01

**\*\*ON A/C A330-700L**



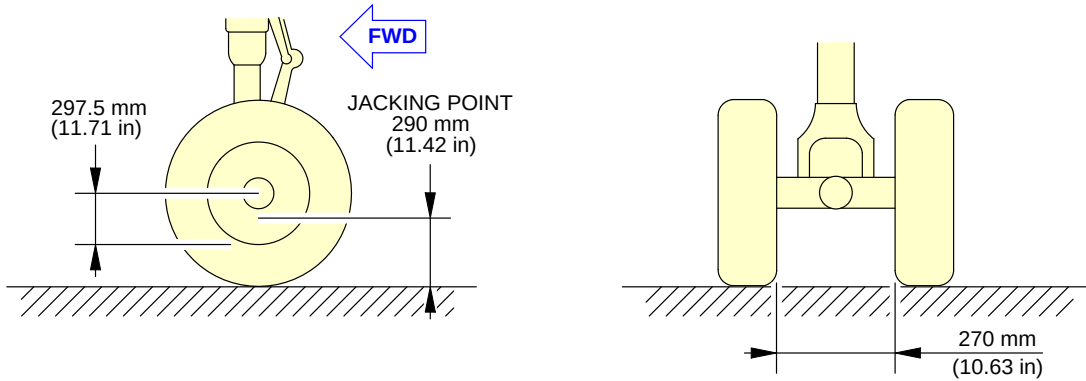
F\_AC\_021401\_1\_0280101\_01\_00

Jacking of the Landing Gear  
(Sheet 1 of 2)

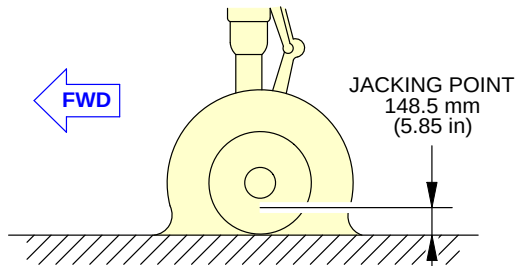
FIGURE-2-14-1-991-028-A01

**\*\*ON A/C A330-700L**

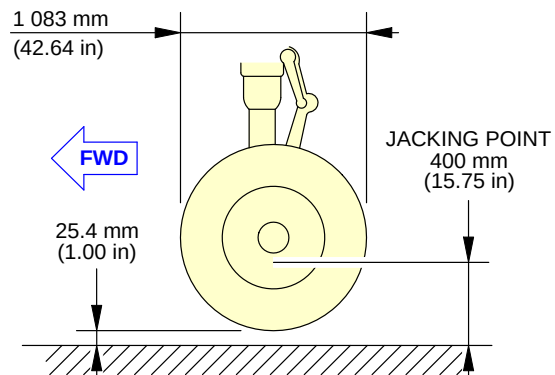
NORMAL ATTITUDE



TWO TIRES DEFLATED OR SHREDDED



HEIGHT OF JACKING POINT TO GROUND  
TO CHANGE/REPLACE THE WHEEL ASSEMBLY



F\_AC\_021401\_1\_0280102\_01\_00

Jacking of the Landing Gear  
(Sheet 2 of 2)

FIGURE-2-14-1-991-028-A01

**AIRCRAFT PERFORMANCE****3-1-0      General Information****\*\*ON A/C A330-700L**General Information

1. Data not published.

**3-3-3 Aerodrome Reference Code****\*\*ON A/C A330-700L**Aerodrome Reference Code

1. A330-700L can operates on aerodromes classified as code 4E as per ICAO Aerodrome Reference Code.

**GROUND MANEUVERING****4-1-0 General Information****\*\*ON A/C A330-700L****General Information**

1. This section gives aircraft turning capability and maneuvering characteristics.

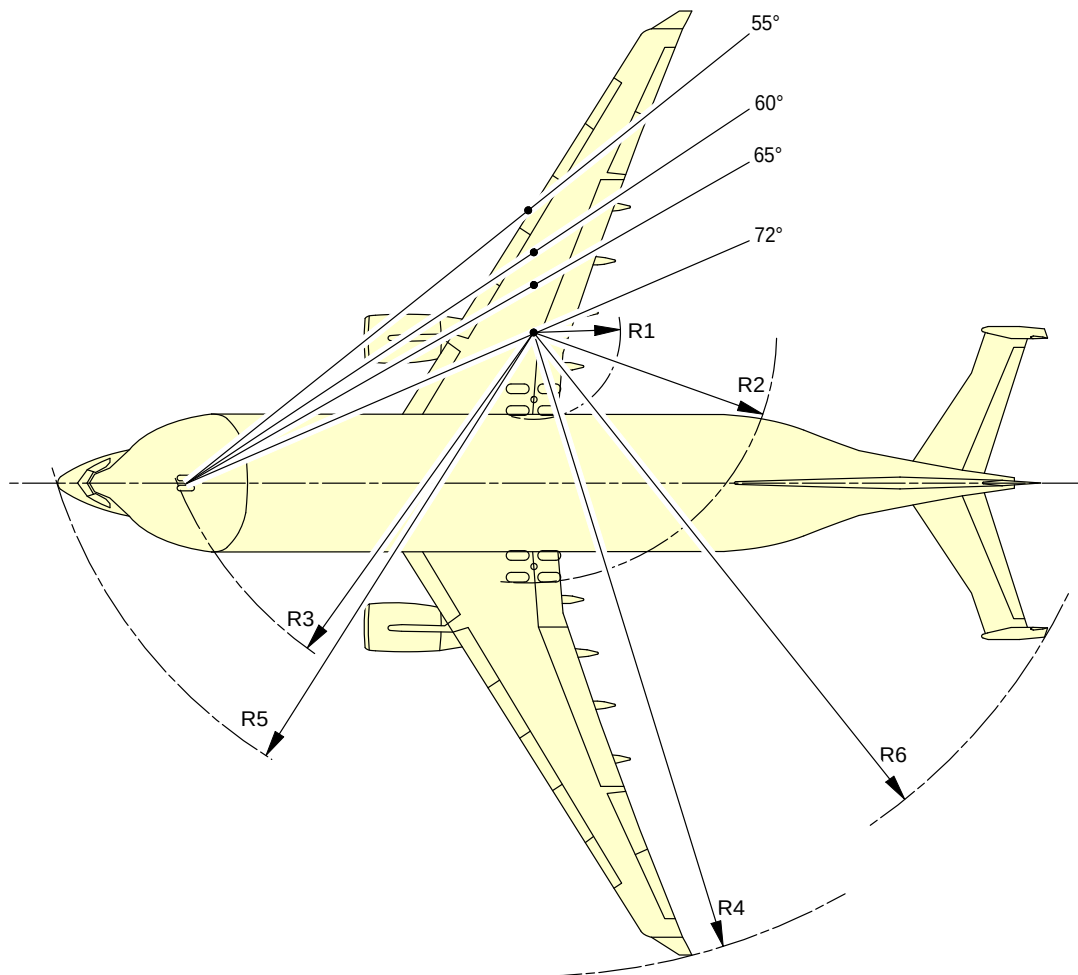
For ease of presentation, this data has been determined from the theoretical limits imposed by the geometry of the aircraft, and where noted, provides for a normal allowance for tire slippage. As such, it reflects the turning capability of the aircraft in favorable operating circumstances. This data should only be used as guidelines for the method of determination of such parameters and for the maneuvering characteristics of this aircraft type.

In the ground operating mode, varying airline practices may demand that more conservative turning procedures be adopted to avoid excessive tire wear and reduce possible maintenance problems. Airline operating techniques will vary in the level of performance, over a wide range of operating circumstances throughout the world. Variations from standard aircraft operating patterns may be necessary to satisfy physical constraints within the maneuvering area, such as adverse grades, limited area or high risk of jet blast damage. For these reasons, ground maneuvering requirements should be coordinated with the airlines in question prior to layout planning.

**4-2-0      Turning Radii****\*\*ON A/C A330-700L**Turning Radii

1. This section gives the turning radii.

**\*\*ON A/C A330-700L**



**NOTE:**  
FOR TURNING RADII VALUES, REFER TO SHEET 2.

F\_AC\_040200\_1\_0180101\_01\_00

Turning Radii  
(Sheet 1)  
FIGURE-4-2-0-991-018-A01



## \*\*ON A/C A330-700L

| BELUGA XL TURNING RADII |                      |                                |    |         |         |        |         |         |         |
|-------------------------|----------------------|--------------------------------|----|---------|---------|--------|---------|---------|---------|
| TYPE OF TURN            | STEERING ANGLE (deg) | EFFECTIVE STEERING ANGLE (deg) |    | R1 RMLG | R2 LMLG | R3 NLG | R4 WING | R5 NOSE | R6 TAIL |
| 2                       | 20                   | 19.2                           | m  | 59.3    | 70.0    | 68.0   | 94.4    | 70.6    | 80.4    |
|                         |                      |                                | ft | 195     | 230     | 223    | 310     | 232     | 264     |
| 2                       | 25                   | 23.9                           | m  | 45.7    | 56.4    | 55.3   | 80.9    | 58.5    | 68.2    |
|                         |                      |                                | ft | 150     | 185     | 181    | 265     | 192     | 224     |
| 2                       | 30                   | 28.6                           | m  | 36.3    | 47.0    | 46.8   | 71.6    | 50.8    | 60.1    |
|                         |                      |                                | ft | 119     | 154     | 154    | 235     | 167     | 197     |
| 2                       | 35                   | 33.3                           | m  | 29.4    | 40.1    | 40.9   | 64.7    | 45.4    | 54.4    |
|                         |                      |                                | ft | 96      | 132     | 134    | 212     | 149     | 178     |
| 2                       | 40                   | 38.0                           | m  | 24.0    | 34.7    | 36.5   | 59.4    | 41.6    | 50.2    |
|                         |                      |                                | ft | 79      | 114     | 120    | 195     | 136     | 165     |
| 2                       | 45                   | 42.5                           | m  | 19.8    | 30.5    | 33.2   | 55.3    | 38.8    | 47.1    |
|                         |                      |                                | ft | 65      | 100     | 109    | 181     | 127     | 155     |
| 2                       | 50                   | 46.9                           | m  | 16.4    | 27.1    | 30.8   | 51.9    | 36.8    | 44.7    |
|                         |                      |                                | ft | 54      | 89      | 101    | 170     | 121     | 147     |
| 2                       | 55                   | 51.2                           | m  | 13.5    | 24.1    | 28.8   | 49.1    | 35.2    | 42.7    |
|                         |                      |                                | ft | 44      | 79      | 94     | 161     | 115     | 140     |
| 2                       | 60                   | 55.1                           | m  | 11.1    | 21.8    | 27.4   | 46.8    | 34.1    | 41.3    |
|                         |                      |                                | ft | 36      | 72      | 90     | 154     | 112     | 135     |
| 2                       | 65                   | 59.6                           | m  | 8.6     | 19.3    | 26.0   | 44.4    | 33      | 39.8    |
|                         |                      |                                | ft | 28      | 63      | 85     | 146     | 108     | 131     |
| 2                       | 72                   | 62.0                           | m  | 7.4     | 18.1    | 25.4   | 43.2    | 32.6    | 39.1    |
|                         |                      |                                | ft | 24      | 59      | 83     | 142     | 107     | 128     |
| 1                       | 50                   | 48.4                           | m  | 15.3    | 26.0    | 30.0   | 50.9    | 36.2    | 43.7    |
|                         |                      |                                | ft | 50      | 85      | 98     | 167     | 119     | 143     |
| 1                       | 55                   | 52.2                           | m  | 12.8    | 23.5    | 28.4   | 48.5    | 34.9    | 42.4    |
|                         |                      |                                | ft | 42      | 77      | 93     | 159     | 115     | 139     |
| 1                       | 60                   | 57.7                           | m  | 9.6     | 20.3    | 26.5   | 45.4    | 33.4    | 40.4    |
|                         |                      |                                | ft | 31      | 67      | 87     | 149     | 110     | 133     |
| 1                       | 65                   | 62.2                           | m  | 7.3     | 18.0    | 25.3   | 43.1    | 32.5    | 39      |
|                         |                      |                                | ft | 24      | 59      | 83     | 141     | 107     | 128     |
| 1                       | 72                   | 68.1                           | m  | 4.5     | 15.2    | 24.1   | 40.4    | 31.6    | 37.6    |
|                         |                      |                                | ft | 15      | 50      | 79     | 133     | 104     | 123     |

### NOTE:

ABOVE 50°, AIRLINES MAY USE TYPE 1 OR TYPE 2 TURNS DEPENDING ON THE SITUATION.

TYPE 1 TURNS USE:

ASYMMETRIC THRUST DURING THE WHOLE TURN AND  
DIFFERENTIAL BRAKING TO INITIATE THE TURN ONLY.

TYPE 2 TURNS USE:

SYMMETRIC THRUST DURING THE WHOLE TURN AND NO DIFFERENTIAL BRAKING AT ALL.

IT IS POSSIBLE TO GET LOWER VALUES THAN THOSE FROM TYPE 1 BY APPLYING  
DIFFERENTIAL BRAKING DURING THE WHOLE TURN.

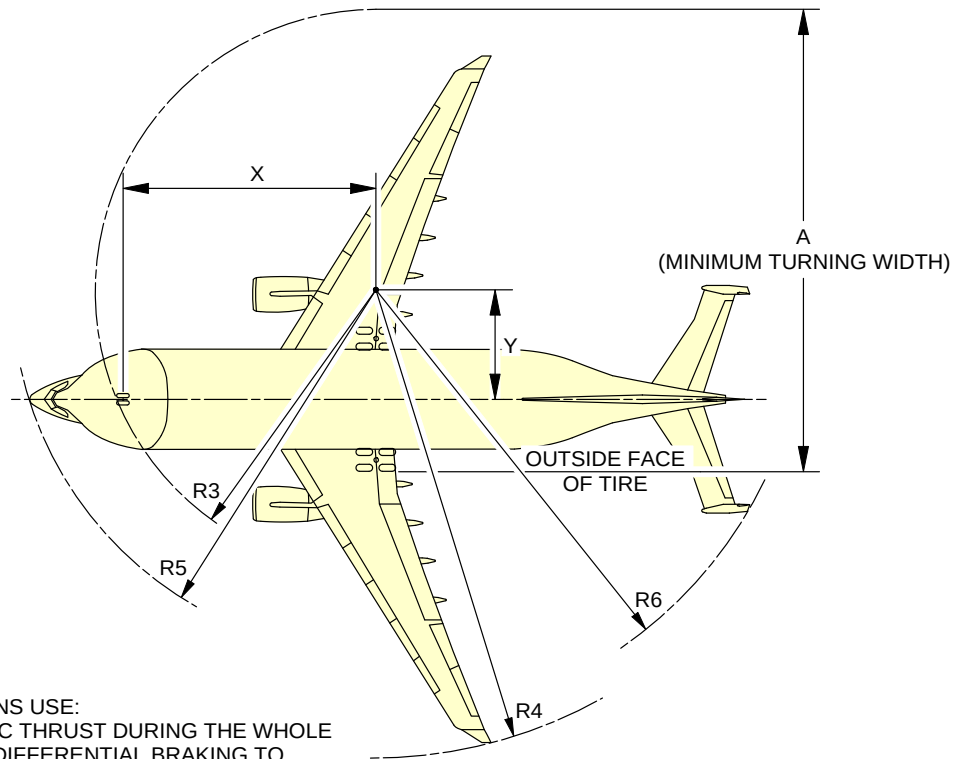
F\_AC\_040200\_1\_0190101\_01\_00

Turning Radii  
(Sheet 2)  
FIGURE-4-2-0-991-019-A01

**4-3-0 Minimum Turning Radii****\*\*ON A/C A330-700L**Minimum Turning Radii

1. This section gives the minimum turning radii.

**\*\*ON A/C A330-700L**



**NOTE:**

TYPE 1 TURNS USE:  
ASYMMETRIC THRUST DURING THE WHOLE  
TURN; AND DIFFERENTIAL BRAKING TO  
INITIATE THE TURN ONLY.  
TYPE 2 TURNS USE:  
SYMMETRIC THRUST DURING THE WHOLE  
TURN; AND NO DIFFERENTIAL BRAKING AT ALL.

| BELUGA XL MINIMUM TURNING RADII |                         |                                   |    |      |      |      |           |            |            |            |
|---------------------------------|-------------------------|-----------------------------------|----|------|------|------|-----------|------------|------------|------------|
| TYPE OF<br>TURN                 | STEERING<br>ANGLE (deg) | EFFECTIVE<br>STEERING ANGLE (deg) |    | X    | Y    | A    | R3<br>NLG | R4<br>WING | R5<br>NOSE | R6<br>TAIL |
| 1                               | 72 (MAX)                | 68.1                              | m  | 22.2 | 8.9  | 39.7 | 24.1      | 40.4       | 31.6       | 37.6       |
|                                 |                         |                                   | ft | 73   | 29   | 130  | 79        | 133        | 104        | 123        |
| 2                               | 72 (MAX)                | 62.0                              | m  | 22.2 | 11.8 | 43.8 | 25.4      | 43.2       | 32.6       | 39.1       |
|                                 |                         |                                   | ft | 73   | 39   | 144  | 83        | 142        | 107        | 128        |
| 1                               | 65 (MAX)                | 62.2                              | m  | 22.2 | 11.7 | 43.6 | 25.3      | 43.1       | 32.5       | 39         |
|                                 |                         |                                   | ft | 73   | 38   | 143  | 83        | 141        | 107        | 128        |
| 2                               | 65 (MAX)                | 59.6                              | m  | 22.2 | 13   | 45.6 | 26        | 44.4       | 33         | 39.8       |
|                                 |                         |                                   | ft | 73   | 43   | 150  | 85        | 146        | 108        | 131        |

**NOTE:**

IT IS POSSIBLE TO GET LOWER VALUES THAN THOSE FROM TYPE 1  
BY APPLYING DIFFERENTIAL BRAKING DURING THE WHOLE TURN.

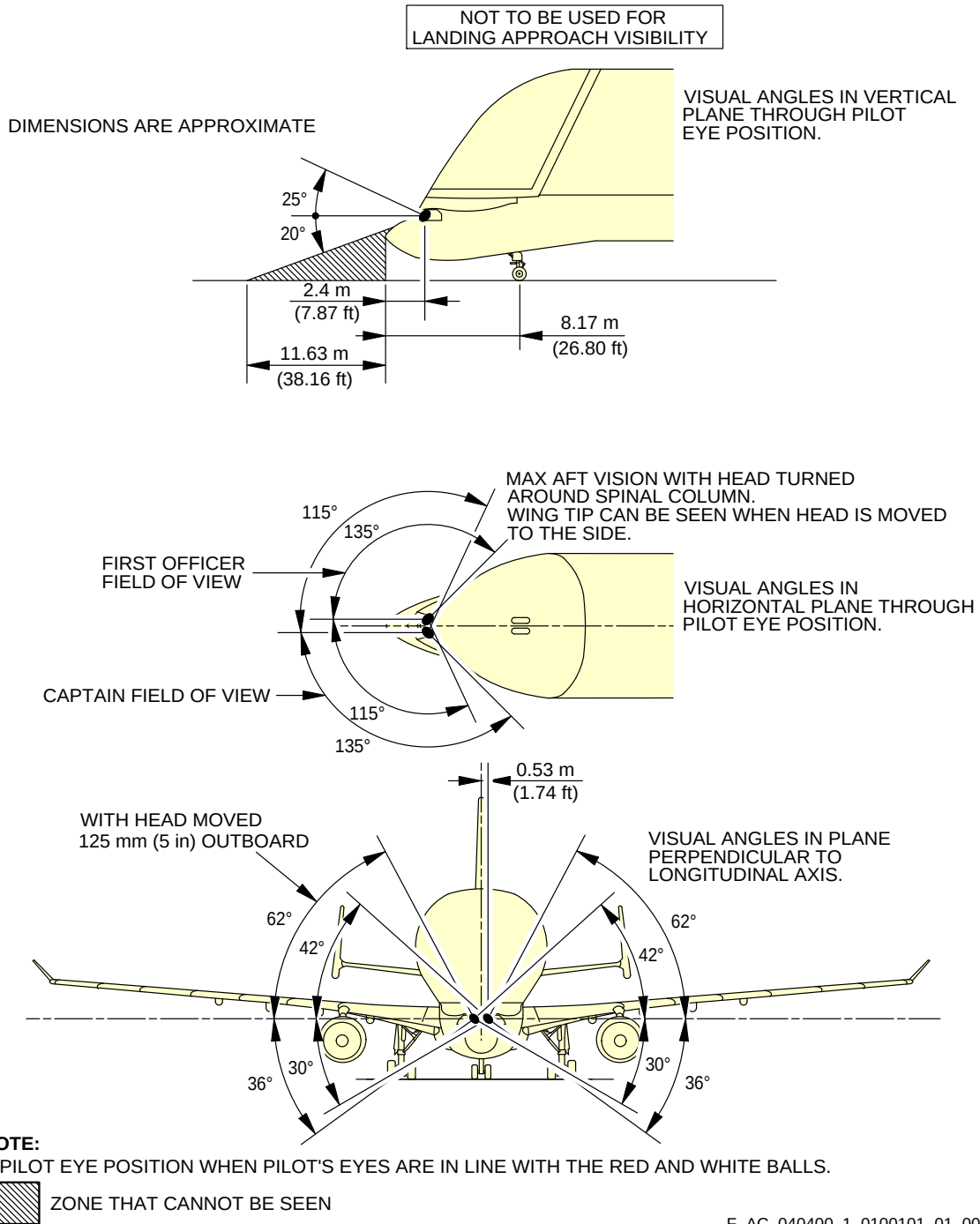
F\_AC\_040300\_1\_0100101\_01\_00

Minimum Turning Radii  
FIGURE-4-3-0-991-010-A01

**4-4-0      Visibility from Cockpit in Static Position****\*\*ON A/C A330-700L**Visibility from Cockpit in Static Position

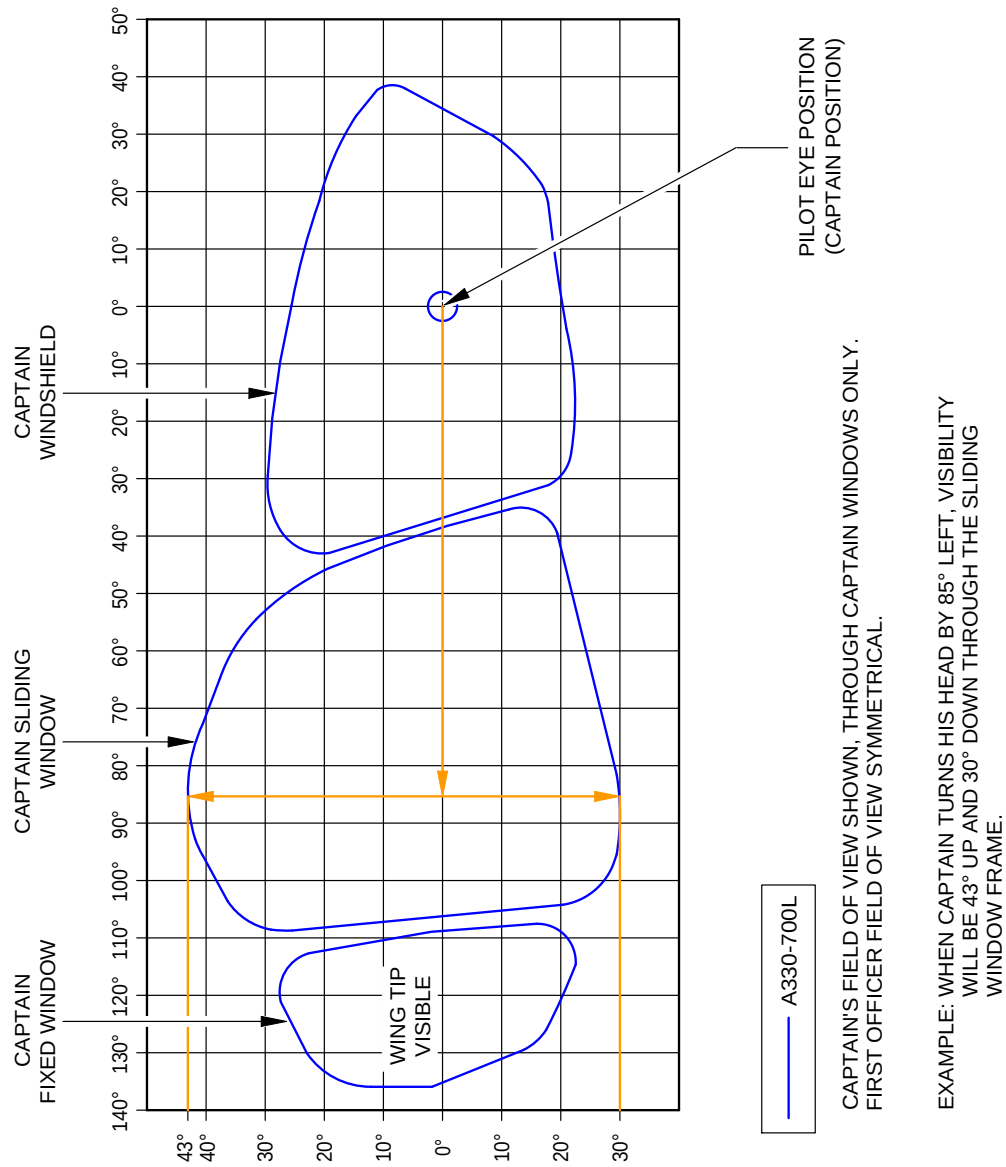
1. This section gives the visibility from cockpit in static position.

**\*\*ON A/C A330-700L**



Visibility from Cockpit in Static Position  
FIGURE-4-4-0-991-010-A01

**\*\*ON A/C A330-700L**



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Binocular Visibility Through Windows from Captain Eye Position  
FIGURE-4-4-0-991-011-A01

**4-5-0 Runway and Taxiway Turn Paths****\*\*ON A/C A330-700L**Runway and Taxiway Turn Paths

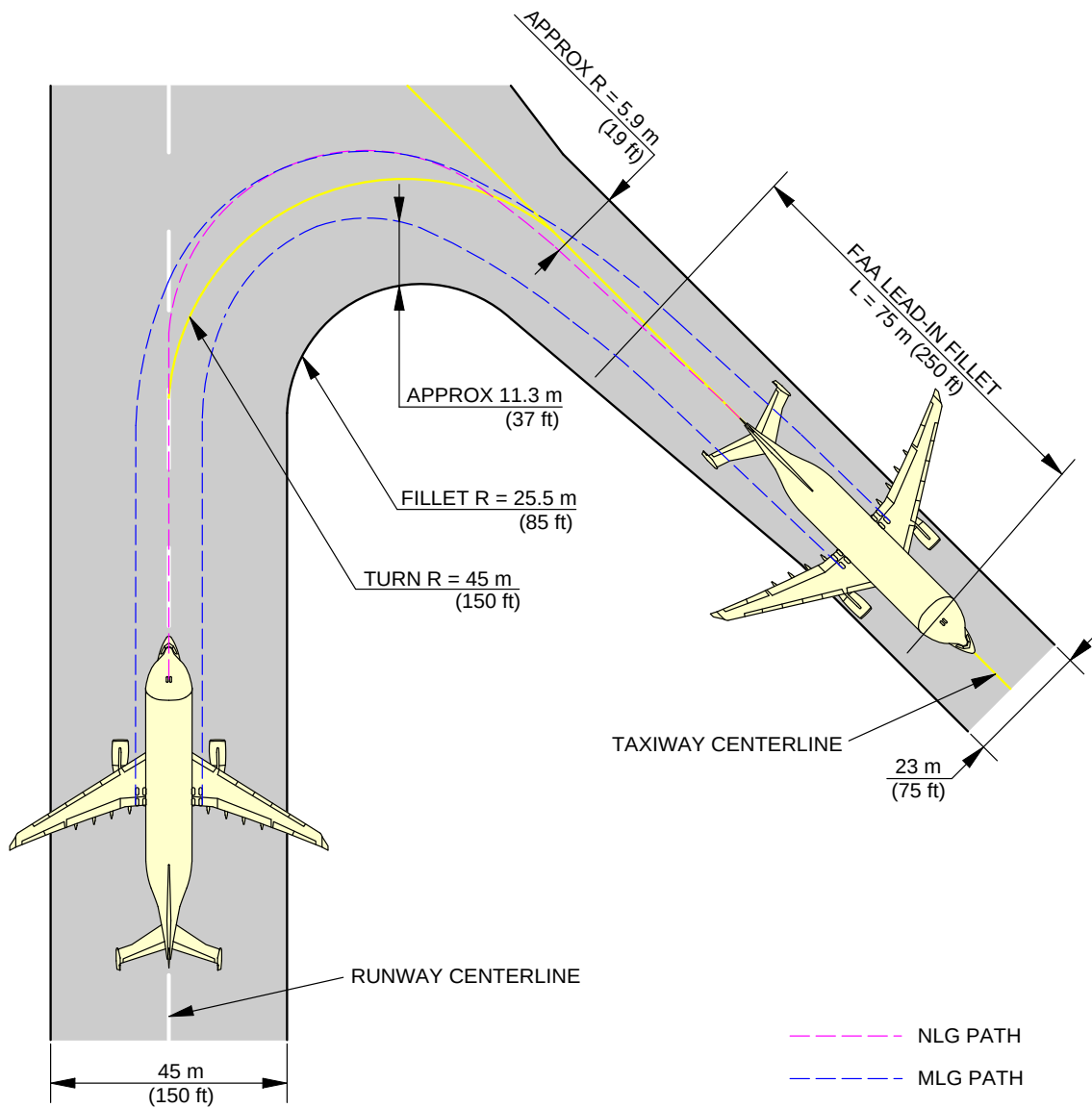
1. Runway and Taxiway Turn Paths.

**4-5-1      135° Turn - Runway to Taxiway****\*\*ON A/C A330-700L**135° Turn - Runway to Taxiway

1. This section gives the 135° turn - runway to taxiway.



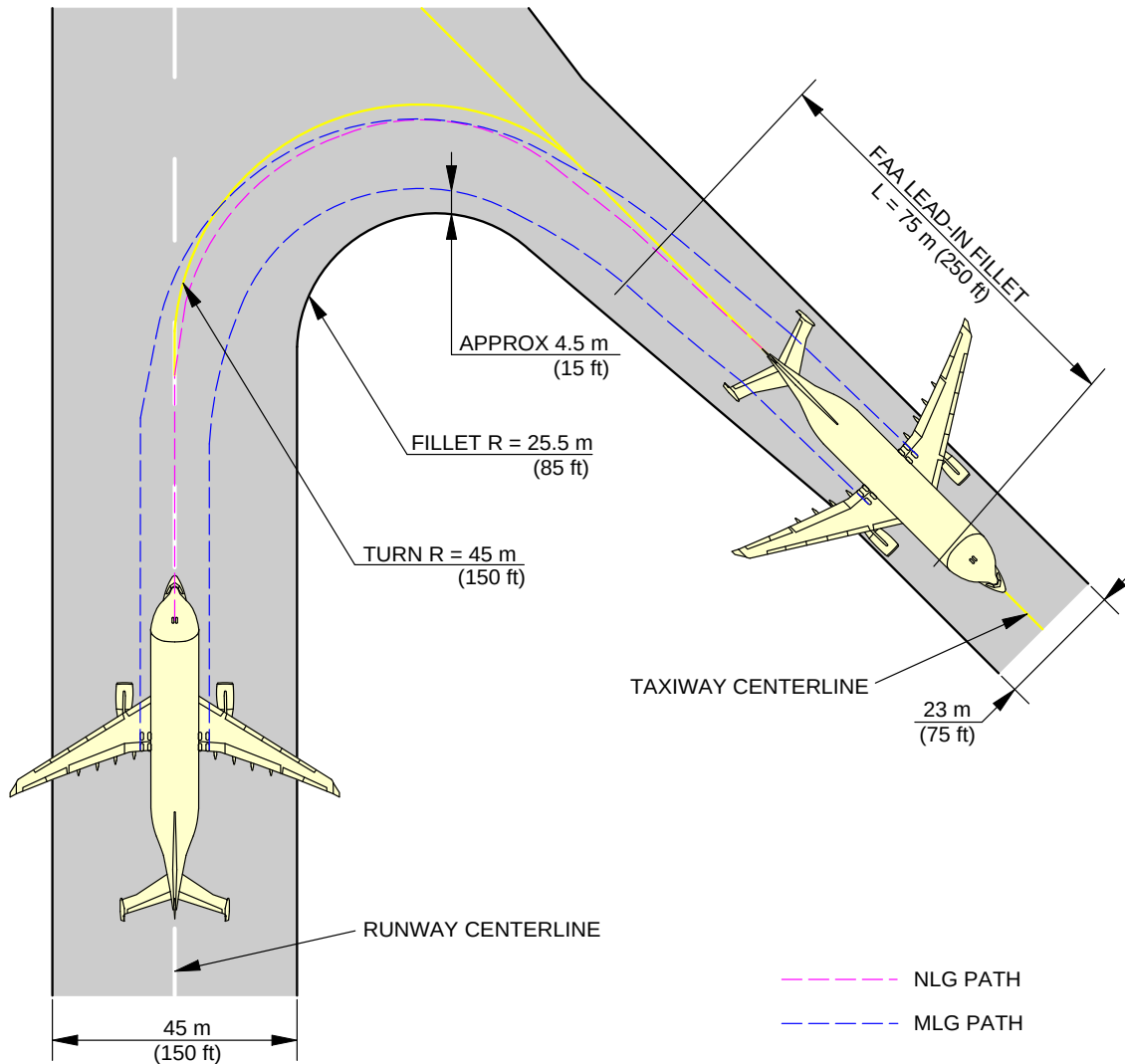
**\*\*ON A/C A330-700L**



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135° Turn - Runway to Taxiway  
Judgemental Oversteer Method  
FIGURE-4-5-1-991-013-A01

**\*\*ON A/C A330-700L**



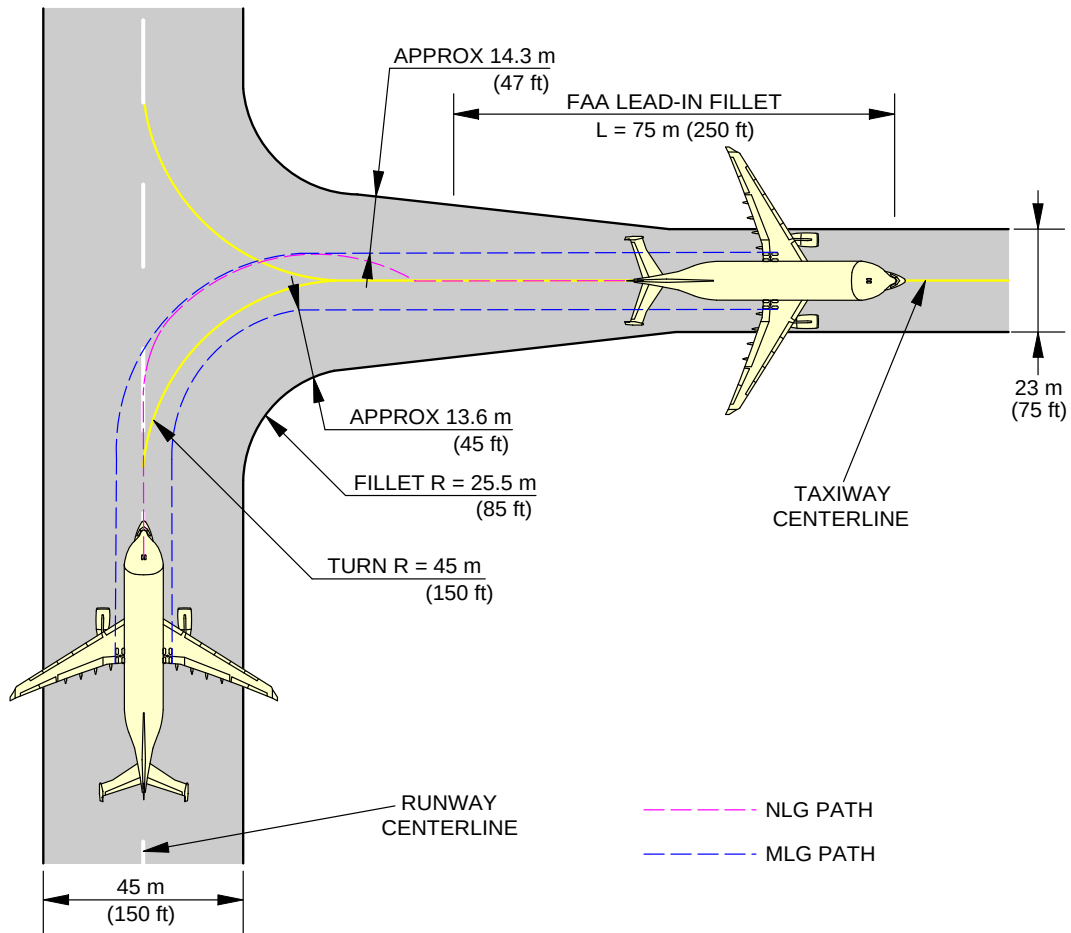
F\_AC\_040501\_1\_0140101\_01\_00

135° Turn - Runway to Taxiway  
Cockpit Over Centerline Method  
FIGURE-4-5-1-991-014-A01

**4-5-2      90° Turn - Runway to Taxiway****\*\*ON A/C A330-700L****90° Turn - Runway to Taxiway**

1. This section gives the 90° turn - runway to taxiway.

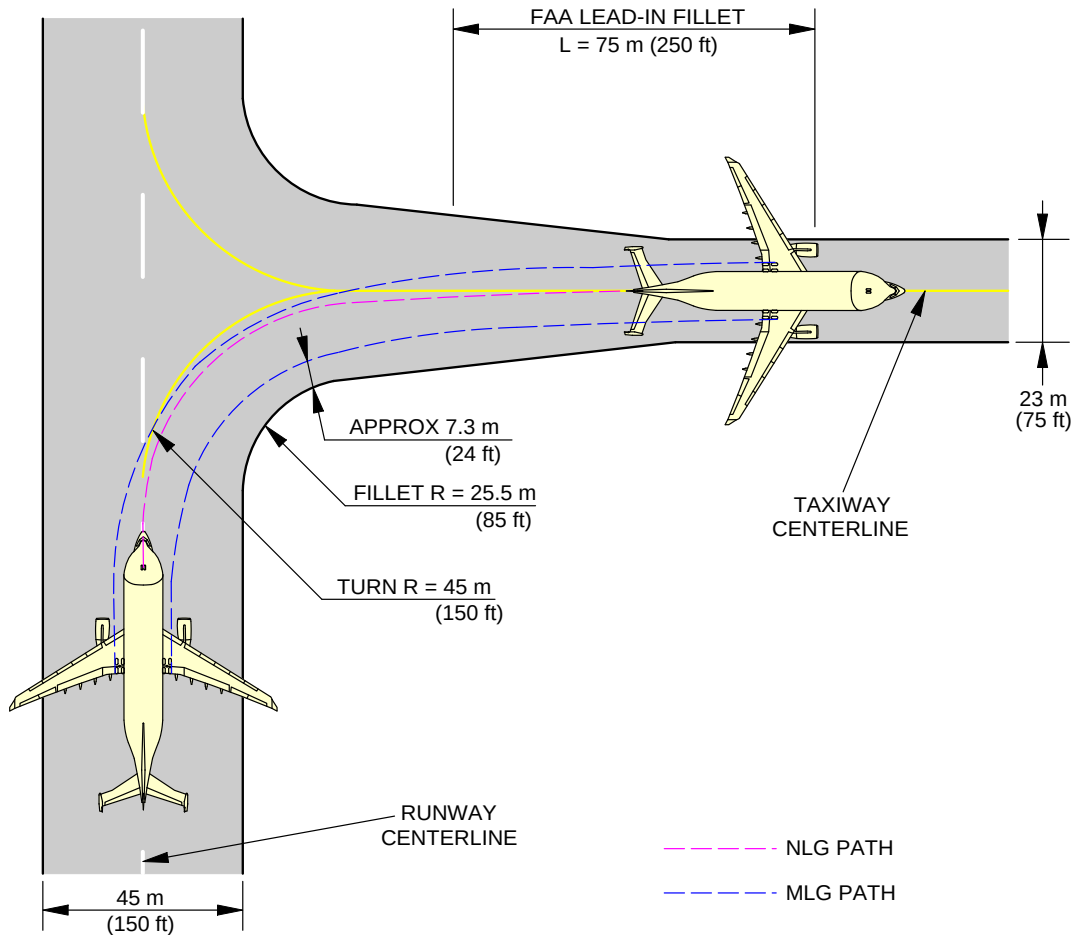
**\*\*ON A/C A330-700L**



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90° Turn - Runway to Taxiway  
Judgemental Oversteer Method  
FIGURE-4-5-2-991-015-A01

**\*\*ON A/C A330-700L**



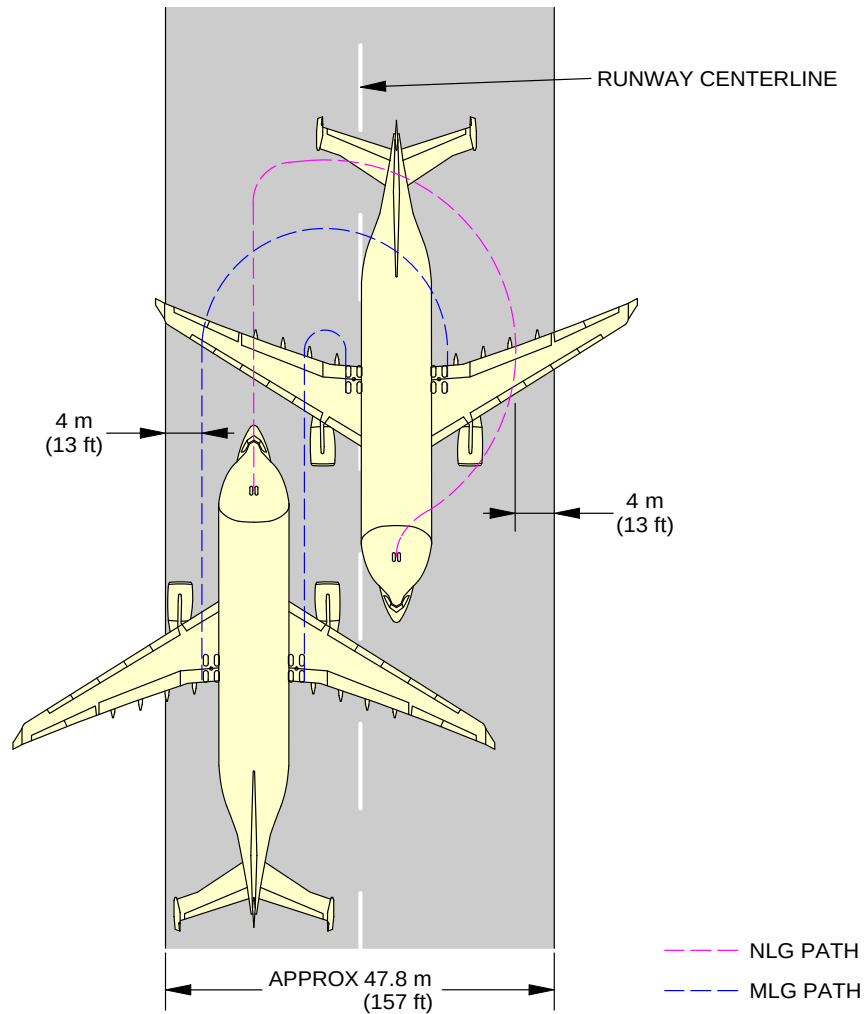
F\_AC\_040502\_1\_0160101\_01\_00

90° Turn - Runway to Taxiway  
Cockpit Over Centerline Method  
FIGURE-4-5-2-991-016-A01

**4-5-3      180° Turn on a Runway****\*\*ON A/C A330-700L****180° Turn on a Runway**

1. This section gives the 180° turn on a runway .

**\*\*ON A/C A330-700L**



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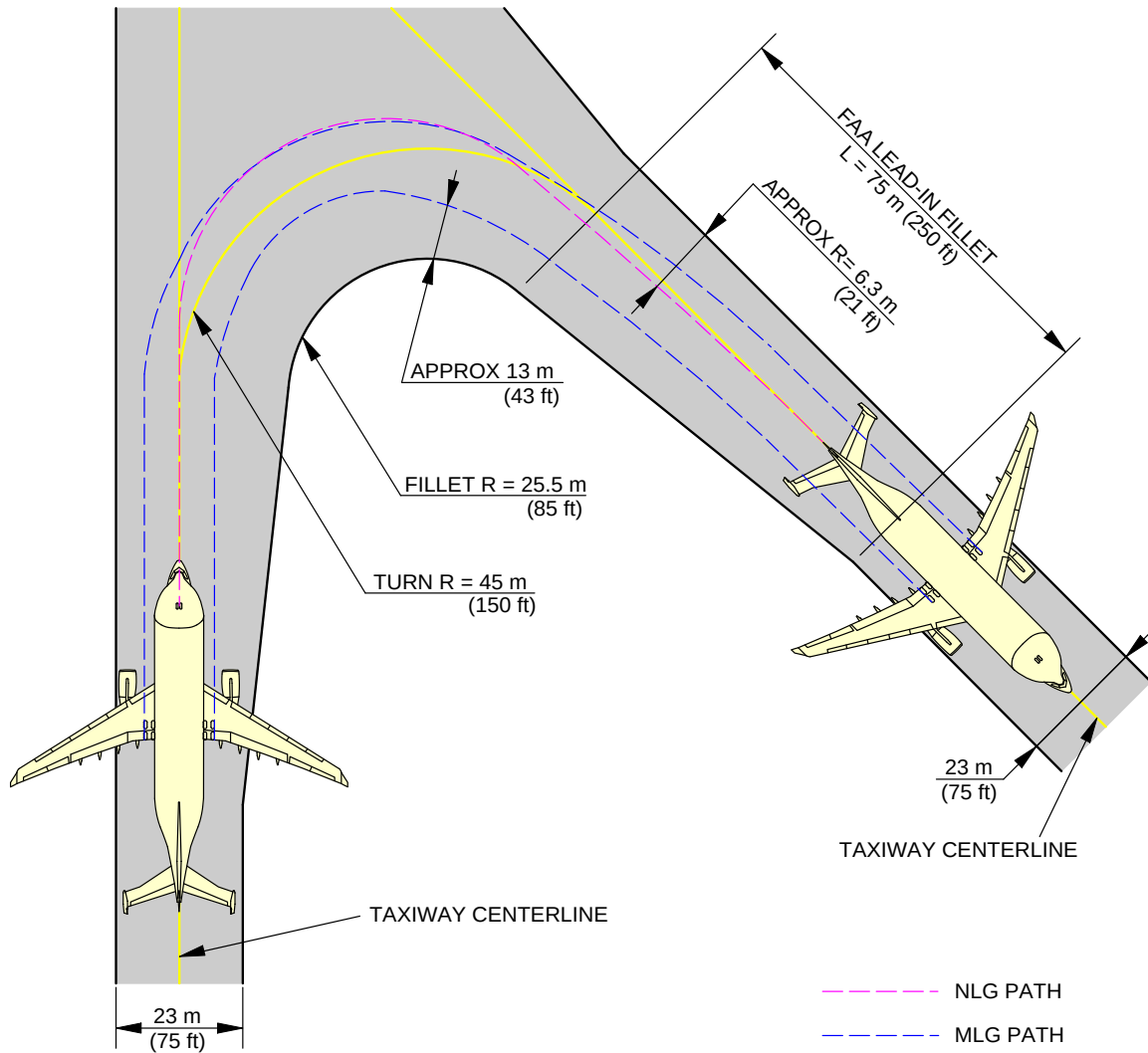
180° Turn on a Runway  
FIGURE-4-5-3-991-017-A01

**4-5-4      135° Turn - Taxiway to Taxiway****\*\*ON A/C A330-700L**135° Turn - Taxiway to Taxiway

1. This section gives the 135° turn - taxiway to taxiway.



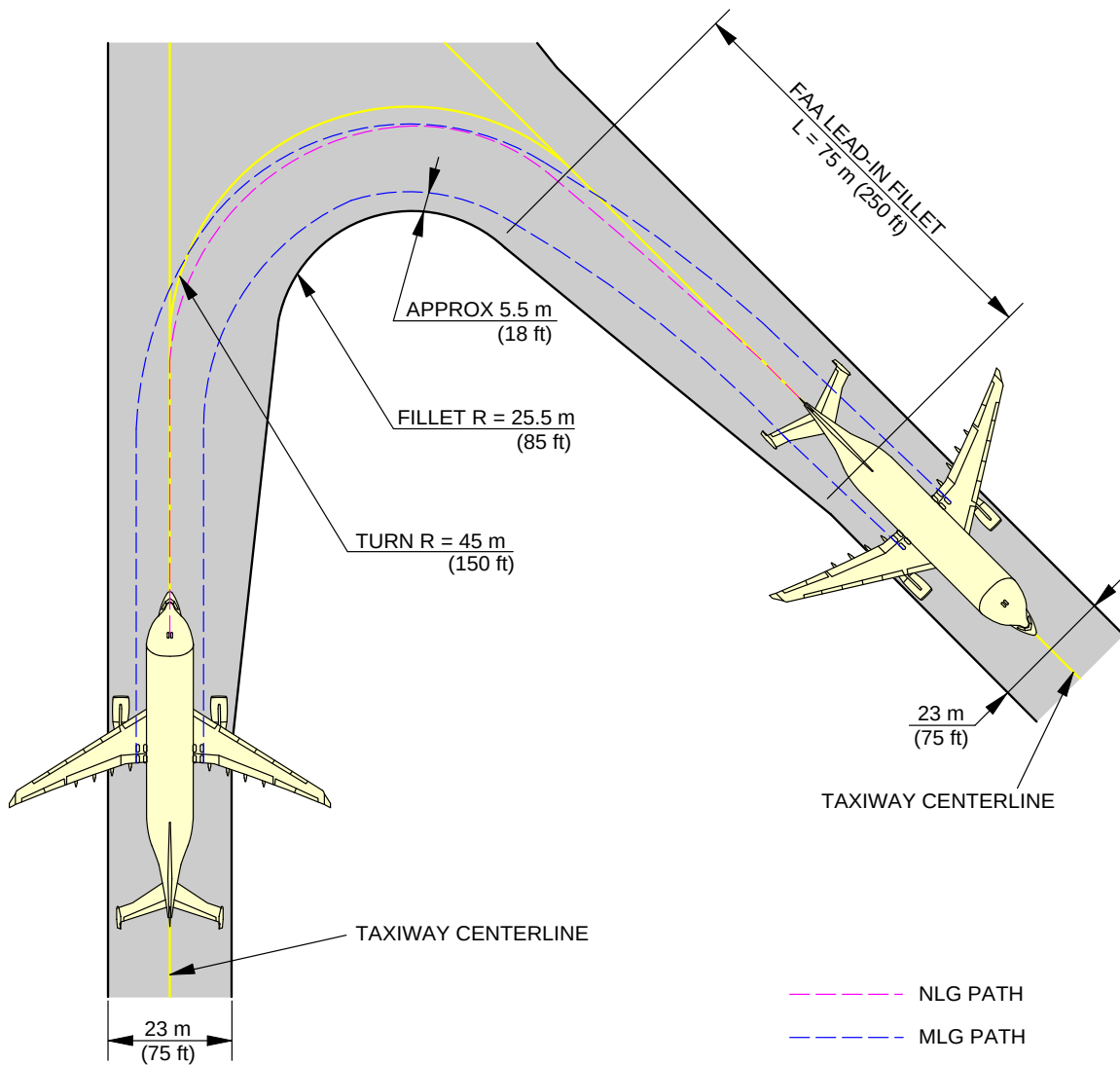
**\*\*ON A/C A330-700L**



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135° Turn - Taxiway to Taxiway  
Judgemental Oversteer Method  
FIGURE-4-5-4-991-015-A01

**\*\*ON A/C A330-700L**



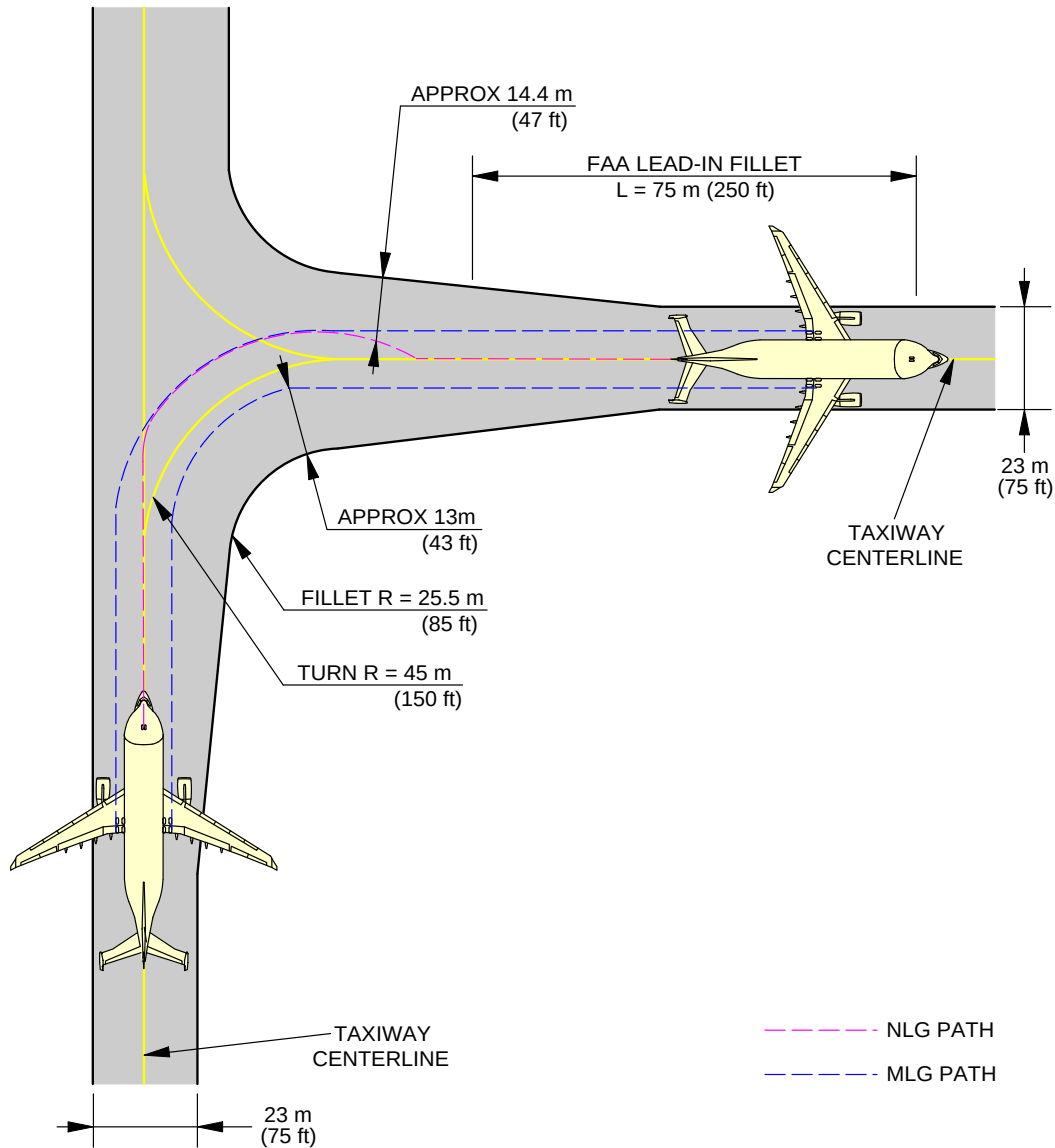
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135° Turn - Taxiway to Taxiway  
Cockpit Over Centerline Method  
FIGURE-4-5-4-991-016-A01

**4-5-5      90° Turn - Taxiway to Taxiway****\*\*ON A/C A330-700L****90° Turn - Taxiway to Taxiway**

1. This section gives the 90° turn - taxiway to taxiway.

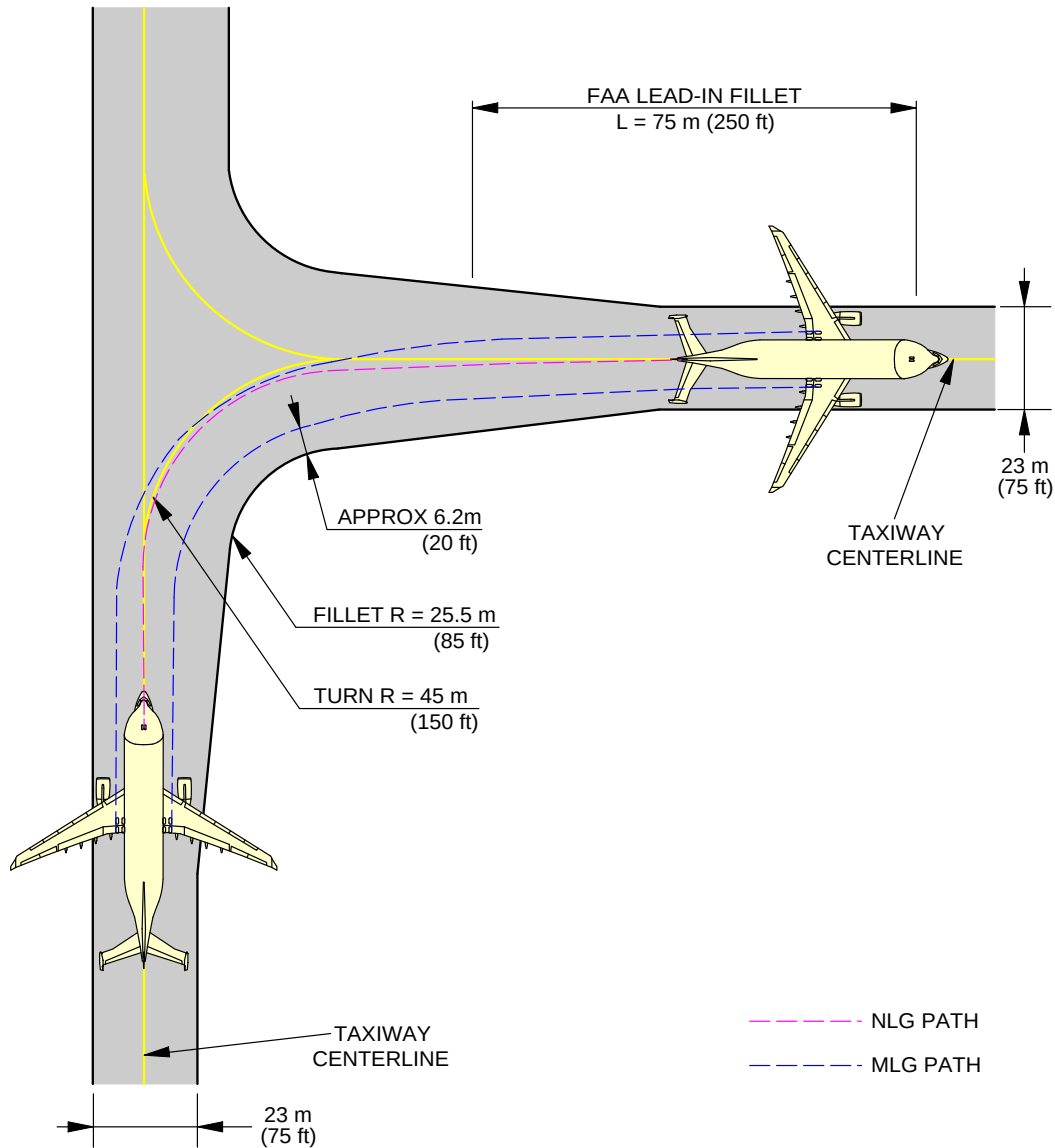
**\*\*ON A/C A330-700L**



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90° Turn - Taxiway to Taxiway  
Judgemental Oversteer Method  
FIGURE-4-5-5-991-017-A01

**\*\*ON A/C A330-700L**



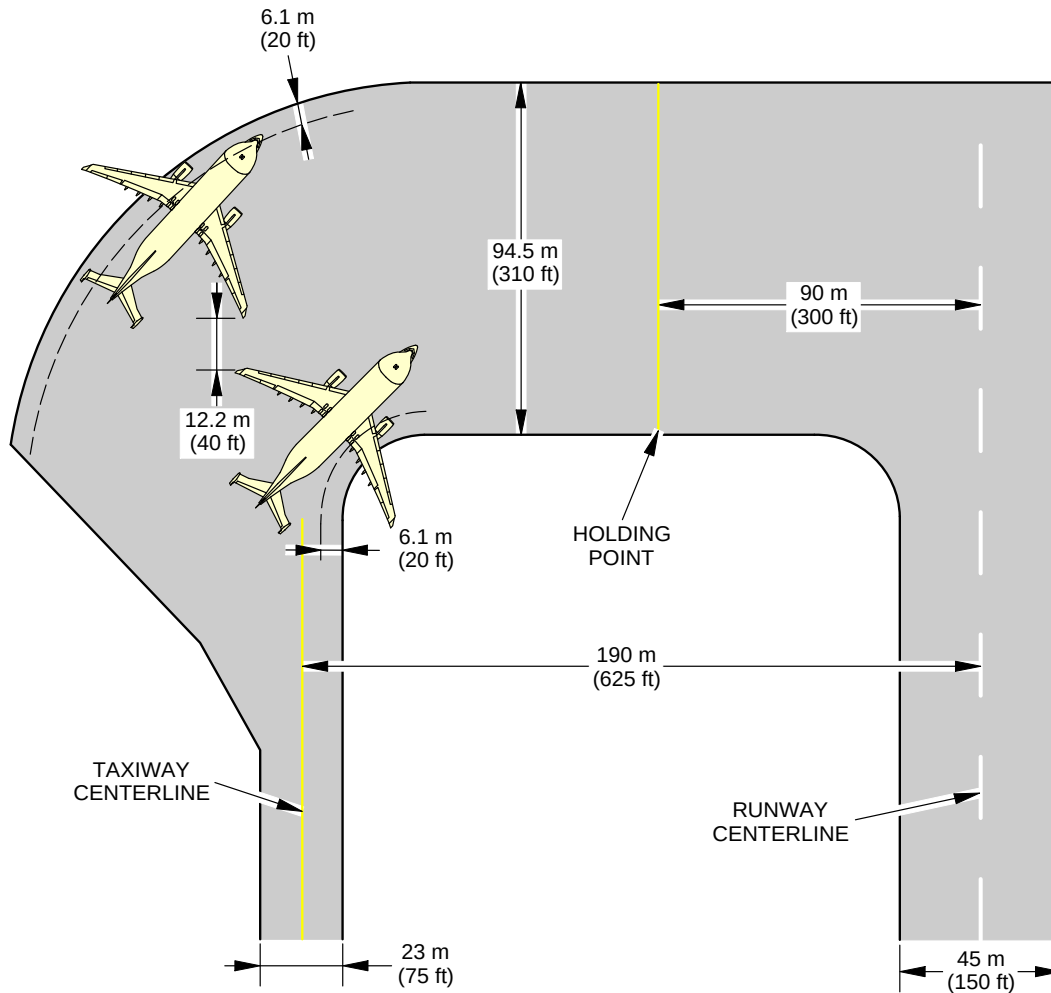
F\_AC\_040505\_1\_0180101\_01\_00

90° Turn - Taxiway to Taxiway  
Cockpit Over Centerline Method  
FIGURE-4-5-5-991-018-A01

**4-6-0      Runway Holding Bay (Apron)****\*\*ON A/C A330-700L**Runway Holding Bay (Apron)

1. This section gives the runway holding bay (Apron).

**\*\*ON A/C A330-700L**



**NOTE:**

COORDINATE WITH USING AIRLINE FOR SPECIFIC PLANNED OPERATING PROCEDURES.

F\_AC\_040600\_1\_0070101\_01\_00

Runway Holding Bay (Apron)  
FIGURE-4-6-0-991-007-A01

**4-7-0 Minimum Line-Up Distance Corrections****\*\*ON A/C A330-700L**Minimum Line-Up Distance Corrections

1. The ground maneuvers were performed using asymmetric thrust and differential-only braking to initiate the turn.

TODA: Take-Off Distance Available

ASDA: Acceleration-Stop Distance Available

2. 90° Turn on Runway Entry

This section gives the minimum line-up distance correction for a 90° turn on runway entry.

This maneuver consists in a 90° turn at minimum turn radius. It starts with the edge of the MLG at a distance of 4.5 m (15 ft) from the taxiway edge, and finishes with the aircraft aligned on the centerline of the runway, see FIGURE 4-7-0-991-028-A.

During the turn, all the clearances must meet the minimum value of 4.5 m (15 ft) for this category of aircraft as recommended in ICAO Annex 14.

3. 180° Turn on Runway Turn Pad

This section gives the minimum line-up distance correction for a 180° turn on the runway turn pad.

This maneuver consists in a 180° turn at minimum turn radius on a runway turn pad with standard ICAO geometry.

It starts with the edge of the MLG at a distance of 4.5 m (15 ft) from the pavement edge, and it finishes with the aircraft aligned on the centerline of the runway, see FIGURE 4-7-0-991-029-A. During the turn, all the clearances must meet the minimum value of 4.5 m (15 ft) for this category of aircraft as recommended in ICAO Annex 14.

4. 180° Turn on Runway Width

This section gives the minimum line-up distance correction for a 180° turn on the runway width. For this maneuver, the pavement width is considered to be the runway width, which is a frozen parameter (45 m (150 ft) and 60 m (200 ft)).

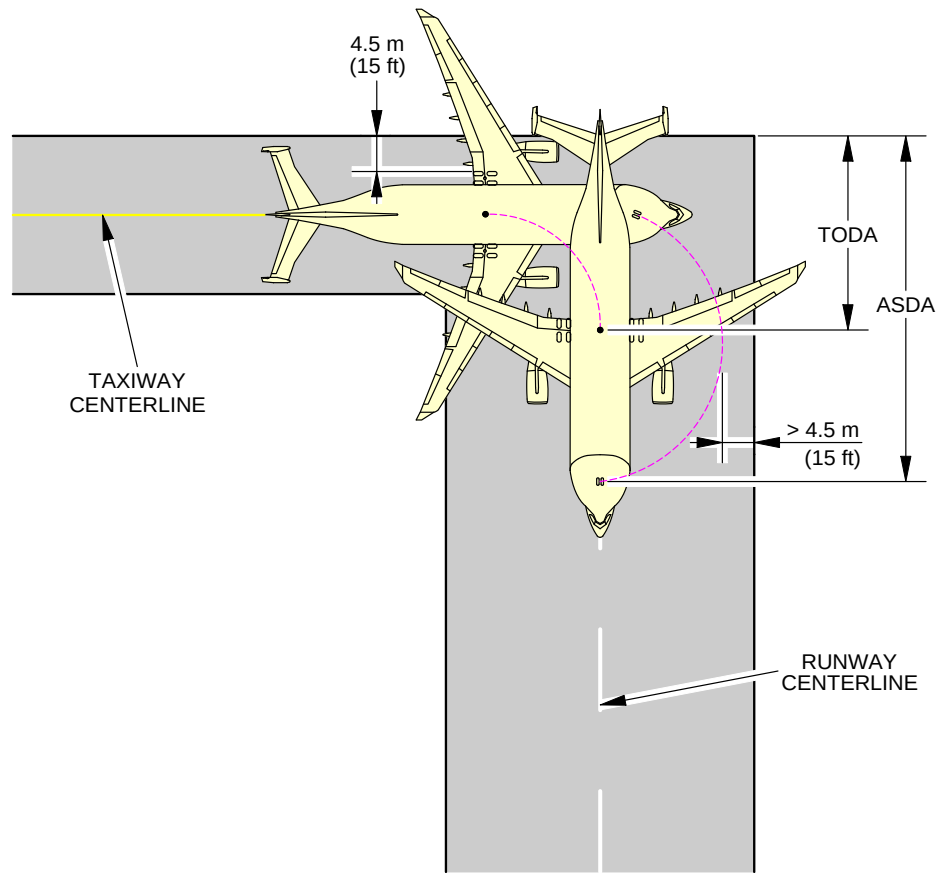
As per the standard operating procedures for the "180° turn on runway" (described in the Flight Crew Operating Manual), the aircraft is initially angled with respect to the runway centerline when starting the 180° turn, see FIGURE 4-7-0-991-030-A.

The value of this angle depends on the aircraft type and is mentioned in the FCOM.

During the turn, all the clearances must meet the minimum value of 4.5 m (15 ft) for this category of aircraft as recommended in ICAO Annex 14.



**\*\*ON A/C A330-700L**



| 90° TURN ON RUNWAY ENTRY |                    |                                         |       |         |        |
|--------------------------|--------------------|-----------------------------------------|-------|---------|--------|
| AIRCRAFT TYPE            | MAX STEERING ANGLE | 45 m (150 ft)/60 m (200 ft) WIDE RUNWAY |       |         |        |
|                          |                    | MINIMUM LINE-UP DISTANCE CORRECTION     |       |         |        |
|                          |                    | ON TODA                                 |       | ON ASDA |        |
| A330-700L                | 65°                | 22.5 m                                  | 74 ft | 44.7 m  | 147 ft |
| A330-700L                | 72°                | 19.7 m                                  | 65 ft | 41.9 m  | 137 ft |

**NOTE:**

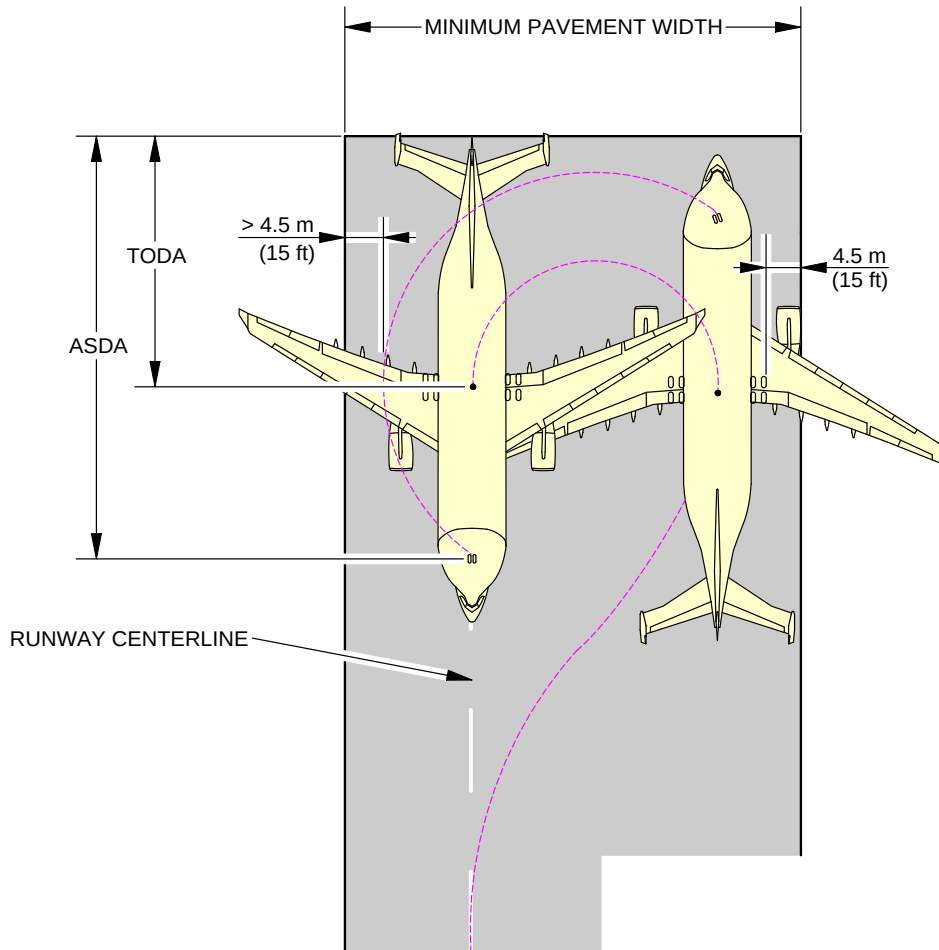
ASDA: ACCELERATION-STOP DISTANCE AVAILABLE

TODA: TAKE-OFF DISTANCE AVAILABLE

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Minimum Line-Up Distance Corrections  
90° Turn on Runway Entry  
FIGURE-4-7-0-991-028-A01

**\*\*ON A/C A330-700L**



| 180° TURN ON RUNWAY TURNPAD |                          |                                         |       |         |        |                                          |        |
|-----------------------------|--------------------------|-----------------------------------------|-------|---------|--------|------------------------------------------|--------|
| AIRCRAFT<br>TYPE            | MAX<br>STEERING<br>ANGLE | 45 m (150 ft)/60 m (200 ft) WIDE RUNWAY |       |         |        | REQUIRED<br>MINIMUM<br>PAVEMENT<br>WIDTH |        |
|                             |                          | MINIMUM LINE-UP<br>DISTANCE CORRECTION  |       |         |        |                                          |        |
|                             |                          | ON TODA                                 |       | ON ASDA |        |                                          |        |
| A330-700L                   | 65°                      | 30.1 m                                  | 99 ft | 52.2 m  | 171 ft | 56.7 m                                   | 186 ft |
| A330-700L                   | 72°                      | 28.9 m                                  | 95 ft | 51.1 m  | 168 ft | 51.1 m                                   | 168 ft |

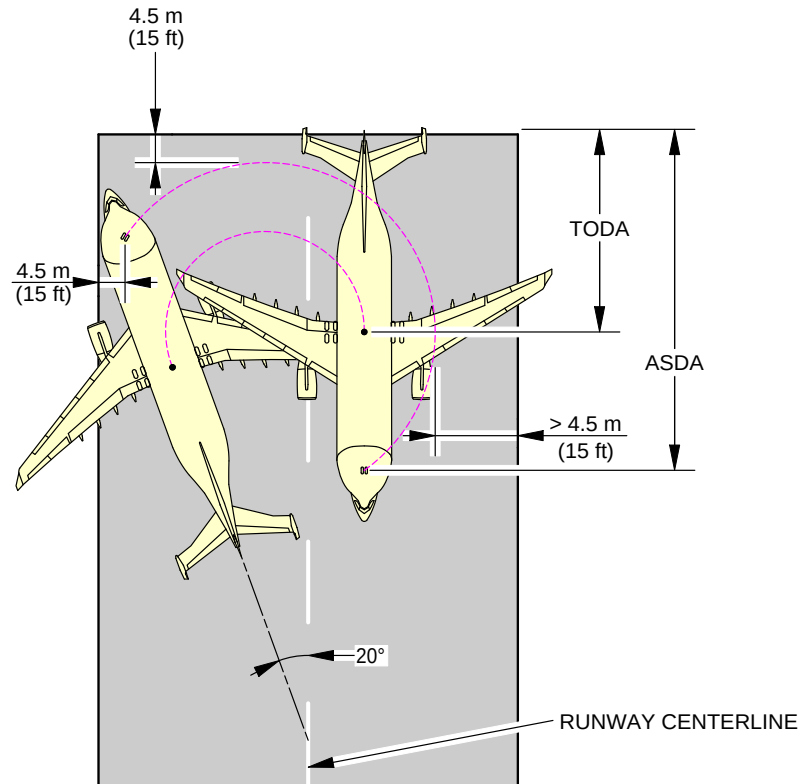
**NOTE:**

ASDA: ACCELERATION-STOP DISTANCE AVAILABLE  
TODA: TAKE-OFF DISTANCE AVAILABLE

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Minimum Line-Up Distance Corrections  
180° Turn on Runway Turn Pad  
FIGURE-4-7-0-991-029-A01

**\*\*ON A/C A330-700L**



| 180° TURN ON RUNWAY WIDTH |                    |                                            |         |                                     |         |         |         |
|---------------------------|--------------------|--------------------------------------------|---------|-------------------------------------|---------|---------|---------|
| AIRCRAFT TYPE             | MAX STEERING ANGLE | 45 m (150 ft) WIDE RUNWAY (STANDARD WIDTH) |         | 60 m (200 ft) WIDE RUNWAY           |         |         |         |
|                           |                    | MINIMUM LINE-UP DISTANCE CORRECTION        |         | MINIMUM LINE-UP DISTANCE CORRECTION |         |         |         |
|                           |                    | ON TODA                                    | ON ASDA | ON TODA                             | ON ASDA | ON TODA | ON ASDA |
| A330-700L                 | 65°                | NOT POSSIBLE                               |         | 44.5 m                              | 146 ft  | 66.6 m  | 219 ft  |
| A330-700L                 | 72°                |                                            |         | 28.9 m                              | 95 ft   | 51.1 m  | 168 ft  |

**NOTE:**

ASDA: ACCELERATION-STOP DISTANCE AVAILABLE

TODA: TAKE-OFF DISTANCE AVAILABLE

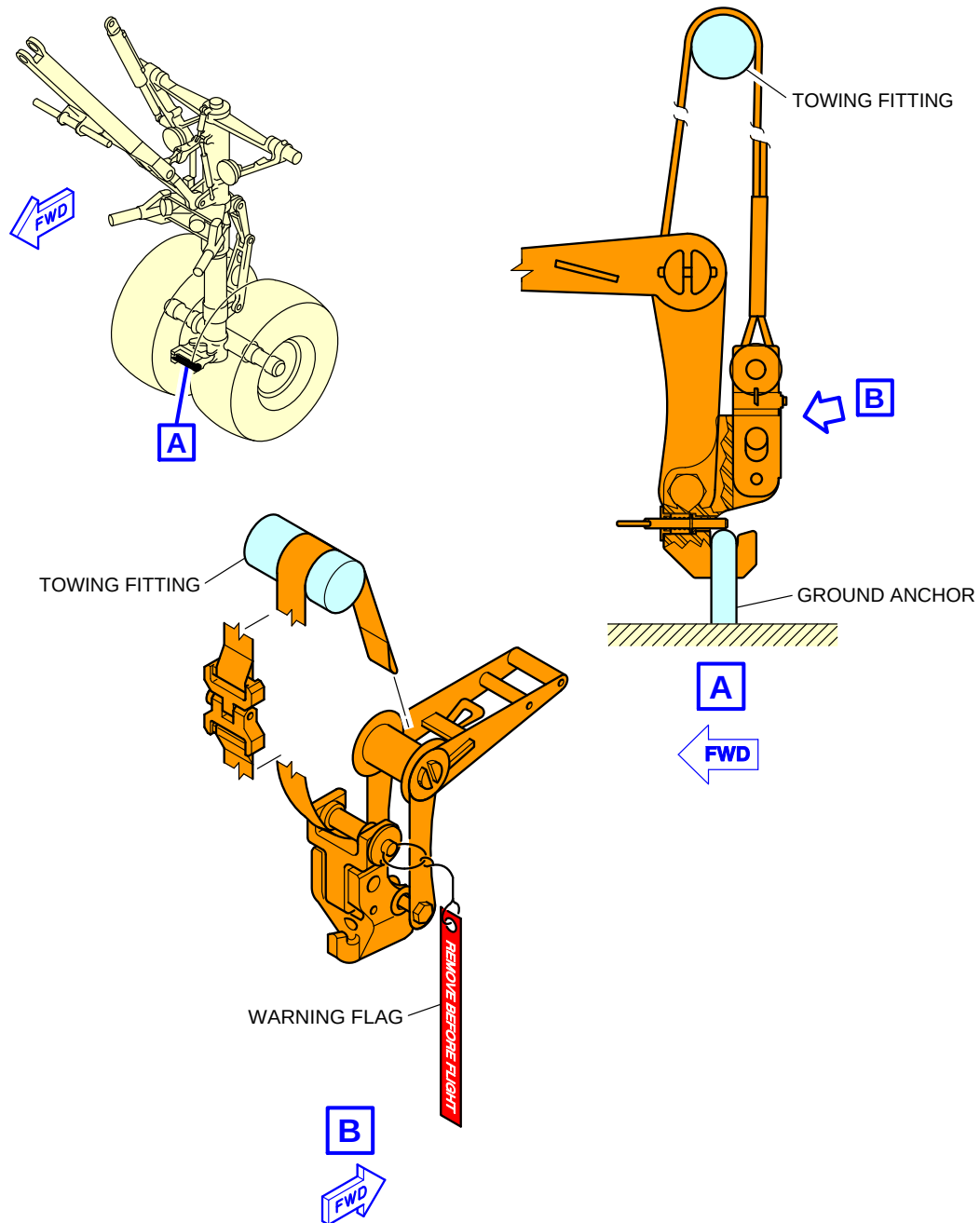
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Minimum Line-Up Distance Corrections  
180° Turn on Runway Width  
FIGURE-4-7-0-991-030-A01

**4-8-0      Aircraft Mooring****\*\*ON A/C A330-700L**Aircraft Mooring

1. This section gives information on aircraft mooring.

**\*\*ON A/C A330-700L**



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Aircraft Mooring  
FIGURE-4-8-0-991-002-A01

**TERMINAL SERVICING****5-1-0 Aircraft Servicing Arrangements****\*\*ON A/C A330-700L****Aircraft Servicing Arrangements****1. Aircraft Servicing Arrangements**

- A. This section provides typical ramp layouts, showing the various GSE items in position during typical turn-round scenarios for cargo aircraft.

These ramp layouts show typical arrangements only. Each operator will have its own specific requirements/regulations for positioning and operation on the ramp.

**5-1-1 Symbols Used on Servicing Diagrams****\*\*ON A/C A330-700L**Symbols Used on Servicing Diagrams

1. This table gives the symbols used on servicing diagrams.

| Ground Support Equipment |                                  |
|--------------------------|----------------------------------|
| AC                       | AIR CONDITIONING UNIT            |
| AS                       | AIR START UNIT                   |
| BULK                     | BULK TRAIN                       |
| CB                       | CONVEYOR BELT                    |
| FUEL                     | FUEL HYDRANT DISPENSER or TANKER |
| GPU                      | GROUND POWER UNIT                |
| ILF                      | INTEGRATED LOADING FACILITIES    |
| LD CL                    | LOWER DECK CARGO LOADER          |
| LV                       | LAVATORY VEHICLE                 |
| TOW                      | TOW TRACTOR                      |
| ULD                      | ULD TRAIN                        |

**5-1-2 Typical Ramp Layout - Open Apron**

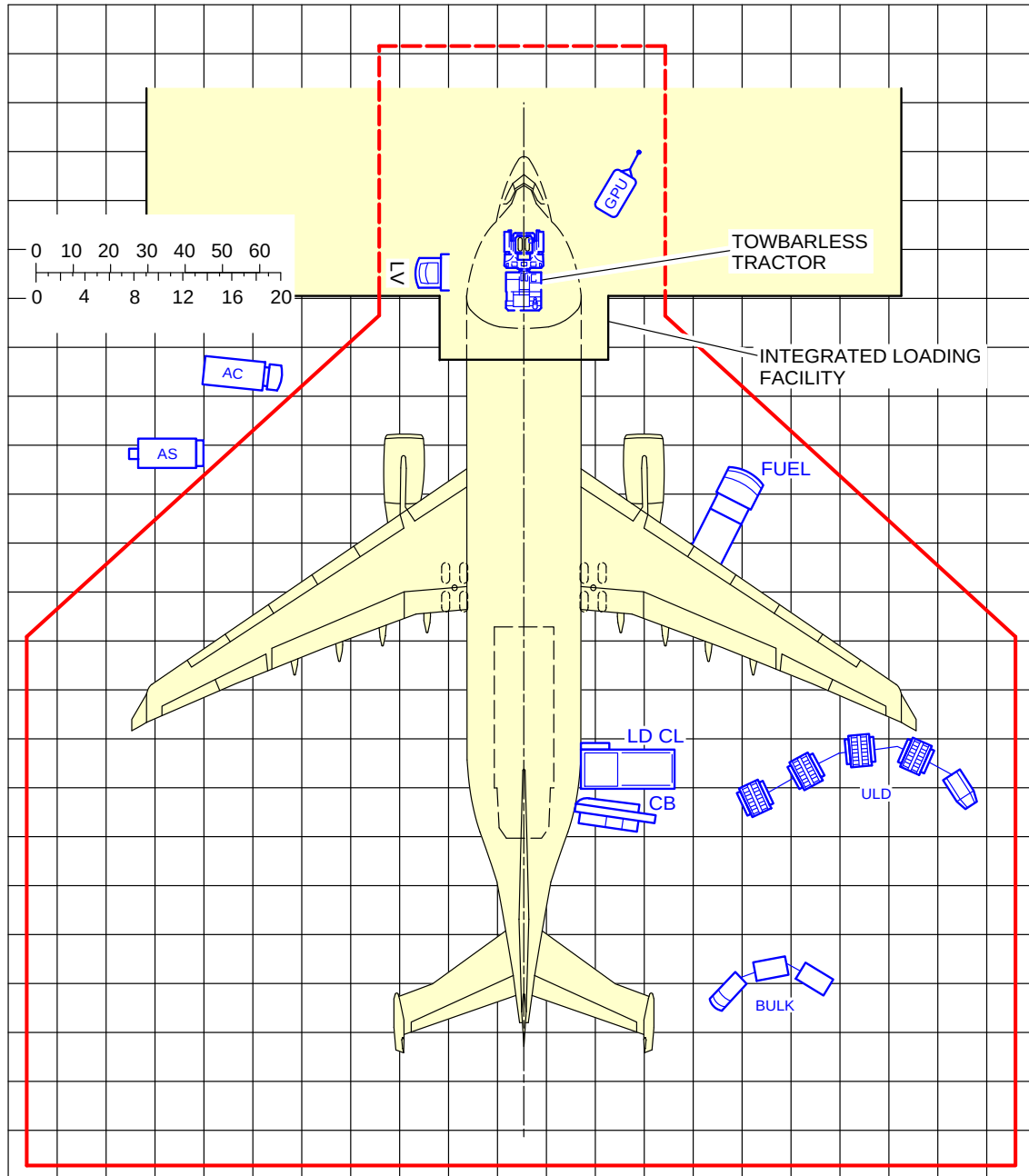
**\*\*ON A/C A330-700L**

05-01-02 Typical Ramp Layout

1. This section gives information about the typical ramp layout.



**\*\*ON A/C A330-700L**



**NOTE:**

— STAND SAFETY LINE

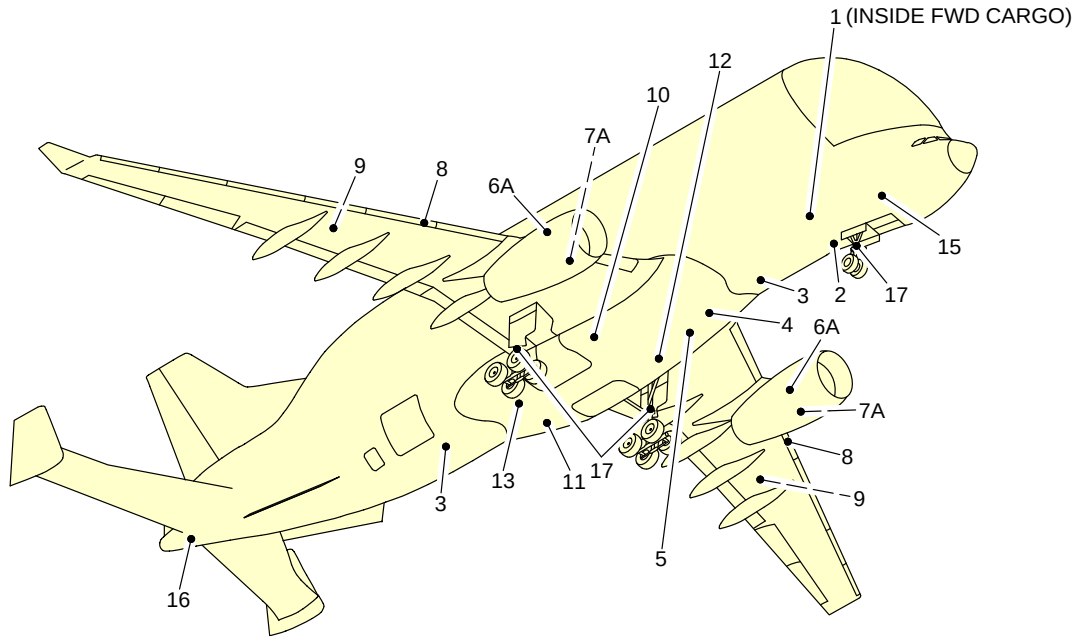
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Typical Ramp Layout  
FIGURE-5-1-2-991-012-A01

**5-4-1 Ground Service Connections Layout****\*\*ON A/C A330-700L**Ground Service Connections Layout

1. This section gives the ground service connections layout.

**\*\*ON A/C A330-700L**



- 1 - OXYGEN SERVICING
- 2 - GROUND ELECTRICAL POWER CONNECTORS
- 3 - POTABLE WATER DRAIN
- 4 - LOW PRESSURE AIR PRE-CONDITIONING
- 5 - HIGH PRESSURE AIR PRE-CONDITIONING AND ENGINE STARTING
- 6A - ENGINE OIL FILLING
- 7A - IDG OIL FILLING
- 8 - PRESSURE REFUEL/DEFUEL COUPLINGS
- 9 - OVERWING REFUEL

- 10 - HYDRAULIC GROUND POWER SUPPLY (YELLOW)
- 11 - HYDRAULIC RESERVOIR FILLING AND GROUND POWER SUPPLY (GREEN)
- 12 - HYDRAULIC RESERVOIR AIR PRESSURIZATION AND GROUND POWER SUPPLY (BLUE)
- 13 - REFUEL/DEFUEL PANEL
- 15 - WASTE WATER SERVICE PANEL
- 16 - APU OIL FILLING
- 17 - GROUNDING (EARTHING) POINT

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Ground Service Connections Layout  
FIGURE-5-4-1-991-007-A01

## 5-4-2 Grounding Points

### \*\*ON A/C A330-700L

#### Grounding (Earthing) Points

#### 1. Grounding (Earthing) Points

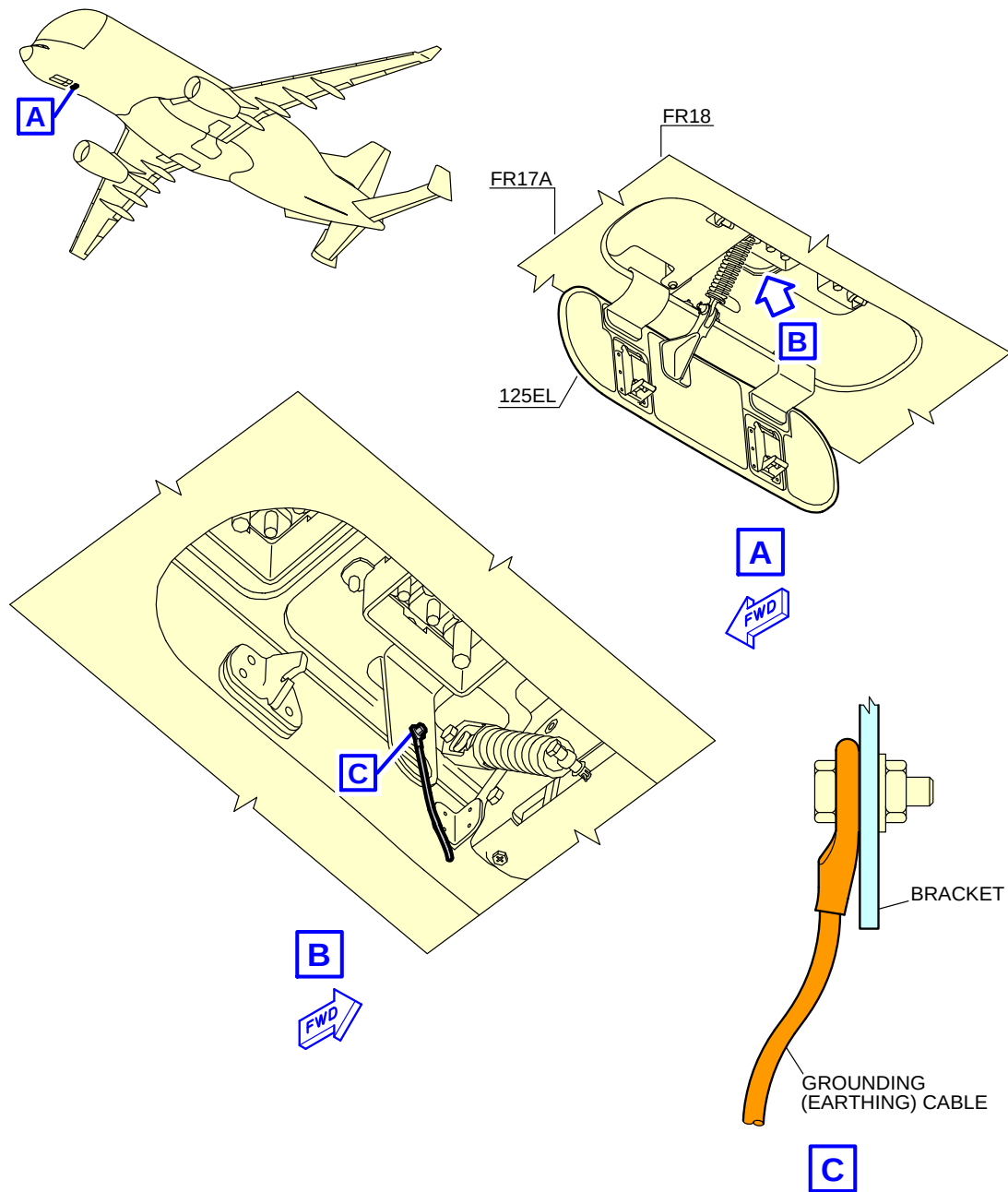
NOTE : The mean height from ground in the below table may change according to the CG position and aircraft weight.

| ACCESS           | DISTANCE              |                          |                      |                            |
|------------------|-----------------------|--------------------------|----------------------|----------------------------|
|                  | AFT OF NOSE           | FROM AIRCRAFT CENTERLINE |                      | MEAN HEIGHT<br>FROM GROUND |
|                  |                       | LH SIDE                  | RH SIDE              |                            |
| On NLG leg       | 8.17 m<br>(26.80 ft)  | On centerline            |                      | 1.40 m<br>(4.59 ft)        |
| On left MLG leg  | 32.1 m<br>(105.31 ft) | 5.34 m<br>(17.52 ft)     |                      | 1.50 m<br>(4.92 ft)        |
| On right MLG leg | 32.1 m<br>(105.31 ft) |                          | 5.34 m<br>(17.52 ft) | 1.50 m<br>(4.92 ft)        |

- A. The grounding (earthing) stud on each landing gear leg is designed for use with a clip-on connector (such as Appleton TGR).
- B. The grounding (earthing) studs are used to connect the aircraft to an approved ground (earth) connection on the ramp or in the hangar for:
  - Refuel/defuel operations
  - Maintenance operations
  - Bad weather conditions
  - Loading/Unloading operations.

NOTE : In all other conditions, the electrostatic discharge through the tire is sufficient.

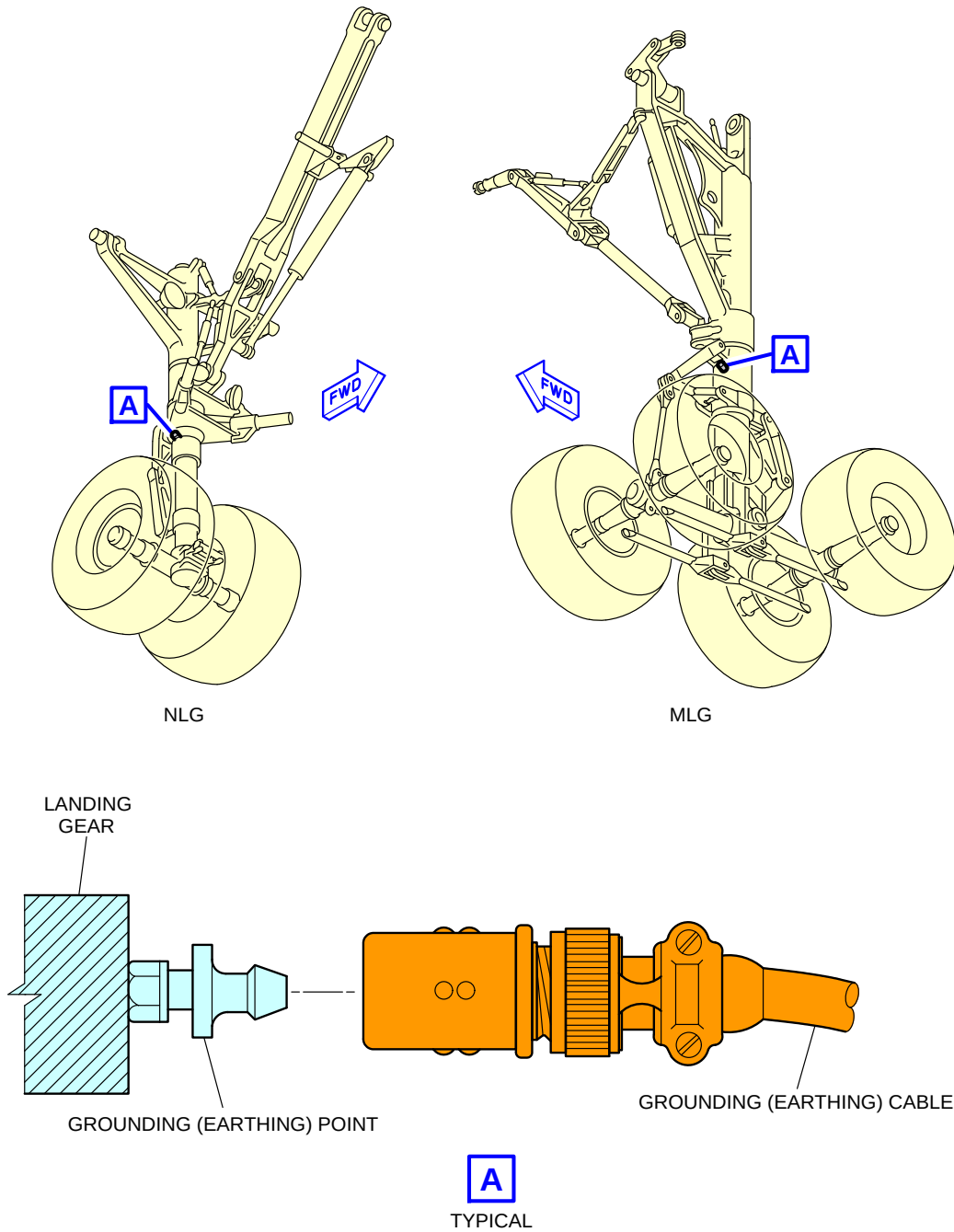
**\*\*ON A/C A330-700L**



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Ground Service Connections  
Grounding (Earthing) Points  
FIGURE-5-4-2-991-006-A01

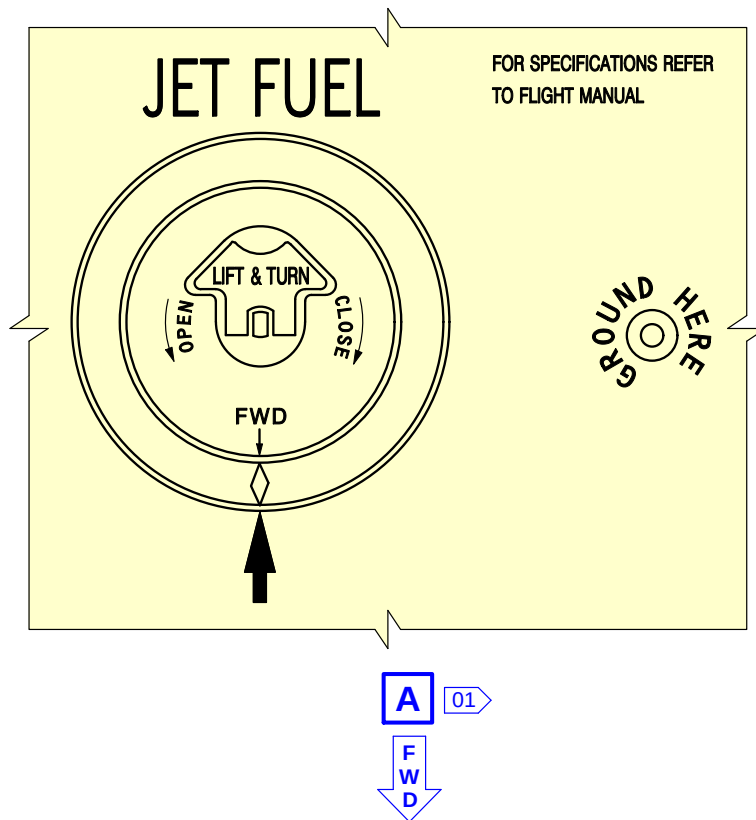
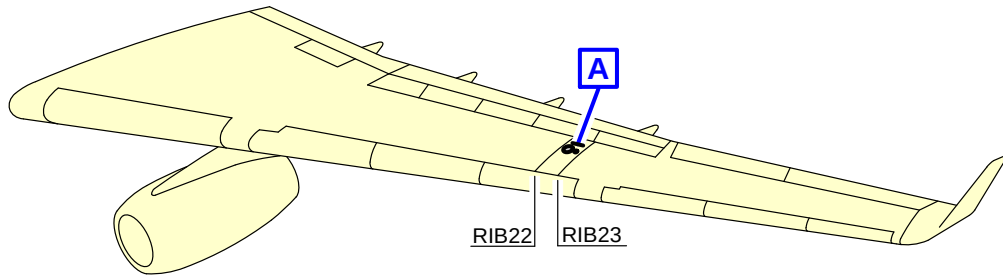
**\*\*ON A/C A330-700L**



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Ground Service Connections  
Grounding (Earthing) Points  
FIGURE-5-4-2-991-007-A01

**\*\*ON A/C A330-700L**



**NOTE:**

**01** LH SHOWN RH SYMMETRICAL

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Ground Service Connections  
Grounding (Earthing) Points  
FIGURE-5-4-2-991-008-A01

**5-4-3 Hydraulic System**
**\*\*ON A/C A330-700L**
Hydraulic System
**1. Ground Service Panels**

NOTE : The mean height from ground in the below table may change according to the CG position and aircraft weight.

| ACCESS                              | DISTANCE               |                          |                     |                         |
|-------------------------------------|------------------------|--------------------------|---------------------|-------------------------|
|                                     | AFT OF NOSE            | FROM AIRCRAFT CENTERLINE |                     | MEAN HEIGHT FROM GROUND |
|                                     |                        | LH SIDE                  | RH SIDE             |                         |
| Green system:<br>Access door 197CB  | 33.10 m<br>(108.60 ft) | 1.28 m<br>(4.20 ft)      |                     | 2.23 m<br>(7.32 ft)     |
| Yellow system:<br>Access door 196BB | 27.30 m<br>(89.57 ft)  |                          | 1.32 m<br>(4.33 ft) | 1.95 m<br>(6.40 ft)     |
| Blue system:<br>Access door 195BB   | 26.30 m<br>(86.29 ft)  | 1.28 m<br>(4.20 ft)      |                     | 1.94 m<br>(6.36 ft)     |

**2. Reservoir Pressurization**

NOTE : The mean height from ground in the below table may change according to the CG position and aircraft weight.

| ACCESS                                                    | DISTANCE              |                          |         |                         |
|-----------------------------------------------------------|-----------------------|--------------------------|---------|-------------------------|
|                                                           | AFT OF NOSE           | FROM AIRCRAFT CENTERLINE |         | MEAN HEIGHT FROM GROUND |
|                                                           |                       | LH SIDE                  | RH SIDE |                         |
| Blue system ground<br>service panel:<br>Access door 195BB | 26.34 m<br>(86.42 ft) | 1.28 m<br>(4.20 ft)      |         | 1.94 m<br>(6.36 ft)     |

**3. Accumulator Charging**

NOTE : The mean height from ground in the below table may change according to the CG position and aircraft weight.



| ACCESS                                        | DISTANCE              |                          |         |                            |
|-----------------------------------------------|-----------------------|--------------------------|---------|----------------------------|
|                                               | AFT OF NOSE           | FROM AIRCRAFT CENTERLINE |         | MEAN HEIGHT<br>FROM GROUND |
|                                               |                       | LH SIDE                  | RH SIDE |                            |
| Blue system accumulator:<br>Access door 195BB | 26.34 m<br>(86.42 ft) | 1.28 m<br>(4.20 ft)      |         | 1.94 m<br>(6.36 ft)        |

#### 4. Reservoir Filling

Two connections (one self-sealing connection for pressurized supply on the Green system ground service panel).

**NOTE :** The mean height from ground in the below table may change according to the CG position and aircraft weight.

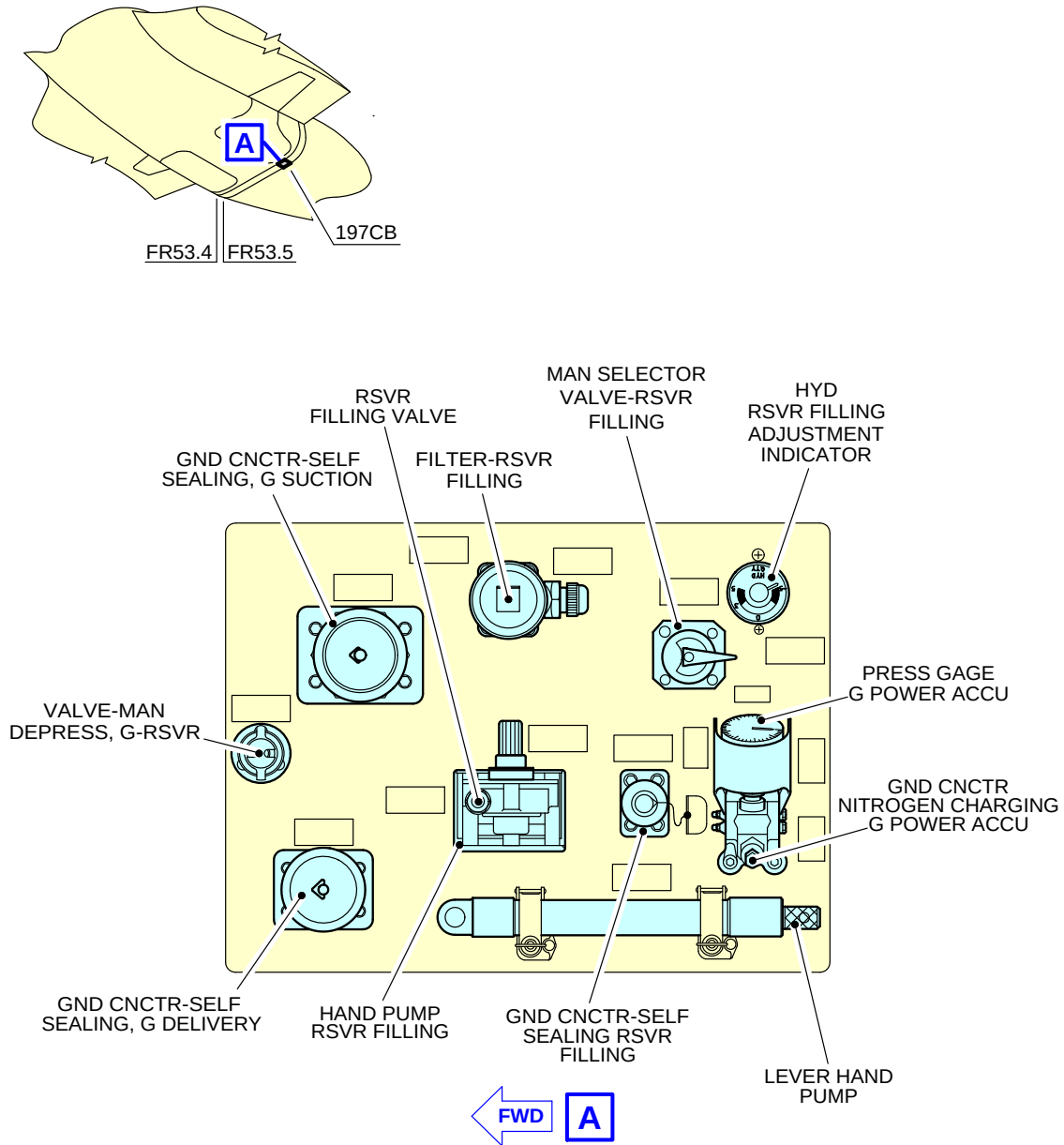
| ACCESS                                                 | DISTANCE               |                          |         |                            |
|--------------------------------------------------------|------------------------|--------------------------|---------|----------------------------|
|                                                        | AFT OF NOSE            | FROM AIRCRAFT CENTERLINE |         | MEAN HEIGHT<br>FROM GROUND |
|                                                        |                        | LH SIDE                  | RH SIDE |                            |
| One handpump filling connection:<br>Access door 197CBB | 33.22 m<br>(108.99 ft) | 1.28 m<br>(4.20 ft)      |         | 2.23 m<br>(7.32 ft)        |

#### 5. A/C Emergency Generation

**NOTE :** The mean height from ground in the below table may change according to the CG position and aircraft weight.

| ACCESS                                                               | DISTANCE               |                          |                       |                            |
|----------------------------------------------------------------------|------------------------|--------------------------|-----------------------|----------------------------|
|                                                                      | AFT OF NOSE            | FROM AIRCRAFT CENTERLINE |                       | MEAN HEIGHT<br>FROM GROUND |
|                                                                      |                        | LH SIDE                  | RH SIDE               |                            |
| Ram Air Turbine (RAT) safety-pin installation:<br>Access panel 633SL | 32.90 m<br>(107.94 ft) |                          | 14.20 m<br>(46.59 ft) | 4.35 m<br>(14.27 ft)       |

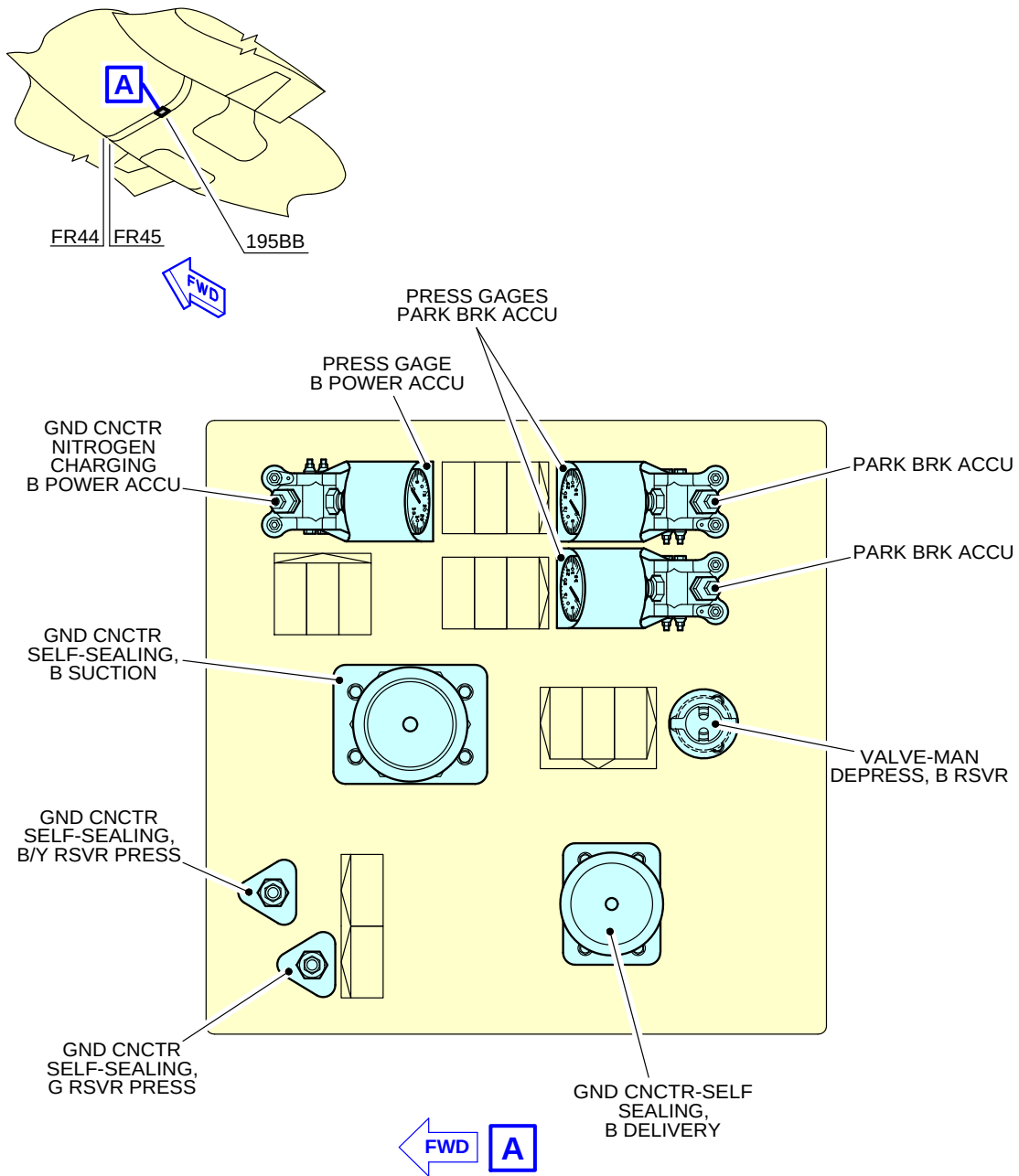
**\*\*ON A/C A330-700L**



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Ground Service Connections  
Green System Ground Service Panel  
FIGURE-5-4-3-991-013-A01

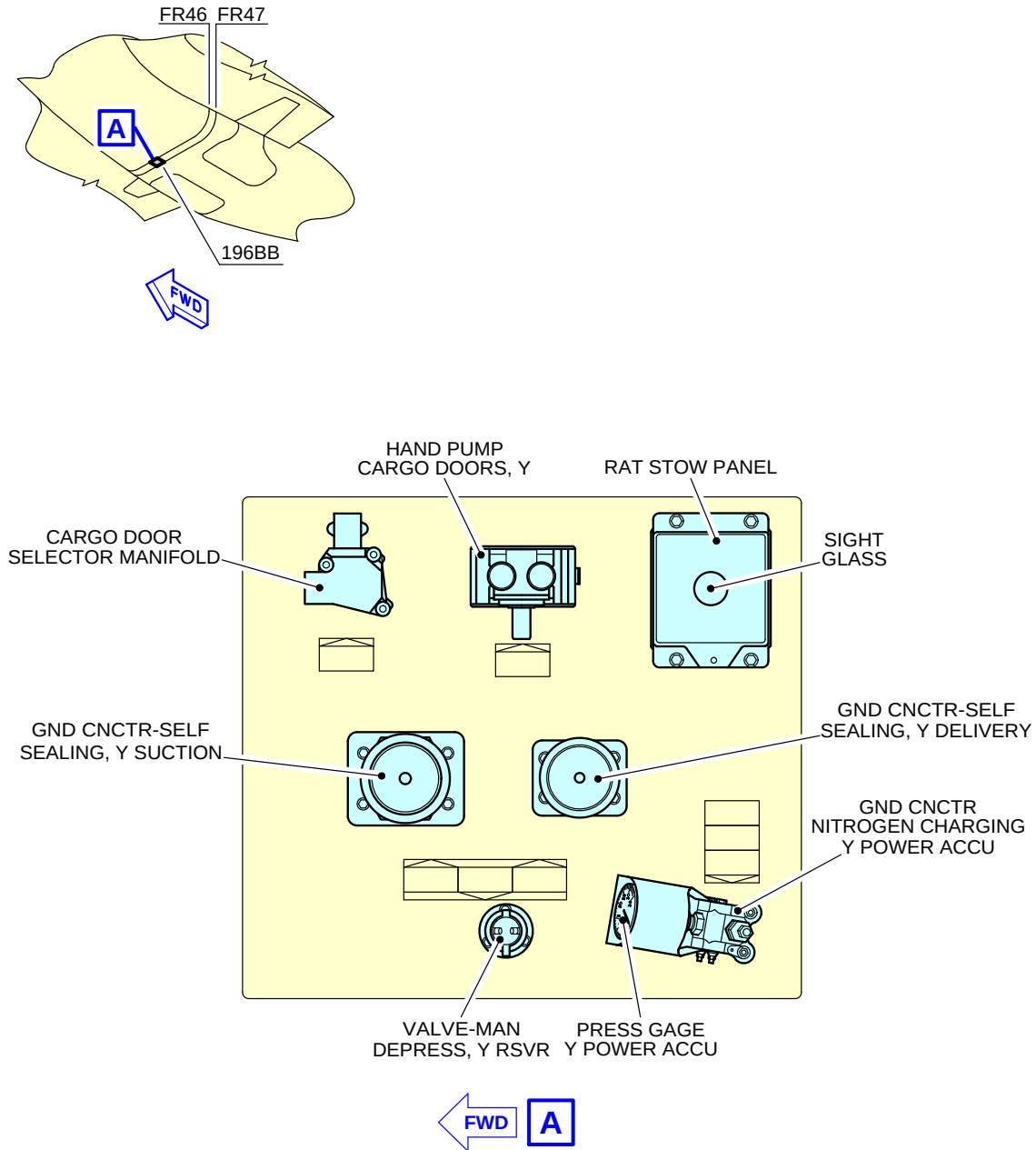
**\*\*ON A/C A330-700L**



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Ground Service Connections  
Blue System Ground Service Panel  
FIGURE-5-4-3-991-014-A01

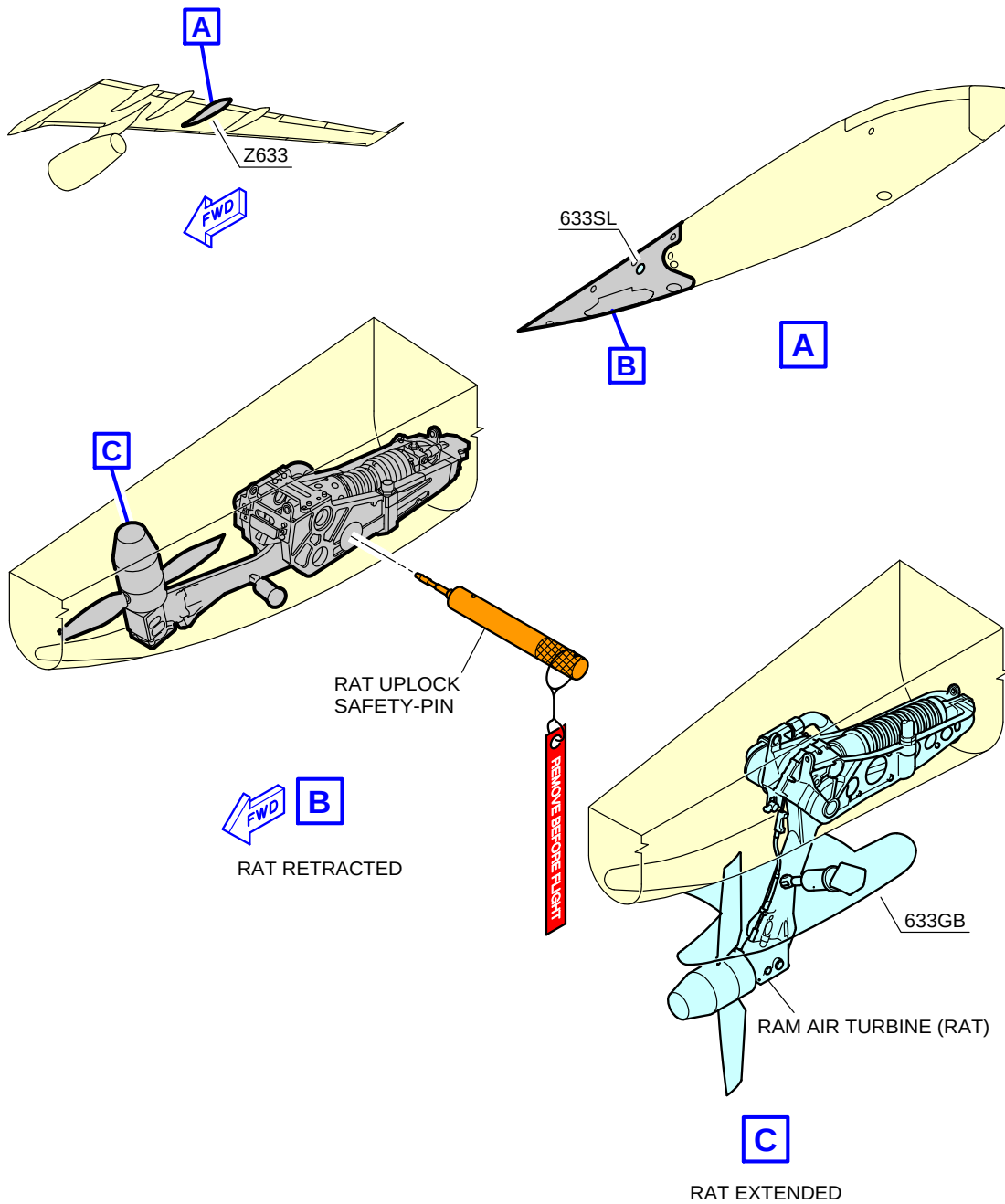
**\*\*ON A/C A330-700L**



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Ground Service Connections  
Yellow System Ground Service Panel  
FIGURE-5-4-3-991-015-A01

**\*\*ON A/C A330-700L**



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Ground Service Connections  
RAT  
FIGURE-5-4-3-991-016-A01

## 5-4-4 Electrical System

### \*\*ON A/C A330-700L

#### Electrical Servicing

##### 1. A/C External Power

NOTE : The mean height from ground in the below table may change according to the CG position and aircraft weight.

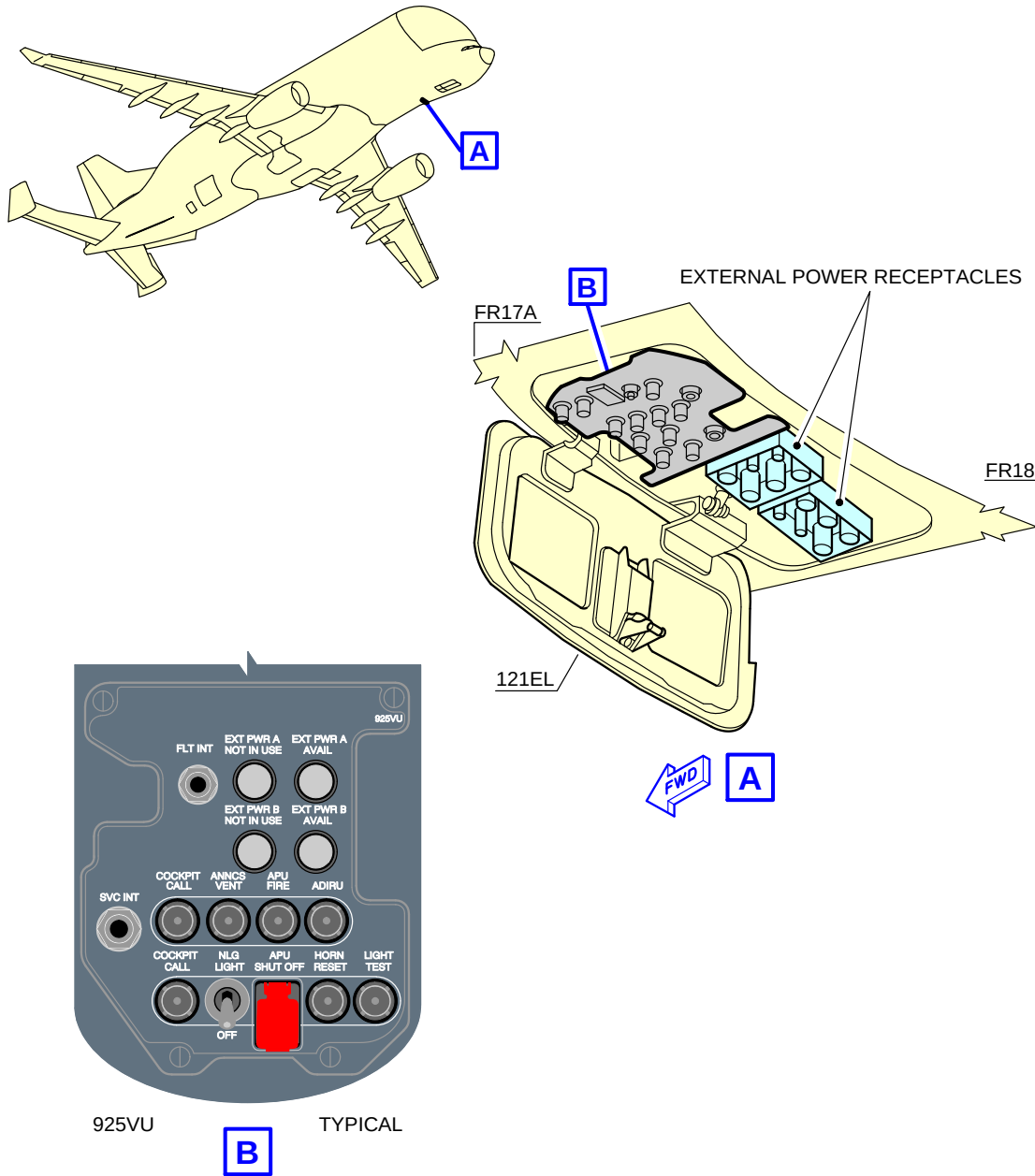
| ACCESS              | DISTANCE    |                          |           |                            |
|---------------------|-------------|--------------------------|-----------|----------------------------|
|                     | AFT OF NOSE | FROM AIRCRAFT CENTERLINE |           | MEAN HEIGHT<br>FROM GROUND |
|                     |             | LH SIDE                  | RH SIDE   |                            |
| A/C external power: | 8.70 m      |                          | 0.10 m    | 1.95 m                     |
| Access door 121EL   | (28.54 ft)  |                          | (0.33 ft) | (6.40 ft)                  |

NOTE : Distances are approximate.

##### 2. Technical Specifications

- A. External power receptacles:
  - Two receptacles according to MS 90362-3 - 90 kVA.
- B. Power supply:
  - Three-phase, 115 V, 400 Hz.
- C. Electrical connectors for servicing:
  - AC outlets: HUBBELL 5258
  - DC outlets: HUBBELL 7472.
- D. Maintenance bus switch:
  - Inside A/C near bulkhead door.

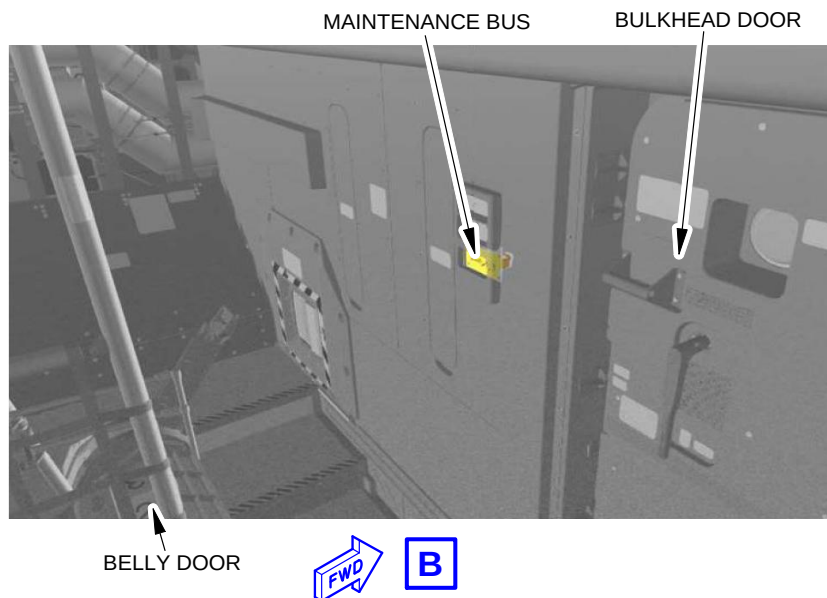
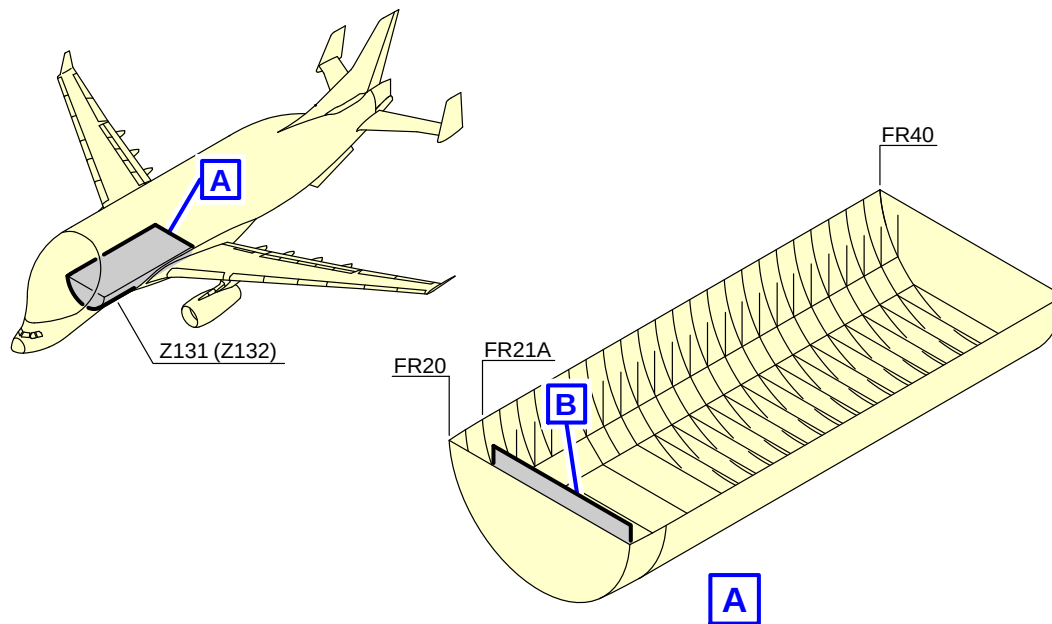
**\*\*ON A/C A330-700L**



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Ground Service Connections  
Electrical Service Panel  
FIGURE-5-4-4-991-007-A01

**\*\*ON A/C A330-700L**



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Ground Service Connections  
Maintenance Bus-Switch Location  
FIGURE-5-4-4-991-008-A01



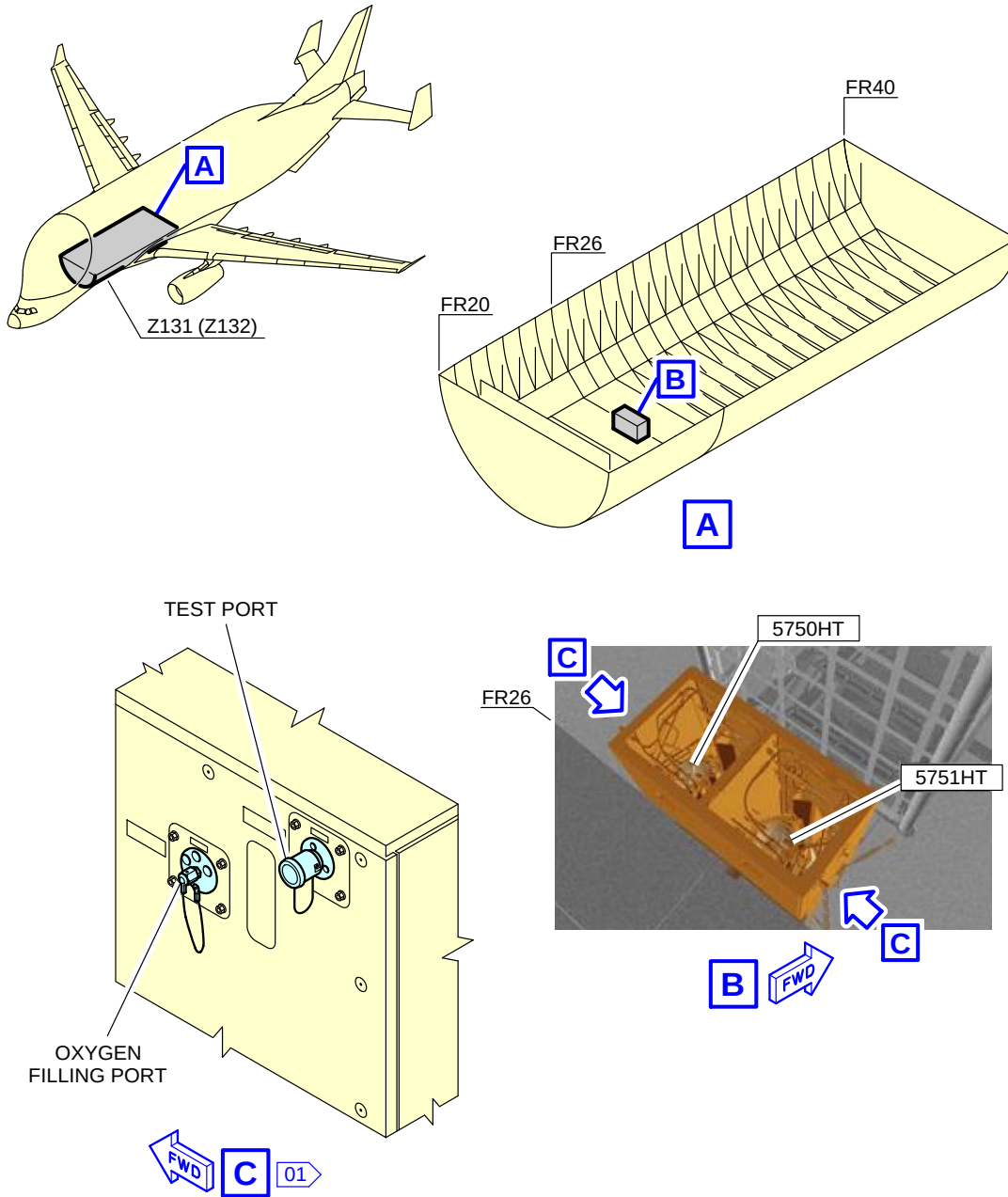
**5-4-5 Oxygen System**
**\*\*ON A/C A330-700L**
Oxygen System
**1. Oxygen Servicing**

NOTE : The mean height from ground in the below table may change according to the CG position and aircraft weight.

| ACCESS                                           | DISTANCE              |                          |                     |                            |
|--------------------------------------------------|-----------------------|--------------------------|---------------------|----------------------------|
|                                                  | AFT OF NOSE           | FROM AIRCRAFT CENTERLINE |                     | MEAN HEIGHT<br>FROM GROUND |
|                                                  |                       | LH SIDE                  | RH SIDE             |                            |
| Oxygen<br>replenishment<br>(option 1):<br>5750HT | 12.54 m<br>(41.14 ft) | 0.44 m<br>(1.44 ft)      |                     | 3.46 m<br>(11.35 ft)       |
| Oxygen<br>replenishment<br>(option 2):<br>5751HT | 12.54 m<br>(41.14 ft) |                          | 0.44 m<br>(1.44 ft) | 3.46 m<br>(11.35 ft)       |

NOTE : Internal charging connection near the belly door, inside aircraft.

**\*\*ON A/C A330-700L**



**NOTE:**

**01** LH SHOWN, RH SYMMETRICAL

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Ground Service Connections  
Oxygen Servicing  
FIGURE-5-4-5-991-005-A01

**5-4-6 Fuel System**
**\*\*ON A/C A330-700L**
Fuel System
**1. Refuel/Defuel Control Panel**

**NOTE :** The mean height from ground in the below table may change according to the CG position and aircraft weight.

| ACCESS                                            | DISTANCE              |                          |                    |                         |
|---------------------------------------------------|-----------------------|--------------------------|--------------------|-------------------------|
|                                                   | AFT OF NOSE           | FROM AIRCRAFT CENTERLINE |                    | MEAN HEIGHT FROM GROUND |
|                                                   |                       | LH SIDE                  | RH SIDE            |                         |
| Refuel/Defuel control panel:<br>Access door 198DB | 32.6 m<br>(106.96 ft) |                          | 0.8 m<br>(2.62 ft) | 1.9 m<br>(6.23 ft)      |

A. Flow rate: 1 580 l/min (417 US gal/min) per connection.

B. Maximum pressure: 50 psi (3.45 bar).

**2. Refuel/Defuel Connectors**

**NOTE :** The mean height from ground in the below table may change according to the CG position and aircraft weight.

| ACCESS                                              | DISTANCE              |                          |                      |                         |
|-----------------------------------------------------|-----------------------|--------------------------|----------------------|-------------------------|
|                                                     | AFT OF NOSE           | FROM AIRCRAFT CENTERLINE |                      | MEAN HEIGHT FROM GROUND |
|                                                     |                       | LH SIDE                  | RH SIDE              |                         |
| Refuel/Defuel coupling, right:<br>Access door 622HB | 28.3 m<br>(92.85 ft)  |                          | 12.6 m<br>(41.34 ft) | 5.1 m<br>(16.73 ft)     |
| Overwing gravity refuel cap                         | 32.8 m<br>(107.61 ft) | 17.2 m<br>(56.43 ft)     | 17.2 m<br>(56.43 ft) | 6.1 m<br>(20.01 ft)     |

A. Four standard 2.5 in. ISO 45 connections.

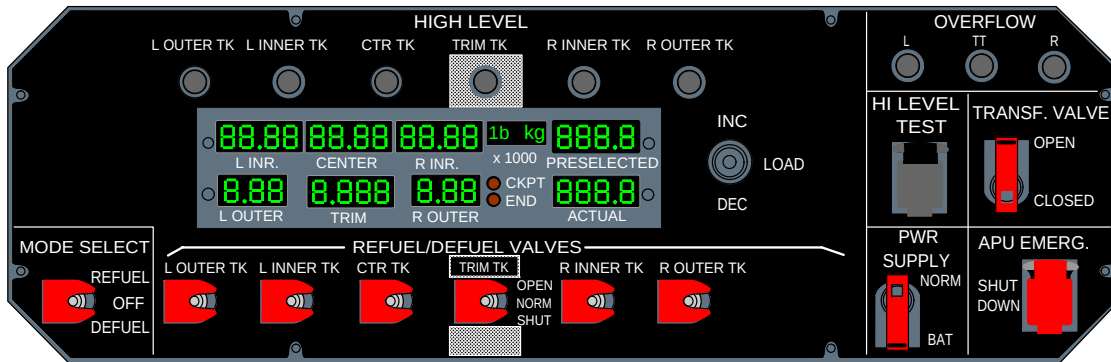
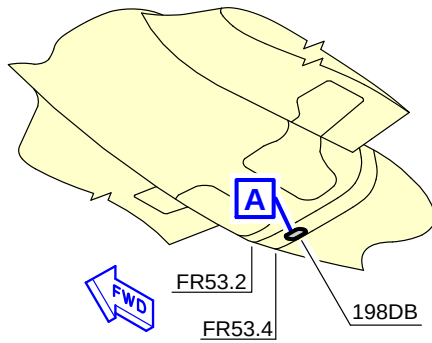
B. Two service connections (gravity refuel).

### 3. Overpressure Protector and NACA Flame Arrestor

**NOTE :** The mean height from ground in the below table may change according to the CG position and aircraft weight.

| ACCESS                                                       | DISTANCE              |                          |                       |                            |
|--------------------------------------------------------------|-----------------------|--------------------------|-----------------------|----------------------------|
|                                                              | AFT OF NOSE           | FROM AIRCRAFT CENTERLINE |                       | MEAN HEIGHT<br>FROM GROUND |
|                                                              |                       | LH SIDE                  | RH SIDE               |                            |
| Overpressure protector (wing):<br>Access panel 550EB (650EB) | 37.8 m<br>(124.02 ft) | 27.17 m<br>(89.14 ft)    | 27.17 m<br>(89.14 ft) | 5.75 m<br>(18.86 ft)       |
| NACA flame arrestor (wing):<br>Access panel 550DB (650DB)    | 37.4 m<br>(122.70 ft) | 26.53 m<br>(87.04 ft)    | 26.53 m<br>(87.04 ft) | 5.7 m<br>(18.70 ft)        |

**\*\*ON A/C A330-700L**

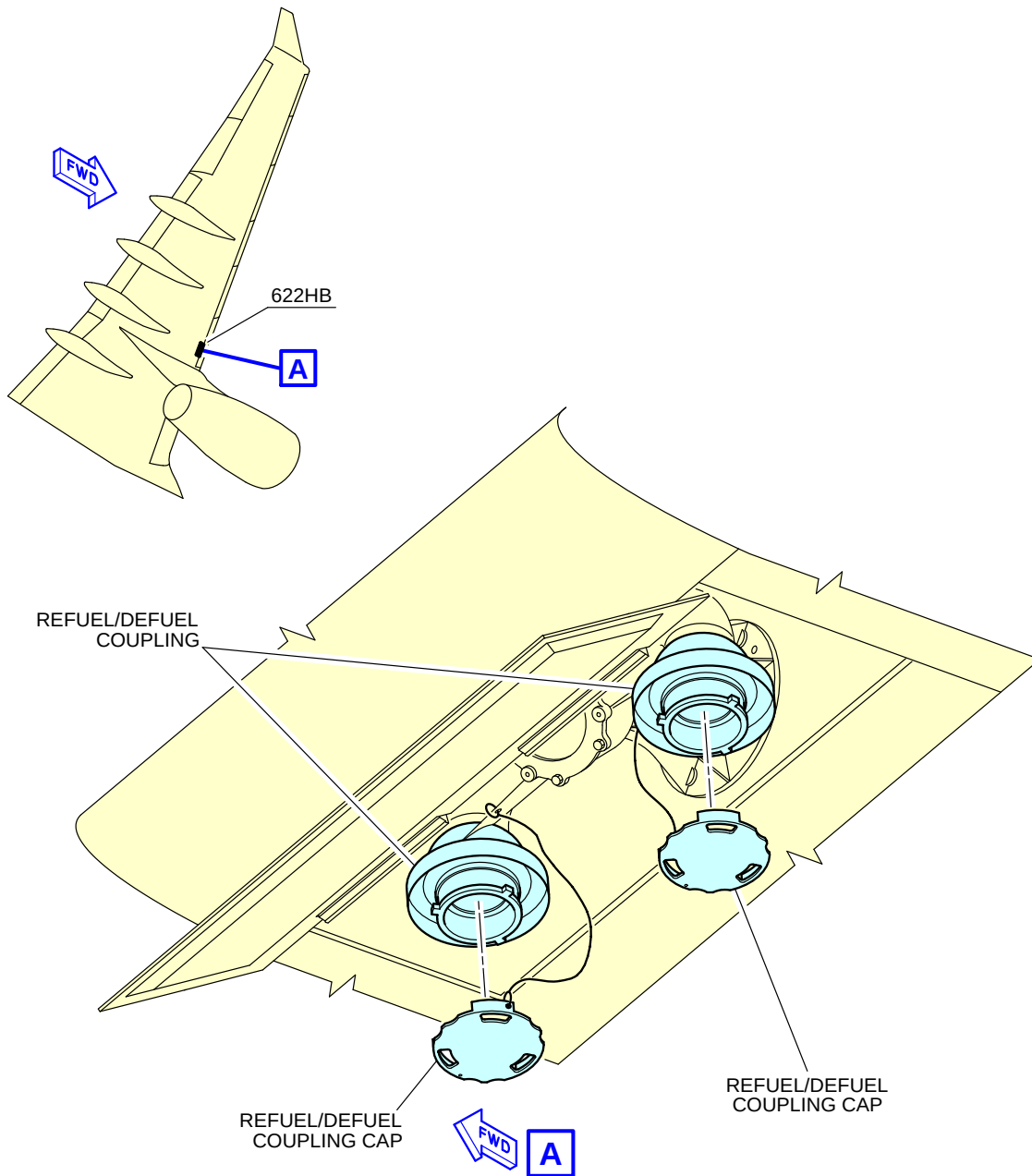


**A**

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Ground Service Connections  
Refuel/Defuel Control Panel  
FIGURE-5-4-6-991-021-A01

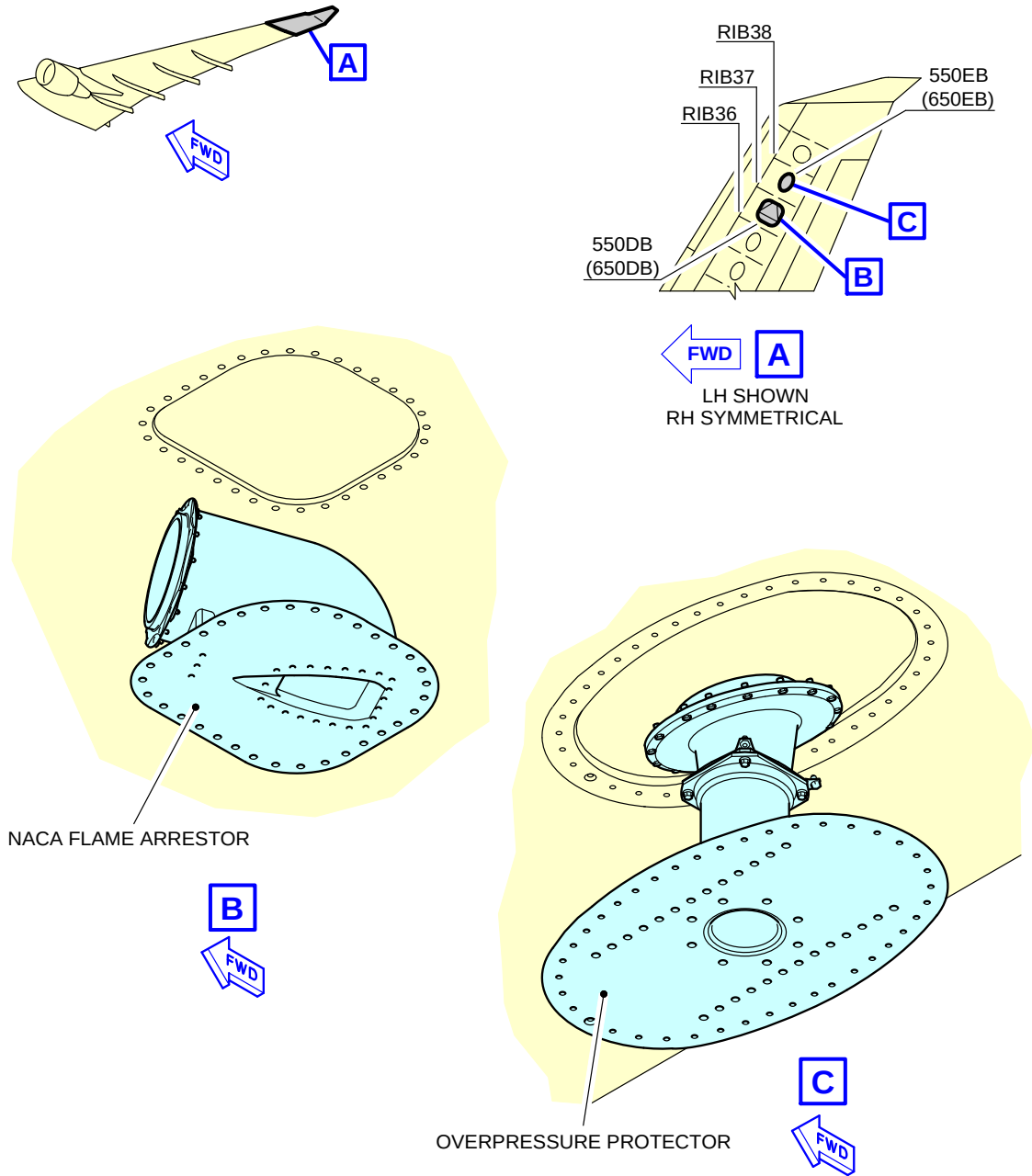
**\*\*ON A/C A330-700L**



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Ground Service Connections  
Refuel/Defuel Coupling  
FIGURE-5-4-6-991-022-A01

**\*\*ON A/C A330-700L**



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Ground Service Connections  
Overpressure Protector and NACA Flame Arrestor - Wing  
FIGURE-5-4-6-991-023-A01

## 5-4-7 Pneumatic System

### \*\*ON A/C A330-700L

#### Pneumatic System

#### 1. High Pressure Air Connection

NOTE : The mean height from ground in the below table may change according to the CG position and aircraft weight.

| ACCESS                              | DISTANCE              |                          |         |                         |
|-------------------------------------|-----------------------|--------------------------|---------|-------------------------|
|                                     | AFT OF NOSE           | FROM AIRCRAFT CENTERLINE |         | MEAN HEIGHT FROM GROUND |
|                                     |                       | LH SIDE                  | RH SIDE |                         |
| HP connectors:<br>Access door 193CB | 20.94 m<br>(68.70 ft) | 0.84 m<br>(2.76 ft)      |         | 1.79 m<br>(5.87 ft)     |

- A. Connectors:
- Two standard 3 in. ISO 2026 connections.

#### 2. Low Pressure Air Connection

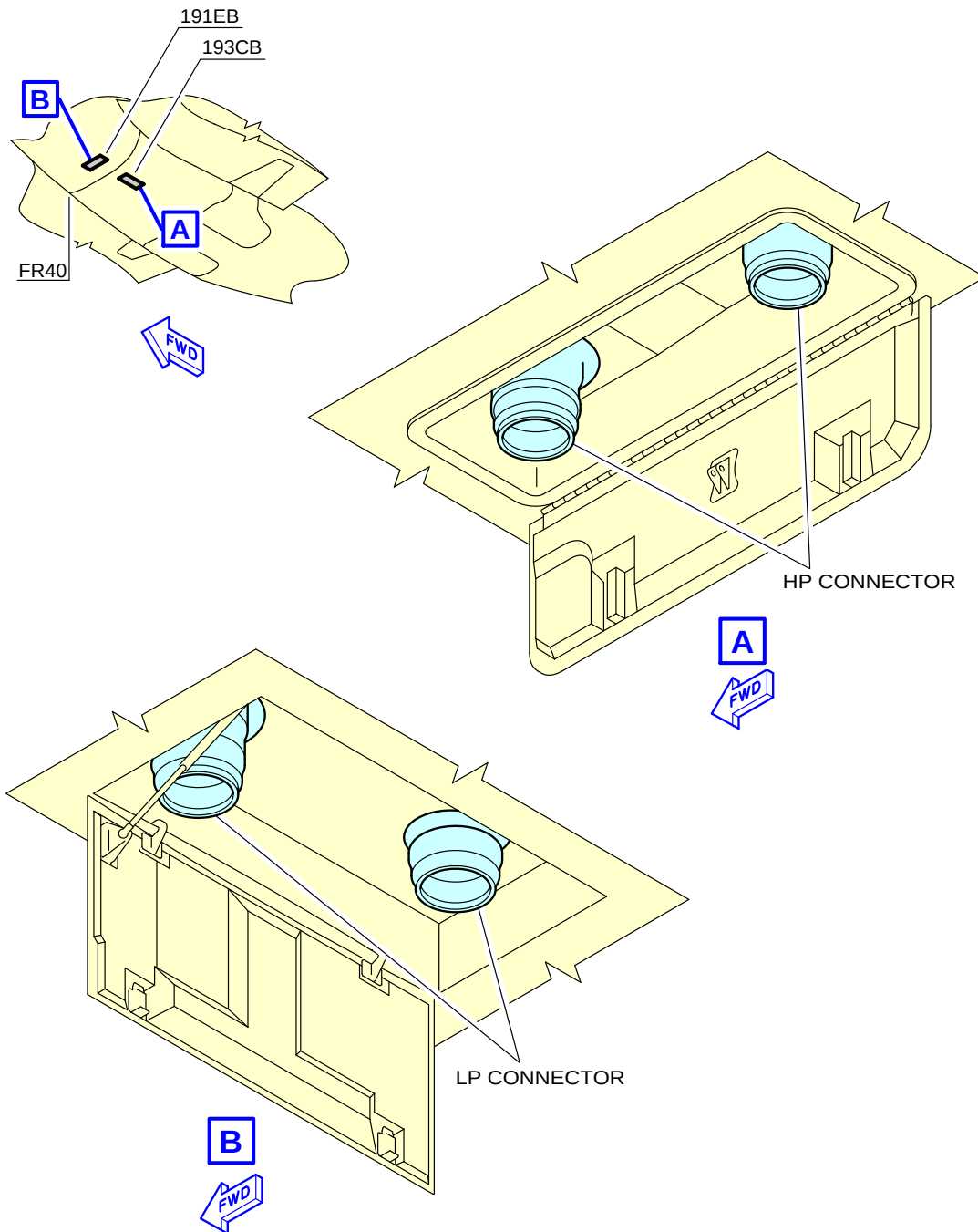
NOTE : The mean height from ground in the below table may change according to the CG position and aircraft weight.

| ACCESS                              | DISTANCE              |                          |         |                         |
|-------------------------------------|-----------------------|--------------------------|---------|-------------------------|
|                                     | AFT OF NOSE           | FROM AIRCRAFT CENTERLINE |         | MEAN HEIGHT FROM GROUND |
|                                     |                       | LH SIDE                  | RH SIDE |                         |
| LP connectors:<br>Access door 191EB | 22.66 m<br>(74.34 ft) | 0.4 m<br>(1.31 ft)       |         | 8 m<br>(26.25 ft)       |

- A. Connectors:
- Two standard 8 in. SAE AS4262 connections.



**\*\*ON A/C A330-700L**



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Ground Service Connections  
LP and HP Ground Connectors  
FIGURE-5-4-7-991-004-A01

## 5-4-8 Oil System

### \*\*ON A/C A330-700L

#### Oil System

#### 1. RR Trent 700 Series Engine

##### A. Engine Oil Replenishment:

One gravity filling cap.

One ozone self-sealing pressure fill and overfill connector per engine.

NOTE : The mean height from ground in the below table may change according to the CG position and aircraft weight.

| ACCESS                         | DISTANCE              |                          |                       | MEAN HEIGHT<br>FROM GROUND |
|--------------------------------|-----------------------|--------------------------|-----------------------|----------------------------|
|                                | AFT OF NOSE           | FROM AIRCRAFT CENTERLINE |                       |                            |
|                                |                       | LH SIDE                  | RH SIDE               |                            |
| Engine 1:<br>Access door 416CR | 25.40 m<br>(83.33 ft) | 7.92 m<br>(25.98 ft)     |                       | 2.05 m<br>(6.73 ft)        |
| Engine 2:<br>Access door 426CR | 25.40 m<br>(83.33 ft) |                          | 10.82 m<br>(35.50 ft) | 2.05 m<br>(6.73 ft)        |

##### (1) Tank capacity:

- Full level: 23.30 l (6.16 US gal).
- Usable: 22.71 l (6.00 US gal).

##### B. IDG Oil Replenishment:

One ozone self-sealing pressure fill and overfill connector per engine.

NOTE : The mean height from ground in the below table may change according to the CG position and aircraft weight.

| ACCESS                         | DISTANCE             |                          |                      |                            |
|--------------------------------|----------------------|--------------------------|----------------------|----------------------------|
|                                | AFT OF NOSE          | FROM AIRCRAFT CENTERLINE |                      | MEAN HEIGHT<br>FROM GROUND |
|                                |                      | LH SIDE                  | RH SIDE              |                            |
| Engine 1:<br>Access door 415CL | 25.9 m<br>(84.97 ft) | 9.65 m<br>(31.66 ft)     |                      | 0.80 m<br>(2.62 ft)        |
| Engine 2:<br>Access door 425CL | 25.9 m<br>(84.97 ft) |                          | 9.09 m<br>(29.82 ft) | 0.80 m<br>(2.62 ft)        |

(1) Max delivery pressure required: 2.76 bar (40 psi).

(2) Max oil capacity of the IDG: 5.50 l (1.45 Us gal).

C. Starter Oil Replenishment:

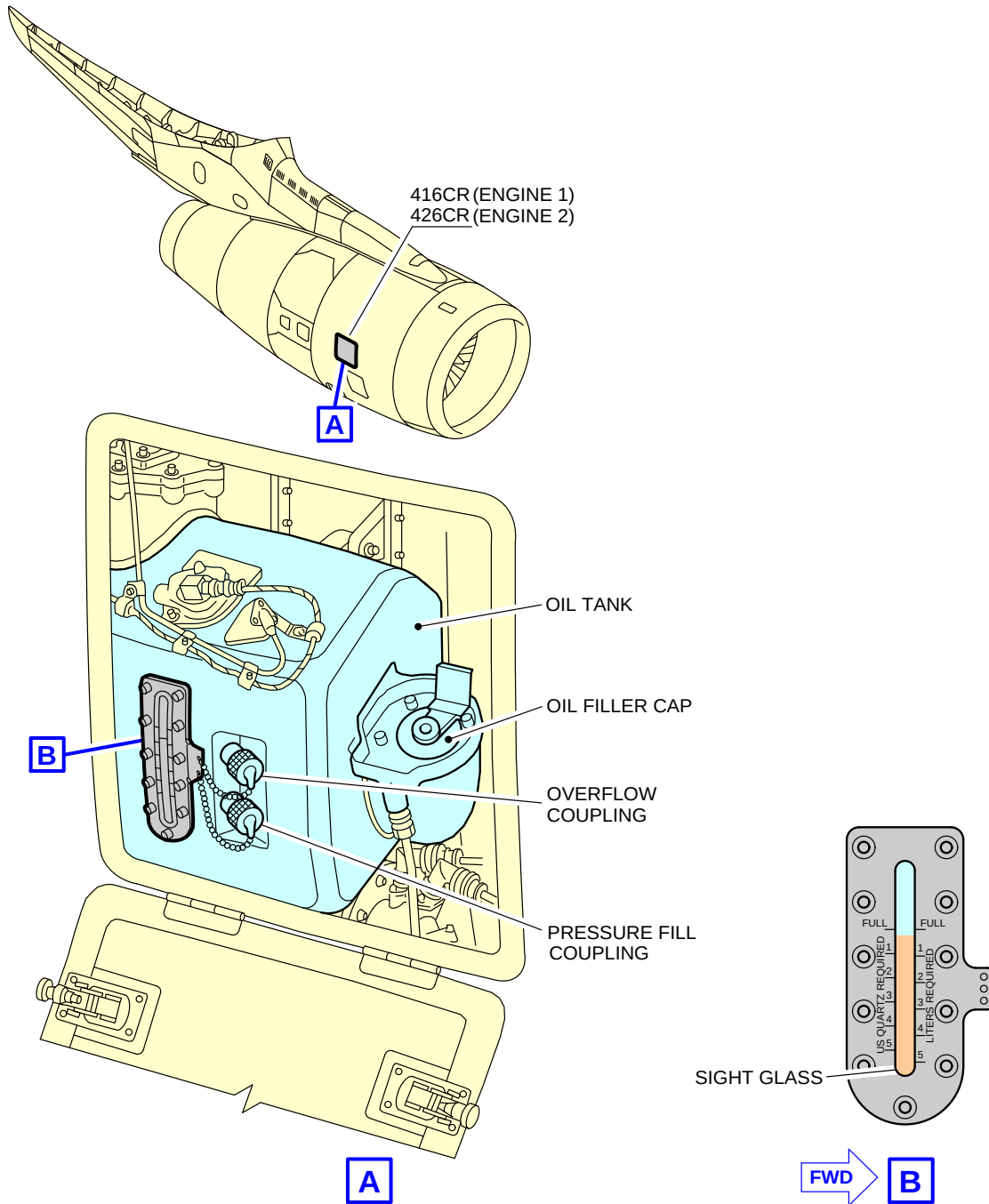
One filling connection per engine.

**NOTE :** The mean height from ground in the below table may change according to the CG position and aircraft weight.

| ACCESS                                    | DISTANCE             |                          |                      |                            |
|-------------------------------------------|----------------------|--------------------------|----------------------|----------------------------|
|                                           | AFT OF NOSE          | FROM AIRCRAFT CENTERLINE |                      | MEAN HEIGHT<br>FROM GROUND |
|                                           |                      | LH SIDE                  | RH SIDE              |                            |
| Engine 1:<br>Access door 415AL<br>(416AR) | 25.9 m<br>(84.97 ft) | 9.65 m<br>(31.66 ft)     |                      | 0.80 m<br>(2.62 ft)        |
| Engine 2:<br>Access door 425AL<br>(426AR) | 25.9 m<br>(84.97 ft) |                          | 9.09 m<br>(29.82 ft) | 0.80 m<br>(2.62 ft)        |

- Max oil capacity of the starter: 0.50 l (0.13 US gal).

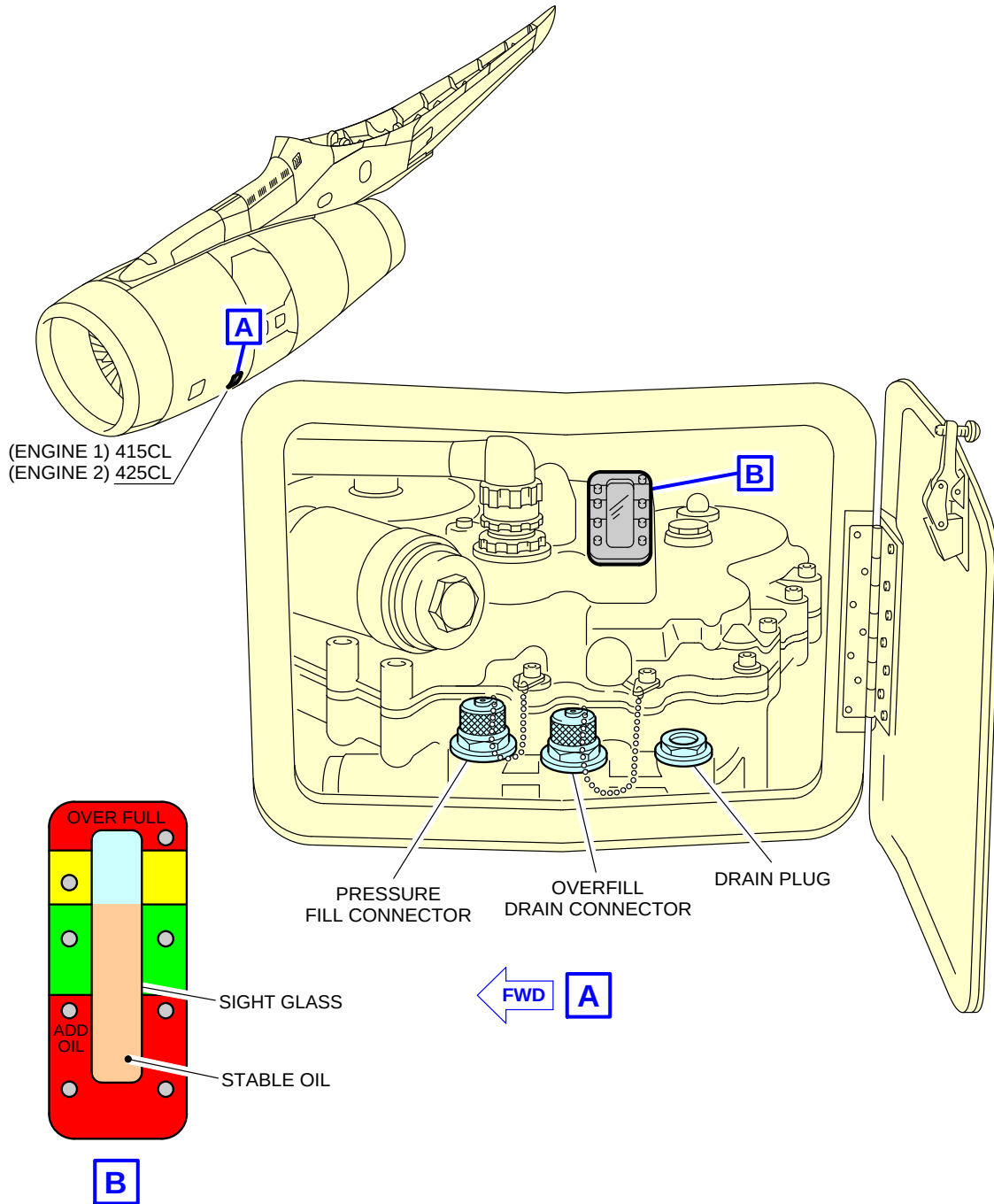
**\*\*ON A/C A330-700L**



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Ground Service Connections  
Engine Oil Tank - RR Trent 700 Series Engine  
FIGURE-5-4-8-991-028-A01

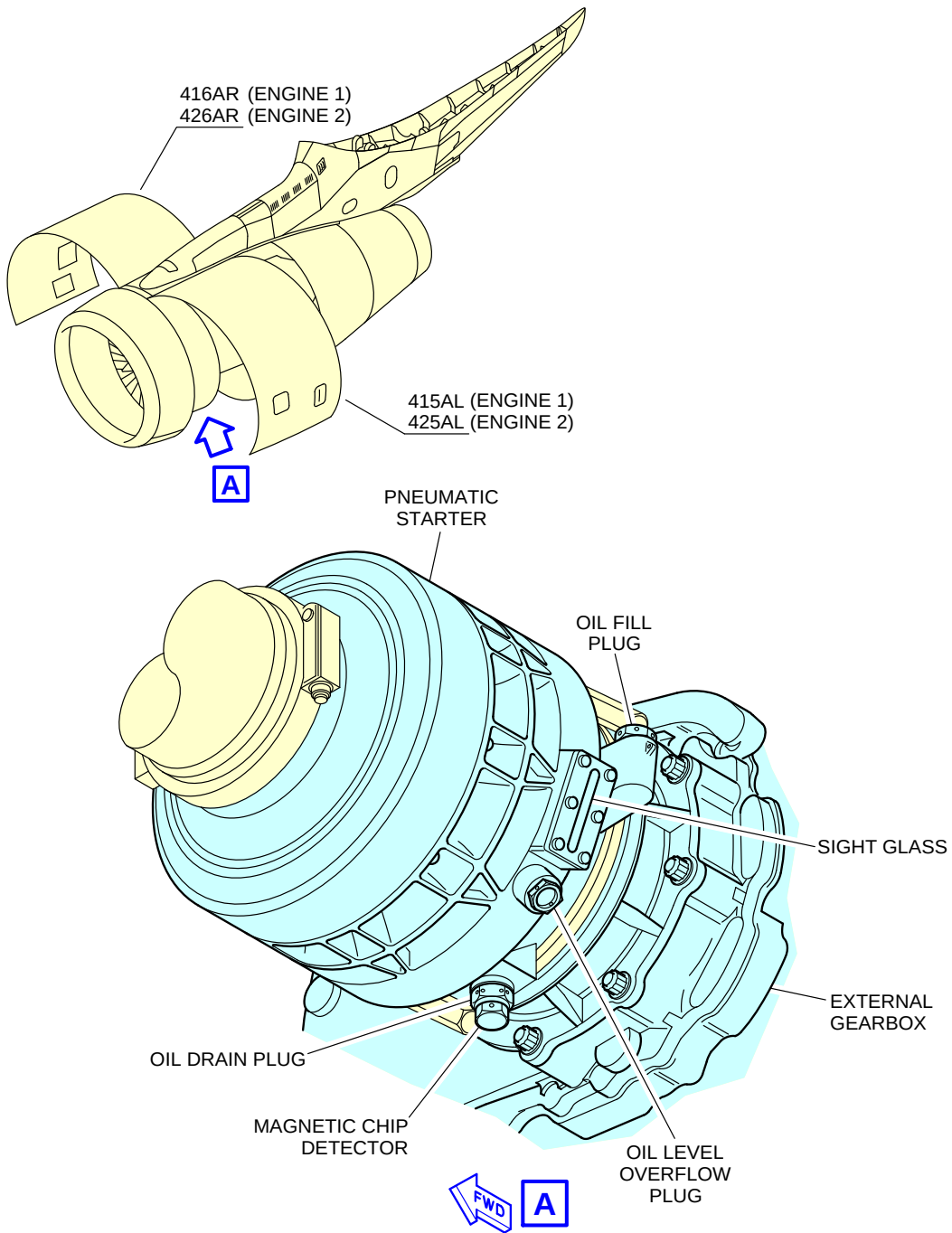
**\*\*ON A/C A330-700L**



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Ground Service Connections  
IDG Oil Tank - RR Trent 700 Series Engine  
FIGURE-5-4-8-991-029-A01

**\*\*ON A/C A330-700L**



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Ground Service Connections  
 Starter Oil Tank - RR Trent 700 Series Engine  
 FIGURE-5-4-8-991-030-A01

## \*\*ON A/C A330-700L

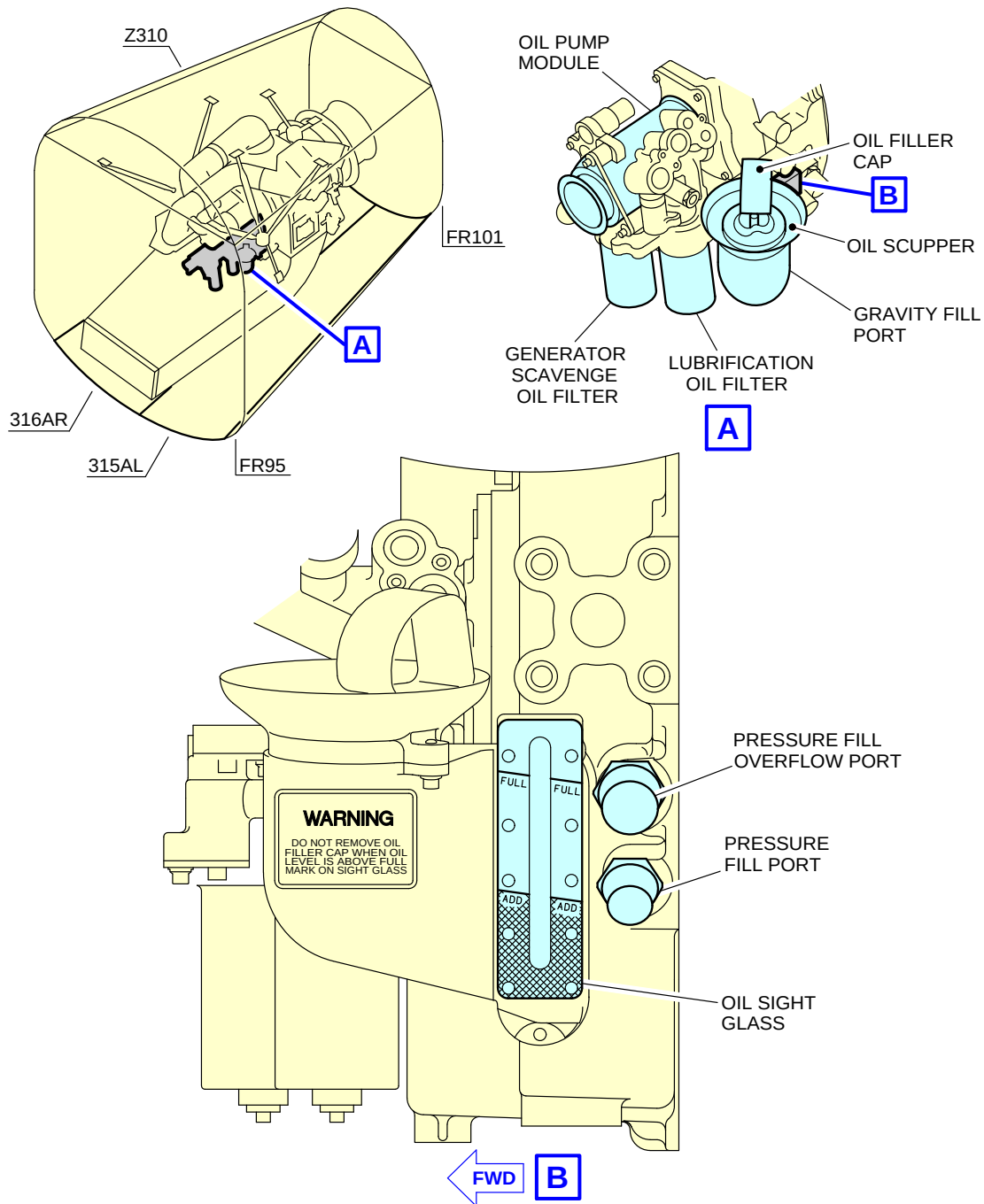
### APU Oil Servicing

1. APU Oil Servicing:  
APU oil gravity filling cap.

NOTE : The mean height from ground in the below table may change according to the CG position and aircraft weight.

| ACCESS                                                       | DISTANCE               |                          |         |                            |
|--------------------------------------------------------------|------------------------|--------------------------|---------|----------------------------|
|                                                              | AFT OF NOSE            | FROM AIRCRAFT CENTERLINE |         | MEAN HEIGHT<br>FROM GROUND |
|                                                              |                        | LH SIDE                  | RH SIDE |                            |
| APU oil<br>replenishment:<br>Access doors<br>315AL and 316AR | 58.94 m<br>(193.37 ft) | 0.4 m<br>(1.31 ft)       |         | 8 m<br>(26.25 ft)          |

**\*\*ON A/C A330-700L**



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Ground Service Connections  
APU Oil Servicing  
FIGURE-5-4-8-991-031-A01



**5-4-10 Waste Water System**
**\*\*ON A/C A330-700L**
Waste Water System
**1. Waste Water Servicing**
**A. There are two waste-water ground-service panels:**

- First panel: One standard connection Roylyn 1 in. (ISO 17775) for flushing and filling
- Second panel: One standard Taco type valve 4 in. (ISO 17775) for draining.

NOTE : Handle used for drainage is located on the first panel.

**B. Capacity waste tanks:**

- Standard: 35 l (9.25 US gal).

NOTE : The waste water drain-system discards the waste water from the galley sink and the lavatory washbasin overboard.

The toilet system moves the waste materials and liquids from the toilet to the waste tank.

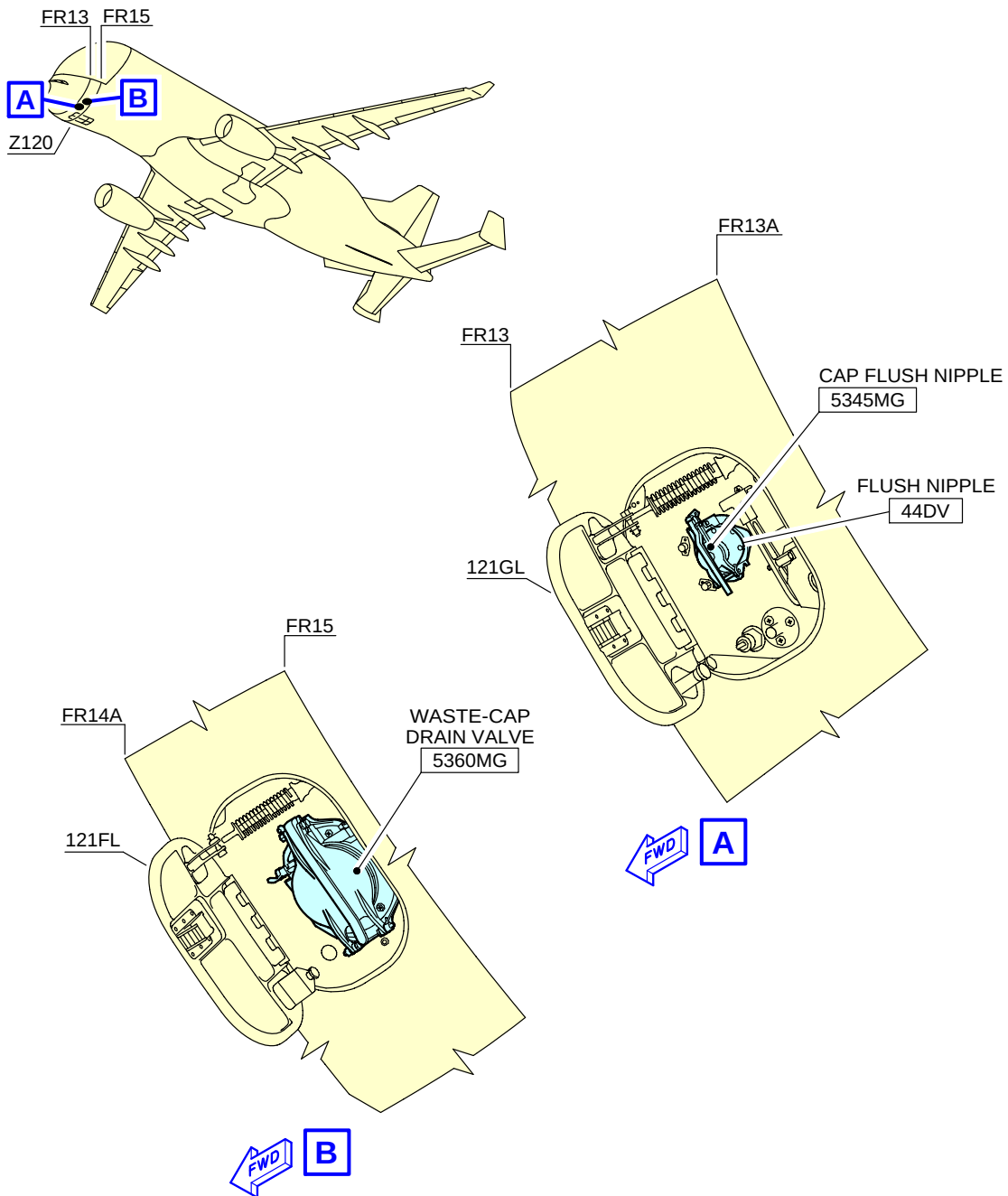
**C. Chemical fluid:**

- Standard: 9.5 l (2.51 US gal).

NOTE : The mean height from ground in the below table may change according to the CG position and aircraft weight.

| ACCESS                                                   | DISTANCE             |                          |         |                         |
|----------------------------------------------------------|----------------------|--------------------------|---------|-------------------------|
|                                                          | AFT OF NOSE          | FROM AIRCRAFT CENTERLINE |         | MEAN HEIGHT FROM GROUND |
|                                                          |                      | LH SIDE                  | RH SIDE |                         |
| Waste-water ground-service-panel 1:<br>Access door 121GL | 6.39 m<br>(20.96 ft) | 2.03 m<br>(6.66 ft)      |         | 2.16 m<br>(7.09 ft)     |
| Waste-water ground-service-panel 2:<br>Access door 121FL | 7.13 m<br>(23.39 ft) | 1.91 m<br>(6.27 ft)      |         | 2.19 m<br>(7.19 ft)     |

**\*\*ON A/C A330-700L**



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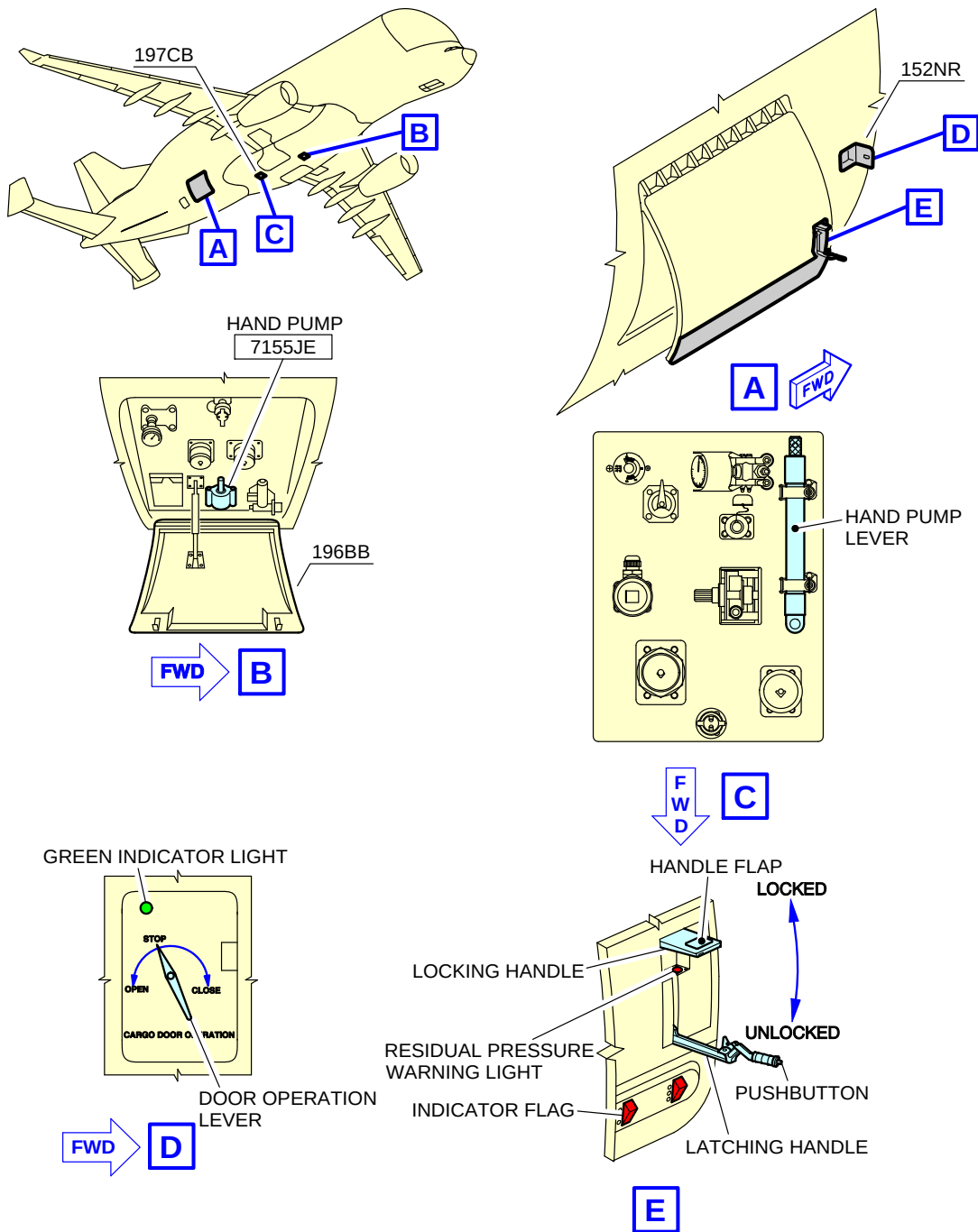
Ground Service Connections  
Waste-Water Ground-Service Panels  
FIGURE-5-4-10-991-008-A01

**5-4-11 Cargo Control Panels**
**\*\*ON A/C A330-700L**
Cargo Control Panels
**1. Cargo Control Panels**

NOTE : The mean height from ground in the below table may change according to the CG position and aircraft weight.

| ACCESS                                            | DISTANCE              |                          |                     |                            |
|---------------------------------------------------|-----------------------|--------------------------|---------------------|----------------------------|
|                                                   | AFT OF NOSE           | FROM AIRCRAFT CENTERLINE |                     | MEAN HEIGHT<br>FROM GROUND |
|                                                   |                       | LH SIDE                  | RH SIDE             |                            |
| AFT cargo door panel:<br>Access door 152NR        | 40.73 m<br>(133.6 ft) |                          | 2.68 m<br>(8.8 ft)  | 4.41 m<br>(14.5 ft)        |
| Main Deck Cargo Door (MDCD) inner operation panel | 9.26 m<br>(30.4 ft)   |                          | 3.23 m<br>(10.6 ft) | 6.09 m<br>(20.0 ft)        |
| MDCD outer operation panel                        | 10.25 m<br>(33.6 ft)  |                          | 3.55 m<br>(11.6 ft) | 6.26 m<br>(20.5 ft)        |

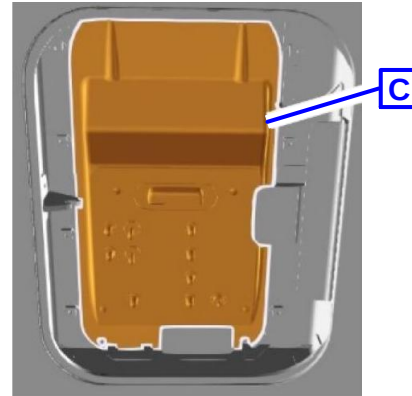
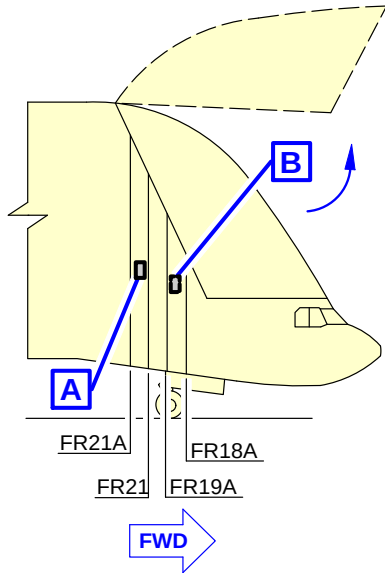
**\*\*ON A/C A330-700L**



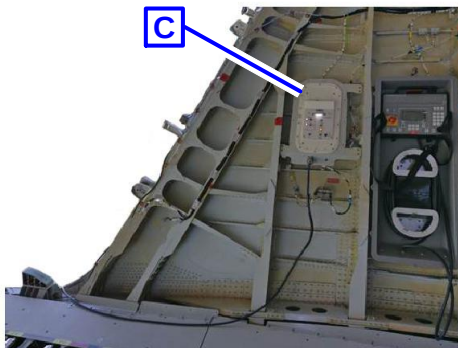
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Lower Cargo Door Operation Panel  
FIGURE-5-4-11-991-008-A01

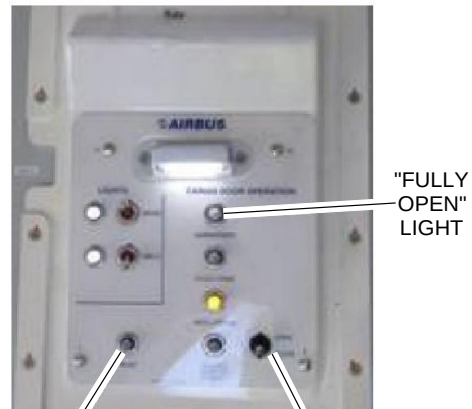
**\*\*ON A/C A330-700L**



**A**  
MDSD OUTER  
OPERATION PANEL



**B**  
MDSD INNER  
OPERATION PANEL



**C**  
TYPICAL  
MDSD OPERATION PANEL

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MDSD Control Panel  
FIGURE-5-4-11-991-009-A01

## 5-5-0 Engine Starting Pneumatic Requirements

### \*\*ON A/C A330-700L

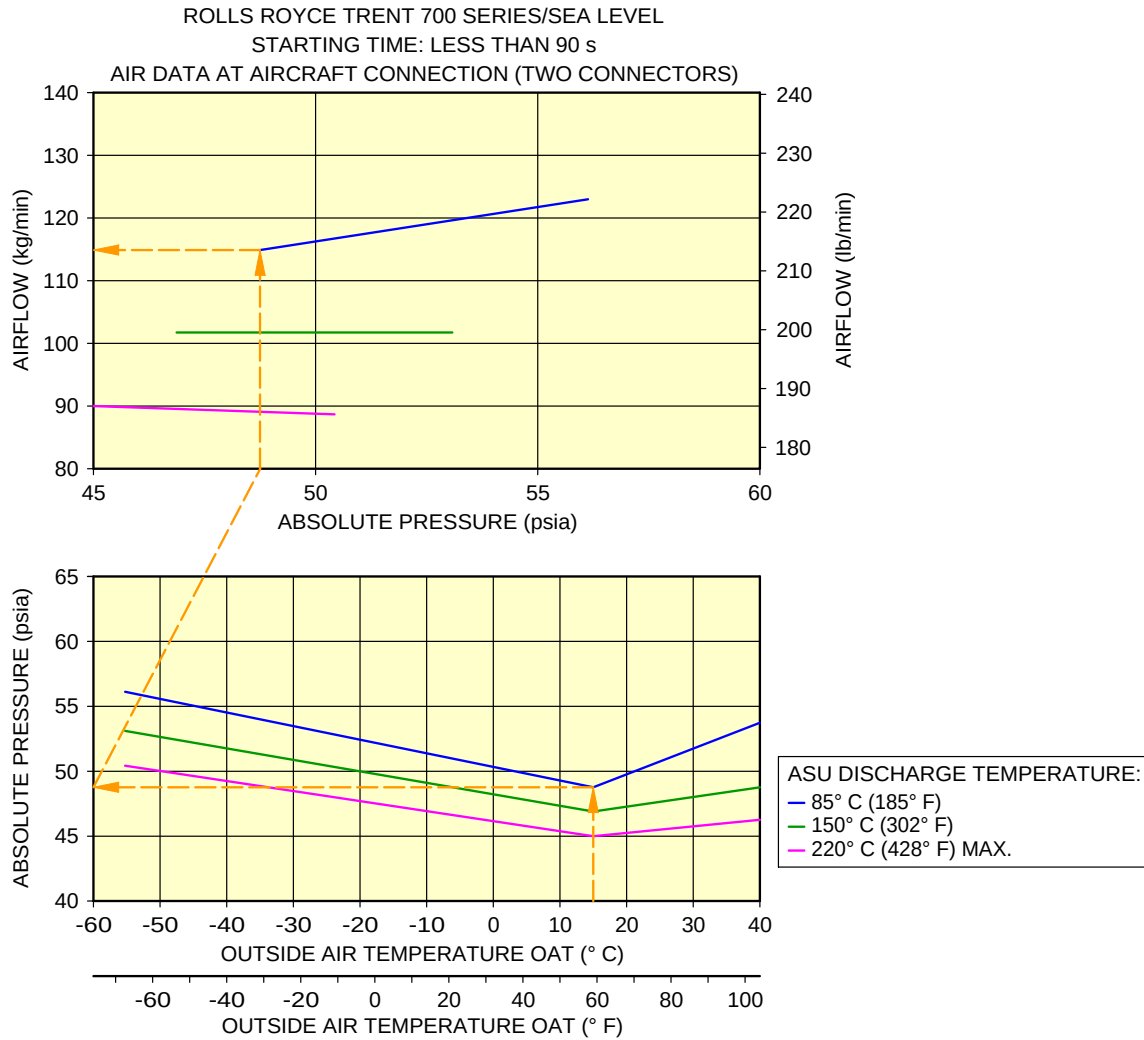
#### Engine Starting Pneumatic Requirements

1. The purpose of this section is to give the minimum air data requirements at the aircraft connection, needed to start the engine within no more than 90 seconds, at sea level (0 feet), for a set of Outside Air Temperatures (OAT).

| ABBREVIATION | DEFINITION                      |
|--------------|---------------------------------|
| A/C          | Aircraft                        |
| ASU          | Air Start Unit                  |
| HPGC         | High Pressure Ground Connection |
| OAT          | Outside Air Temperature         |

- A. Air data (discharge temperature, absolute discharge pressure) are given at the HPGC.
- B. For the requirements below, the configuration with two HPGC is used. Using one connector only (for a given mass flow rate and discharge pressure from the ASU) will increase the pressure loss in the ducts of the bleed system and therefore lower the performances at the engine starter.
- C. For a given OAT the following charts are used to determine an acceptable combination for air discharge temperature, absolute discharge pressure and mass flow rate.
- D. This section is addressing requirements for the ASU only, and is not representative of the start performance of the aircraft using the APU or engine cross bleed procedure.
- E. To protect the A/C, the charts feature, if necessary:
  - The maximum discharge pressure at the HPGC
  - The maximum discharge temperature at the HPGC.

## \*\*ON A/C A330-700L



### EXAMPLE:

FOR AN OAT OF 15° C (59° F) AND AN ASU PROVIDING A DISCHARGE TEMPERATURE OF 85° C (185° F) AT HPGC:

- THE REQUIRED PRESSURE AT HPGC IS 48.85 psia
- THE REQUIRED AIRFLOW AT HPGC IS 114.9 kg/min.

### NOTE:

IN CASE THE ACTUAL DISCHARGE TEMPERATURE OF THE ASU DIFFERS SUBSTANTIALLY FROM THE ONES GIVEN IN THE CHARTS, A SIMPLE INTERPOLATION (LINEAR) IS SUFFICIENT TO DETERMINE THE REQUIRED AIR DATA.

### EXAMPLE:

FOR AN OAT OF 15° C (59° F) AND AN ASU PROVIDING A DISCHARGE TEMPERATURE OF 117.5° C (243.5° F) AT HPGC, INTERPOLATING BETWEEN THE LINES 85° C (185° F) AND 150° C (302° F) RESULTS IN:

- A REQUIRED PRESSURE AT HPGC OF 47.84 psia
- A REQUIRED AIRFLOW AT HPGC OF 113.9 kg/min.

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Engine Starting Pneumatic Requirements  
Rolls Royce Trent 700 Series Engine  
FIGURE-5-5-0-991-011-A01

**5-6-0      Ground Pneumatic Power Requirements****\*\*ON A/C A330-700L**Ground Pneumatic Power Requirements

1. Not applicable.



**5-7-0      Preconditioned Airflow Requirements****\*\*ON A/C A330-700L**Preconditioned Airflow Requirements

1. Not applicable.

**5-8-0 Ground Towing Requirements****\*\*ON A/C A330-700L**Ground Towing Requirements**1. This section gives information on aircraft towing.**

The A330-700L is designed with means for conventional or towbarless towing. Information/ procedures can be found for both in chapter 9 of the Aircraft Maintenance Manual.

Status on towbarless towing equipment qualification can be found in ISI 09.11.00001.

It is possible to tow or push the aircraft, at maximum ramp weight with engines at zero or up to idle thrust, using a towbar attached to the NLG. One towbar fitting is installed at the front of the leg (optional towing fitting for towing from the rear of the NLG available).

The main landing gears have attachment points for towing or debogging.

This section shows the chart to determine the drawbar pull and tow tractor mass requirements as a function of the following physical characteristics:

- Aircraft weight
- Number of engines at idle
- Slope.

The chart is based on the A330 engine type with the highest idle thrust.

The chart is therefore valid for A330-700L model.

**2. Towbar design guidelines**

The aircraft towbar shall comply with the following standards:

- ISO 8267-1, "Aircraft - Towbar Attachment Fitting - Interface Requirements - Part 1: Main Line Aircraft"
- ISO 9667, "Aircraft Ground Support Equipment - Towbars"
- IATA Airport Handling Manual AHM 958, "Functional Specification for an Aircraft Towbar".

A conventional type towbar is required which should be equipped with a damping system (to protect the NLG against jerks), a rotating toweye and with towing shear pins:

- A traction shear pin calibrated at 28 620 daN (64 340 lbf),
- A torsion pin calibrated at 3 130 m.daN (276 991 lbf.in).

The towing head is designed according to ISO 8267-1, cat. III.

**3. This section gives information on aircraft towing with towbarless tractors.**

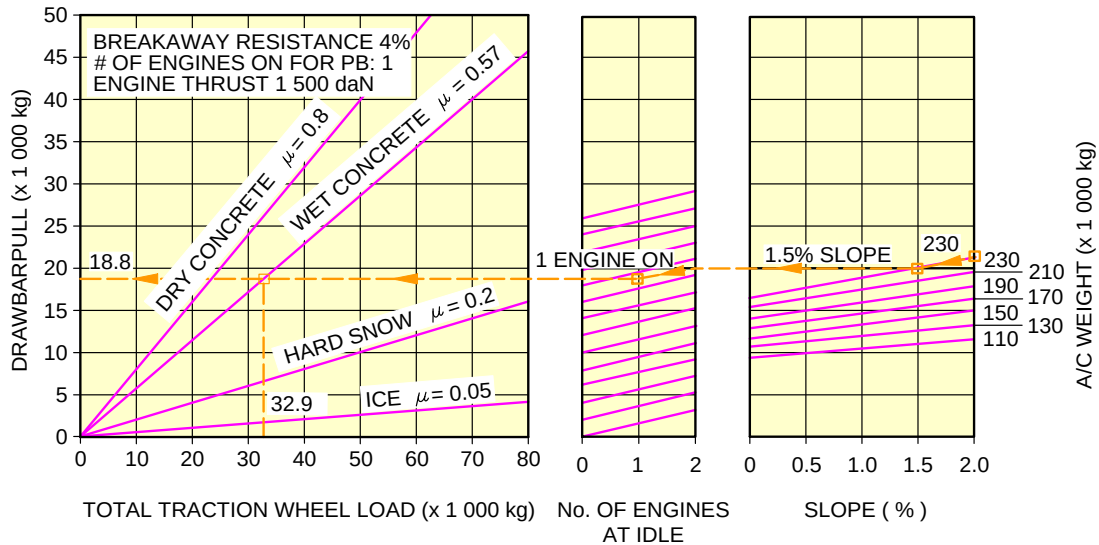
There are special approval procedures for towbarless tractors. Before towing, make sure that the towbarless tractor is approved for towing A330-700L.

The list of towbarless towing vehicles that are specially accepted for A330-700L are given in Airbus In-Service Information, refer to ISI 09.11.00001.

Ram Air Outlet (RAO) 1 and RAO 2 must be closed when a towbarless tractor approaches the aircraft.

If RAO 2 (RH) is stuck open, the towbarless towing is permitted but special attention is necessary because of the space between RAO and the tractor is very less in this configuration. For safe approach of the tractor for the aircraft towing, it is recommended to follow the aircraft center-line trajectory.

## \*\*ON A/C A330-700L



EXAMPLE HOW TO DETERMINE THE TRACTION WHEEL LOAD REQUIREMENT TO TOW A A330 AT 230 000 kg, AT 1.5% SLOPE, 1 ENGINE AT IDLE AND FOR WET TARMAC CONDITIONS:

- ON THE RIGHT HAND SIDE OF THE GRAPH, CHOOSE THE RELEVANT AIRCRAFT WEIGHT (230 000 kg),
- FROM THIS POINT DRAW A PARALLEL LINE TO THE REQUIRED SLOPE PERCENTAGE (1.5%),
- FROM THE POINT OBTAINED DRAW A STRAIGHT HORIZONTAL LINE UNTIL No. OF ENGINES AT IDLE = 2,
- FROM THIS POINT DRAW A PARALLEL LINE TO THE REQUESTED No. OF ENGINES (1),
- FROM THIS POINT DRAW A STRAIGHT HORIZONTAL LINE TO THE DRAWBAR PULL AXIS,
- THE Y-COORDINATE OBTAINED IS THE NECESSARY DRAWBAR PULL FOR THE TRACTOR (18 800 kg),
- SEARCH THE INTERSECTION WITH THE "WET CONCRETE" LINE.
- THE OBTAINED X-COORDINATE IS THE RECOMMENDED TOTAL TRACTION WHEEL LOAD (32 900 kg).

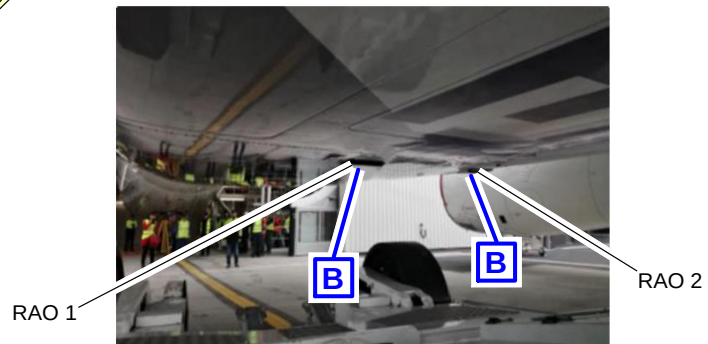
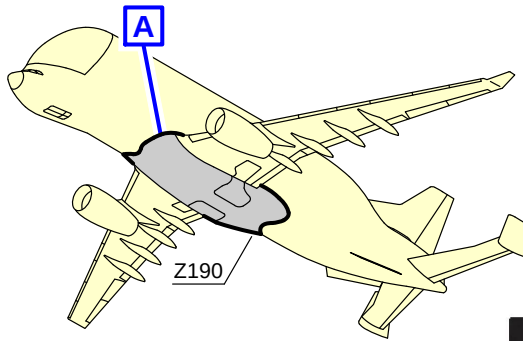
### NOTE:

FOR ALL WHEEL-DRIVEN VEHICLES, THE TOTAL TRACTION WHEEL LOAD IS THE TRACTOR WEIGHT.

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Ground Towing Requirements  
FIGURE-5-8-0-991-015-A01

**\*\*ON A/C A330-700L**



RAO LOCATION



RAO IN CLOSED POSITION



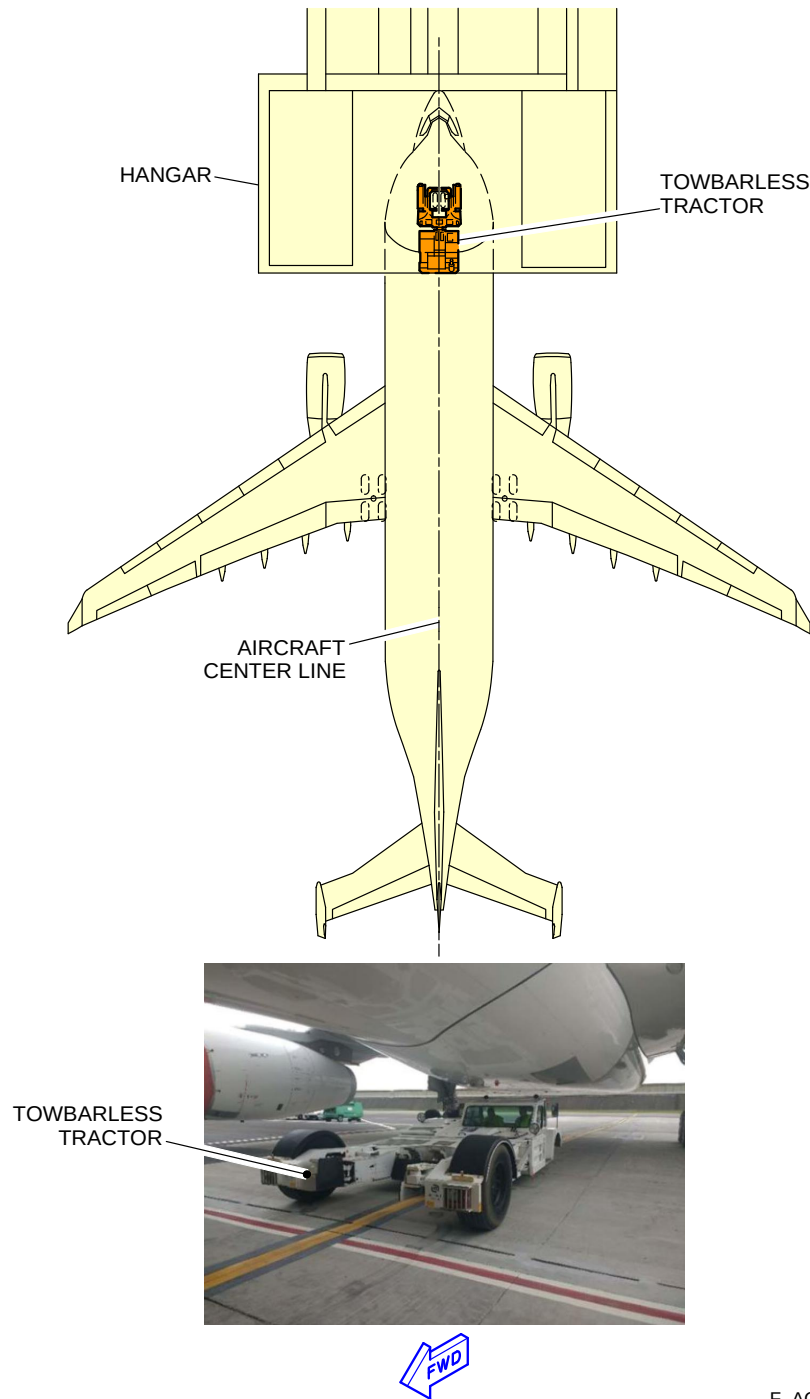
RAO IN OPENED POSITION



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Ground Towing Requirements  
Location of the RAOs (Sheet 1 of 2)  
FIGURE-5-8-0-991-018-A01

**\*\*ON A/C A330-700L**



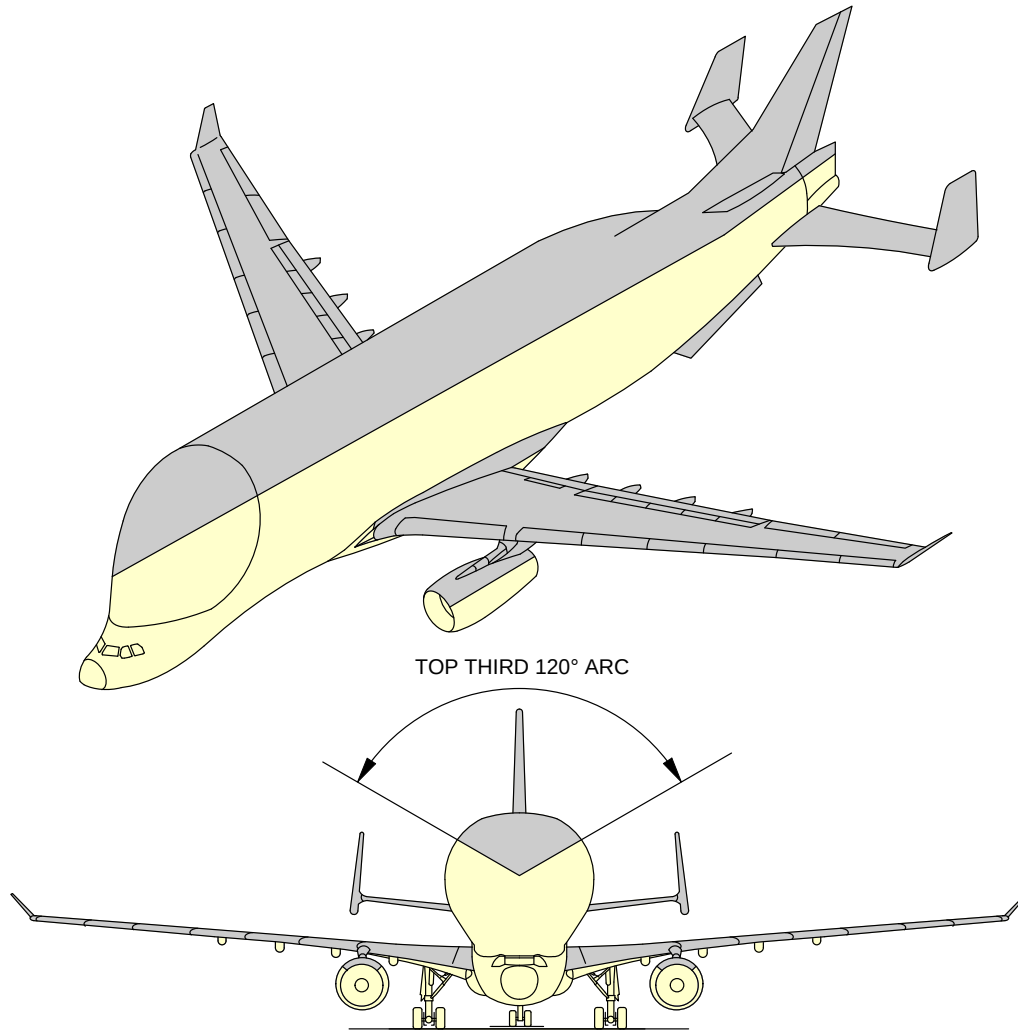
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Ground Towing Requirements  
Tractor Trajectory for Towing Operations (Sheet 2 of 2)  
FIGURE-5-8-0-991-018-A01

**5-9-0 De-Icing and External Cleaning****\*\*ON A/C A330-700L**De-Icing and External Cleaning**1. De-Icing and External Cleaning on Ground**

The mobile equipment for aircraft de-icing and external cleaning must be capable of reaching heights up to approximately 20 m (65.62 ft).

**\*\*ON A/C A330-700L**



| AREA                                 | WING TOP SURFACE (BOTH SIDES) | WING TIP DEVICES (BOTH INSIDE AND OUTSIDE SURFACES) (BOTH SIDES) | HTP TOP SURFACE (BOTH SIDES) | VTP SURFACE (BOTH SIDES) | FUSELAGE TOP SURFACE (TOP THIRD-120° ARC) | NACELLE, PYLON TOP SURFACE (TOP THIRD-120° ARC) (ALL ENGINES) | TOTAL DE-ICED AREA |
|--------------------------------------|-------------------------------|------------------------------------------------------------------|------------------------------|--------------------------|-------------------------------------------|---------------------------------------------------------------|--------------------|
| m <sup>2</sup><br>(ft <sup>2</sup> ) | 306<br>(3 294)                | 11<br>(118)                                                      | 65<br>(700)                  | 210<br>(2 260)           | 410<br>(4 413)                            | 46<br>(495)                                                   | 1 048<br>(11 281)  |

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De-Icing and External Cleaning  
FIGURE-5-9-0-991-001-A01



**OPERATING CONDITIONS****6-1-0 Engine Exhaust Velocities and Temperatures****\*\*ON A/C A330-700L****Engine Exhaust Velocities and Temperatures****1. General**

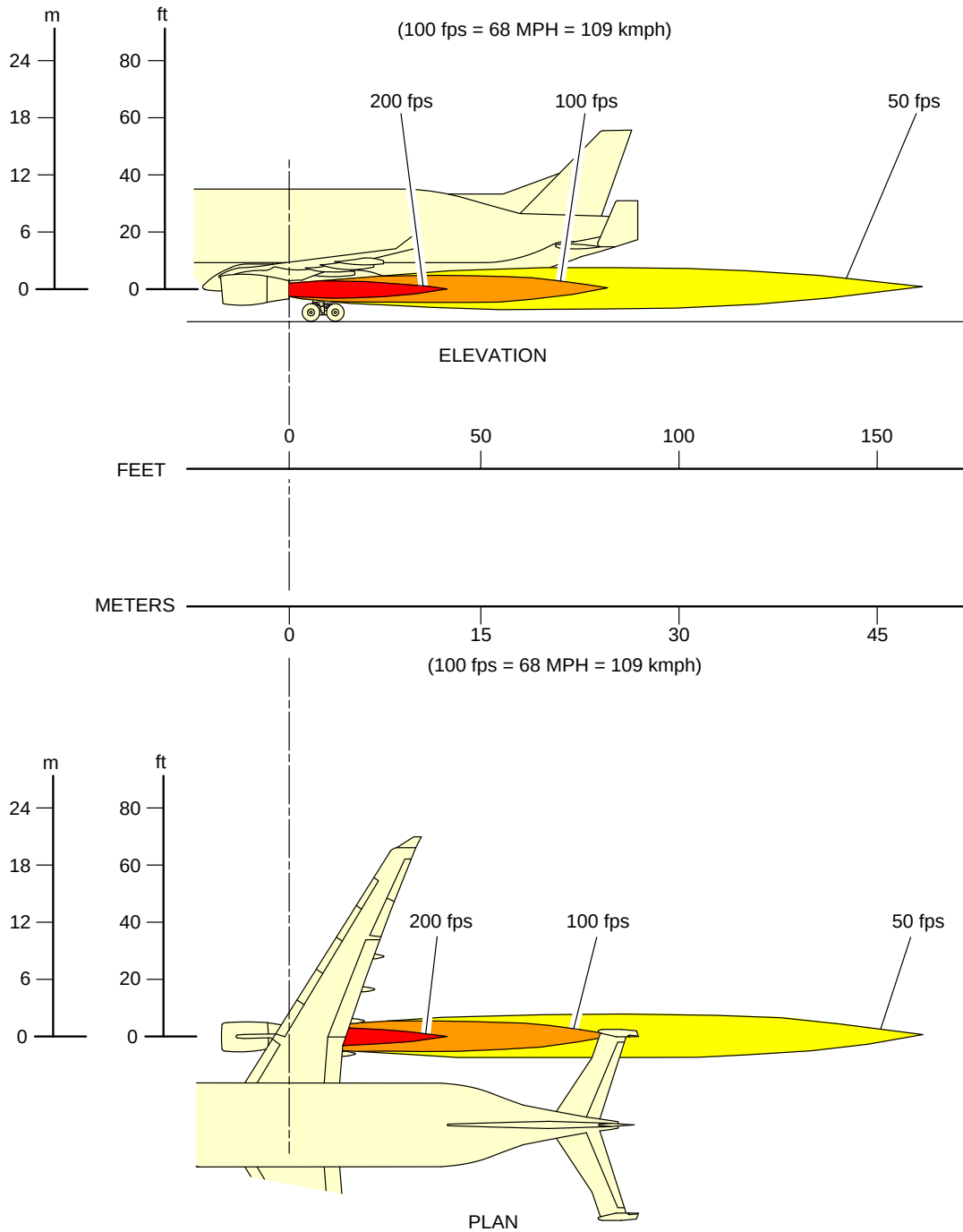
This section shows the estimated engine exhaust efflux velocities and temperatures contours for ground idle, breakaway and maximum takeoff conditions.

**6-1-1 Engine Exhaust Velocities Contours - Ground Idle Power****\*\*ON A/C A330-700L**Engine Exhaust Velocities Contours - Ground Idle Power

1. This section gives engine exhaust velocities contours at ground idle power.

NOTE : The three values give velocities isolines values at the borders of colored areas.

**\*\*ON A/C A330-700L**



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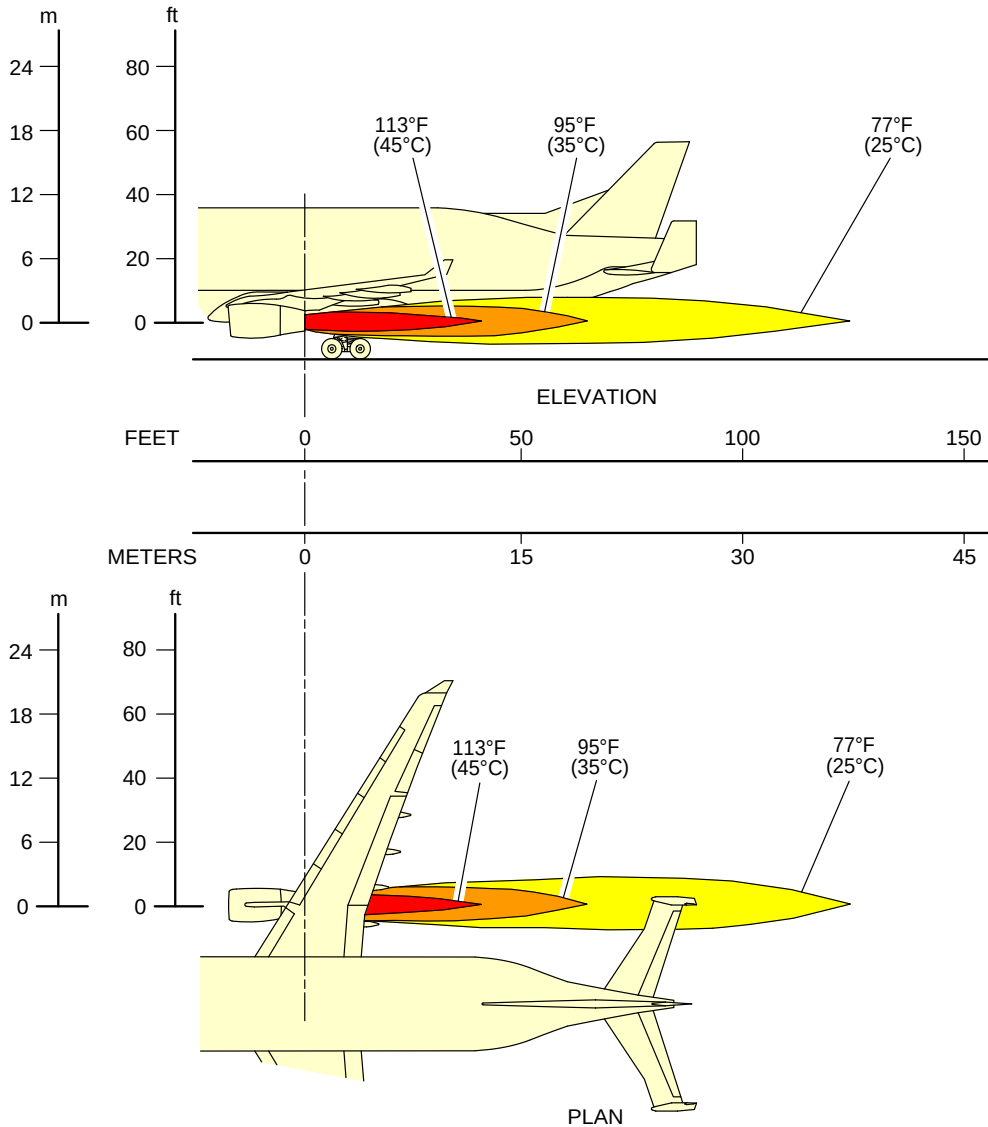
Engine Exhaust Velocities  
Ground Idle Power - RR Trent 700 Series Engine  
FIGURE-6-1-1-991-007-A01

**6-1-2 Engine Exhaust Temperatures Contours - Ground Idle Power****\*\*ON A/C A330-700L**Engine Exhaust Temperatures Contours - Ground Idle Power

1. This section gives engine exhaust temperatures contours at ground idle power.

NOTE : The three values give temperature isolines values at the borders of colored areas.

**\*\*ON A/C A330-700L**



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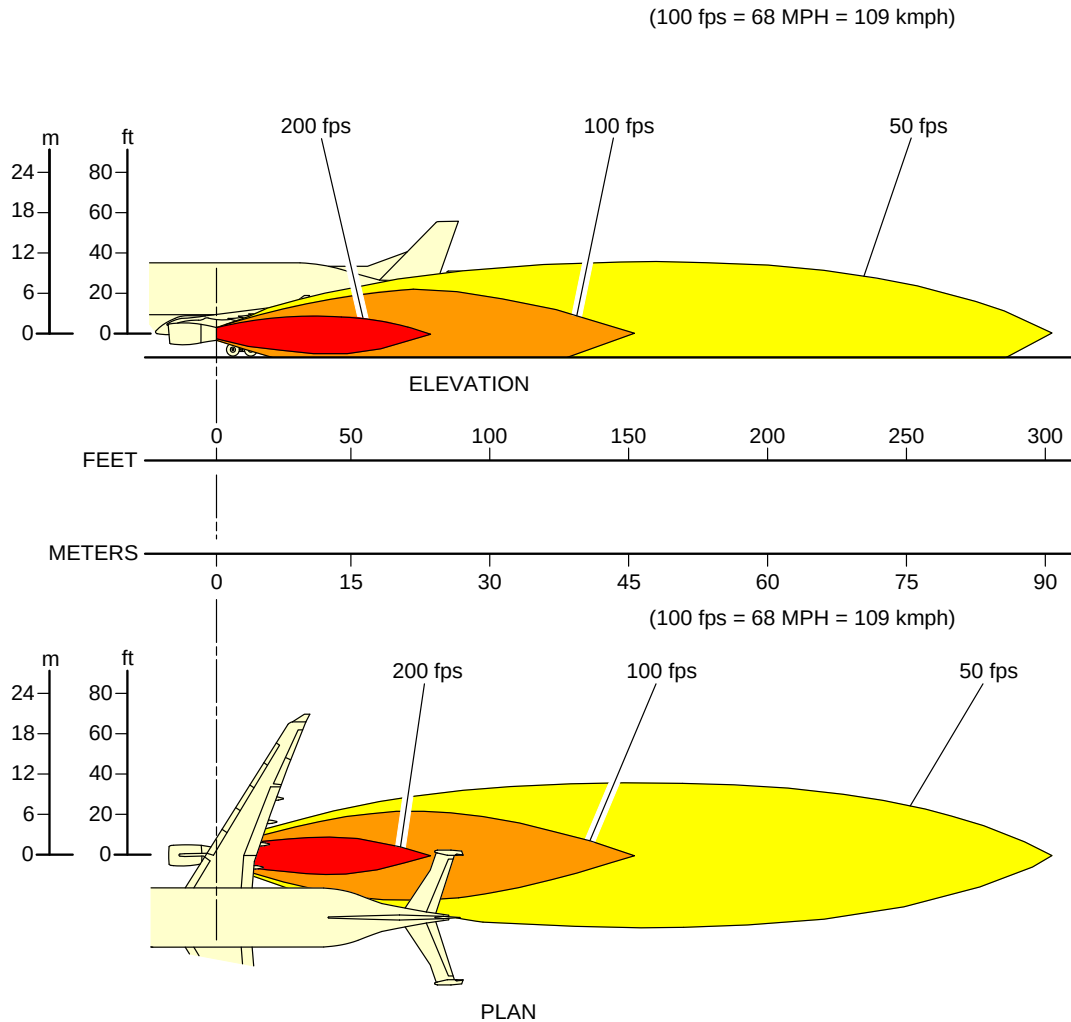
Engine Exhaust Temperatures  
Ground Idle Power - RR Trent 700 Series Engine  
FIGURE-6-1-2-991-007-A01

**6-1-3 Engine Exhaust Velocities Contours - Breakaway Power****\*\*ON A/C A330-700L**Engine Exhaust Velocities Contours - Breakaway Power

1. This section gives engine exhaust velocities contours at breakaway power.

NOTE : The three values give velocities isolines values at the borders of colored areas.

**\*\*ON A/C A330-700L**



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Engine Exhaust Velocities  
Breakaway Power - RR Trent 700 Series Engine  
FIGURE-6-1-3-991-009-A01

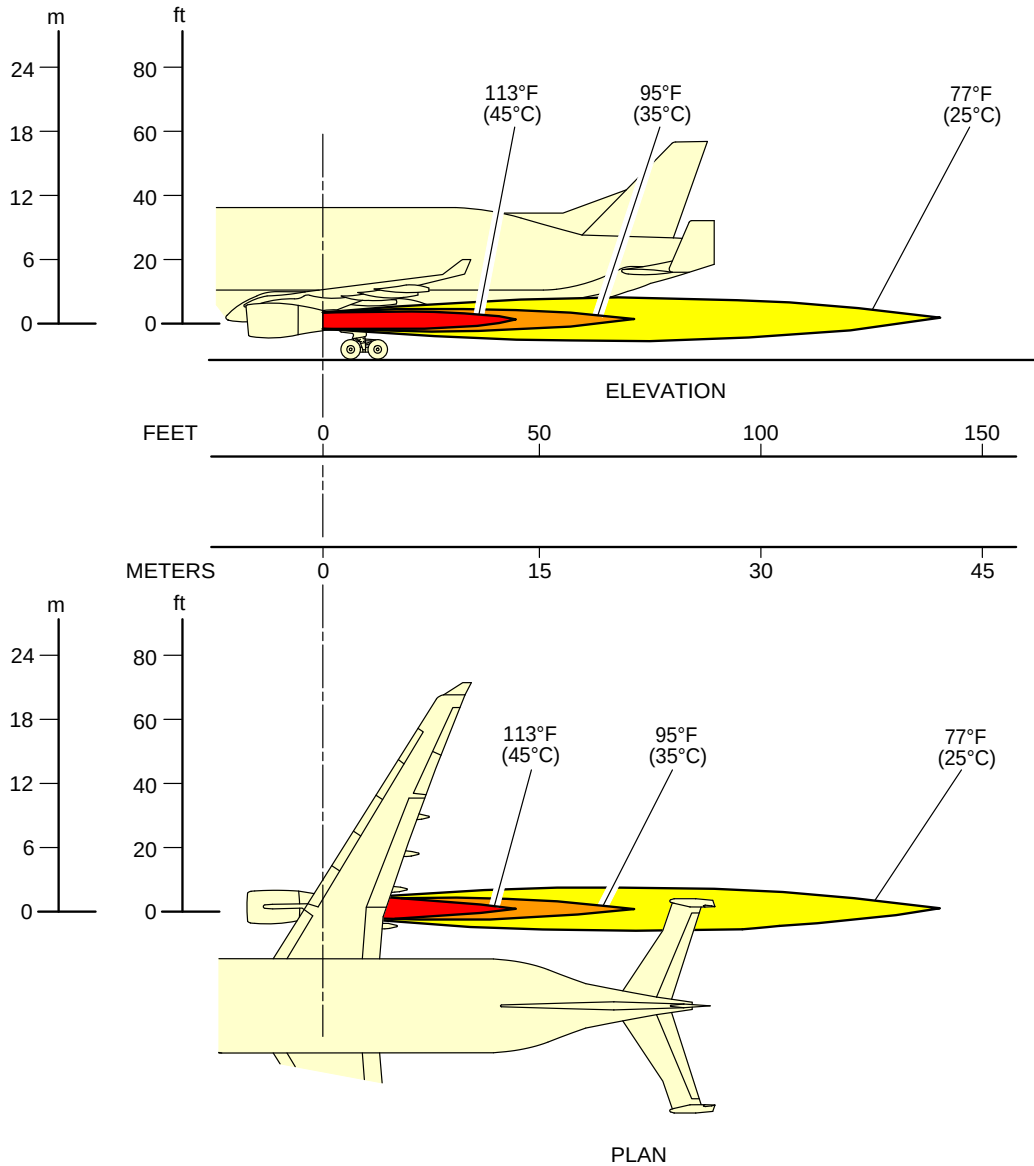
**6-1-4 Engine Exhaust Temperatures Contours - Breakaway Power****\*\*ON A/C A330-700L**Engine Exhaust Temperatures Contours - Breakaway Power

1. This section gives engine exhaust temperatures contours at breakaway power.

NOTE : The three values give temperature isolines values at the borders of colored areas.



**\*\*ON A/C A330-700L**



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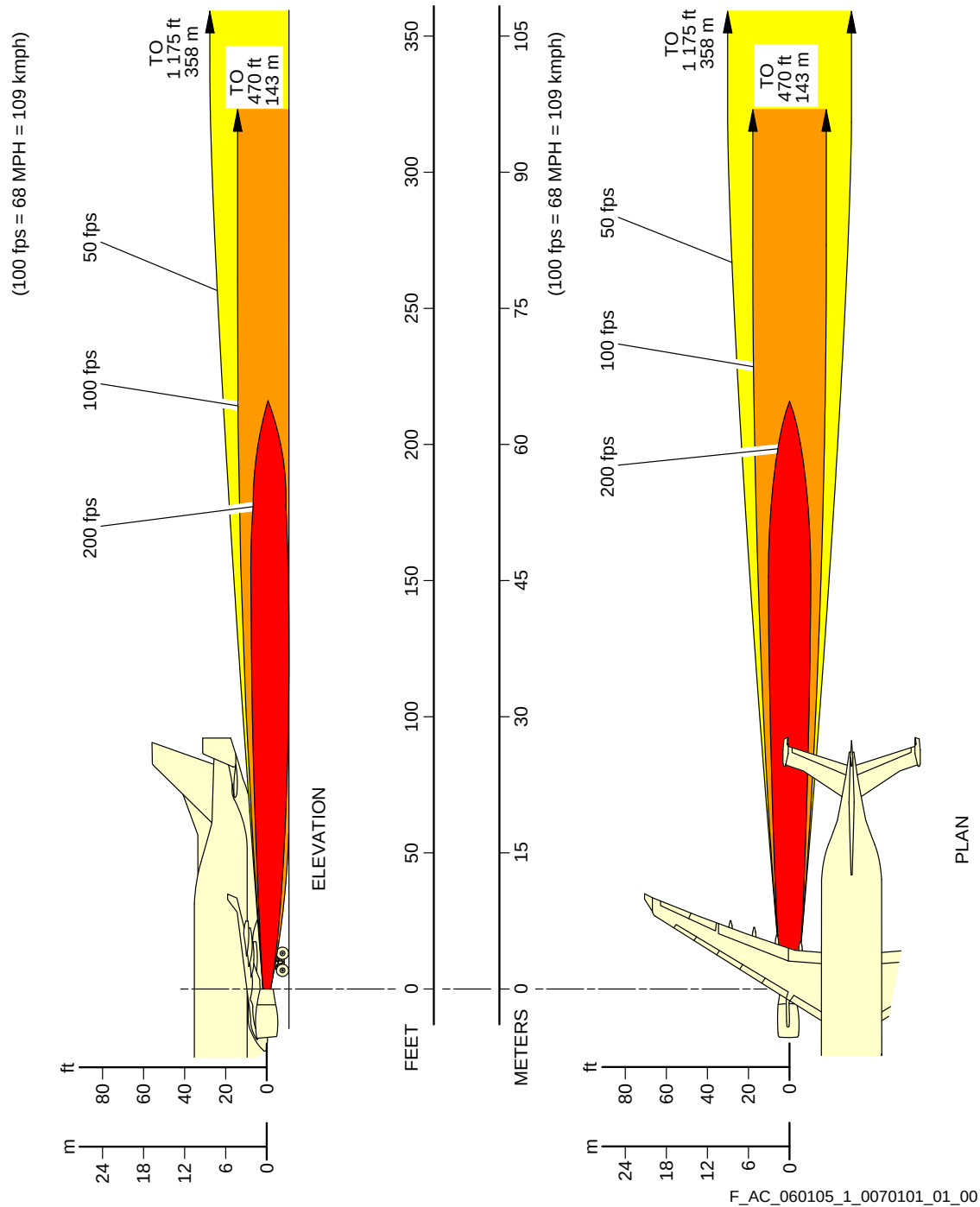
Engine Exhaust Temperatures  
Breakaway Power - RR Trent 700 Series Engine  
FIGURE-6-1-4-991-009-A01

**6-1-5 Engine Exhaust Velocities Contours - Takeoff Power****\*\*ON A/C A330-700L**Engine Exhaust Velocities Contours - Takeoff Power

1. This section gives engine exhaust velocities contours at takeoff power.

NOTE : The three values give velocities isolines values at the borders of colored areas.

**\*\*ON A/C A330-700L**



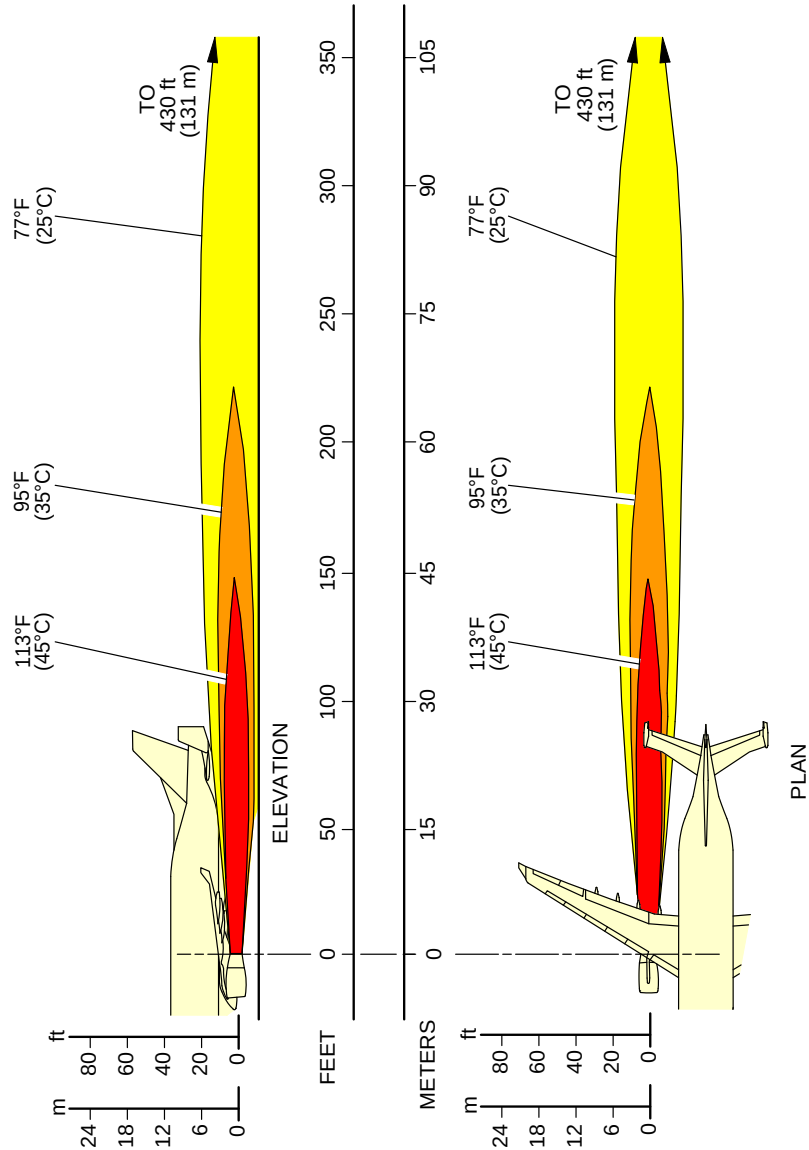
Engine Exhaust Velocities  
Takeoff Power - RR Trent 700 Series Engine  
FIGURE-6-1-5-991-007-A01

**6-1-6 Engine Exhaust Temperatures Contours - Takeoff Power****\*\*ON A/C A330-700L**Engine Exhaust Temperatures Contours - Takeoff Power

1. This section gives engine exhaust temperatures contours at takeoff power.

NOTE : The three values give temperature isolines values at the borders of colored areas.

**\*\*ON A/C A330-700L**



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Engine Exhaust Temperatures  
Takeoff Power - RR Trent 700 Series Engine  
FIGURE-6-1-6-991-007-A01

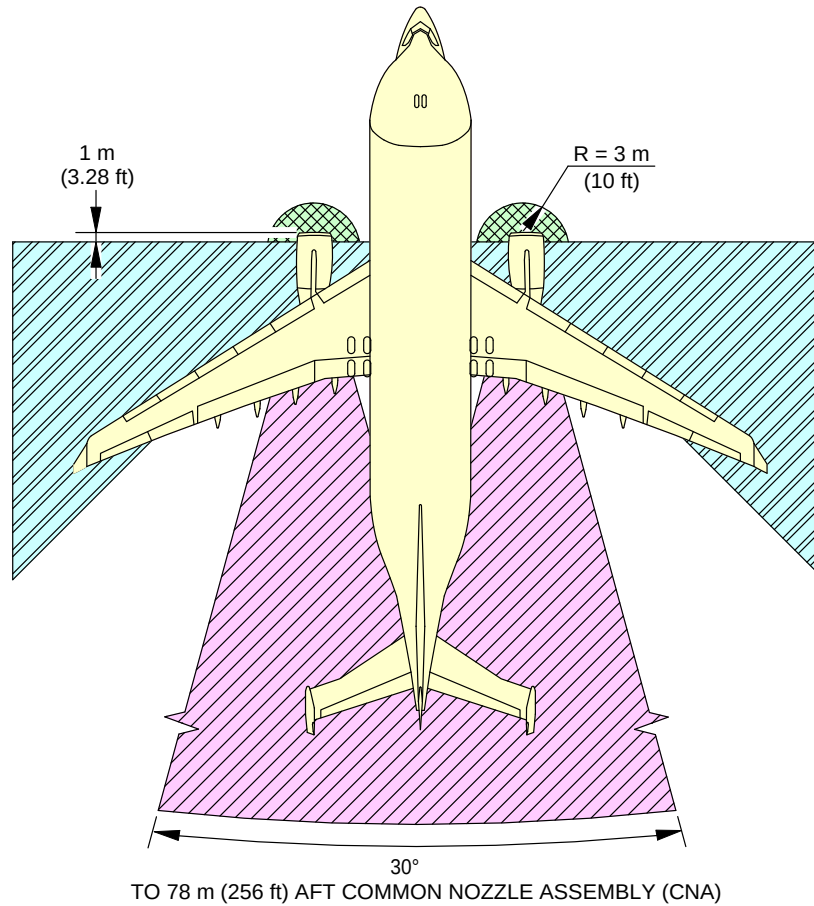
**6-3-0      Danger Areas of Engines****\*\*ON A/C A330-700L**Danger Areas of Engines

1. Danger Areas of the Engines.

**6-3-1      Ground Idle Power****\*\*ON A/C A330-700L**Ground Idle Power

1. This section gives danger areas of the engines at ground idle power conditions.

**\*\*ON A/C A330-700L**



**NOTE:**

 INTAKE SUCTION DANGER AREA MINIMUM POWER

 ENTRY CORRIDOR

 EXHAUST DANGER AREA

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Danger Areas of Engines  
RR Trent 700 Series Engine  
FIGURE-6-3-1-991-007-A01



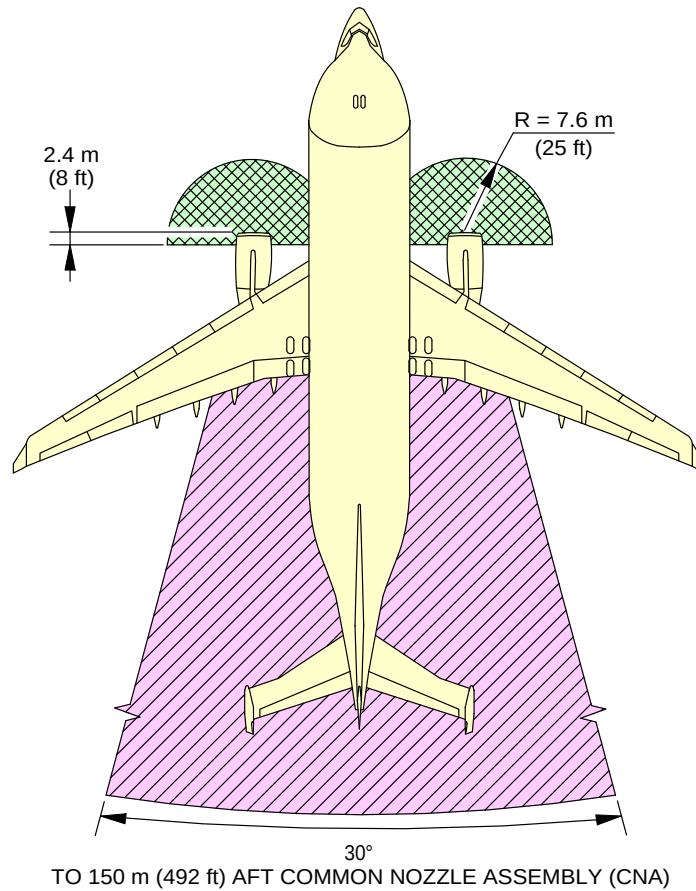
**6-3-2 Breakaway Power**

**\*\*ON A/C A330-700L**

Breakaway Power

1. This section gives danger areas of the engines at breakaway power conditions.

**\*\*ON A/C A330-700L**



**NOTE:**



INTAKE SUCTION DANGER AREA BREAKAWAY POWER



EXHAUST DANGER AREA

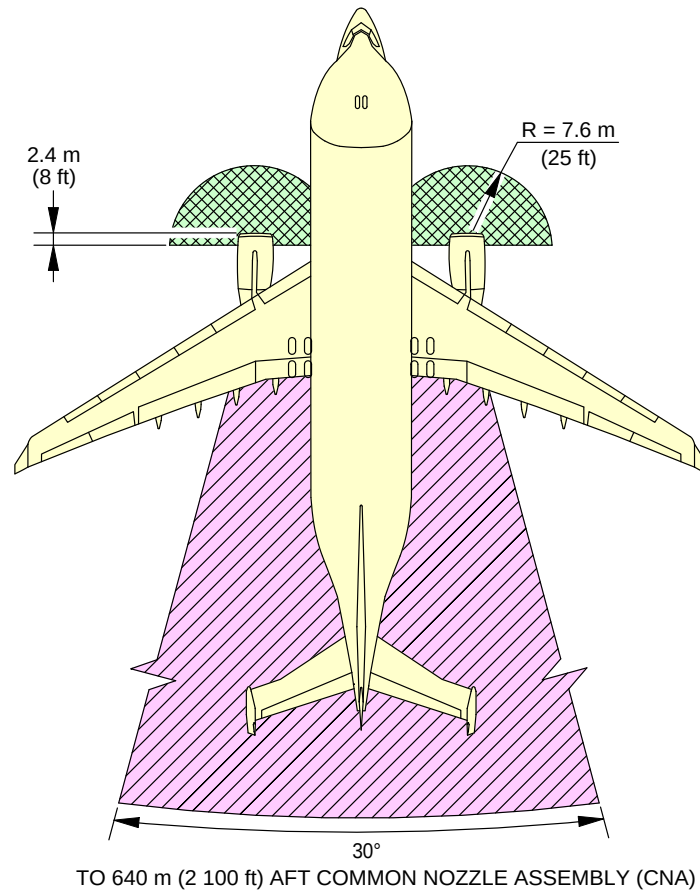
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Danger Areas of Engines  
RR Trent 700 Series Engine  
FIGURE-6-3-2-991-007-A01

**6-3-3 Takeoff Power****\*\*ON A/C A330-700L**Take-Off Power

1. This section gives danger areas of the engines at maximum take-off power conditions.

**\*\*ON A/C A330-700L**



**NOTE:**



INTAKE SUCTION DANGER AREA TAKE-OFF POWER



EXHAUST DANGER AREA

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Danger Areas of Engines  
RR Trent 700 Series Engine  
FIGURE-6-3-3-991-007-A01

**6-4-0      APU Exhaust Velocities and Temperatures****\*\*ON A/C A330-700L**APU Exhaust Velocities and Temperatures

1. APU Exhaust Velocities and Temperatures.

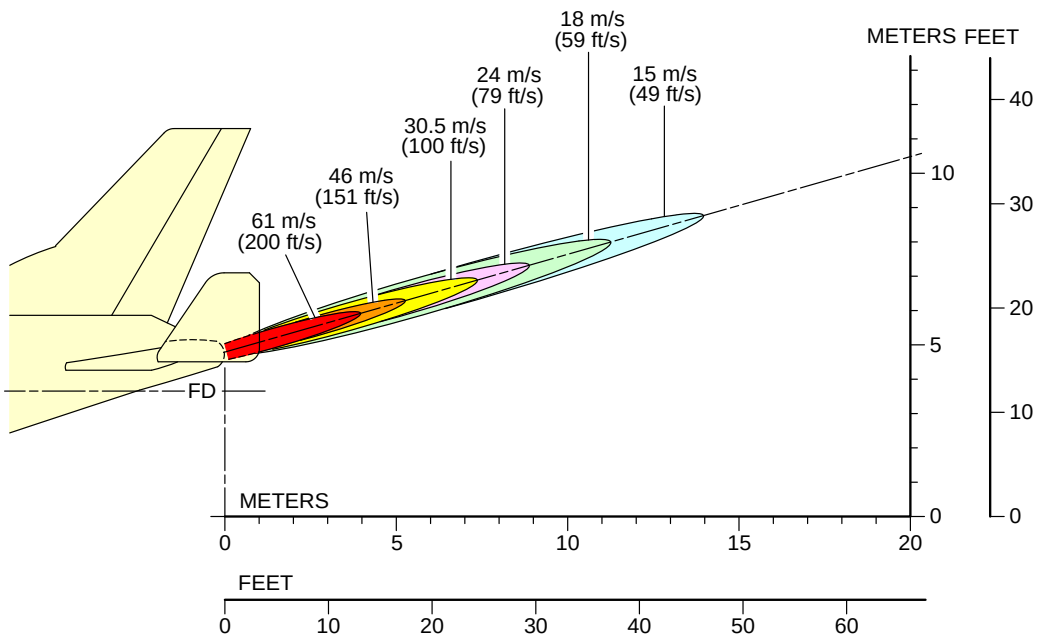
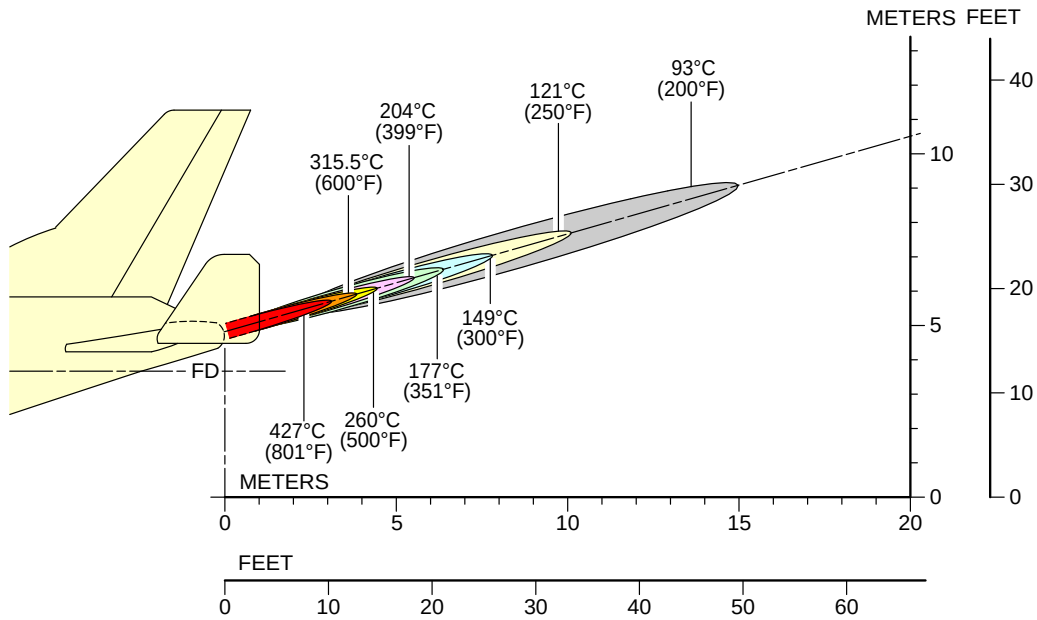
**6-4-1      APU**

**\*\*ON A/C A330-700L**

APU - GARRETT

1. This section gives APU exhaust velocities and temperatures.

**\*\*ON A/C A330-700L**



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Exhaust Velocities and Temperatures  
APU – GARRETT GTCP 331-350  
FIGURE-6-4-1-991-004-A01

## 6-5-0 Pilot Visibility in Approach

**\*\*ON A/C A330-700L**

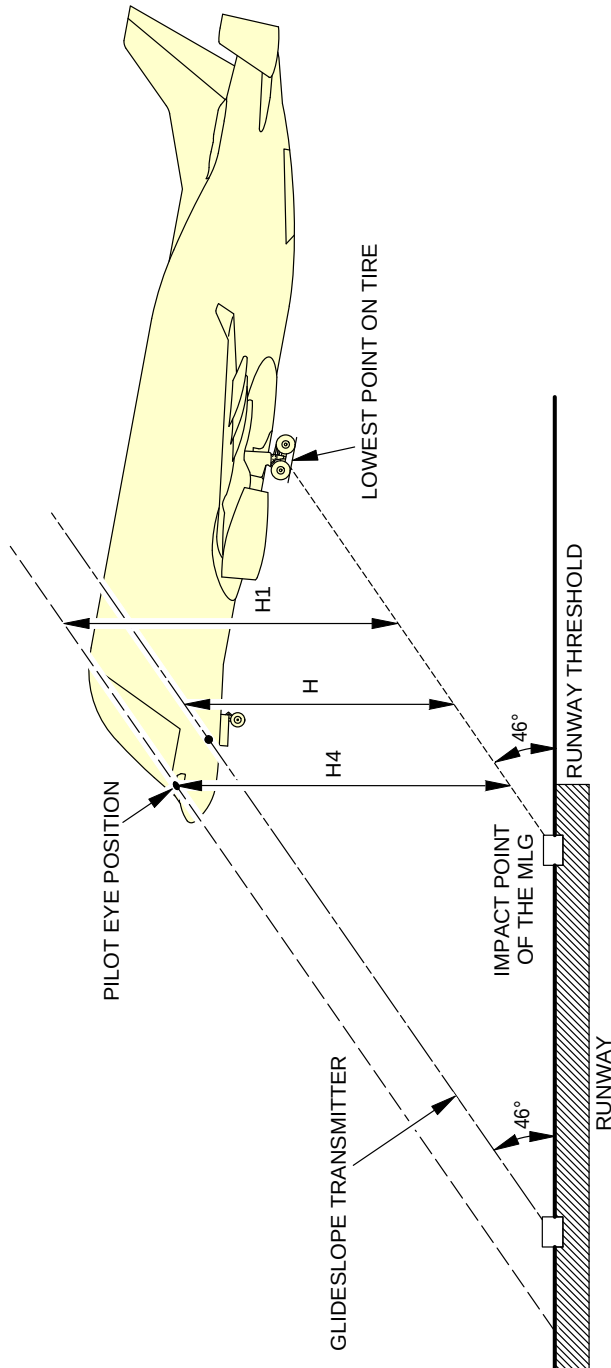
### Pilot Visibility in Approach

1. This section gives data about pilot visibility in approach.

| Distance | Value m (ft)      |
|----------|-------------------|
| H        | 7.14 m (23.43 ft) |
| H1       | 8.91 m (29.23 ft) |
| H4       | 7.42 m (24.34 ft) |



**\*\*ON A/C A330-700L**



**NOTE:**  
H IS THE DISTANCE BETWEEN THE INSTRUMENT LANDING SYSTEM (ILS) BEAM AND THE WHEEL PATH.  
H1 IS THE DISTANCE BETWEEN THE PILOT EYE PATH AND THE WHEEL PATH.  
H4 IS THE DISTANCE OF THE HEIGHT OF THE PILOT'S EYE RELATIVE TO THE LOW POINT OF THE LANDING GEAR.

- PILOT EYE PATH
- GLIDESLOPE ILS
- MLG PATH
- GLIDESLOPE RECEIVER

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Pilot Visibility in Approach  
Navigation Aids  
FIGURE-6-5-0-991-001-A01

**PAVEMENT DATA****7-1-0 General Information****\*\*ON A/C A330-700L****General Information**

1. A brief description of the pavement charts that follow will help in airport planning.

To aid in the interpolation between the discrete values shown, each aircraft configuration is shown with a minimum range of five loads on the Main Landing Gear (MLG).

All curves on the charts represent data at a constant specified tire pressure with:

- The aircraft loaded to the Maximum Ramp Weight (MRW),
- The CG at its maximum permissible aft position.

Pavement requirements for commercial aircraft are derived from the static analysis of loads imposed on the MLG struts.

**Landing Gear Footprint:**

Section 07-02-00 presents basic data on the landing gear footprint configuration, MRW and tire sizes and pressures.

**Maximum Pavement Loads:**

Section 07-03-00 shows maximum vertical and horizontal pavement loads for certain critical conditions at the tire-ground interfaces.

**Landing Gear Loading on Pavement:**

The curves related to the landing gear loading on pavement are not given in section 07-04-00. Because the relationship between the aircraft weight, the center of gravity and the landing gear loading on the pavement is not strictly linear, it cannot be shown in chart format. But you can find in section 07-03-00 the maximum vertical and horizontal pavement loads for some critical conditions at the tire/ground interfaces for all the operational weight variants of the aircraft.

For questions that are related to landing gear loading on pavement, contact Airbus.

Flexible Pavement Requirements - US Army Corps of Engineers Design Method:

The flexible pavement requirements curves as per as U.S. Army Corps of Engineers Design Method are not given in section 07-05-00 since the related data is available through free software.

Sections 07-02-00 and 07-03-00 give all the inputs data required for the use of such software. For questions that are related to the flexible pavement requirements, contact Airbus.

**Flexible Pavement Requirements - LCN Conversion Method:**

The Load Classification Number (LCN) curves are not given in section 07-06-00 since the LCN system for the reporting pavement strength is old and are replaced by the ICAO recommended ACN/PCN system in 1983 and ACR/PCR system in 2020.

For questions that are related to the LCN system, contact Airbus.

**Rigid Pavement Requirements - PCA (Portland Cement Association) Design Method:**

The rigid pavement requirements curves as per as Portland Cement Association Design Method are not given in section 07-07-00 since the related data is available through free software.

Sections 07-02-00 and 07-03-00 give all the inputs data required for the use of such software. For questions that are related to the rigid pavement requirements, contact Airbus.

**Rigid Pavement Requirements - LCN Conversion:**

The LCN curves are not given in section 07-08-00 since the LCN system for reporting pavement strength is old and are replaced by the ICAO recommended ACN/PCN system in 1983 and ACR/PCR system in 2020.

For questions that are related to the LCN system, contact Airbus.

**ACN/PCN Reporting System:**

Section 07-09-00 gives ACN data prepared according to the ACN/PCN system as referenced in ICAO Annex 14, "Aerodromes", Volume 1 "Aerodrome Design and Operations".

Eighth Edition July 2018, incorporating Amendments 1 to 14 and ICAO doc 9157, "Aerodrome Design Manual", part 3 "Pavements" Second Edition 1983.

The ACN/PCN system is applicable until November 2024.

ACN is the Aircraft Classification Number and PCN is the corresponding Pavement Classification Number.

An aircraft having an ACN less than or equal to the PCN can operate without restriction on the pavement.

Numerically the ACN is two times the derived single wheel load expressed in thousands of kilograms.

The derived single wheel load is calculated as the load on a single tire inflated to 1.25 MPa (181 psi) that would have the same pavement requirements as the aircraft.

Computationally the ACN/PCN system uses PCA program PDILB for rigid pavements and S-77-1 for flexible pavements to calculate ACN values.

The Airport Authority must decide on the method of pavement analysis.  
The results of their evaluation should be reported with the following format:

| PCN           |                   |                           |                    |
|---------------|-------------------|---------------------------|--------------------|
| PAVEMENT TYPE | SUBGRADE CATEGORY | TIRE PRESSURE CATEGORY    | EVALUATION METHOD  |
| R - Rigid     | A - High          | W - No limit              | T - Technical      |
| F - Flexible  | B - Medium        | X - To 1.75 MPa (254 psi) | U - Using Aircraft |
|               | C - Low           | Y - To 1.25 MPa (181 psi) |                    |
|               | D - Ultra Low     | Z - To 0.5 MPa (73 psi)   |                    |

Section 07-09-00 shows the aircraft ACN values.

For flexible pavements, the four subgrade categories are:

|                       |        |
|-----------------------|--------|
| A. High Strength      | CBR 15 |
| B. Medium Strength    | CBR 10 |
| C. Low Strength       | CBR 6  |
| D. Ultra Low Strength | CBR 3  |

For rigid pavements, the four subgrade categories are:

|                       |                                              |
|-----------------------|----------------------------------------------|
| A. High Strength      | Subgrade k = 150 MN/m <sup>3</sup> (550 pci) |
| B. Medium Strength    | Subgrade k = 80 MN/m <sup>3</sup> (300 pci)  |
| C. Low Strength       | Subgrade k = 40 MN/m <sup>3</sup> (150 pci)  |
| D. Ultra Low Strength | Subgrade k = 20 MN/m <sup>3</sup> (75 pci)   |

ACR/PCR Reporting System:

Section 07-10-00 gives ACR data prepared according to the ACR/PCR system as referenced in ICAO Annex 14, "Aerodromes", Volume 1 "Aerodrome Design and Operations".

Eight Edition July 2018, incorporating Amendments 1 to 15 and ICAO doc 9157, "Aerodrome Design Manual", part 3 "Pavements" Third Edition 2021.

The ACR/PCR system is effective from November 2020 and will be applicable in November 2024.

ACR is the Aircraft Classification Rating and PCR is the related Pavement Classification Rating.

An aircraft with an ACR less than or equal to the PCR can operate without restriction on the pavement.

Numerically the ACR is two times the derived single-wheel load expressed in hundreds of kilograms.

The derived single-wheel load is calculated as the load on a single tire inflated to 1.50 MPa (218 psi) that can have the same pavement requirements as the aircraft.

Computationally the ACR/PCR system relies on the Linear Elastic Analysis (LEA).

The ACR are computed with the official ICAO-ACR software.

States can start their own methods for PCR determination, which agree with the overall parameters of the ACR/PCR method.

The results of their analysis should be reported with the following format:

| PCR           |                   |                           |                    |
|---------------|-------------------|---------------------------|--------------------|
| PAVEMENT TYPE | SUBGRADE CATEGORY | TIRE PRESSURE CATEGORY    | EVALUATION METHOD  |
| R - Rigid     | A - High          | W - No limit              | T - Technical      |
| F - Flexible  | B - Medium        | X - To 1.75 MPa (254 psi) | U - Using Aircraft |
|               | C - Low           | Y - To 1.25 MPa (181 psi) |                    |
|               | D - Ultra Low     | Z - To 0.5 MPa (73 psi)   |                    |

Section 07-10-00 shows the aircraft ACR values.

For flexible and rigid pavement, the four subgrade categories are defined based on the subgrade modulus of elasticity (E):

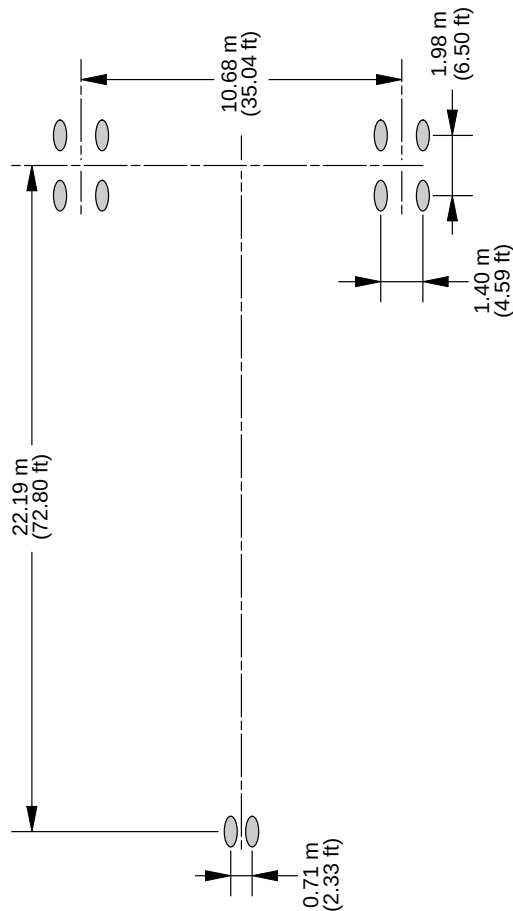
|                       |                          |
|-----------------------|--------------------------|
| A. High Strength      | E = 200 MPa (29 008 psi) |
| B. Medium Strength    | E = 120 MPa (17 405 psi) |
| C. Low Strength       | E = 80 MPa (11 603 psi)  |
| D. Ultra Low Strength | E = 50 MPa (7 252 psi)   |

**7-2-0 Landing Gear Footprint****\*\*ON A/C A330-700L**Landing Gear Footprint

1. This section gives data about the landing gear footprint in relation to the aircraft MRW and tire sizes and pressures.

The landing-gear footprint information is given for all the operational weight variants of the aircraft.

**\*\*ON A/C A330-700L**



| WEIGHT<br>VARIANT                | MAXIMUM<br>RAMP<br>WEIGHT  | PERCENTAGE<br>OF WEIGHT<br>ON MAIN<br>GEAR GROUP | NOSE GEAR TIRE<br>SIZE | NOSE GEAR<br>TIRE<br>PRESSURE | MAIN GEAR TIRE<br>SIZE | MAIN GEAR<br>TIRE<br>PRESSURE |
|----------------------------------|----------------------------|--------------------------------------------------|------------------------|-------------------------------|------------------------|-------------------------------|
| A330-700L<br>WV000<br>ZCG +0.7 m | 227 900 kg<br>(502 425 lb) | 90.1%                                            | 1050x395R16 28PR       | 12.7 bar<br>(184 psi)         | 1400x530R23 36PR       | 14.2 bar<br>(206 psi)         |
| A330-700L<br>WV000<br>ZCG +0.6 m | 227 900 kg<br>(502 425 lb) | 90.4%                                            | 1050x395R16 28PR       | 12.7 bar<br>(184 psi)         | 1400x530R23 36PR       | 14.2 bar<br>(206 psi)         |
| A330-700L<br>WV000<br>ZCG +0.5 m | 227 900 kg<br>(502 425 lb) | 90.7%                                            | 1050x395R16 28PR       | 12.7 bar<br>(184 psi)         | 1400x530R23 36PR       | 14.2 bar<br>(206 psi)         |
| A330-700L<br>WV000<br>ZCG +0.4 m | 227 900 kg<br>(502 425 lb) | 90.9%                                            | 1050x395R16 28PR       | 12.7 bar<br>(184 psi)         | 1400x530R23 36PR       | 14.2 bar<br>(206 psi)         |
| A330-700L<br>WV000<br>ZCG +0.3 m | 227 900 kg<br>(502 425 lb) | 91.2%                                            | 1050x395R16 28PR       | 12.7 bar<br>(184 psi)         | 1400x530R23 36PR       | 14.2 bar<br>(206 psi)         |
| A330-700L<br>WV000<br>ZCG +0.2 m | 227 900 kg<br>(502 425 lb) | 91.5%                                            | 1050x395R16 28PR       | 12.7 bar<br>(184 psi)         | 1400x530R23 36PR       | 14.2 bar<br>(206 psi)         |
| A330-700L<br>WV000<br>ZCG +0.1 m | 227 900 kg<br>(502 425 lb) | 91.7%                                            | 1050x395R16 28PR       | 12.7 bar<br>(184 psi)         | 1400x530R23 36PR       | 14.2 bar<br>(206 psi)         |

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Landing Gear Footprint  
(Sheet 1 of 3)  
FIGURE-7-2-0-991-056-A01

**\*\*ON A/C A330-700L**

| WEIGHT<br>VARIANT                | MAXIMUM<br>RAMP<br>WEIGHT  | PERCENTAGE<br>OF WEIGHT<br>ON MAIN<br>GEAR GROUP | NOSE GEAR TIRE<br>SIZE | NOSE GEAR<br>TIRE<br>PRESSURE | MAIN GEAR TIRE<br>SIZE | MAIN GEAR<br>TIRE<br>PRESSURE |
|----------------------------------|----------------------------|--------------------------------------------------|------------------------|-------------------------------|------------------------|-------------------------------|
| A330-700L<br>WV000<br>ZCG 0 m    | 227 900 kg<br>(502 425 lb) | 91.7%                                            | 1050x395R16 28PR       | 12.7 bar<br>(184 psi)         | 1400x530R23 36PR       | 14.2 bar<br>(206 psi)         |
| A330-700L<br>WV000<br>ZCG -0.1 m | 227 900 kg<br>(502 425 lb) | 91.7%                                            | 1050x395R16 28PR       | 12.7 bar<br>(184 psi)         | 1400x530R23 36PR       | 14.2 bar<br>(206 psi)         |
| A330-700L<br>WV000<br>ZCG -0.2 m | 227 900 kg<br>(502 425 lb) | 91.7%                                            | 1050x395R16 28PR       | 12.7 bar<br>(184 psi)         | 1400x530R23 36PR       | 14.2 bar<br>(206 psi)         |
| A330-700L<br>WV000<br>ZCG -0.3 m | 227 900 kg<br>(502 425 lb) | 91.7%                                            | 1050x395R16 28PR       | 12.7 bar<br>(184 psi)         | 1400x530R23 36PR       | 14.2 bar<br>(206 psi)         |
| A330-700L<br>WV000<br>ZCG -0.4 m | 227 900 kg<br>(502 425 lb) | 91.7%                                            | 1050x395R16 28PR       | 12.7 bar<br>(184 psi)         | 1400x530R23 36PR       | 14.2 bar<br>(206 psi)         |
| A330-700L<br>WV000<br>ZCG -0.5 m | 227 900 kg<br>(502 425 lb) | 91.7%                                            | 1050x395R16 28PR       | 12.7 bar<br>(184 psi)         | 1400x530R23 36PR       | 14.2 bar<br>(206 psi)         |
| A330-700L<br>WV001<br>ZCG +0.7 m | 205 900 kg<br>(453 925 lb) | 91.2%                                            | 1050x395R16 28PR       | 12.7 bar<br>(184 psi)         | 1400x530R23 36PR       | 14.2 bar<br>(206 psi)         |
| A330-700L<br>WV001<br>ZCG +0.6 m | 205 900 kg<br>(453 925 lb) | 91.5%                                            | 1050x395R16 28PR       | 12.7 bar<br>(184 psi)         | 1400x530R23 36PR       | 14.2 bar<br>(206 psi)         |
| A330-700L<br>WV001<br>ZCG +0.5 m | 205 900 kg<br>(453 925 lb) | 91.8%                                            | 1050x395R16 28PR       | 12.7 bar<br>(184 psi)         | 1400x530R23 36PR       | 14.2 bar<br>(206 psi)         |
| A330-700L<br>WV001<br>ZCG +0.4 m | 205 900 kg<br>(453 925 lb) | 92.1%                                            | 1050x395R16 28PR       | 12.7 bar<br>(184 psi)         | 1400x530R23 36PR       | 14.2 bar<br>(206 psi)         |
| A330-700L<br>WV001<br>ZCG +0.3 m | 205 900 kg<br>(453 925 lb) | 92.4%                                            | 1050x395R16 28PR       | 12.7 bar<br>(184 psi)         | 1400x530R23 36PR       | 14.2 bar<br>(206 psi)         |
| A330-700L<br>WV001<br>ZCG +0.2 m | 205 900 kg<br>(453 925 lb) | 92.7%                                            | 1050x395R16 28PR       | 12.7 bar<br>(184 psi)         | 1400x530R23 36PR       | 14.2 bar<br>(206 psi)         |
| A330-700L<br>WV001<br>ZCG +0.1 m | 205 900 kg<br>(453 925 lb) | 92.9%                                            | 1050x395R16 28PR       | 12.7 bar<br>(184 psi)         | 1400x530R23 36PR       | 14.2 bar<br>(206 psi)         |
| A330-700L<br>WV001<br>ZCG 0 m    | 205 900 kg<br>(453 925 lb) | 92.9%                                            | 1050x395R16 28PR       | 12.7 bar<br>(184 psi)         | 1400x530R23 36PR       | 14.2 bar<br>(206 psi)         |
| A330-700L<br>WV001<br>ZCG -0.1 m | 205 900 kg<br>(453 925 lb) | 92.9%                                            | 1050x395R16 28PR       | 12.7 bar<br>(184 psi)         | 1400x530R23 36PR       | 14.2 bar<br>(206 psi)         |

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Landing Gear Footprint  
(Sheet 2 of 3)  
FIGURE-7-2-0-991-056-A01



**\*\*ON A/C A330-700L**

| WEIGHT<br>VARIANT                | MAXIMUM<br>RAMP<br>WEIGHT  | PERCENTAGE<br>OF WEIGHT<br>ON MAIN<br>GEAR GROUP | NOSE GEAR TIRE<br>SIZE | NOSE GEAR<br>TIRE<br>PRESSURE | MAIN GEAR TIRE<br>SIZE | MAIN GEAR<br>TIRE<br>PRESSURE |
|----------------------------------|----------------------------|--------------------------------------------------|------------------------|-------------------------------|------------------------|-------------------------------|
| A330-700L<br>WV001<br>ZCG -0.2 m | 205 900 kg<br>(453 925 lb) | 92.9%                                            | 1050x395R16 28PR       | 12.7 bar<br>(184 psi)         | 1400x530R23 36PR       | 14.2 bar<br>(206 psi)         |
| A330-700L<br>WV001<br>ZCG -0.3 m | 205 900 kg<br>(453 925 lb) | 92.9%                                            | 1050x395R16 28PR       | 12.7 bar<br>(184 psi)         | 1400x530R23 36PR       | 14.2 bar<br>(206 psi)         |
| A330-700L<br>WV001<br>ZCG -0.4 m | 205 900 kg<br>(453 925 lb) | 92.9%                                            | 1050x395R16 28PR       | 12.7 bar<br>(184 psi)         | 1400x530R23 36PR       | 14.2 bar<br>(206 psi)         |
| A330-700L<br>WV001<br>ZCG -0.5 m | 205 900 kg<br>(453 925 lb) | 92.9%                                            | 1050x395R16 28PR       | 12.7 bar<br>(184 psi)         | 1400x530R23 36PR       | 14.2 bar<br>(206 psi)         |

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Landing Gear Footprint  
(Sheet 3 of 3)  
FIGURE-7-2-0-991-056-A01

**7-3-0 Maximum Pavement Loads****\*\*ON A/C A330-700L**Maximum Pavement Loads

1. This section gives maximum vertical and horizontal pavement loads for some critical conditions at the tire-ground interfaces.

The maximum pavement loads are given for all the operational weight variants of the aircraft.

$V_{(NG)}$  MAXIMUM VERTICAL NOSE GEAR GROUND LOAD AT MOST FORWARD CG  
 $V_{(MG)}$  MAXIMUM VERTICAL MAIN GEAR GROUND LOAD AT MOST AFT CG  
 $H$  MAXIMUM HORIZONTAL GROUND LOAD FROM BRAKING

| 1                          | 2                       | 3                          | 4                                                   |                           | 5                                                   | 6                                          |
|----------------------------|-------------------------|----------------------------|-----------------------------------------------------|---------------------------|-----------------------------------------------------|--------------------------------------------|
|                            |                         | V (NG)                     |                                                     | V (MG) (PER STRUT)        | H (PER STRUT)                                       |                                            |
| MODEL                      | MAXIMUM RAMP WEIGHT     | STATIC LOAD AT MOST FWD CG | STATIC BRAKING AT 10 ft/s <sup>2</sup> DECELERATION | STATIC LOAD AT MAX AFT CG | STEADY BRAKING AT 10 ft/s <sup>2</sup> DECELERATION | AT INSTANTANEOUS BRAKING COEFFICIENT = 0.8 |
| A330-700L WV000 ZCG +0.7 m | 227 900 kg (502 425 lb) | 24 700 kg (54 450 lb)      | 20.6% MAC (a)                                       | 102 720 kg (226 450 lb)   | 23.6% MAC (a)                                       | 82 180 kg (181 175 lb) (b)                 |
| A330-700L WV000 ZCG +0.6 m | 227 900 kg (502 425 lb) | 24 700 kg (54 450 lb)      | 20.6% MAC (a)                                       | 103 020 kg (227 125 lb)   | 24.4% MAC (a)                                       | 82 410 kg (181 700 lb) (b)                 |
| A330-700L WV000 ZCG +0.5 m | 227 900 kg (502 425 lb) | 24 700 kg (54 450 lb)      | 20.6% MAC (a)                                       | 103 320 kg (227 775 lb)   | 25.2% MAC (a)                                       | 82 650 kg (182 225 lb) (b)                 |
| A330-700L WV000 ZCG +0.4 m | 227 900 kg (502 425 lb) | 24 700 kg (54 450 lb)      | 20.6% MAC (a)                                       | 103 610 kg (228 425 lb)   | 26% MAC (a)                                         | 82 890 kg (182 750 lb) (b)                 |
| A330-700L WV000 ZCG +0.3 m | 227 900 kg (502 425 lb) | 24 700 kg (54 450 lb)      | 20.6% MAC (a)                                       | 103 950 kg (229 175 lb)   | 26.9% MAC (a)                                       | 83 160 kg (183 325 lb) (b)                 |
| A330-700L WV000 ZCG +0.2 m | 227 900 kg (502 425 lb) | 24 700 kg (54 450 lb)      | 20.6% MAC (a)                                       | 104 250 kg (229 825 lb)   | 27.7% MAC (a)                                       | 83 400 kg (183 875 lb) (b)                 |
| A330-700L WV000 ZCG +0.1 m | 227 900 kg (502 425 lb) | 24 700 kg (54 450 lb)      | 20.6% MAC (a)                                       | 104 440 kg (230 250 lb)   | 28.2% MAC (a)                                       | 83 550 kg (184 200 lb) (b)                 |
| A330-700L WV000 ZCG 0 m    | 227 900 kg (502 425 lb) | 24 700 kg (54 450 lb)      | 20.6% MAC (a)                                       | 104 440 kg (230 250 lb)   | 28.2% MAC (a)                                       | 83 550 kg (184 200 lb) (b)                 |

(b) BRAKED MAIN GEAR.

Maximum Pavement Loads  
(Sheet 1 of 3)  
FIGURE-7-3-0-991-022-A01

**\*\*ON A/C A330-700L**

| 1                          | 2                       | 3                                      | 4                                                   | 5                                        | 6                                                   |                                            |
|----------------------------|-------------------------|----------------------------------------|-----------------------------------------------------|------------------------------------------|-----------------------------------------------------|--------------------------------------------|
|                            |                         | V (NG)                                 |                                                     | V (MG) (PER STRUT)                       | H (PER STRUT)                                       |                                            |
| MODEL                      | MAXIMUM RAMP WEIGHT     | STATIC LOAD AT MOST FWD CG             | STATIC BRAKING AT 10 ft/s <sup>2</sup> DECELERATION | STATIC LOAD AT MAX AFT CG                | STEADY BRAKING AT 10 ft/s <sup>2</sup> DECELERATION | AT INSTANTANEOUS BRAKING COEFFICIENT = 0.8 |
| A330-700L WV000 ZCG -0.1 m | 227 900 kg (502 425 lb) | 24 700 kg (54 450 lb)<br>20.6% MAC (a) | 40 850 kg (90 050 lb)                               | 104 440 kg (230 250 lb)<br>28.2% MAC (a) | 35 420 kg (78 075 lb) (b)                           | 83 550 kg (184 200 lb) (b)                 |
| A330-700L WV000 ZCG -0.2 m | 227 900 kg (502 425 lb) | 24 700 kg (54 450 lb)<br>20.6% MAC (a) | 40 530 kg (89 350 lb)                               | 104 440 kg (230 250 lb)<br>28.2% MAC (a) | 35 420 kg (78 075 lb) (b)                           | 83 550 kg (184 200 lb) (b)                 |
| A330-700L WV000 ZCG -0.3 m | 227 900 kg (502 425 lb) | 24 700 kg (54 450 lb)<br>20.6% MAC (a) | 40 210 kg (88 650 lb)                               | 104 440 kg (230 250 lb)<br>28.2% MAC (a) | 35 420 kg (78 075 lb) (b)                           | 83 550 kg (184 200 lb) (b)                 |
| A330-700L WV000 ZCG -0.4 m | 227 900 kg (502 425 lb) | 24 700 kg (54 450 lb)<br>20.6% MAC (a) | 39 890 kg (87 950 lb)                               | 104 440 kg (230 250 lb)<br>28.2% MAC (a) | 35 420 kg (78 075 lb) (b)                           | 83 550 kg (184 200 lb) (b)                 |
| A330-700L WV000 ZCG -0.5 m | 227 900 kg (502 425 lb) | 24 700 kg (54 450 lb)<br>20.6% MAC (a) | 39 570 kg (87 250 lb)                               | 104 440 kg (230 250 lb)<br>28.2% MAC (a) | 35 420 kg (78 075 lb) (b)                           | 83 550 kg (184 200 lb) (b)                 |
| A330-700L WV001 ZCG +0.7 m | 205 900 kg (453 925 lb) | 24 070 kg (53 075 lb)<br>18% MAC (a)   | 40 970 kg (90 325 lb)                               | 93 920 kg (207 050 lb)<br>26.9% MAC (a)  | 32 000 kg (70 550 lb) (b)                           | 75 130 kg (165 650 lb) (b)                 |
| A330-700L WV001 ZCG +0.6 m | 205 900 kg (453 925 lb) | 24 070 kg (53 075 lb)<br>18% MAC (a)   | 40 680 kg (89 675 lb)                               | 94 220 kg (207 725 lb)<br>27.8% MAC (a)  | 32 000 kg (70 550 lb) (b)                           | 75 380 kg (166 175 lb) (b)                 |
| A330-700L WV001 ZCG +0.5 m | 205 900 kg (453 925 lb) | 24 070 kg (53 075 lb)<br>18% MAC (a)   | 40 390 kg (89 050 lb)                               | 94 520 kg (208 400 lb)<br>28.7% MAC (a)  | 32 000 kg (70 550 lb) (b)                           | 75 620 kg (166 700 lb) (b)                 |
| A330-700L WV001 ZCG +0.4 m | 205 900 kg (453 925 lb) | 24 070 kg (53 075 lb)<br>18% MAC (a)   | 40 100 kg (88 400 lb)                               | 94 830 kg (209 050 lb)<br>29.6% MAC (a)  | 32 000 kg (70 550 lb) (b)                           | 75 860 kg (167 250 lb) (b)                 |
| A330-700L WV001 ZCG +0.3 m | 205 900 kg (453 925 lb) | 24 070 kg (53 075 lb)<br>18% MAC (a)   | 39 810 kg (87 775 lb)                               | 95 130 kg (209 725 lb)<br>30.5% MAC (a)  | 32 000 kg (70 550 lb) (b)                           | 76 100 kg (167 775 lb) (b)                 |
| A330-700L WV001 ZCG +0.2 m | 205 900 kg (453 925 lb) | 24 070 kg (53 075 lb)<br>18% MAC (a)   | 39 520 kg (87 125 lb)                               | 95 430 kg (210 400 lb)<br>31.4% MAC (a)  | 32 000 kg (70 550 lb) (b)                           | 76 350 kg (168 325 lb) (b)                 |
| A330-700L WV001 ZCG +0.1 m | 205 900 kg (453 925 lb) | 24 070 kg (53 075 lb)<br>18% MAC (a)   | 39 240 kg (86 500 lb)                               | 95 640 kg (210 850 lb)<br>32% MAC (a)    | 32 000 kg (70 550 lb) (b)                           | 76 510 kg (168 675 lb) (b)                 |
| A330-700L WV001 ZCG 0 m    | 205 900 kg (453 925 lb) | 24 070 kg (53 075 lb)<br>18% MAC (a)   | 38 950 kg (85 875 lb)                               | 95 640 kg (210 850 lb)<br>32% MAC (a)    | 32 000 kg (70 550 lb) (b)                           | 76 510 kg (168 675 lb) (b)                 |

**NOTE:**

(a) LOADS CALCULATED USING AIRCRAFT AT MRW.

(b) BRAKED MAIN GEAR.

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Maximum Pavement Loads  
(Sheet 2 of 3)  
FIGURE-7-3-0-991-022-A01

**\*\*ON A/C A330-700L**

| 1                                | 2                          | 3                          | 4                                                   | 5                             | 6                                                   |                                            |
|----------------------------------|----------------------------|----------------------------|-----------------------------------------------------|-------------------------------|-----------------------------------------------------|--------------------------------------------|
|                                  |                            | V <sub>(NG)</sub>          |                                                     | V <sub>(MG)</sub> (PER STRUT) | H (PER STRUT)                                       |                                            |
| MODEL                            | MAXIMUM RAMP WEIGHT        | STATIC LOAD AT MOST FWD CG | STATIC BRAKING AT 10 ft/s <sup>2</sup> DECELERATION | STATIC LOAD AT MAX AFT CG     | STEADY BRAKING AT 10 ft/s <sup>2</sup> DECELERATION | AT INSTANTANEOUS BRAKING COEFFICIENT = 0.8 |
| A330-700L<br>WV001<br>ZCG -0.1 m | 205 900 kg<br>(453 925 lb) | 24 070 kg<br>(53 075 lb)   | 38 660 kg<br>(85 225 lb)                            | 95 640 kg<br>(210 850 lb)     | 32 000 kg<br>(70 550 lb)                            | 76 510 kg<br>(168 675 lb)                  |
| A330-700L<br>WV001<br>ZCG -0.2 m | 205 900 kg<br>(453 925 lb) | 24 070 kg<br>(53 075 lb)   | 38 370 kg<br>(84 600 lb)                            | 95 640 kg<br>(210 850 lb)     | 32 000 kg<br>(70 550 lb)                            | 76 510 kg<br>(168 675 lb)                  |
| A330-700L<br>WV001<br>ZCG -0.3 m | 205 900 kg<br>(453 925 lb) | 24 070 kg<br>(53 075 lb)   | 38 080 kg<br>(83 950 lb)                            | 95 640 kg<br>(210 850 lb)     | 32 000 kg<br>(70 550 lb)                            | 76 510 kg<br>(168 675 lb)                  |
| A330-700L<br>WV001<br>ZCG -0.4 m | 205 900 kg<br>(453 925 lb) | 24 070 kg<br>(53 075 lb)   | 37 790 kg<br>(83 325 lb)                            | 95 640 kg<br>(210 850 lb)     | 32 000 kg<br>(70 550 lb)                            | 76 510 kg<br>(168 675 lb)                  |
| A330-700L<br>WV001<br>ZCG -0.5 m | 205 900 kg<br>(453 925 lb) | 24 070 kg<br>(53 075 lb)   | 37 510 kg<br>(82 675 lb)                            | 95 640 kg<br>(210 850 lb)     | 32 000 kg<br>(70 550 lb)                            | 76 510 kg<br>(168 675 lb)                  |

**NOTE:**

(a) LOADS CALCULATED USING AIRCRAFT AT MRW.

(b) BRAKED MAIN GEAR.

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Maximum Pavement Loads  
(Sheet 3 of 3)  
FIGURE-7-3-0-991-022-A01

**7-4-0 Landing Gear Loading on Pavement****\*\*ON A/C A330-700L**Landing Gear Loading on Pavement

1. The curves related to the landing gear loading on pavement are not given in section 07-04-00. Because the relationship between the aircraft weight, the center of gravity and the landing gear loading on the pavement is not strictly linear, it cannot be shown in chart format. But you can find in section 07-03-00 the maximum vertical and horizontal pavement loads for some critical conditions at the tire/ground interfaces for all the operational weight variants of the aircraft.

For questions that are related to the landing gear loading on pavement, contact Airbus.

**7-5-0 Flexible Pavement Requirements - U.S. Army Corps of Engineers Design Method****\*\*ON A/C A330-700L**Flexible Pavement Requirements - US Army Corps of Engineers Design Method

1. The flexible pavement requirements curves as per as U.S. Army Corps of Engineers Design Method are not given in section 07-05-00 since the related data is available through free software.  
Sections 07-02-00 and 07-03-00 give all the inputs data required for the use of such software.

NOTE : The U.S. Army Corps of Engineers Design Method for flexible pavements is being gradually superseded by mechanistic-empirical design methods mostly relying on Linear Elastic Analysis (LEA). The number of parameters considered by such methods is not applicable for a chart format and the use of dedicated pavement-design software is necessary.

For questions that are related to the flexible pavement requirements, contact Airbus.

**7-6-0 Flexible Pavement Requirements - LCN Conversion****\*\*ON A/C A330-700L**Flexible Pavement Requirements - LCN Conversion

1. The Load Classification Number (LCN) curves are not given in section 07-06-00 since the LCN system for the reporting pavement strength is old and are replaced by the ICAO recommended ACN/PCN system in 1983 and ACR/PCR system in 2020.

For questions that are related to the LCN system, contact Airbus.



**7-7-0 Rigid Pavement Requirements - Portland Cement Association Design Method****\*\*ON A/C A330-700L**Rigid Pavement Requirements - Portland Cement Association Design Method

1. The rigid-pavement requirements curves by Portland Cement Association Design Method are not given in section 07-07-00 since the related data is available through free software. Sections 07-02-00 and 07-03-00 give all the inputs data required for the use of such software.

NOTE : The Portland Cement Association Design Method for rigid pavements is being gradually superseded by mechanistic-empirical design methods mostly relying on Finite Element Analysis (FEM). The number of parameters considered by such methods is not applicable for a chart format and the use of dedicated pavement-design is necessary.

For questions that are related to the rigid pavement requirements, contact Airbus.

**7-8-0 Rigid Pavement Requirements - LCN Conversion****\*\*ON A/C A330-700L**Rigid Pavement Requirements - LCN Conversion

1. The Load Classification Number (LCN) curves are not given in section 07-08-00 since the LCN system for the reporting pavement strength is old and are replaced by the ICAO recommended ACN/PCN system in 1983 and ACR/PCR system in 2020.

For questions that are related to the LCN system, contact Airbus.

**7-9-0 ACN/PCN Reporting System - Flexible and Rigid Pavements****\*\*ON A/C A330-700L**ACN/PCN Reporting System - Flexible and Rigid Pavements

1. This section gives data about the Aircraft Classification Number (ACN) for an aircraft gross weight in relation with standard subgrade strength values for flexible and rigid pavement.  
To find the ACN of an aircraft on flexible and rigid pavement, you must know the aircraft gross weight and the subgrade strength.

NOTE : An aircraft with an ACN equal to or less than the reported PCN can operate on that pavement, subject to any limitation on the tire pressure.  
(Ref: ICAO Aerodrome Design Manual, Part 3, Chapter 1, Second Edition 1983).

2. Aircraft Classification Number - ACN table

The tables in figure FIGURE 7-9-0-991-028-A gives ACN data in tabular format for all the operational weight variants of the aircraft.

As an approximation, use a linear interpolation in order to get the ACN at the required operating weight using the following equation:

$$\text{ACN} = \text{ACN min} + (\text{ACN max} - \text{ACN min}) \times (\text{Operating Weight} - 130\,000 \text{ kg}) / (\text{MRW} - 130\,000 \text{ kg})$$

Please note that the interpolation error may reach 5% to 10%.

As an approximation, also use a linear interpolation in order to get the aircraft weight at the pavement PCN using the following equation:

$$\text{Operating weight} = 130\,000 \text{ kg} + (\text{MRW} - 130\,000 \text{ kg}) \times (\text{PCN} - \text{ACN min}) / (\text{ACN max} - \text{ACN min})$$

Please note that the interpolation error may reach up to 5%.

With ACN max = ACN calculated at the MRW in the table and with ACN min = ACN calculated at 130 000 kg.

For questions or specific calculation related to ACN/PCN Reporting System, contact Airbus.

**\*\*ON A/C A330-700L**

| AIRCRAFT TYPE                    | ALL UP<br>MASS (kg) | LOAD ON<br>ONE MAIN<br>GEAR LEG<br>(%) | TIRE<br>PRESSURE<br>(MPa) | ACN FOR<br>RIGID PAVEMENT<br>SUBGRADES - MN/m <sup>3</sup> |              |           |                     | ACN FOR<br>FLEXIBLE PAVEMENT<br>SUBGRADES - CBR |              |          |                    |
|----------------------------------|---------------------|----------------------------------------|---------------------------|------------------------------------------------------------|--------------|-----------|---------------------|-------------------------------------------------|--------------|----------|--------------------|
|                                  |                     |                                        |                           | HIGH<br>150                                                | MEDIUM<br>80 | LOW<br>40 | ULTRA<br>-LOW<br>20 | HIGH<br>15                                      | MEDIUM<br>10 | LOW<br>6 | ULTRA<br>-LOW<br>3 |
| A330-700L<br>WV000<br>ZCG +0.7 m | 227 900             | 45.1                                   | 1.42                      | 48                                                         | 56           | 66        | 78                  | 53                                              | 57           | 65       | 88                 |
|                                  | 130 000             | 45.1                                   |                           | 28                                                         | 28           | 32        | 37                  | 27                                              | 28           | 31       | 38                 |
| A330-700L<br>WV000<br>ZCG +0.6 m | 227 900             | 45.2                                   | 1.42                      | 48                                                         | 56           | 67        | 78                  | 53                                              | 57           | 66       | 89                 |
|                                  | 130 000             | 45.2                                   |                           | 28                                                         | 28           | 32        | 37                  | 27                                              | 28           | 31       | 38                 |
| A330-700L<br>WV000<br>ZCG +0.5 m | 227 900             | 45.3                                   | 1.42                      | 48                                                         | 57           | 67        | 78                  | 53                                              | 57           | 66       | 89                 |
|                                  | 130 000             | 45.3                                   |                           | 29                                                         | 28           | 32        | 37                  | 27                                              | 28           | 31       | 39                 |
| A330-700L<br>WV000<br>ZCG +0.4 m | 227 900             | 45.5                                   | 1.42                      | 49                                                         | 57           | 67        | 79                  | 53                                              | 58           | 66       | 89                 |
|                                  | 130 000             | 45.5                                   |                           | 29                                                         | 28           | 32        | 37                  | 27                                              | 29           | 31       | 39                 |
| A330-700L<br>WV000<br>ZCG +0.3 m | 227 900             | 45.6                                   | 1.42                      | 49                                                         | 57           | 68        | 79                  | 54                                              | 58           | 66       | 90                 |
|                                  | 130 000             | 45.6                                   |                           | 29                                                         | 28           | 32        | 37                  | 27                                              | 29           | 31       | 39                 |
| A330-700L<br>WV000<br>ZCG +0.2 m | 227 900             | 45.7                                   | 1.42                      | 49                                                         | 57           | 68        | 79                  | 54                                              | 58           | 67       | 90                 |
|                                  | 130 000             | 45.7                                   |                           | 29                                                         | 28           | 32        | 37                  | 28                                              | 29           | 31       | 39                 |
| A330-700L<br>WV000<br>ZCG +0.1 m | 227 900             | 45.8                                   | 1.42                      | 49                                                         | 57           | 68        | 79                  | 54                                              | 58           | 67       | 90                 |
|                                  | 130 000             | 45.8                                   |                           | 29                                                         | 29           | 32        | 37                  | 28                                              | 29           | 31       | 39                 |
| A330-700L<br>WV000<br>ZCG 0 m    | 227 900             | 45.8                                   | 1.42                      | 49                                                         | 57           | 68        | 79                  | 54                                              | 58           | 67       | 90                 |
|                                  | 130 000             | 45.8                                   |                           | 29                                                         | 29           | 32        | 37                  | 28                                              | 29           | 31       | 39                 |
| A330-700L<br>WV000<br>ZCG -0.1 m | 227 900             | 45.8                                   | 1.42                      | 49                                                         | 57           | 68        | 79                  | 54                                              | 58           | 67       | 90                 |
|                                  | 130 000             | 45.8                                   |                           | 29                                                         | 29           | 32        | 37                  | 28                                              | 29           | 31       | 39                 |
| A330-700L<br>WV000<br>ZCG -0.2 m | 227 900             | 45.8                                   | 1.42                      | 49                                                         | 57           | 68        | 79                  | 54                                              | 58           | 67       | 90                 |
|                                  | 130 000             | 45.8                                   |                           | 29                                                         | 29           | 32        | 37                  | 28                                              | 29           | 31       | 39                 |
| A330-700L<br>WV000<br>ZCG -0.3 m | 227 900             | 45.8                                   | 1.42                      | 49                                                         | 57           | 68        | 79                  | 54                                              | 58           | 67       | 90                 |
|                                  | 130 000             | 45.8                                   |                           | 29                                                         | 29           | 32        | 37                  | 28                                              | 29           | 31       | 39                 |
| A330-700L<br>WV000<br>ZCG -0.4 m | 227 900             | 45.8                                   | 1.42                      | 49                                                         | 57           | 68        | 79                  | 54                                              | 58           | 67       | 90                 |
|                                  | 130 000             | 45.8                                   |                           | 29                                                         | 29           | 32        | 37                  | 28                                              | 29           | 31       | 39                 |
| A330-700L<br>WV000<br>ZCG -0.5 m | 227 900             | 45.8                                   | 1.42                      | 49                                                         | 57           | 68        | 79                  | 54                                              | 58           | 67       | 90                 |
|                                  | 130 000             | 45.8                                   |                           | 29                                                         | 29           | 32        | 37                  | 28                                              | 29           | 31       | 39                 |

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Aircraft Classification Number  
ACN Table (Sheet 1 of 2)  
FIGURE-7-9-0-991-028-A01

## \*\*ON A/C A330-700L

| AIRCRAFT TYPE                    | ALL UP<br>MASS (kg) | LOAD ON<br>ONE MAIN<br>GEAR LEG<br>(%) | TIRE<br>PRESSURE<br>(MPa) | ACN FOR<br>RIGID PAVEMENT<br>SUBGRADES - MN/m³ |              |           |                     | ACN FOR<br>FLEXIBLE PAVEMENT<br>SUBGRADES - CBR |              |          |                    |
|----------------------------------|---------------------|----------------------------------------|---------------------------|------------------------------------------------|--------------|-----------|---------------------|-------------------------------------------------|--------------|----------|--------------------|
|                                  |                     |                                        |                           | HIGH<br>150                                    | MEDIUM<br>80 | LOW<br>40 | ULTRA<br>-LOW<br>20 | HIGH<br>15                                      | MEDIUM<br>10 | LOW<br>6 | ULTRA<br>-LOW<br>3 |
| A330-700L<br>WV001<br>ZCG +0.7 m | 205 900             | 45.6                                   | 1.42                      | 44                                             | 50           | 59        | 69                  | 47                                              | 51           | 58       | 77                 |
|                                  | 130 000             | 45.6                                   |                           | 29                                             | 28           | 32        | 37                  | 27                                              | 29           | 31       | 39                 |
| A330-700L<br>WV001<br>ZCG +0.6 m | 205 900             | 45.8                                   | 1.42                      | 44                                             | 50           | 59        | 69                  | 48                                              | 51           | 58       | 78                 |
|                                  | 130 000             | 45.8                                   |                           | 29                                             | 28           | 32        | 37                  | 28                                              | 29           | 31       | 39                 |
| A330-700L<br>WV001<br>ZCG +0.5 m | 205 900             | 45.9                                   | 1.42                      | 44                                             | 50           | 59        | 69                  | 48                                              | 51           | 58       | 78                 |
|                                  | 130 000             | 45.9                                   |                           | 29                                             | 29           | 33        | 37                  | 28                                              | 29           | 32       | 39                 |
| A330-700L<br>WV001<br>ZCG +0.4 m | 205 900             | 46.1                                   | 1.42                      | 45                                             | 51           | 60        | 70                  | 48                                              | 51           | 58       | 78                 |
|                                  | 130 000             | 46.1                                   |                           | 29                                             | 29           | 33        | 38                  | 28                                              | 29           | 32       | 39                 |
| A330-700L<br>WV001<br>ZCG +0.3 m | 205 900             | 46.2                                   | 1.42                      | 45                                             | 51           | 60        | 70                  | 48                                              | 52           | 59       | 79                 |
|                                  | 130 000             | 46.2                                   |                           | 29                                             | 29           | 33        | 38                  | 28                                              | 29           | 32       | 40                 |
| A330-700L<br>WV001<br>ZCG +0.2 m | 205 900             | 46.3                                   | 1.42                      | 45                                             | 51           | 60        | 70                  | 48                                              | 52           | 59       | 79                 |
|                                  | 130 000             | 46.3                                   |                           | 29                                             | 29           | 33        | 38                  | 28                                              | 29           | 32       | 40                 |
| A330-700L<br>WV001<br>ZCG +0.1 m | 205 900             | 46.4                                   | 1.42                      | 45                                             | 51           | 60        | 70                  | 48                                              | 52           | 59       | 79                 |
|                                  | 130 000             | 46.4                                   |                           | 29                                             | 29           | 33        | 38                  | 28                                              | 29           | 32       | 40                 |
| A330-700L<br>WV001<br>ZCG 0 m    | 205 900             | 46.4                                   | 1.42                      | 45                                             | 51           | 60        | 70                  | 48                                              | 52           | 59       | 79                 |
|                                  | 130 000             | 46.4                                   |                           | 29                                             | 29           | 33        | 38                  | 28                                              | 29           | 32       | 40                 |
| A330-700L<br>WV001<br>ZCG -0.1 m | 205 900             | 46.4                                   | 1.42                      | 45                                             | 51           | 60        | 70                  | 48                                              | 52           | 59       | 79                 |
|                                  | 130 000             | 46.4                                   |                           | 29                                             | 29           | 33        | 38                  | 28                                              | 29           | 32       | 40                 |
| A330-700L<br>WV001<br>ZCG -0.2 m | 205 900             | 46.4                                   | 1.42                      | 45                                             | 51           | 60        | 70                  | 48                                              | 52           | 59       | 79                 |
|                                  | 130 000             | 46.4                                   |                           | 29                                             | 29           | 33        | 38                  | 28                                              | 29           | 32       | 40                 |
| A330-700L<br>WV001<br>ZCG -0.3 m | 205 900             | 46.4                                   | 1.42                      | 45                                             | 51           | 60        | 70                  | 48                                              | 52           | 59       | 79                 |
|                                  | 130 000             | 46.4                                   |                           | 29                                             | 29           | 33        | 38                  | 28                                              | 29           | 32       | 40                 |
| A330-700L<br>WV001<br>ZCG -0.4 m | 205 900             | 46.4                                   | 1.42                      | 45                                             | 51           | 60        | 70                  | 48                                              | 52           | 59       | 79                 |
|                                  | 130 000             | 46.4                                   |                           | 29                                             | 29           | 33        | 38                  | 28                                              | 29           | 32       | 40                 |
| A330-700L<br>WV001<br>ZCG -0.5 m | 205 900             | 46.4                                   | 1.42                      | 45                                             | 51           | 60        | 70                  | 48                                              | 52           | 59       | 79                 |
|                                  | 130 000             | 46.4                                   |                           | 29                                             | 29           | 33        | 38                  | 28                                              | 29           | 32       | 40                 |

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Aircraft Classification Number  
ACN Table (Sheet 2 of 2)  
FIGURE-7-9-0-991-028-A01

**7-10-0 ACR/PCR Reporting System - Flexible And Rigid Pavement****\*\*ON A/C A330-700L**ACR/PCR Reporting System - Flexible and Rigid Pavements

1. The ACR/PCR system has been developed by the ICAO to overcome the deficiencies of the ACN/PCN system. Significant advances in pavement design methods had occurred since its development in the late 1970s early 1980s, leading to inconsistencies with the pavement-strength-rating system.

The ACR/PCR system entails new procedures for the determination of both the ACR and the PCR that are consistent with the current pavement design procedures. This allows to capture the effects of the improved characteristics of new pavement materials as well as modern landing gear configurations, thus leading to an improved accuracy.

This section give data about the Aircraft Classification Rating (ACR) for the maximum ramp weight in relation with standard subgrade strength values for flexible and rigid pavement. To determine the ACR at other aircraft gross weight, use the official ICAO-ACR software.

NOTE : An aircraft with an ACR equal to or less than the reported PCR can operate on that pavement, subject to any limitation on the tire pressure. (Ref: ICAO Aerodrome Design Manual, Part 3, Third Edition 2020).

2. Aircraft Classification Rating - ACR Table

The table FIGURE 7-10-0-991-010-A gives ACR data in tabular format for all the operational weight variants of the aircraft.

For questions or specific calculation related to ACR/PCR reporting system, contact Airbus.

**\*\*ON A/C A330-700L**

| AIRCRAFT<br>TYPE                 | ALL UP<br>MASS<br>(kg) | LOAD ON<br>ONE MAIN<br>GEAR LEG<br>(%) | TIRE<br>PRESSURE<br>(MPa) | ACR FOR<br>RIGID PAVEMENT<br>SUBGRADES - MPa |               |           |                     | ACR FOR<br>FLEXIBLE PAVEMENT<br>SUBGRADES - MPa |               |           |                     |
|----------------------------------|------------------------|----------------------------------------|---------------------------|----------------------------------------------|---------------|-----------|---------------------|-------------------------------------------------|---------------|-----------|---------------------|
|                                  |                        |                                        |                           | HIGH<br>200                                  | MEDIUM<br>120 | LOW<br>80 | ULTRA<br>-LOW<br>50 | HIGH<br>200                                     | MEDIUM<br>120 | LOW<br>80 | ULTRA<br>-LOW<br>50 |
| A330-700L<br>WV000<br>ZCG +0.7 m | 227 900                | 45.1                                   | 1.42                      | 550                                          | 630           | 710       | 800                 | 520                                             | 540           | 590       | 710                 |
| A330-700L<br>WV000<br>ZCG +0.6 m | 227 900                | 45.2                                   | 1.42                      | 560                                          | 640           | 710       | 810                 | 520                                             | 540           | 590       | 710                 |
| A330-700L<br>WV000<br>ZCG +0.5 m | 227 900                | 45.3                                   | 1.42                      | 560                                          | 640           | 720       | 810                 | 530                                             | 550           | 590       | 710                 |
| A330-700L<br>WV000<br>ZCG +0.4 m | 227 900                | 45.5                                   | 1.42                      | 560                                          | 640           | 720       | 810                 | 530                                             | 550           | 590       | 710                 |
| A330-700L<br>WV000<br>ZCG +0.3 m | 227 900                | 45.6                                   | 1.42                      | 560                                          | 640           | 720       | 820                 | 530                                             | 550           | 600       | 720                 |
| A330-700L<br>WV000<br>ZCG +0.2 m | 227 900                | 45.7                                   | 1.42                      | 570                                          | 650           | 720       | 820                 | 530                                             | 550           | 600       | 720                 |
| A330-700L<br>WV000<br>ZCG +0.1 m | 227 900                | 45.8                                   | 1.42                      | 570                                          | 650           | 730       | 820                 | 530                                             | 550           | 600       | 720                 |
| A330-700L<br>WV000<br>ZCG 0 m    | 227 900                | 45.8                                   | 1.42                      | 570                                          | 650           | 730       | 820                 | 530                                             | 550           | 600       | 720                 |
| A330-700L<br>WV000<br>ZCG -0.1 m | 227 900                | 45.8                                   | 1.42                      | 570                                          | 650           | 730       | 820                 | 530                                             | 550           | 600       | 720                 |
| A330-700L<br>WV000<br>ZCG -0.2 m | 227 900                | 45.8                                   | 1.42                      | 570                                          | 650           | 730       | 820                 | 530                                             | 550           | 600       | 720                 |
| A330-700L<br>WV000<br>ZCG -0.3 m | 227 900                | 45.8                                   | 1.42                      | 570                                          | 650           | 730       | 820                 | 530                                             | 550           | 600       | 720                 |
| A330-700L<br>WV000<br>ZCG -0.4 m | 227 900                | 45.8                                   | 1.42                      | 570                                          | 650           | 730       | 820                 | 530                                             | 550           | 600       | 720                 |
| A330-700L<br>WV000<br>ZCG -0.5 m | 227 900                | 45.8                                   | 1.42                      | 570                                          | 650           | 730       | 820                 | 530                                             | 550           | 600       | 720                 |
| A330-700L<br>WV001<br>ZCG +0.7 m | 205 900                | 45.6                                   | 1.42                      | 500                                          | 560           | 620       | 710                 | 480                                             | 490           | 530       | 620                 |
| A330-700L<br>WV001<br>ZCG +0.6 m | 205 900                | 45.8                                   | 1.42                      | 500                                          | 560           | 630       | 710                 | 480                                             | 490           | 530       | 620                 |

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ACR Table  
(Sheet 1 of 2)  
FIGURE-7-10-0-991-010-A01

**\*\*ON A/C A330-700L**

| AIRCRAFT<br>TYPE                 | ALL UP<br>MASS<br>(kg) | LOAD ON<br>ONE MAIN<br>GEAR LEG<br>(%) | TIRE<br>PRESSURE<br>(MPa) | ACR FOR<br>RIGID PAVEMENT<br>SUBGRADES - MPa |               |           |                     | ACR FOR<br>FLEXIBLE PAVEMENT<br>SUBGRADES - MPa |               |           |                     |
|----------------------------------|------------------------|----------------------------------------|---------------------------|----------------------------------------------|---------------|-----------|---------------------|-------------------------------------------------|---------------|-----------|---------------------|
|                                  |                        |                                        |                           | HIGH<br>200                                  | MEDIUM<br>120 | LOW<br>80 | ULTRA<br>-LOW<br>50 | HIGH<br>200                                     | MEDIUM<br>120 | LOW<br>80 | ULTRA<br>-LOW<br>50 |
| A330-700L<br>WV001<br>ZCG +0.5 m | 205 900                | 45.9                                   | 1.42                      | 500                                          | 560           | 630       | 710                 | 480                                             | 500           | 530       | 620                 |
| A330-700L<br>WV001<br>ZCG +0.4 m | 205 900                | 46.1                                   | 1.42                      | 500                                          | 570           | 630       | 720                 | 480                                             | 500           | 530       | 620                 |
| A330-700L<br>WV001<br>ZCG +0.3 m | 205 900                | 46.2                                   | 1.42                      | 500                                          | 570           | 640       | 720                 | 480                                             | 500           | 530       | 630                 |
| A330-700L<br>WV001<br>ZCG +0.2 m | 205 900                | 46.3                                   | 1.42                      | 510                                          | 570           | 640       | 720                 | 480                                             | 500           | 540       | 630                 |
| A330-700L<br>WV001<br>ZCG +0.1 m | 205 900                | 46.4                                   | 1.42                      | 510                                          | 570           | 640       | 730                 | 490                                             | 500           | 540       | 630                 |
| A330-700L<br>WV001<br>ZCG 0 m    | 205 900                | 46.4                                   | 1.42                      | 510                                          | 570           | 640       | 730                 | 490                                             | 500           | 540       | 630                 |
| A330-700L<br>WV001<br>ZCG -0.1 m | 205 900                | 46.4                                   | 1.42                      | 510                                          | 570           | 640       | 730                 | 490                                             | 500           | 540       | 630                 |
| A330-700L<br>WV001<br>ZCG -0.2 m | 205 900                | 46.4                                   | 1.42                      | 510                                          | 570           | 640       | 730                 | 490                                             | 500           | 540       | 630                 |
| A330-700L<br>WV001<br>ZCG -0.3 m | 205 900                | 46.4                                   | 1.42                      | 510                                          | 570           | 640       | 730                 | 490                                             | 500           | 540       | 630                 |
| A330-700L<br>WV001<br>ZCG -0.4 m | 205 900                | 46.4                                   | 1.42                      | 510                                          | 570           | 640       | 730                 | 490                                             | 500           | 540       | 630                 |
| A330-700L<br>WV001<br>ZCG -0.5 m | 205 900                | 46.4                                   | 1.42                      | 510                                          | 570           | 640       | 730                 | 490                                             | 500           | 540       | 630                 |

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ACR Table  
(Sheet 2 of 2)  
FIGURE-7-10-0-991-010-A01

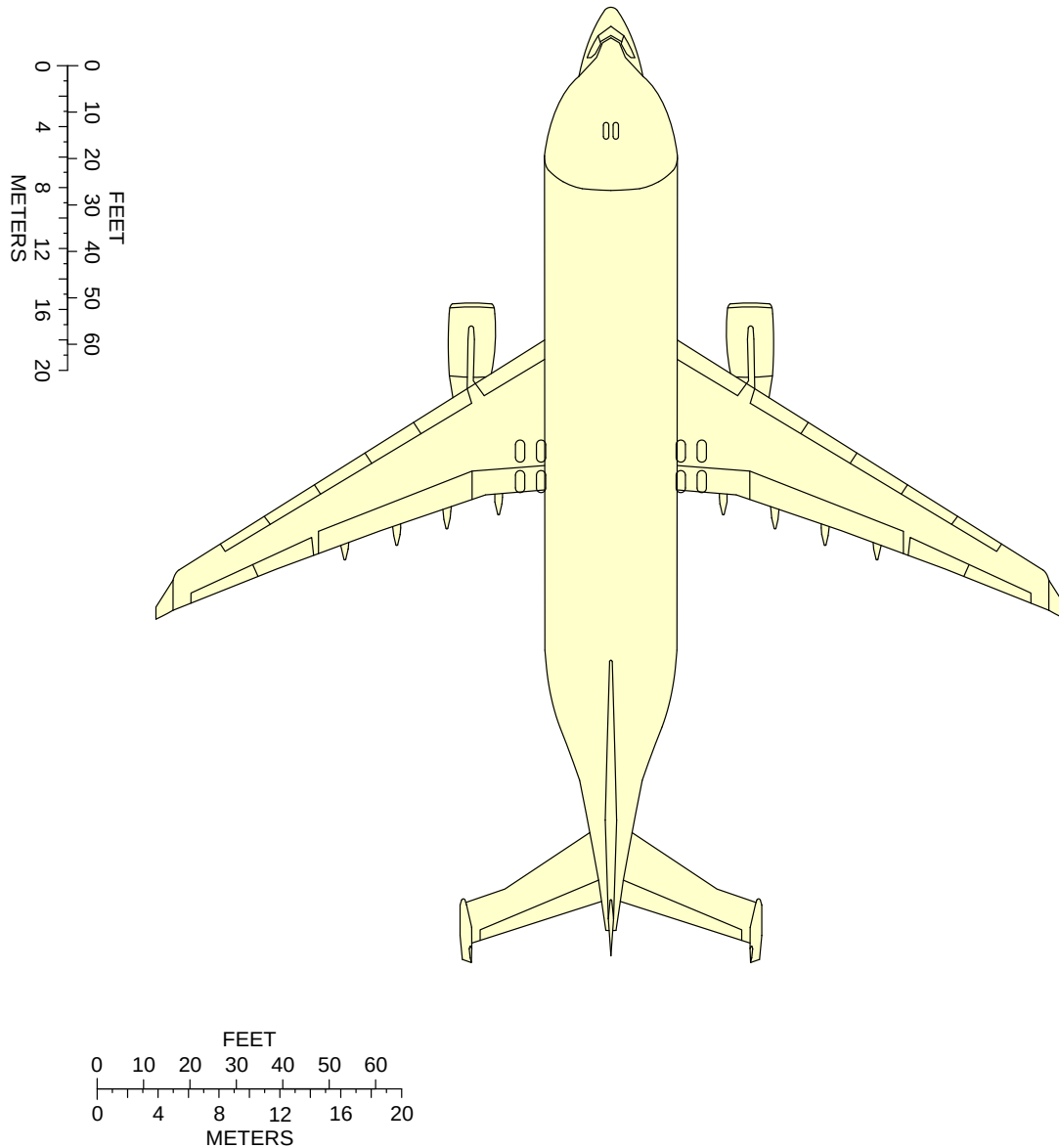


**SCALED DRAWINGS****8-0-0      SCALED DRAWINGS****\*\*ON A/C A330-700L****Scaled Drawings**

1. This section gives the scaled drawings.

**NOTE** : When printing this drawing, make sure to adjust for proper scaling.

**\*\*ON A/C A330-700L**



**NOTE:**  
WHEN PRINTING THIS DRAWING, MAKE SURE TO ADJUST FOR  
PROPER SCALING.

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Scaled Drawing  
FIGURE-8-0-0-991-007-A01

**AIRCRAFT RESCUE AND FIRE FIGHTING****10-0-0      AIRCRAFT RESCUE AND FIRE FIGHTING****\*\*ON A/C A330-700L****Aircraft Rescue and Fire Fighting****1.    Aircraft Rescue and Fire Fighting Charts**

This sections gives the data related to aircraft rescue and fire fighting.

The figures contained in this section are the figures that are in the Aircraft Rescue and Fire Fighting Charts poster available for download on AIRBUSWorld and the Airbus website.

**\*\*ON A/C A330-700L**

# AIRBUS

# A330-700L

## Aircraft Rescue and Fire Fighting Chart

## ARFC

**NOTE:**

THIS CHART GIVES THE GENERAL LAYOUT OF THE A330-700L STANDARD VERSION.  
THE NUMBER AND ARRANGEMENT OF THE INDIVIDUAL ITEMS VARY WITH THE CUSTOMERS.  
FIGURES CONTAINED IN THIS POSTER ARE AVAILABLE SEPARATELY IN THE CHAPTER 10 OF THE  
"AIRCRAFT CHARACTERISTICS - AIRPORT AND MAINTENANCE PLANNING" DOCUMENT.

**ISSUED BY:**

AIRBUS S.A.S  
CUSTOMER SERVICES  
TECHNICAL DATA SUPPORT AND SERVICES  
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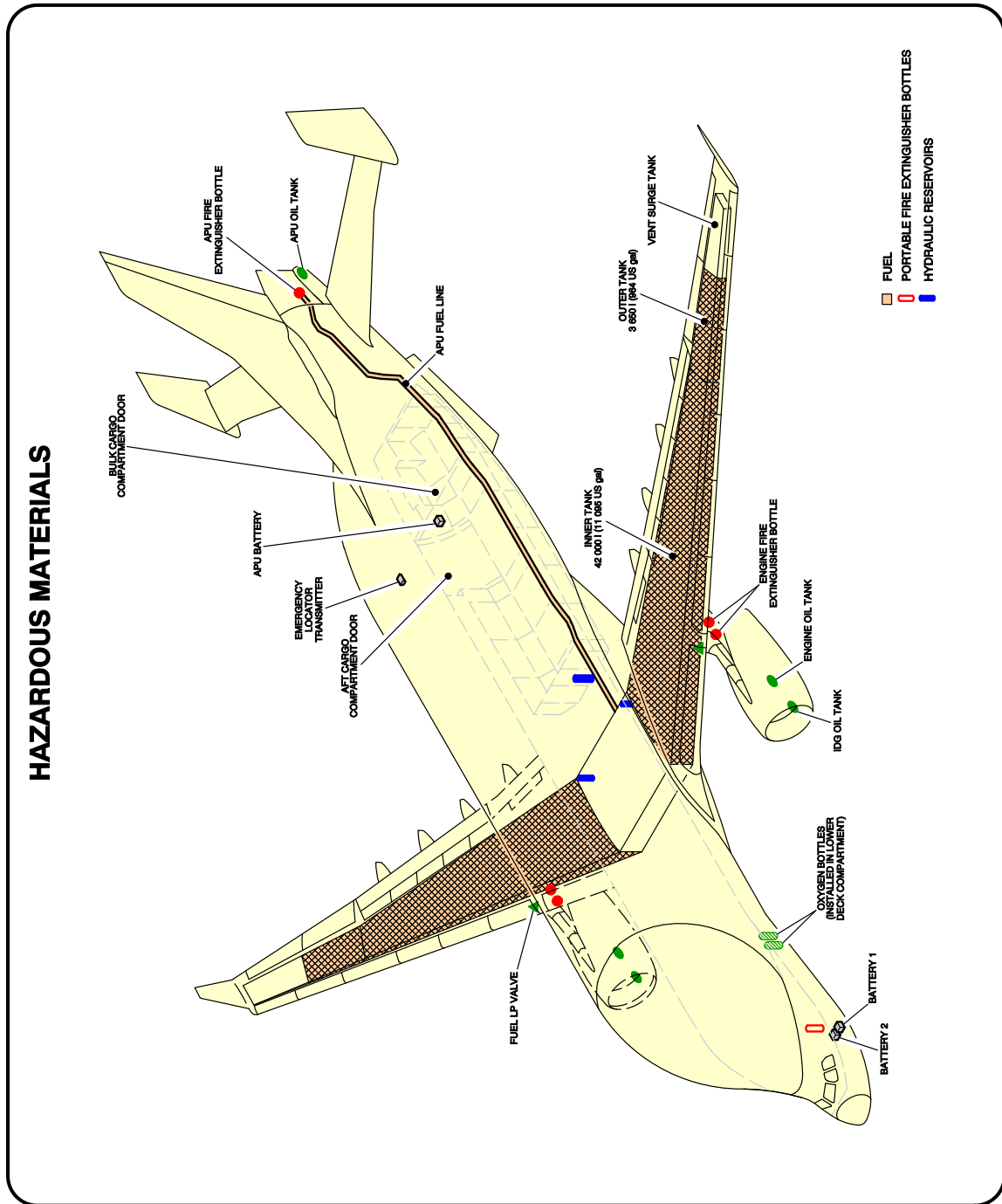
REVISION DATE: JAN 2020  
REFERENCE : F\_RF\_000000\_1\_A33070L  
SHEET 1/2

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Front Page  
FIGURE-10-0-0-991-086-A01

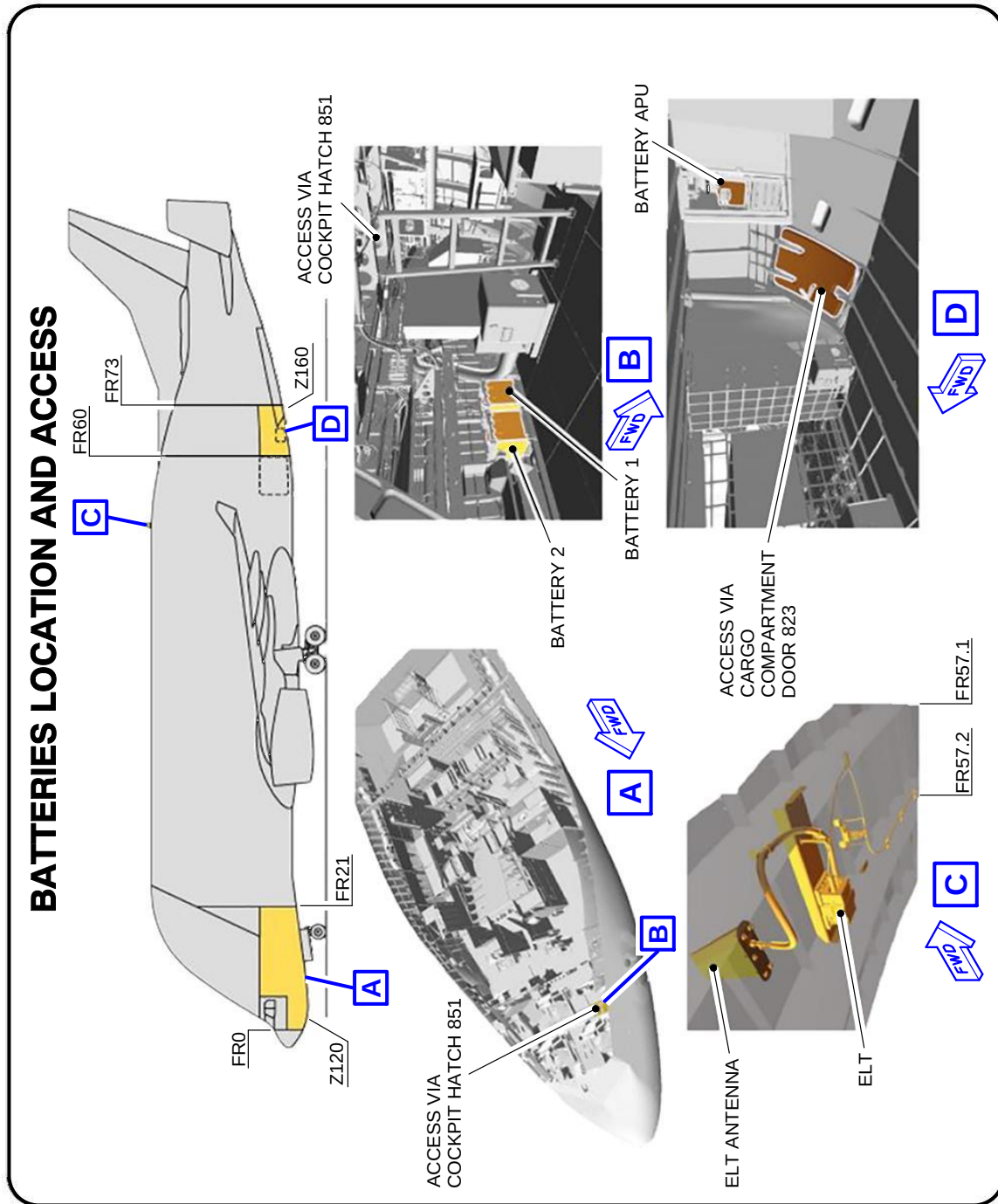
\*\*ON A/C A330-700L



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Highly Flammable and Hazardous Materials and Components  
FIGURE-10-0-0-991-087-A01

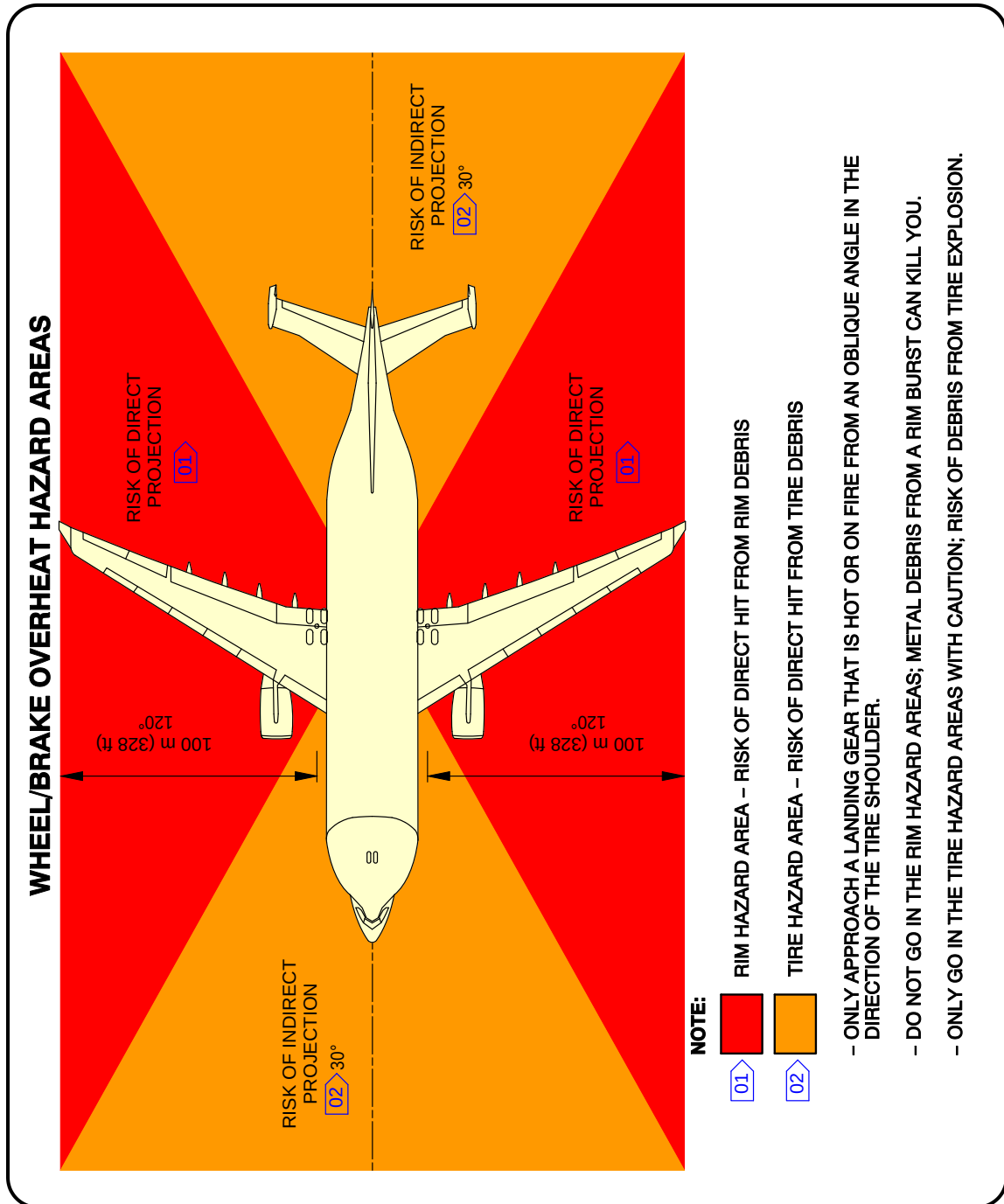
**\*\*ON A/C A330-700L**



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Batteries Location and Access  
FIGURE-10-0-0-991-088-A01

\*\*ON A/C A330-700L



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Wheel/Brake Overheat  
Wheel Safety Area (Sheet 1 of 2)  
FIGURE-10-0-0-991-089-A01

**\*\*ON A/C A330-700L**

## **BRAKE OVERHEAT AND LANDING GEAR FIRE**

**WARNING:** BE VERY CAREFUL WHEN THERE IS A BRAKE OVERHEAT AND/OR LANDING GEAR FIRE. THERE IS A RISK OF TIRE EXPLOSION AND/OR WHEEL RIM BURST THAT CAN CAUSE DEATH OR INJURY. MAKE SURE THAT YOU OBEY THE SAFETY PRECAUTIONS THAT FOLLOW.

THE PROCEDURES THAT FOLLOW GIVE RECOMMENDATIONS AND SAFETY PRECAUTIONS FOR THE COOLING OF VERY HOT BRAKES AFTER ABNORMAL OPERATIONS SUCH AS A REJECTED TAKE-OFF OR OVERWEIGHT LANDING. FOR THE COOLING OF BRAKES AFTER NORMAL TAXI-IN, REFER TO YOUR COMPANY PROCEDURES.

### **BRAKE OVERHEAT:**

- 1 - GET THE BRAKE TEMPERATURE FROM THE COCKPIT OR USE A REMOTE MEASUREMENT TECHNIQUE. THE REAL TEMPERATURE OF THE BRAKES CAN BE MUCH HIGHER THAN THE TEMPERATURE SHOWN ON THE ECAM.  
**NOTE:** AT HIGH TEMPERATURES (>800°C), THERE IS A RISK OF WARPING OF THE LANDING GEAR STRUTS AND AXLES.
- 2 - APPROACH THE LANDING GEAR WITH EXTREME CAUTION AND FROM AN OBLIQUE ANGLE IN THE DIRECTION OF THE TIRE SHOULDER. DO NOT GO INTO THE RIM HAZARD AREA AND ONLY GO IN THE TIRE HAZARD AREA WITH CAUTION. (REF FIG. WHEEL/BRAKE OVERHEAT HAZARD AREAS). IF POSSIBLE, STAY IN A VEHICLE.
- 3 - LOOK AT THE CONDITION OF THE TIRES:  
IF THE TIRES ARE STILL INFLATED (FUSE PLUGS NOT MELTED), THERE IS A RISK OF TIRE EXPLOSION AND RIM BURST. DO NOT USE COOLING FANS BECAUSE THEY CAN PREVENT OPERATION OF THE FUSE PLUGS.
- 4 - USE WATER MIST TO DECREASE THE TEMPERATURE OF THE COMPLETE WHEEL AND BRAKE ASSEMBLY. USE A TECHNIQUE THAT PREVENTS SUDDEN COOLING. SUDDEN COOLING CAN CAUSE WHEEL CRACKS OR RIM BURST. DO NOT APPLY WATER, FOAM OR CO<sub>2</sub>. THESE COOLING AGENTS (AND ESPECIALLY CO<sub>2</sub>, WHICH HAS A VERY STRONG COOLING EFFECT) CAN CAUSE THERMAL SHOCKS AND BURST OF HOT PARTS.

### **LANDING GEAR FIRE:**

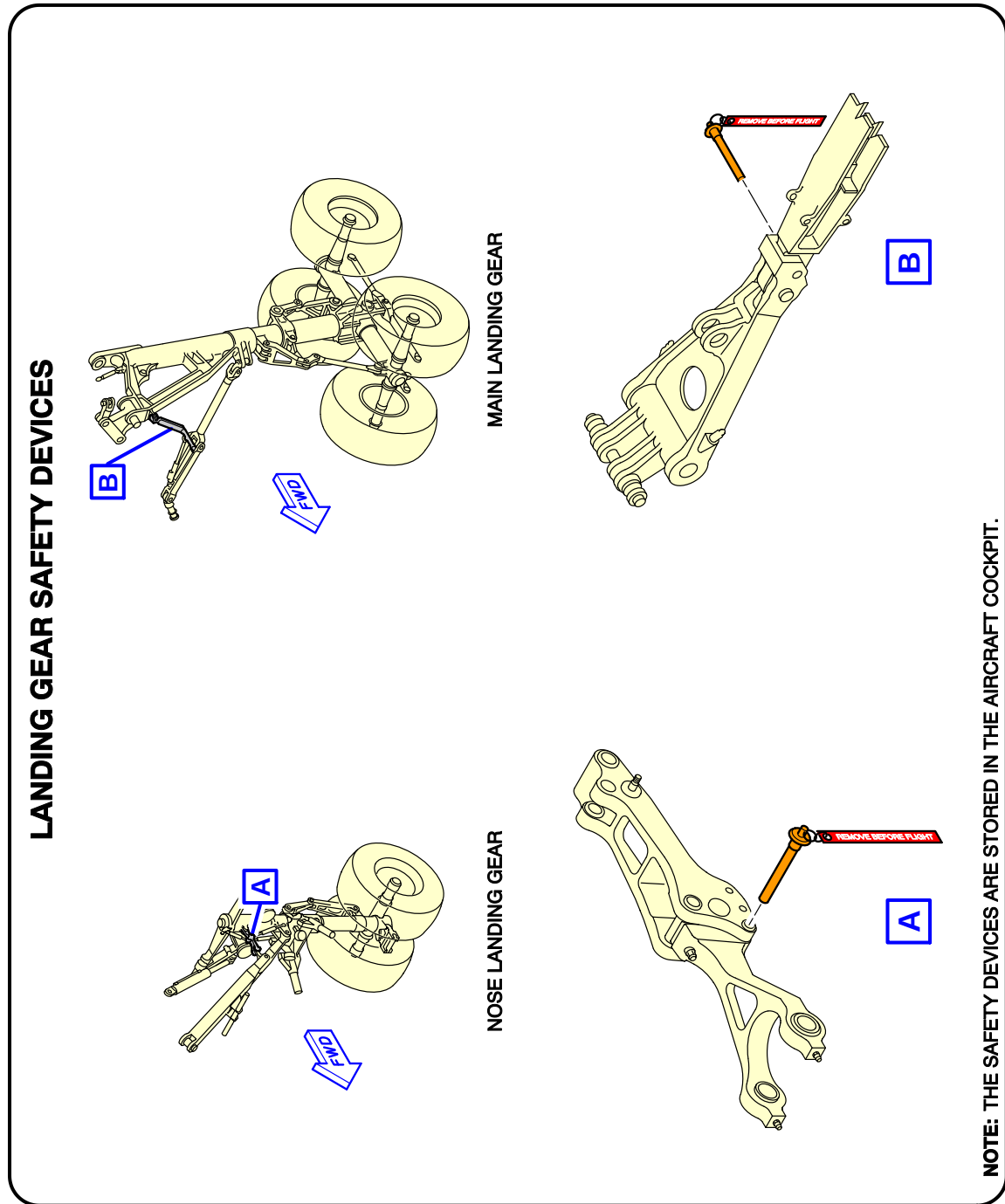
- CAUTION:** AIRBUS RECOMMENDS THAT YOU DO NOT USE DRY POWDERS OR DRY CHEMICALS ON HOT BRAKES OR TO EXTINGUISH LANDING GEAR FIRES. THESE AGENTS CAN CHANGE INTO SOLID OR ENAMELED DEPOSITS. THEY CAN DECREASE THE SPEED OF HEAT DISSIPATION WITH A POSSIBLE RISK OF PERMANENT STRUCTURAL DAMAGE TO THE BRAKES, WHEELS OR WHEEL AXLES.
- 1 - IMMEDIATELY STOP THE FIRE:
    - A) APPROACH THE LANDING GEAR WITH EXTREME CAUTION FROM AN OBLIQUE ANGLE IN THE DIRECTION OF THE TIRE SHOULDER. DO NOT GO INTO THE RIM HAZARD AREA AND ONLY GO IN THE TIRE HAZARD AREA WITH CAUTION. IF POSSIBLE, STAY IN A VEHICLE.
    - B) USE LARGE AMOUNTS OF WATER, WATER MIST; IF THE FUEL TANKS ARE AT RISK, USE FOAM. USE A TECHNIQUE THAT PREVENTS SUDDEN COOLING. SUDDEN COOLING CAN CAUSE WHEEL CRACKS OR RIM BURST.
    - C) DO NOT USE FANS OR BLOWERS.

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Wheel/Brake Overheat  
Recommendations (Sheet 2 of 2)  
FIGURE-10-0-0-991-089-A01



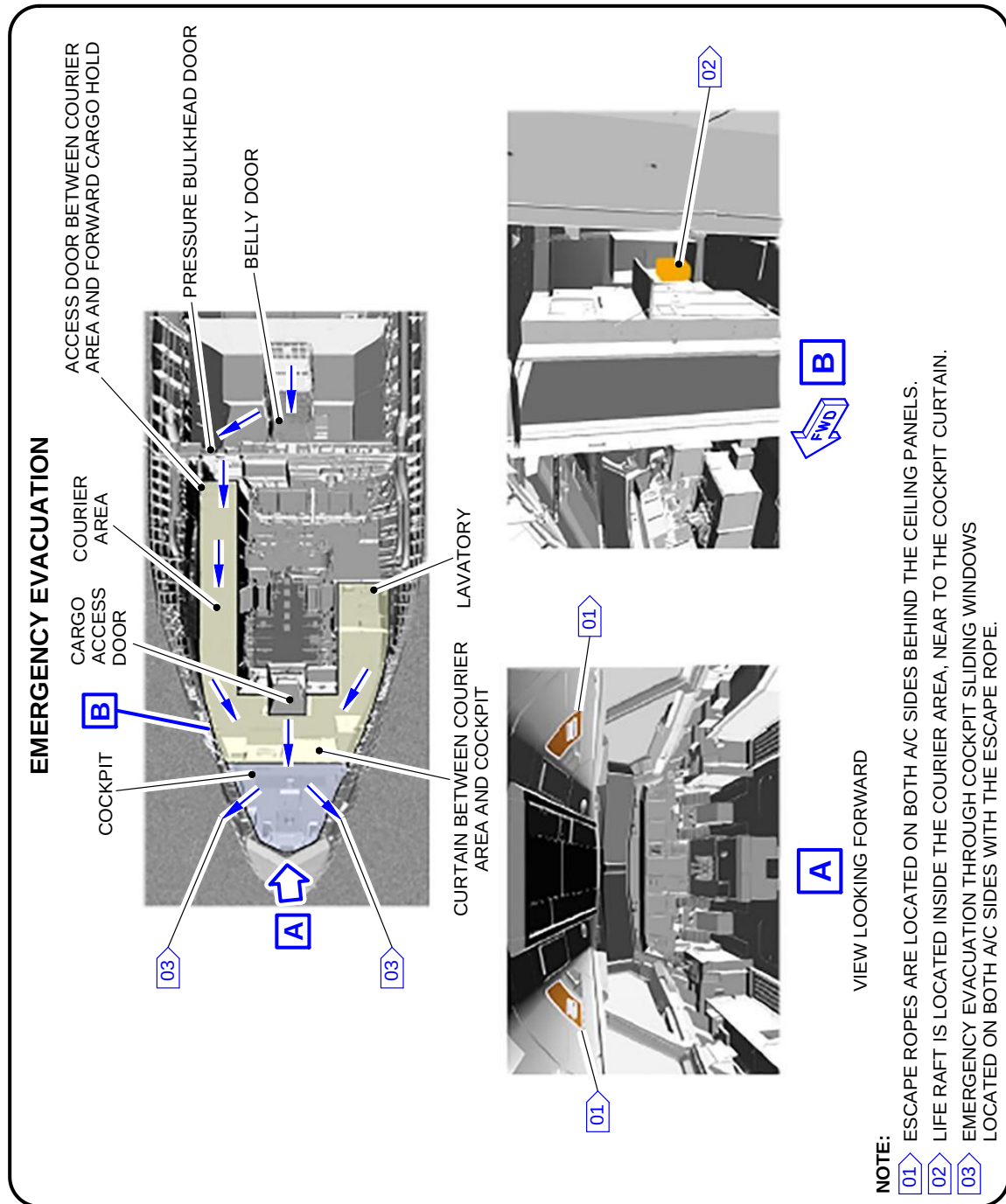
**\*\*ON A/C A330-700L**



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Ground Lock Safety-Devices  
FIGURE-10-0-0-991-091-A01

**\*\*ON A/C A330-700L**

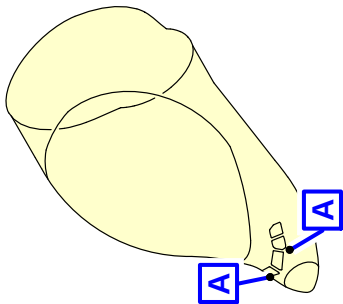


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Emergency Evacuation Devices  
FIGURE-10-0-0-991-092-A01

**\*\*ON A/C A330-700L**

## SLIDING WINDOWS EVACUATION



ESCAPE ROPE STORAGE



HANDLE BUTTON

### EVACUATION FROM THE INTERIOR BY SLIDING WINDOWS

- PUSH THE BUTTON AND PULL THE HANDLE UNTIL IT STOP TO OPEN THE SLIDING WINDOW
- USE THE ESCAPE ROPE TO GET DOWN



HANDLE

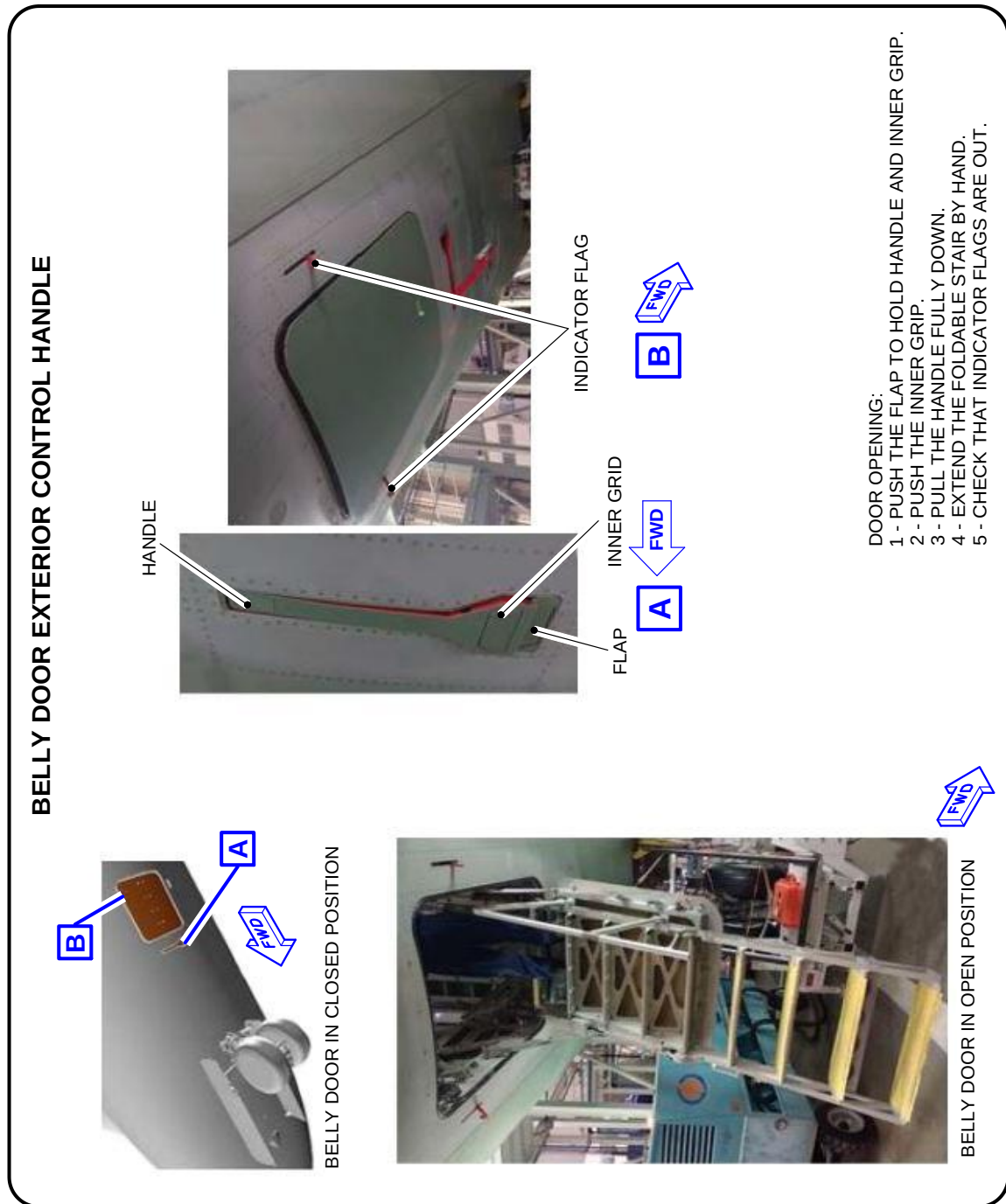
### ACCESS FROM THE EXTERIOR BY SLIDING WINDOWS

- TURN THE HANDLE TO OPEN
- PUSH WINDOW INWARD THEN SLIDE AFTWARD

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Sliding Windows Evacuation  
FIGURE-10-0-0-991-101-A01

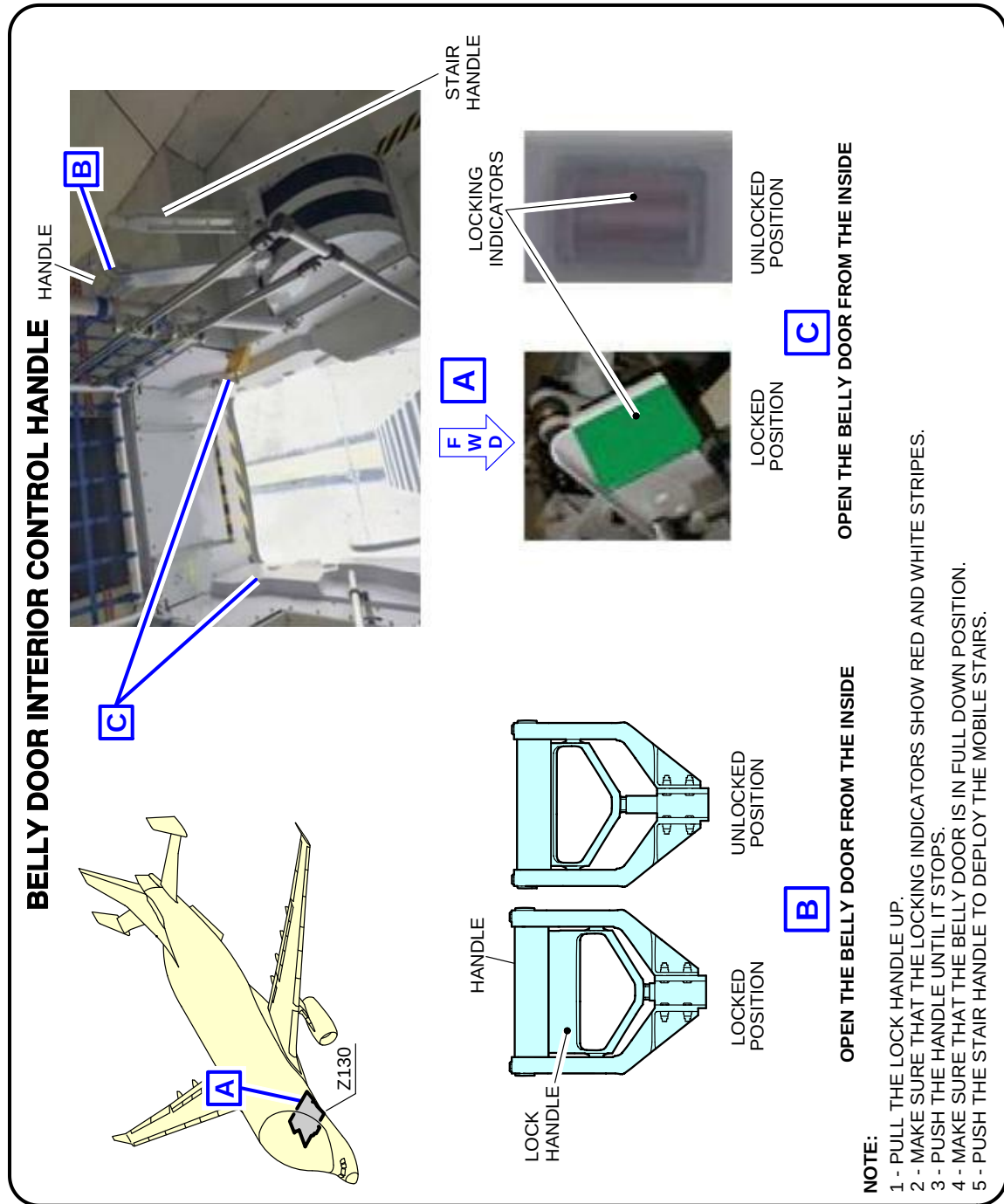
**\*\*ON A/C A330-700L**



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Belly-Door Exterior Control-Handle  
FIGURE-10-0-0-991-093-A01

**\*\*ON A/C A330-700L**



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Belly-Door Interior Control-Handle  
FIGURE-10-0-0-991-102-A01

**\*\*ON A/C A330-700L**

### PRESSURE BULKHEAD DOOR



FR21

Z132

A

B

C

HANDLE

#### [B] OPENING OF THE PRESSURE BULKHEAD DOOR FROM COURIER AREA



- 1 - TURN THE HANDLE COUNTERCLOCKWISE.
- 2 - MAKE SURE THAT GREEN LINE ON HANDLE AND RED LINE ON DOOR LINING ARE ALIGNED.
- 3 - MAKE SURE THAT THE VISUAL INDICATOR SHOWS A RED TARGET.
- 4 - PULL THE PRESSURE BULKHEAD DOOR.

#### [C] OPENING OF THE PRESSURE BULKHEAD DOOR FROM FORWARD COMPARTMENT AREA



- 1 - TURN THE HANDLE CLOCKWISE.
- 2 - MAKE SURE THAT GREEN LINE ON HANDLE AND RED LINE ON DOOR LINING ARE ALIGNED.
- 3 - PUSH THE PRESSURE BULKHEAD DOOR.

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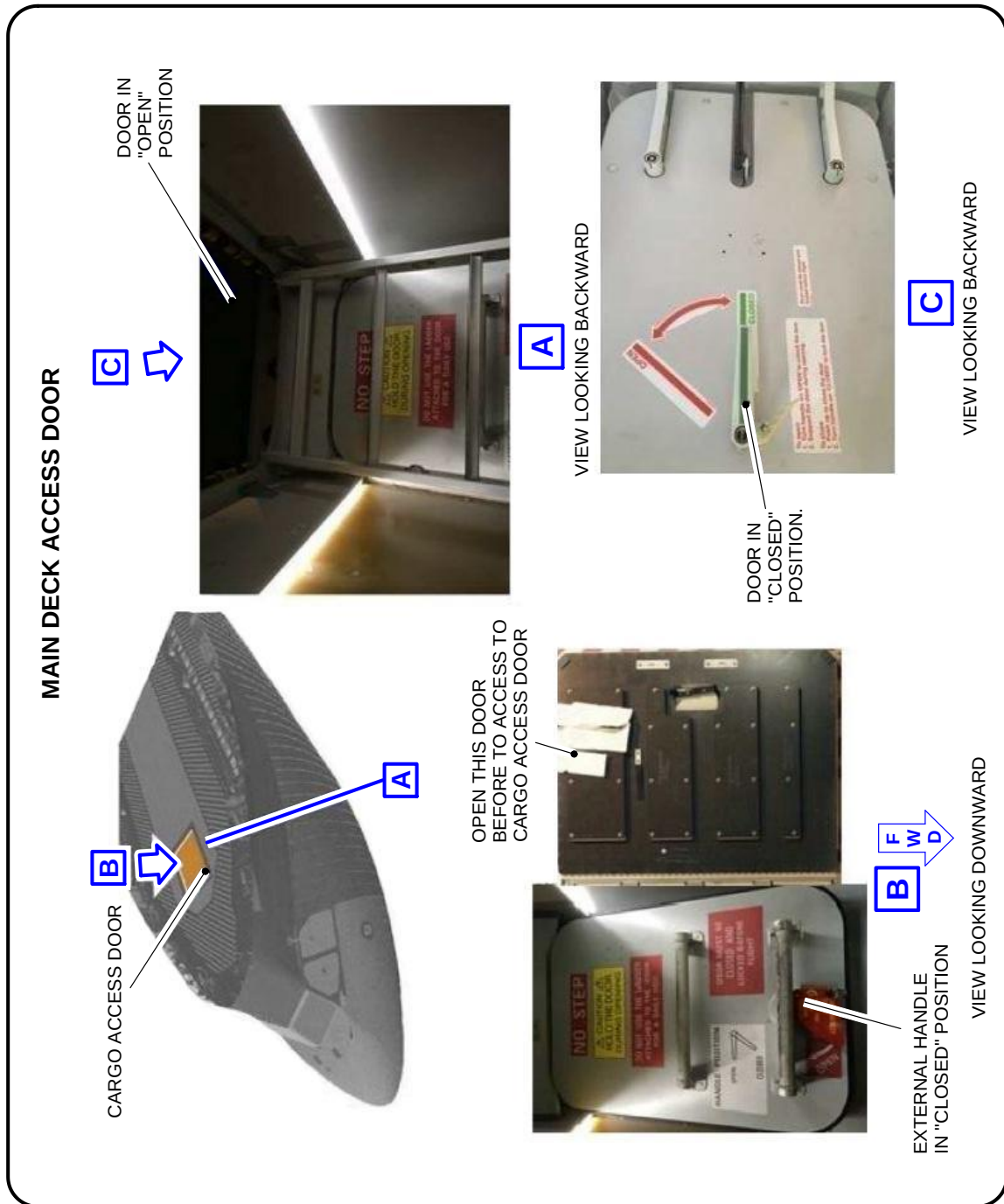
Pressure Bulkhead Door  
FIGURE-10-0-0-991-103-A01

## 10-0-0

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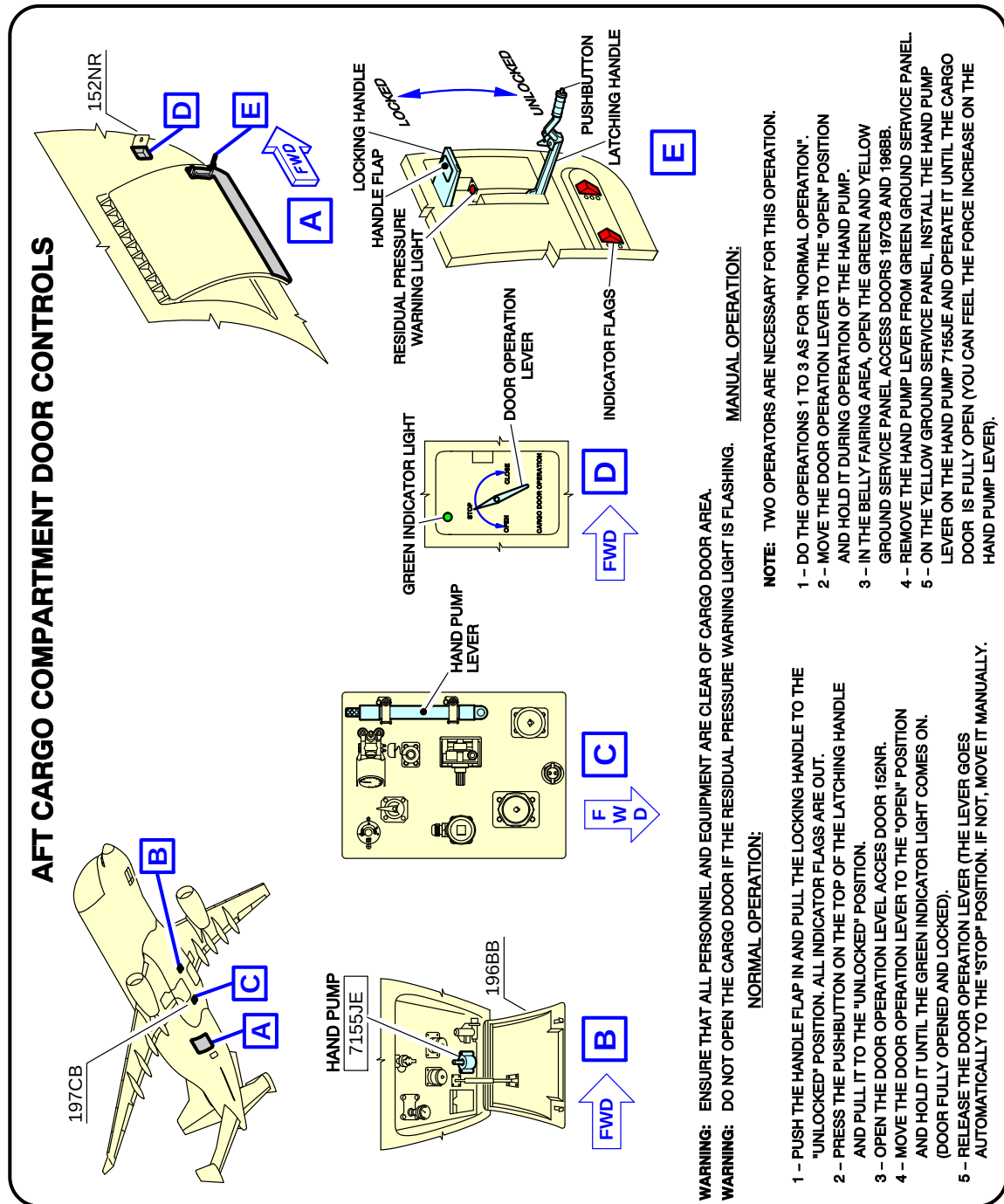
\*\*ON A/C A330-700L



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Main-Deck Access Door  
FIGURE-10-0-0-991-094-A01

\*\*ON A/C A330-700L

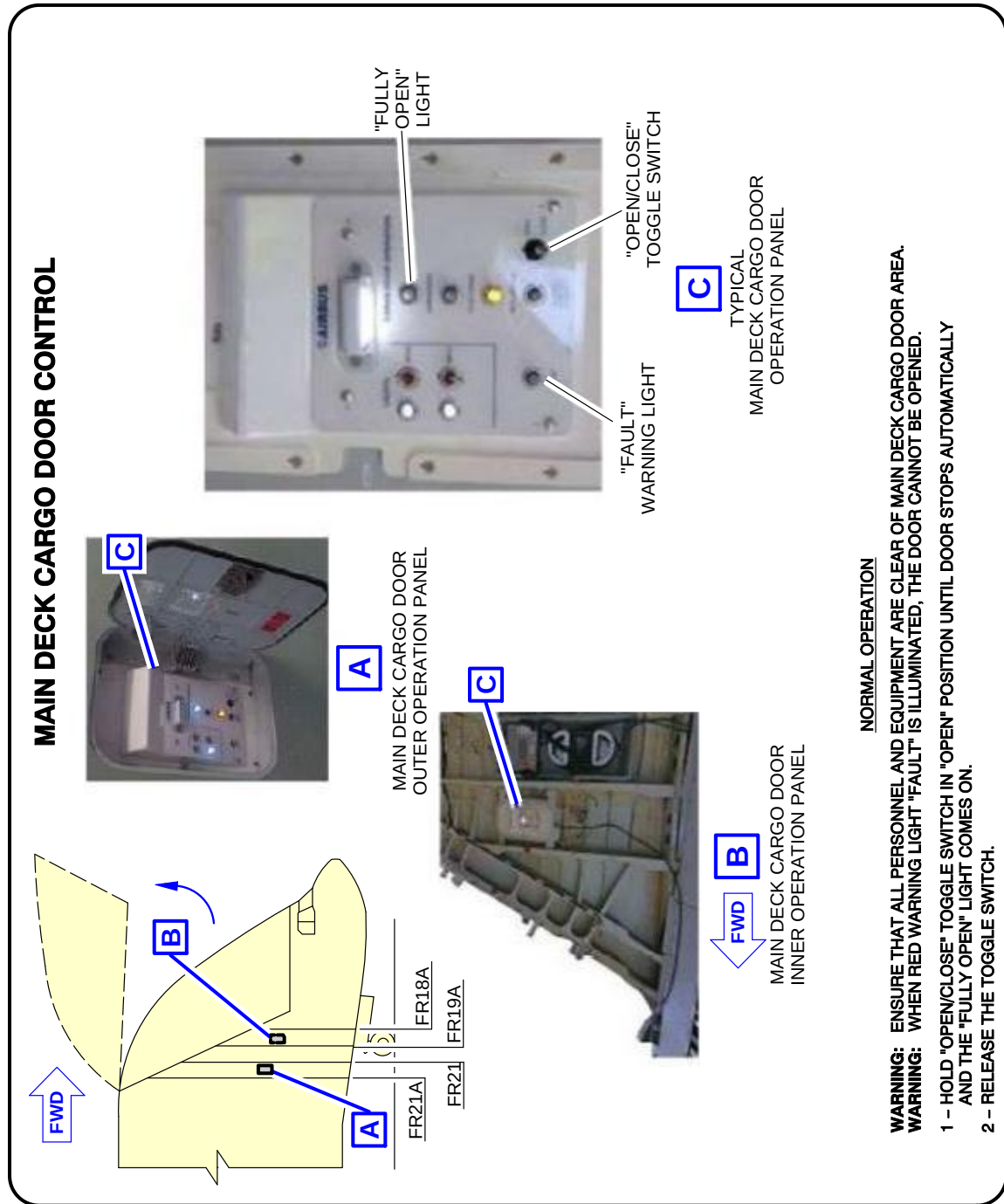


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Aft-Cargo-Compartment Door Controls  
 FIGURE-10-0-0-991-095-A01



**\*\*ON A/C A330-700L**



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Main-Deck Cargo Door-Control  
FIGURE-10-0-0-991-096-A01

The image is a composite showing the location of control panels on the F-35 aircraft and four detailed views of the panels labeled A, B, C, and D.

**Top Left:** A line drawing of the F-35 aircraft. A callout line points to the cockpit area, labeled "121EL". Another callout line points to the rear fuselage area, labeled "198DB".

**Top Right:** A photograph of the cockpit interior, showing the control panels. A blue box with the letter "D" is overlaid on the right side of the image.

**Bottom Left:** A photograph of the cockpit interior, showing the control panels. A blue box with the letter "A" is overlaid on the left side of the image. A blue box with the letter "C" is overlaid on the right side of the image. A blue box with the letter "B" is overlaid on the top right side of the image.

**Bottom Center:** A photograph of the control panel labeled "A". It features two red "FIRE" buttons, each with a "FIRE" label and a "PULL" label. The panel also includes "ENG1" and "ENG2" labels, "FIRE" labels, and "TEST" buttons.

**Bottom Right:** A photograph of the control panel labeled "C". It features a red "FIRE" button with a "FIRE" label and a "PULL" label. The panel also includes "APU" and "AGENT" labels, and "DISCH" and "INCH" buttons.

**Bottom Center (Right):** A photograph of the control panel labeled "B". It features a red "FIRE" button with a "FIRE" label and a "PULL" label. The panel also includes "GAB" and "ATE ON" labels, and "TEST" and "DISCH" buttons.

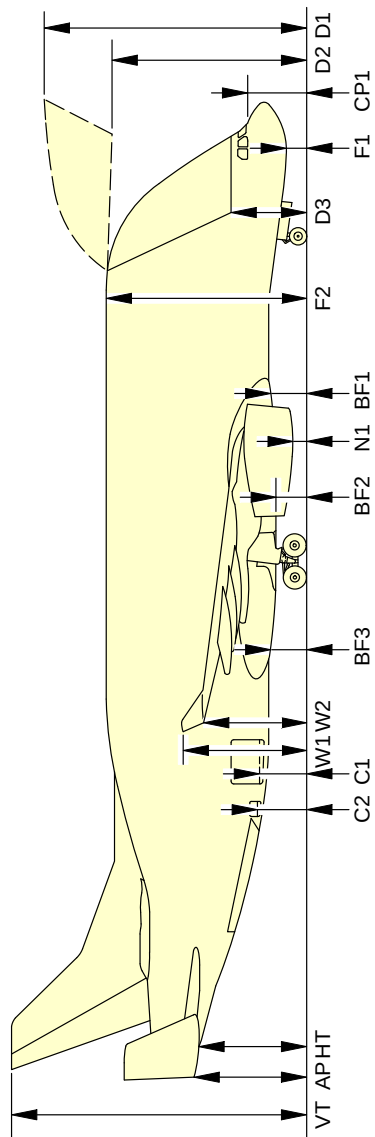
Control Panels  
FIGURE-10-0-0-991-097-A01

## APU AIR INTAKE



APU Air Intake  
FIGURE-10-0-0-991-104-A01

## GROUND CLEARANCES



## DIMENSIONS ARE RELATED TO AIRCRAFT WEIGHT AND CG CONFIGURATION

| A/C CONFIGURATION |    | 135 000 kg (297 625 lb) |       |              |       |
|-------------------|----|-------------------------|-------|--------------|-------|
|                   |    | FWD CG (18%)            |       | AFT CG (25%) |       |
|                   |    | m                       | ft    | m            | ft    |
| DOORS             | D1 | 17.18                   | 56.36 | 17.25        | 56.59 |
|                   | D2 | 12.37                   | 40.58 | 12.43        | 40.78 |
|                   | D3 | 5.09                    | 16.70 | 5.16         | 16.93 |
|                   | C1 | 3.31                    | 10.86 | 3.27         | 10.73 |
|                   | C2 | 3.36                    | 11.02 | 3.30         | 10.83 |
| TAIL PLANE        | VT | 19.21                   | 63.02 | 19.10        | 62.66 |
|                   | HT | 7.16                    | 23.49 | 7.06         | 23.16 |
|                   | AP | 7.12                    | 23.36 | 7.02         | 23.03 |

**NOTE:**  
CARGO DOOR GROUND CLEARANCES ARE MEASURED FROM THE CENTER OF THE DOOR SILL AND FROM FLOOR LEVEL.

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Ground Clearances  
FIGURE-10-0-0-991-099-A01

**\*\*ON A/C A330-700L**

## BREAK-IN POINT-PRINCIPLE



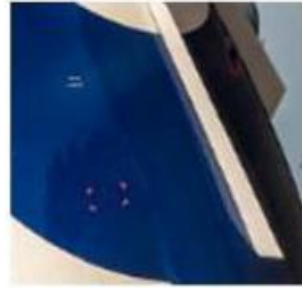
**A**



**B**



**A**



**B**

**FWD**

### NOTE:

- 1 - THE BREAK IN SURFACE SHALL PERMIT ACCESS FROM OUTSIDE IN ORDER TO RESCUE PERSONNEL TO GET IN AND TAKE VICTIMS OUT OF THE DAMAGED AIRCRAFT.
- 2 - RESCUE TEAM WOULD USE AN AXE OR A GRINDER TO CLEAR A FREE - AISLE THROUGH THE FUSELAGE IN BREAK - IN MARKED AREA.

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Structural Break-in Points  
FIGURE-10-0-0-991-100-A01